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Patient-specific cutting guides for total knee arthroplasty (Review)

Zhao C, Jefferies JG, Marimuthu K, Kumar VS, Chen D, Harris IA, MacDessi SJ

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Patient-specific cutting guides for total knee arthroplasty (Review)

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[Intervention Review]

Patient-specific cutting guides for total knee arthroplasty

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ABSTRACT

Background

Accurate alignment of components is considered important for improved outcomes when restoring lower limb alignment in total knee arthroplasty (TKA). Patient-specific guides were developed with the intention of improving surgical efficiency, accuracy of component positioning and overall limb alignment. Currently, the benefits of patient-specific guides in achieving these goals and whether this has an impact on clinical and functional outcomes remains unclear.

Objectives

To assess the benefits and harms of patient-specific cutting guides versus conventional cutting guides or computer-assisted surgical navigation in people undergoing primary TKA.

Search methods

We searched the Cochrane Central Register of Controlled Trials (CENTRAL), MEDLINE (Ovid), Embase (Ovid) and trial registers up to 21st January 2025, unrestricted by language.

Selection criteria

We included randomised controlled trials (RCTs) comparing patient-specific cutting guides (PSGs) to conventional instrumentation (CON) or computer-assisted surgical navigation (NAV) in TKA. Major outcomes were survival of implant (risk of revision), function, radiographic lower limb alignment, pain, global assessment, total adverse events and re-operation rate.

Data collection and analysis

We used standard methods recommended by Cochrane.

Main results

Forty-four studies with 3664 participants were identified. Out of these, 40 studies with 3134 participants compared PSG to CON, two studies with 140 participants compared PSG to NAV and two studies with 390 participants compared PSG to NAV and CON. Regarding the imaging modality, 27 trials used magnetic resonance imaging (MRI)-based PSGs, 18 trials used computed tomographic (CT)-based PSGs and 2 trials used both CT-based and MR-based PSGs. The mean age of participants ranged from 63 years to 74 years. Fifty-three per cent of participants were male, with more than 90% of participants having knee osteoarthritis. Most of the included trials were at risk of bias; 25/44 (56.8%) studies were at risk of selection bias, and 39/44 (88.6%) were at risk of performance and detection biases.

1. PSGs compared to conventional instrumentation

Compared to conventional instrumentation, PSGs may result in little to no difference in survival of the implant. At 26 months, 7/347 participants (20 per 1000) in the conventional instrumentation group reported survival of implant (risk of revision) compared to 4/342 participants (16 per 1000) in the PSG group (RR 0.79 (95% CI 0.25 to 2.52); $I^2 = 0\%$; 689 participants; 9 studies; low-certainty evidence downgraded for bias and imprecision).

Compared to conventional instrumentation, PSGs may result in little to no improvement in function. Mean function (KSS, OKS, WOMAC, KOOS) was 15.1 points with conventional instrumentation and 13.44 points with PSG (SMD -0.11, 95% CI -0.25 to 0.03 back-translated to MD 1.66 points lower (95% CI 3.77 points lower, 0.45 points higher; $I^2 = 56\%$; 23 studies, 1913 participants; low-certainty evidence downgraded for bias and indirectness)) at short-term (up to two years).

PSGs may result in little to no difference in precision in lower limb alignment as measured by the proportion of outliers for femorotibial coronal angle (FTCA). At up to two years, 266/1208 participants (220 per 1000) in the conventional instrumentation group reported radiographic outlier events compared to 234/1204 participants (189 per 1000) in the PSG group (RR 0.86, 95% CI 0.68 to 1.10, $I^2 = 49\%$; 29 studies, 2412 participants; low-certainty evidence downgraded for bias and imprecision).

Low-certainty evidence, downgraded for bias and indirectness, showed that PSGs may result in little to no difference in pain. Mean pain (0 to 100 scale, 0 no pain) was 17.4 points with conventional instrumentation and 15.8 points with PSG (SMD -0.09, 95% CI -0.23 to 0.06, back-translated to MD 1.52 points lower (95% CI 3.88 lower, 1.01 higher; $I^2 = 0\%$, 10 studies, 715 participants)).

None of the included studies provided data on global assessment.

Low-certainty evidence, downgraded for bias and imprecision, showed that PSGs may result in little to no difference in adverse events. Total adverse events (infection, thrombosis) were reported in 60/590 participants (102 per 1000) in the conventional instrumentation group compared to 61/579 participants (99 per 1000) in the PSG group (RR 0.97, 95% CI 0.68 to 1.39, $I^2 = 5\%$, 14 studies, 1169 participants).

Low-certainty evidence, downgraded for bias and imprecision, showed that PSGs may result in little to no difference in re-operation rates. Re-operations were reported in 21/424 participants (50 per 1000) in the conventional instrumentation group compared to 16/419 participants (44 per 1000) in the PSG group (RR 0.87, 95% CI 0.45 to 1.68, $I^2 = 0\%$, 10 studies, 843 participants).

2. PSGs compared to computer-assisted navigation

Low-certainty evidence (downgraded for bias and imprecision) showed that, compared to computer-assisted navigation, PSGs may result in little to no difference in function. Mean function (0 to 100, 100 best function) was 52.5 points with computer-assisted navigation compared to 57.5 points with PSGs at the short term (less than two years): MD 5.00 points higher (95% CI 1.31 lower to 11.31 higher, $I^2 = 0\%$, 2 studies, 120 participants).

Low-certainty evidence (downgraded for bias and imprecision) showed that use of PSGs may result in worse precision in lower limb alignment compared to computer-assisted navigation. At the short term (less than two years), 13/135 participants (96 per 1000) in the computer-assisted navigation group reported radiographic outlier events compared to 28/135 participants (196 per 1000) in the PSG group (RR 2.04, 95% CI 0.87 to 4.82, $I^2 = 44\%$, 3 studies, 270 participants).

None of the included studies provided data on survival of the implant, global assessment, pain, re-operations or adverse events.

Authors' conclusions

Low-certainty evidence showed no meaningful differences with regard to survival of implant, re-operation rate, adverse events, function, pain, and global assessment when comparing patient-specific cutting guides to conventional instrumentation or to computer-assisted navigation in total knee arthroplasty.

PLAIN LANGUAGE SUMMARY

Patient-specific cutting guides (PSG) in total knee replacement

Key messages:

- Patient-specific cutting guides (a technique in which a patient undergoes a scan of the knee which is used to make a customised cutting block to perform a total knee replacement) do not help in producing more accurately aligned components compared to conventional cutting guides.
- No meaningful differences between patient-specific guides against conventional or computer navigation have been found regarding implant survival, re-operation rate, complications, function or pain.
- Computer-assisted navigation (using software during knee replacement) may improve the precision of overall lower-limb alignment and tibiofemoral sagittal alignment (alignment looking side on at the knee) compared to patient-specific guides.

What is Total Knee Replacement (TKR)?

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TKR is a treatment for terminal, painful arthritis of the knee joint.

During TKR, the worn bony ends of the knee joint are cut away and replaced with metal and plastic components designed to smoothly move and function like a normal, healthy knee joint.

During surgery, how surgeons perform the bone cuts and then position or align the new components is vital to the success of a TKR.

What is a patient-specific cutting guide or computer-assisted navigation in TKR?

Traditionally in TKR, metal rods placed within or outside the femur and tibia are used to place conventional cutting guides (CON) to facilitate bone cuts and implant placement.

Computer-assisted knee replacement uses intraoperative mapping of the patient's knee joint and computer-assisted navigation (NAV) for bone cuts.

In patient-specific cutting guides (PSG), a preoperative investigation (CT or MRI scan) is used to create a 3D knee model. Cutting guides are then made specifically for each patient. PSG are postulated to help better align implants, reduce operative time and blood loss, and produce better function after TKR.

What did we want to find out?

We wanted to find out if PSG improves survival of implant, pain, function and precision for overall lower limb alignment compared to conventional guides and computer-assisted navigation in TKR.

What did we do?

We searched for studies comparing PSG with conventional surgical cutting guides or computer-assisted navigation used in TKR surgery. We compared and summarised the results of these studies, rating our confidence in the evidence based on factors such as study methodology and size.

What did we find?

We included 44 randomised controlled trials with 3664 people (64.3% female) aged between 63 and 74 years.

38 out of 42 randomised control trials compared PSG to conventional TKR cutting guides, 2 compared PSG to computer-assisted navigation and 2 trials compared PSG to both conventional cutting guides and computer-assisted navigation. Outcomes were largely reported around two years after the surgery.

Main results

Conventional instrumentation (CON)

Compared to CON, PSG had little benefit:

Survival of implant:

1 out of 100 people underwent revision of the implant following PSG.

2 out of 100 people underwent revision of the implant following CON.

Function (12-60, 12 is best function):

People who had PSG rated their function as 13.44 points.

People who had CON rated their function as 15.1 points.

Precision in lower limb alignment as measured by the proportion of outliers for femorotibial coronal angle: (a radiographic outlier is an x-ray measure of thigh and shinbone alignment after TKR looking at how components and bones are lined up if looking from the front of a patient):

19 out of 100 people reported radiographic outlier events with PSG.

22 out of 100 people reported radiographic outlier events with CON.

Pain (lower scores mean less pain):

People who had PSG rated their pain as 15.8 points.

People who had CON rated their pain as 17.4 points.

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Total adverse events (infection, blood clots):

10 out of 100 people reported adverse events with PSG.

10 out of 100 people reported adverse events with CON.

Re-operations:

4 out of 100 people reported re-operations with PSG.

5 out of 100 people reported re-operations with CON.

Computer-assisted navigation (NAV)

Compared to NAV, PSG had little benefit:

Function (higher scores mean better function):

People who had PSG rated their function as 57.5 points.

People who had NAV rated their function as 52.5 points.

Precision in lower limb alignment (proportion of outliers for femorotibial coronal angle):

19 out of 100 people reported radiographic outlier events with PSG.

9 out of 100 people reported radiographic outlier events with NAV.

What are the limitations of the evidence?

PSGs do not appear to offer a clear advantage in producing more accurately aligned components in TKR, or in improving precision of lower limb alignment except for improving the rotation of the femoral component. PSG may result in no difference in pain, function, adverse events, or re-operation compared to conventional cutting guides.

NAV may improve the precision of lower limb alignment compared to PSG. We are uncertain of clinically meaningful differences between PSG, CON and NAV when analysing implant survival, re-operations performed and complications.

How up to date is this evidence?

The evidence is up-to-date as of 21st January 2025.

SUMMARY OF FINDINGS

Summary of findings 1. Patient-specific cutting guides compared to conventional instrumentation for total knee arthroplasty

Patient-specific cutting guides compared to conventional instrumentation for total knee arthroplasty

Patient or population: Patients undergoing total knee arthroplasty

Setting: Hospitals

Intervention: Patient-specific cutting guides

Comparison: Conventional instrumentation

Outcomes	Anticipated absolute effects* (95% CI)			Relative effect (95% CI)	Nº of participants (studies)	Certainty of the evidence (GRADE)	Comments
	Conventional instrumentation	Patient-specific cutting guides	Difference				
Survival of implant Follow up: 26 months	20 per 1000	16 per 1000 (5 to 51)	4 fewer per 1000 (15 fewer to 31 more)	RR 0.79 (0.25 to 2.52)	689 (9 RCTs)	⊕⊕⊕⊕ LOW ^{1,4}	PSGs may result in little to no difference in the survival of the implant.
Function Scale: various scales (KOOS, WOMAC, OKS, KSS) Follow up: short-term (< 2 years)	Mean function 15.1 points ²	Mean function 13.44 points	MD 1.66 points lower (3.77 points lower to 0.45 points higher)	-	1913 (23 RCTs)	⊕⊕⊕⊕ LOW ^{1,3}	PSGs may result in little to no improvement in function. SMD 0.11 lower (0.25 lower to 0.03 higher) ²
Radiographic lower limb alignment Outliers for femorotibial coronal alignment (FTCA) Follow-up: short-term (< 2 years)	220 per 1000	189 per 1000 (150 to 242)	31 fewer per 1000 (70 fewer to 22 more)	RR 0.86 (0.68 to 1.10)	2412 (29 RCTs)	⊕⊕⊕⊕ LOW ^{1,4}	PSGs may result in little to no difference in precision in lower limb alignment.
Pain Scale: 0 to 100, 0 no pain Follow-up: short-term (< 2 years)	Mean pain 17.4 points ²	Mean pain 15.88 points	MD 1.52 points lower (3.88 points lower to 1.01 points higher)	-	715 (10 RCTs)	⊕⊕⊕⊕ LOW ^{1,3}	PSGs may result in little to no difference in pain. SMD 0.09 lower (0.23 lower to 0.06 higher) ²

Global Assessment	-	-	-	-	-	-	No studies provided data
Total adverse events	102 per 1000	99 per 1000	3 fewer per 1000	RR 0.97 (0.68 to 1.39)	1169 (14 RCTs)	⊕⊕⊕⊕ LOW ^{1,4}	PSGs may result in little to no difference in adverse events.
Follow-up: end of study		(69 to 141)	(33 fewer to 40 more)				
Re-operation rate	50 per 1000	44 per 1000	6 fewer per 1000	RR 0.87 (0.45 to 1.68)	843 (10 RCTs)	⊕⊕⊕⊕ LOW ^{1,4}	PSGs may result in little to no difference in re-operation rates.
Follow-up: end of study		(22 to 83)	(27 fewer to 34 more)				

***The risk in the intervention group** (and its 95% confidence interval) is based on the assumed risk in the comparison group and the **relative effect** of the intervention (and its 95% CI).

CI: Confidence interval; **RR:** Risk ratio; **KOOS:** Knee Injury and Osteoarthritis outcome score; **WOMAC:** Western Ontario and McMaster Universities osteoarthritis index; **OKS:** Oxford Knee Score; **KSS:** Knee Society Score

GRADE Working Group grades of evidence

High certainty: We are very confident that the true effect lies close to that of the estimate of the effect

Moderate certainty: We are moderately confident in the effect estimate: the true effect is likely to be close to the estimate of the effect, but there is a possibility that it is substantially different

Low certainty: Our confidence in the effect estimate is limited: the true effect may be substantially different from the estimate of the effect

Very low certainty: We have very little confidence in the effect estimate: the true effect is likely to be substantially different from the estimate of effect

¹ Downgraded due to lack of participant blinding and high risk of detection and performance biases

² Control group risk was estimated from the conventional instrumentation group at follow-up for pain, knee function in [Boonen 2013](#). SMD was back-translated using control group baseline SD for function (7.5) and pain (16.9) in [Boonen 2013](#).

³ Downgraded due to indirectness; limitations in outcome measurement (diversity in scales and timing of measurement)

⁴ Downgraded due to imprecision; wide CI with very few events

Summary of findings 2. Patient-specific cutting guides compared to computer-assisted navigation

Patient-specific cutting guides compared with computer-assisted surgical navigation for total knee arthroplasty

Patient or population: Patients undergoing total knee arthroplasty

Settings: Hospitals

Intervention: Patient-specific cutting guides

Comparison: Computer-assisted navigation

Outcomes	Anticipated absolute effects* (95% CI)	Relative effect	No of participants	Quality of the evidence	Comments
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	Comput- er-assisted navigation	Patient-spe- cific cutting guides	Difference	(95% CI)	(studies)	(GRADE)	
Survival of implant	-	-	-	-	-	-	No studies provided data.
Function Scale: KSS (0 to 100, 100 best function) Follow-up: short-term (< 2 years)	Mean function 52.5 points ¹	Mean function 57.5	MD 5.00 points higher (1.31 points lower to 11.31 points higher)		120 (2 RCTs)	⊕⊕⊕⊕ LOW ^{2,3}	PSGs may result in little to no difference in function.
Radiographic lower limb align- ment Outliers for femorotibial coronal alignment (FTCA) Follow-up: short-term (< 2 years)	96 per 1000	196 per 1000 (84 to 464)	100 more per 1000 (13 fewer to 368 more)	RR 2.04 (0.87 to 4.82)	270 (3 RCTs)	⊕⊕⊕⊕ LOW ^{2, 4}	PSGs may result in worse precision in lower limb alignment.
Pain	-	-		-	-	-	No studies provided data.
Global assessment	-	-		-	-	-	No studies provided data.
Total adverse events	-	-		-	-	-	No studies provided data.
Re-operation rate	-	-		-	-	-	No studies provided data.

***The risk in the intervention group** (and its 95% confidence interval) is based on the assumed risk in the comparison group and the **relative effect** of the intervention (and its 95% CI).

CI: Confidence interval; **RR:** Risk ratio; **OR:** Odds ratio; **KSS:** Knee Society Score

GRADE Working Group grades of evidence

High quality: Further research is very unlikely to change our confidence in the estimate of effect.

Moderate quality: Further research is likely to have an important impact on our confidence in the estimate of effect and may change the estimate.

Low quality: Further research is very likely to have an important impact on our confidence in the estimate of effect and is likely to change the estimate.

Very low quality: We are very uncertain about the estimate.

¹ Control group risk was estimated from the computer-assisted navigation group at follow-up for function in [Yan 2014](#).

² Downgraded due to lack of participant blinding and high risk of detection and performance biases

³ Downgraded due to imprecision; low sample size and wide CI

⁴ Downgraded due to imprecision; wide CI with few events

BACKGROUND

Description of the condition

Osteoarthritis (OA), also known as degenerative arthritis, involves mechanical and age-related degradation of the articular surfaces of diarthrodial joints. OA is the most common disease of weight-bearing joints in adults and the knee joint is the most commonly affected major joint (Heijink 2012). A variety of causes, including hereditary, metabolic, developmental and mechanical conditions, can initiate the process of articular cartilage degeneration. Advancing age, obesity, female sex and a low level of education are other significant risk factors for the development of osteoarthritis (Andrianakos 2006).

Individuals with OA usually present with pain and loss of motion of the affected joint. The pain is usually a dull ache that occurs during weightbearing activities such as walking, and gradually progresses with time. As the articular cartilage degenerates, the underlying subchondral bone and associated nerve endings are exposed, resulting in pain. The cartilage debris also initiates inflammation within the joint. In advanced stages, the underlying bone becomes eburnated and deformed; bony outgrowths called osteophytes develop in response to chemotactic factors in an attempt to increase the surface area of the joint. Advanced OA can result in limb deformity, joint contractures and muscle atrophy (O'Reilly 1998; Zhang 2010).

Rheumatoid arthritis is an autoimmune, inflammatory disorder that can affect multiple tissues and organs but predominantly affects synovial joints. The body's immune system, which protects human health by attacking foreign particles such as bacteria and viruses, mistakenly attacks the body's own tissues (Firestein 2003). Usually the disease affects the small joints of hands and feet. However, knee joint involvement is common, and if left untreated can lead to severe deformity and disability. Other uncommon causes of inflammatory arthritis of the knee include ankylosing spondylitis, Reiter's disease and seronegative spondyloarthropathy.

Post-traumatic arthritis can occur following trauma to the articular cartilage, menisci or ligaments forming the knee joint. Trauma that results in an irregular joint surface can lead to abnormal loading of the joint, cartilage degeneration and finally osteoarthritis.

Description of the intervention

Total knee arthroplasty (TKA), also known as total knee replacement, is a surgical procedure whereby the surfaces of the knee joint are replaced with metal and plastic implants. It is the procedure of choice when more than one compartment of the knee is irreversibly damaged due to arthritis. The number of primary TKAs performed in the United States doubled in the last decade, with further growth of 85% projected by 2030 (Sloan 2018; Weinstein 2013).

TKA is successful in relieving pain and improving function in individuals with advanced arthritis (Carr 2012). However, complications such as infection, instability, loosening or fracture can occur in around 5% of patients in the initial month following the procedure (Belmont 2014). The lifetime risk of revision arthroplasty has been found to be around 2% in recent registry data (Ackerman 2022). It has been shown that suboptimal alignment of prosthetic

components may increase the risk of failure and subsequent need for revision surgery (Fang 2009).

In TKA surgery, calibrated bone resections are made to the distal end of the femur and proximal end of the tibia to allow placement of the femoral and tibial prostheses. Accurately aligning these cuts determines alignment of the prosthesis in the sagittal, coronal and axial planes. A number of techniques are available to assist in aligning the components in TKA, including conventional cutting guides (standard cutting guides), computer-assisted surgical navigation and patient-specific cutting guides (MacDessi 2014).

In conventional techniques (CON), bony resections and subsequent alignment of the prosthesis are determined by using visual anatomic landmarks and the anatomic axes of femur and tibia to place intra and extra-medullary cutting guides. Advantages of CON include its established satisfaction rates based upon long-term clinical data and widespread use around the world. Disadvantages include that CON may be prone to error, as it is not based on an accurate assessment of the overall alignment of the mechanical axes of the lower limb (Iorio 2013; Laskin 2003). In addition, conventional guides are difficult to use on knees with extra-articular deformities (deformity above or below the knee joint) (Catani 2012; Lee 2014; Tigani 2012) and actually may not be possible if patient geometry differs greatly from the boundaries of the manufactured instrumentation design with limits of resection. Finally, canal instrumentation may increase the risk of blood loss (Howell 2008).

Computer-assisted surgical navigation (NAV) involves the creation of a digital image using intraoperative optical trackers and a camera that guides the bone resections for component positioning during surgery (Desai 2011). The advantages of computer-assisted techniques include dynamic assessment of deformity during surgery, calculation of soft tissue tension and accurate component positioning (Chauhan 2004), and possible reduced alignment outliers compared to CON (MacDessi 2014). However, disadvantages such as prolonged operative time, high cost, extra instrumentation and associated learning curve have prevented this strategy from gaining widespread acceptance (Bauwens 2007).

Patient-specific cutting guides (PSGs) or custom-made cutting guides are a technology used in TKA whereby preoperative imaging (magnetic resonance imaging (MRI), computerised tomography (CT) scans and plain radiographs) is used to create anatomic three-dimensional models of the patient's knee (Ast 2012). PSGs are manufactured based on these models and are fitted to the bone surfaces during surgery for the execution of bone resections based on preoperatively planned resection angles of the femur and tibia. Advantages of PSG include obviating the need to place pins in the shaft of the femur and tibia (as used in NAV) and the need to instrument the intramedullary canal of bone (as used in CON). They may reduce operating times, kit and resource requirements for theatres and hospitals. The patient-specific nature of the cutting blocks also allows for other personalised alignment strategies to be considered beyond those of CON and its traditional mechanical axis alignment. Disadvantages of PSG include manufacturing costs and resources for pre-procedure radiological investigations (Nunley 2012a).

A contemporary technique that has emerged after CON, PSG and NAV is robotically-assisted TKA. This involves a preoperative

component and an intraoperative, robotic component. A patient undergoes preoperative imaging (CT-based) of the lower limbs to create a 3D model of the knee. This 3D model forms the basis of planning for bone cuts, alignment correction and implant positioning. Intraoperatively, following registration of the patient's bony landmarks to the 3D virtual model, a surgeon-controlled robotic arm is used to assist the bony cuts as planned (Kayani 2019, Saber 2023). Several types of robotic-assisted devices are available, using saws or burrs and crucially offering varying levels of surgeon control. They are a source of recent research interest. Advantages include increased precision, intraoperative planning adjustments and the potential for safety with saw cutting 'haptic boundaries' (Hampp 2019). Disadvantages include increased operating time, cost, potential technical issues with conversion to CON and a learning curve (Kayani 2019; Khan 2016).

How the intervention might work

The proposed main benefit of using either PSGs or computer-assisted navigation in TKA is improved precision in restoring lower limb alignment compared to conventional techniques (Bäthis 2004; Ng 2012). Achieving mechanical alignment of components has been correlated in some studies with improved patient-reported outcome and survivorship in total knee arthroplasty (Jeffery 1991; Longstaff 2009). While PSGs may assist in improving component positioning, they are not a substitute for other predictors of successful TKA, such as careful preoperative planning, appropriate soft tissue balancing and precise implantation techniques (Ng 2012).

Patient-specific cutting guides have been shown to reduce the duration of surgery and incision length (Chareancholvanich 2013; Noble 2012; Nunley 2012a). Both PSGs and computer-assisted navigation may reduce surgical blood loss by avoiding penetration of the intramedullary canal (Cucchi 2019; Kalairajah 2005). The potential harm of producing an embolic load of fat and marrow elements generated while penetrating the intramedullary canal may also be avoided with PSG (Parvizi 2001; Restrepo 2007). However, the clinical relevance of these proposed advantages of PSI compared to conventional cutting guides requires further study.

Potential disadvantages of patient-specific cutting guides include the high cost of manufacturing, the need for preoperative MRI or CT and the influence of a learning curve (Nunley 2012a; Victor 2014). Computer-assisted navigation may increase surgical time and may have complications associated with pin placement such as pin tract infection and fracture (Bäthis 2004). Furthermore, some studies have concluded that PSG and NAV may not offer any additional benefit compared to conventional cutting guides in restoring neutral lower limb alignment (Chareancholvanich 2013; Kim 2007; Marimuthu 2014; Victor 2014). Several authors have reported a need to switch to conventional technique intraoperatively when PSG-based resections deviated excessively from the resections based on conventional or navigation guides (Abane 2015; Ollivier 2016; Roh 2013; Woolson 2014).

Why it is important to do this review

Despite inconsistent evidence validating its effectiveness, there is continued use of PSGs for TKA (Noble 2012; Victor 2014). PSGs have been shown to improve the accuracy of alignment of components by reducing outliers in some studies (Ng 2012). However, studies comparing PSGs to conventional instrumented techniques differ

widely in their findings (MacDessi 2014; Marimuthu 2014; Noble 2012; Victor 2014).

In a retrospective comparative analysis, Ng 2012 found that the neutral mechanical axis of the limb was restored more often with PSGs compared to CON with fewer outliers in the PSG group. Daniilidis 2013 also found in a retrospective study that PSGs produced fewer outliers in the Hip-Knee-Ankle (HKA) angle than in the conventional group. Heyse 2012a conducted a retrospective analysis showing that PSGs resulted in fewer outliers in femoral component rotation than the conventional group.

Multiple RCTS have compared the rate of outliers in both overall coronal alignment and individual femoral and tibial component alignment between PSGs and CON with conflicting results (Boonen 2013; Chanalithichai 2019; Chareancholvanich 2013; De Vloo 2017; Parratte 2013; Roh 2013; Van Leeuwen 2018; Victor 2014). Chareancholvanich 2013 reported a reduction in the number of femoral coronal alignment outliers using PSGs, whilst Victor 2014 reported a reduction in tibial coronal outliers with CON TKA. De Vloo 2017 also reported femoral component positioning was improved in the coronal plane for PSI, but significantly more tibial slope was introduced. Boonen 2013 reported a significantly lower number of sagittal femoral outliers when using a conventional method. Whereas Roh 2013 and Parratte 2013 did not find any significant difference between the two groups in terms of sagittal outliers. Chanalithichai 2019 reported no difference in femoral component rotation between groups and Van Leeuwen 2018 did not show a difference in outlier frequency for any alignment parameters between groups.

RCTs comparing clinical and functional outcomes between PSG and CON have also shown widely differing results. Sariali 2019 showed improved radiological and clinical outcomes in the PSG group compared to CON. However, Zomar 2021 showed no difference in clinical or functional outcomes between the groups. Whether there is a correlation between radiological outcomes and clinical outcomes also remains to be seen.

While there are numerous studies comparing computer-assisted surgical navigation with conventional techniques (Allen 2014; Kim 2012), there are only a small number of studies in the literature comparing PSG with NAV in total knee arthroplasty (Lionberger 2014; Ollivier 2016; Zahn 2020).

This systematic review was undertaken to determine whether the use of PSGs in total knee arthroplasty offers any significant advantage over CON or NAV.

OBJECTIVES

To assess the benefits and risks of patient-specific cutting guides compared to conventional instruments or computer-assisted surgery in total knee arthroplasty.

This review has been conducted in accordance with the guidelines recommended by the Cochrane Musculoskeletal Group Editorial Board (Ghogomu 2014). The protocol for this review has been published.

METHODS

Criteria for considering studies for this review

Types of studies

We included randomised controlled trials (RCTs). We included studies reported as full text, those published as abstracts only, and unpublished data. There was no language restriction.

Types of participants

We included adults undergoing primary TKA for advanced arthritis due to any cause, including osteoarthritis, post-traumatic arthritis, or inflammatory arthritis. We excluded participants receiving TKA for tumours, unicompartmental replacement or revision knee replacement. We excluded participants undergoing TKA following either a previous osteotomy or previous unicompartmental replacement arthroplasty on the same knee.

Types of interventions

We included RCTs comparing total knee arthroplasty performed using patient-specific guides versus conventional instrumentation or computer-assisted surgery.

The technique, patient-specific cutting guides (PSGs), is a technology used in TKA whereby preoperative imaging (magnetic resonance imaging (MRI), computerised tomography (CT) scans and plain radiographs) is used to create anatomic three-dimensional models of the patient's knee (Ast 2012). PSGs are then manufactured based on these personalised models. These models are fitted to the bone surfaces during surgery to permit the facilitation of bone resections based on preoperatively planned resection angles of the femur and tibia. Using PSG to perform TKA differs from both NAV and CON. There is no requirement to place reference pins in the shaft of the femur and tibia as seen in NAV, whilst it negates the need to instrument the intramedullary canal of femur or tibia as seen in CON.

Different implant designs (posterior stabilised, cruciate retaining and cruciate sacrificing) and different bearing mechanisms (fixed bearing and mobile bearing) used for total knee arthroplasty were documented. We also included all modes of prosthetic fixation, including cemented, non-cemented and hybrid fixation. Studies were included irrespective of whether patellar resurfacing was performed.

Types of outcome measures

Major outcomes

1. Survival rate of the implant (any revision surgery to change a prosthetic component, i.e. the proportion of knees that undergo revision surgery)
2. Functional outcome measures with validated instruments (such as participant-reported measures: Western Ontario & McMaster Universities Osteoarthritis Index (WOMAC), Knee Injury and Osteoarthritis Outcome Score (KOOS), International Knee Documentation Committee (IKDC) score, Oxford Knee score (OKS) and Knee Society Score (KSS))
3. Radiographic lower limb alignment: recorded as the number of outliers for femorotibial coronal angle (FTCA) also called the hip-knee-ankle (HKA) angle measured following surgery. In trials where the investigators performed these measurements at more

than one time point, the earliest postoperative measurement was used for analysis. "Outliers" were defined as alignments outside $\pm 3^\circ$ range for FTCA measured as a dichotomous variable.

4. Pain: mean overall pain measured on a visual analogue scale (VAS) or numerical rating scale (NRS), or other measures of pain
5. Participant-reported global assessment (e.g. better, not better, worse, dichotomised into better and not better or worse)
6. Total adverse events (including infection, thrombosis, palsy, death, etc.)
7. Re-operation rate (not involving implant change)

Minor outcomes

1. Quality of life
2. Operative time
3. Blood loss (including transfusion rate)
4. Range of motion
5. Length of hospital stay
6. Incision length
7. Radiographic outliers in alignment for individual prosthetic components including; femoral component coronal alignment, tibial component coronal alignment, femoral component sagittal alignment, tibial component sagittal alignment and femoral component rotational alignment. "Outliers" were defined as alignments outside $\pm 3^\circ$ range for all individual component alignments (Victor 2014); we will measure these as dichotomous variables.

Timing of outcome assessment

1. Short term: Less than two years from index procedure (primary time point)
2. Intermediate term: two to 10 years from index procedure
3. Long term: more than 10 years from index procedure

A short term of less than two years is chosen as it is in chronological keeping with the period when most early revisions are performed for arthroplasty (a major outcome measure); backed up by national joint registry data (<https://www.njrcentre.org.uk/>) and evidence-based recommendations in the literature (Ahmad 2021). Furthermore, clinical complications and radiological outcome measurements are typically performed or recorded in the weeks following surgery up to this time point over follow-up. The majority of studies evaluating radiological post-surgical outcomes as a primary outcome measurement report data out to this time period.

Intermediate or 'medium'-term outcomes vary in the orthopaedic literature from two to over five, seven or 10 years (Ahmad 2021). This time period generally may allow for early estimates of survivorship and patient-reported outcome measures in particular. The time period of two to five years was therefore chosen in order to make allowance for these across the range of included studies.

Long-term outcomes of more than five years form a benchmark of time frames in national joint registry data and arthroplasty literature. At this point, this is established to give a more precise estimate of survivorship (<https://www.njrcentre.org.uk/https://aoanjrr.sahmri.com/annual-reports-2024>).

Search methods for identification of studies

Electronic searches

We searched the following databases up to 21st January 2025:

- Cochrane Central Register of Controlled Trials (CENTRAL) in the Cochrane Library (up to 21st January 2025, Issue 1);
- MEDLINE Ovid (1946 to 21st January 2025);
- Embase Ovid (1974 to 21st January 2025).

We conducted a search of ClinicalTrials.gov (www.ClinicalTrials.gov) and the World Health Organization International Clinical Trials Registry Platform (ICTRP) Search Portal (apps.who.int/trialsearch/) for ongoing or unpublished trials.

We searched all databases from their inception to the present and imposed no restrictions on language of publication.

See [Appendix 1](#) for search strategies. The search strategy has been adapted from a Cochrane review by Pacheco-Brousseau and colleagues ([Pacheco-Brousseau 2022](#)). We searched all databases from 2010 to the present because of major practice changes in TKA surgical techniques ([Blasier 2009](#)).

Searching other resources

We checked reference lists of all primary studies and review articles for additional references. We searched for errata or retractions from included studies published in full text on PubMed (<https://pubmed.ncbi.nlm.nih.gov/>).

Data collection and analysis

Selection of studies

Four review authors (CZ, JJ, VSK and KM) independently screened titles and abstracts for inclusion of all the potentially relevant studies identified as a result of the search and coded them as 'retrieve' (eligible or potentially eligible/unclear) or 'do not retrieve'. We retrieved the full-text study reports/publications, and four review authors (CZ, JJ, VSK and KM) independently screened the full text and identified studies for inclusion. The same authors also identified and recorded reasons for exclusion of the ineligible studies. Any disagreement was resolved through discussion or, if required, through discussion with a fourth person (SM). We identified and excluded duplicates and collated multiple reports of the same study under a single reference ID so that each study, rather than each report, was the unit of interest in the review. We recorded the selection process in sufficient detail to complete a Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) flow diagram ([PRISMA 2020](#)) and a [Characteristics of excluded studies](#) table.

Data extraction and management

We used a data collection form for study characteristics and outcome data, which was piloted on at least 10 studies in the review. Four review authors (CZ, JJ, VSK and KM) extracted study characteristics from included studies. A separate review author (DC) spot-checked the study characteristics for accuracy against the trial report. We extracted the following study characteristics:

1. Methods: study design, total duration of study, details of any 'run-in' period, study setting, withdrawals, and date of study.

2. Participants: N, mean age, age range, sex, BMI, disease duration, severity of condition; inclusion criteria, and exclusion criteria.

3. Interventions: intervention (patient-specific guides) and comparators (conventional instrumentation or computer-assisted surgical navigation); including type of prosthesis used (cruciate-retaining or cruciate-sacrificing); type of bearing surface (fixed bearing or mobile bearing); type of imaging modality used to manufacture PSG (CT-based or MRI based); patella resurfacing.

4. Outcomes: major and minor outcomes specified and collected, and time points reported.

5. Characteristics of the design of the trial as outlined below in the 'Assessment of risk of bias in included studies' section.

6. Notes: funding for trial, and notable declarations of interest of trial authors.

7. Information needed to assess GRADE (e.g. baseline risk in the control group for key outcomes).

Four review authors (CZ, JJ, VSK and KM) independently extracted outcome data from the included studies. We extracted the number of events and number of participants per treatment group for dichotomous outcomes and means and standard deviations and number of participants per treatment group for continuous outcomes. We noted in the '[Characteristics of included studies](#)' table if outcome data were not reported in a usable way and when data were transformed or estimated from a graph. We resolved disagreements by consensus or involved a fourth person (SM). Three review authors (CZ, JJ and KM) transferred data into the RevMan Web ([RevMan Web 2020](#)) file. We double-checked that data were entered correctly by comparing the data presented in the systematic review with the study reports.

In studies reporting more than one measure for physical function (patient-reported), we chose a continuous scale over an ordinal scale and used one of the following scores, selecting the measure highest on the list if two or more were reported in a single study. However, we also reported all scores used.

1. Knee injury and Osteoarthritis Outcome Score (KOOS score);
2. Oxford knee score (OKS);
3. WOMAC disability subscore;
4. Composite disability scores other than WOMAC;
5. Disability other than walking;
6. WOMAC global scale;
7. Lequesne Osteoarthritis Index global score;
8. Other algofunctional scale.

In trials where more than one measure of pain was reported, we used the following hierarchy, using the highest measure from the following list (e.g. we chose overall pain over pain on walking).

1. Overall pain (global pain), on a visual analogue scale (VAS) score, or numerical rating scale (NRS);
2. Pain on walking;
3. WOMAC Osteoarthritis Index pain subscore;
4. Pain on activities other than walking;
5. Rest pain;
6. Other algofunctional scale.

For quality of life, we accepted the following scores, taking the highest on the list if two or more were reported in a single study.

1. EuroQoL-5 Dimensions (EQ-5D);
2. 36-Item Short Form Survey (SF-36);
3. 12-Item Short Form Survey (SF-12);
4. Other scale system.

If a trial reported both final values and change from baseline values of the same outcome, we extracted the change from baseline values, provided the standard deviation (SD) for the change score was also reported. If both unadjusted and adjusted values for the same outcome were reported, we extracted the adjusted values.

If data were analysed based on an intention-to-treat (ITT) principle and per-protocol (as-treated), we extracted ITT data for both benefits and harms.

If there were multiple time points, we extracted the data for three time frames:

1. Short term: Less than 2 years from index procedure;
2. Intermediate term: 2 to 10 years from index procedure;
3. Long term: more than 10 years from index procedure.

If there were multiple time points within the above-mentioned timeframes, we included the data from the latest time point within the period. For example, if a particular study reported outcomes at 12, 24 and 36 months of follow-up, we considered the data from 24 months as short term and those at 36 months as intermediate term, and we did not extract the 12-month data. We have noted in the [Characteristics of included studies](#) tables, which time frames we have excluded from the review.

Assessment of risk of bias in included studies

Four review authors (CZ, JJ, VSK and KM) independently assessed the risk of bias for each study. We resolved any disagreements by discussion or by involving another author (IAH). We contacted the corresponding study authors for clarification if information was unclear. We assessed the risk of bias using the original Cochrane risk of bias tool ([Higgins 2024a](#)) according to the following domains:

1. Random sequence generation;
2. Allocation concealment;
3. Blinding of participants and personnel;
4. Blinding of outcome assessment;
5. Incomplete outcome data;
6. Selective outcome reporting;
7. Other bias.

We graded each potential source of bias as high, low or unclear risk, and provided a quote from the study report together with a justification for our judgement in the risk of bias table. We summarised the risk of bias judgements across different studies for each of the domains listed. We considered blinding separately for different key outcomes where necessary (e.g. for unblinded outcome assessment, risk of bias may differ for patient-reported

outcomes and objective outcomes such as re-operation rate). As well, we considered the impact of missing data on key outcomes.

Information on risk of bias related to unpublished data or correspondence with a trialist was noted in the risk of bias table. When considering treatment effects, we considered the risk of bias for the studies that contributed to that outcome. We presented the figures generated by the risk of bias tool to provide summary assessments of the risk of bias.

Overall risk of bias

We made explicit judgements about the level of risk of bias:

- Overall low risk of bias (low risk of bias for all key domains within the study);
- Overall high risk of bias (high risk of bias for one or more key domains within the study);
- Overall unclear risk of bias (unclear risk of bias for one or more key domains within the study if no other bias domains were judged high risk).

Assessment of bias in conducting the systematic review

We conducted the review according to the published protocol and reported any deviations from it in the [Differences between protocol and review](#) section of the systematic review.

Measures of treatment effect

We used the Cochrane Collaboration statistical software, Review Manager 5.4 ([RevMan Web 2020](#)), to perform data analysis. We analysed dichotomous data (number of outliers in radiological alignment of component, adverse events, survival of implant) as risk ratios and used 95% confidence intervals (CIs). Continuous data (patient-reported outcome measures, pain, quality of life, operative time, blood loss) were analysed as mean difference (MD) or standardised mean difference (SMD), depending on whether the same scale was used to measure an outcome, and 95% CIs. We entered data presented as a scale with a consistent direction of effect across studies.

When different scales were used to measure the same conceptual outcome (e.g. pain score), SMDs were calculated instead, with corresponding 95% CIs. SMDs were back-translated to a typical scale (e.g. 0 to 10 for pain) by multiplying the SMD by a typical among-person standard deviation (e.g. the standard deviation of the control group at baseline from the most representative trial) ([Higgins 2024a](#)).

We used 1.5 points on a 0 to 10-point VAS pain scale and 10 points on a 0 to 100-point function scale as the minimal clinically important difference (MCID).

Unit of analysis issues

Where multiple trial arms were reported in a single trial, we included only the relevant arms. The participant was the unit of analysis wherever possible. In studies reporting results for bilateral surgery without adjustment for the non-independence between limbs, there may be potential errors, and we reported this as a potential source of other bias. We attempted to re-analyse such studies by calculating sample sizes where possible, as outlined in the *Cochrane Handbook for Systematic Reviews of Interventions* ([Higgins 2024](#)).

Dealing with missing data

We contacted investigators in order to verify key study characteristics and obtain missing numerical outcome data where possible (e.g. when a study was identified as abstract only or when data were not available for all participants). Where this was not possible, and the missing data were thought to introduce serious bias, we explored the impact of including such studies in the overall assessment of results by a sensitivity analysis. Any assumptions and imputations to handle missing data were described and the effect of imputation explored by sensitivity analyses.

For dichotomous outcomes (e.g. number of alignment outliers), the withdrawal rate was calculated using the number of patients randomised in the group as the denominator.

For continuous outcomes (e.g. mean change in pain score), we calculated the MD or SMD based on the number of patients analysed at that time point. If the number of patients analysed was not presented for each time point, the number of randomised patients in each group at baseline was used.

Standard deviations were imputed from a representative study when there were missing standard deviation data (data from studies with most weight or least bias were used) according to the methods recommended in Chapter 6 of the *Cochrane Handbook for Systematic Reviews of Interventions* (Deeks 2024).

Assessment of heterogeneity

Clinical and methodological diversity was assessed in terms of participants, interventions, outcomes and study characteristics (e.g. study design, outcome measurement tools, etc.) for the included studies to determine whether a meta-analysis was appropriate. This was conducted by observing these data from the 'Characteristics of included studies' table. Statistical heterogeneity was assessed by visual inspection of the forest plot to assess the direction and magnitude of effects and the degree of overlap between CIs.

We used the I^2 statistic to quantify inconsistency among the trials in each analysis. We also considered the P value from the χ^2 test. If we identified substantial heterogeneity, we reported it and explored possible causes by prespecified subgroup analysis. When there were few studies, we used caution in applying the thresholds below to interpret statistical heterogeneity.

We used this approximate guide for the interpretation of an I^2 value (Deeks 2024):

- 0% to 40% might 'not be important';
- 30% to 60% may represent 'moderate' heterogeneity;
- 50% to 90% may represent 'substantial' heterogeneity;
- 75% to 100% represents 'considerable' heterogeneity.

We kept in mind that the observed value of I^2 depends on: (i) magnitude and direction of effects and (ii) strength of evidence for heterogeneity (e.g. P value from the χ^2 test, or a CI for the I^2 statistic: uncertainty in the value of the I^2 statistic is substantial when the number of studies is small).

The χ^2 test was interpreted where a P value ≤ 0.10 indicated evidence of statistical heterogeneity.

If we identified substantial heterogeneity, we reported it and investigated possible causes by following the recommendations in section 10.10 of the *Cochrane Handbook* (Deeks 2024).

Assessment of reporting biases

To assess outcome reporting bias, we checked trial protocols against published reports. For studies published after 1 July 2005, we screened ClinicalTrials.gov and WHO ICTRP trial registers for the trial protocol and evaluated whether selective reporting of outcomes was present.

Data synthesis

Our primary comparison was patient-specific guides (PSGs) versus conventional instrumentation (CON) for TKA.

Our other main comparison was patient-specific guides (PSGs) versus computer-assisted surgical navigation (NAV) for TKA.

We undertook meta-analyses only where this was meaningful, i.e. if the treatments, participants and the underlying clinical question were similar enough for pooling to make sense. Results of the studies were analysed using RevMan Web 2020. The primary analysis included all trials regardless of their risk of bias.

The fixed-effect model was compared against the random-effects model to assess the possible presence of small sample bias in the included studies (i.e. in which the intervention effect is more beneficial in smaller studies). If we considered the studies to be sufficiently clinically homogenous, we pooled data in a meta-analysis using a random-effects model, irrespective of the I^2 values, and performed a sensitivity analysis with the fixed-effects model.

Subgroup analysis and investigation of heterogeneity

We carried out the following subgroup analyses:

1. PSG (CT-based) versus PSG (MRI-based): Radiological alignment of components

We performed an additional subgroup analysis for the difference in alignment outliers between MRI-based PSGs and CT-based PSGs. It was postulated that the type of imaging modality used to manufacture the PSG could impact on accuracy of alignment achieved.

At the time of conceptualisation of this review, we planned for a subgroup analysis between Cruciate Retaining (CR) versus Posterior Stabilised (PS) knees as well as between patellar resurfacing versus non-resurfacing. These subgroup analyses were considered to be meaningful in studies reporting patient outcome measures, as implant design and patellar resurfacing may be potential confounders affecting functional outcome. The patient's BMI and severity of preoperative deformity was also postulated to be a possible confounder for producing alignment outliers. We did not perform any of these subgroup analyses due to insufficient data.

We used the formal test for subgroup interactions in Review Manager (RevMan Web 2020) and used caution in the interpretation of subgroup analyses as advised in section 10 of the *Cochrane*

Handbook (Deeks 2024). The magnitude of the effects were compared between the subgroups by means of assessing the overlap of the CIs of the summary estimate. Non-overlap of the CIs indicates statistical significance.

Sensitivity analysis

We performed sensitivity analyses to assess the impact of any selection bias attributable to inclusion of trials with inadequate or unclear treatment allocation concealment. We also undertook a second sensitivity analysis to assess the robustness of the outcomes to detection bias (by restricting it to studies with adequate blinding of outcome assessors).

Summary of findings and assessment of the certainty of the evidence

We followed the guidelines in the *Cochrane Handbook for Systematic Reviews of Interventions*, Chapters 14 and 15 (Schünemann 2024; Schünemann 2024a), for interpreting results, and are aware of distinguishing a lack of evidence of effect from a lack of effect. We based our conclusions only on findings from the quantitative or narrative synthesis of included studies for this review. We avoided making recommendations for practice, and our implications for research suggest priorities for future research and outline what the remaining uncertainties are in the area.

We created a summary of findings (SoF) table using the following (major) outcomes at short-term (up to two years).

1. Survival rate of the implant (any revision surgery to change a prosthetic component, i.e. the proportion of knees that underwent revision surgery);
2. Functional outcome measures with validated instruments (overall high risk of bias (high risk of bias for one or more key domains within the study));
3. Overall unclear risk of bias (unclear risk of bias for one or more key domains within the study) for participant-reported measures;
4. Radiographic lower limb alignment: recorded as the number of outliers for femorotibial coronal angle (FTCA) also called the hip-knee-ankle (HKA) angle;
5. Pain: mean overall pain measured on a visual analogue scale (VAS) or numerical rating scale (NRS);

6. Global assessment (participant-reported);
7. Total adverse events (including infection, thrombosis, palsy, death, etc.);
8. Re-operation rate (not involving implant change).

The first SoF table (Summary of findings 1) compared PSGs with CON and the second SoF (Summary of findings 2) compared PSGs with NAV.

Three reviewers (CZ, JJ and KM) independently assessed the certainty of the evidence with disagreements resolved by discussion or involving a third review author (SM). We used the five GRADE considerations (study limitations (overall risk of bias), consistency of effect, imprecision, indirectness and publication bias) to assess the certainty of a body of evidence as it related to the studies which contributed data to the analyses for the prespecified outcomes, and reported the certainty of evidence as high, moderate, low, or very low. We justified, documented and incorporated judgements into reporting of results for each outcome.

We used GRADEpro software to prepare the SoF tables (GRADEpro 2015). We justified all decisions to downgrade the certainty of evidence for each outcome using footnotes, and we made comments to aid the reader's understanding of the review where necessary.

RESULTS

Description of studies

Results of the search

Overall, there are 44 included studies, 83 excluded studies, two studies awaiting classification and no ongoing studies.

The results of the search are shown in Figure 1. The search identified 17,924 records (MEDLINE 5551, Embase 8102 and Central 4271). After removal of duplicates, 8954 records were screened. We retrieved 136 studies for full-text screening with 44 trials (53 records) included, and 83 records excluded (61 due to ineligible study design, five due to ineligible intervention and 15 had an ineligible comparator). Two studies are awaiting classification as the full texts are available in Chinese (Wang 2021) and Czech (Musli 2016).

Figure 1. Prisma flow diagram

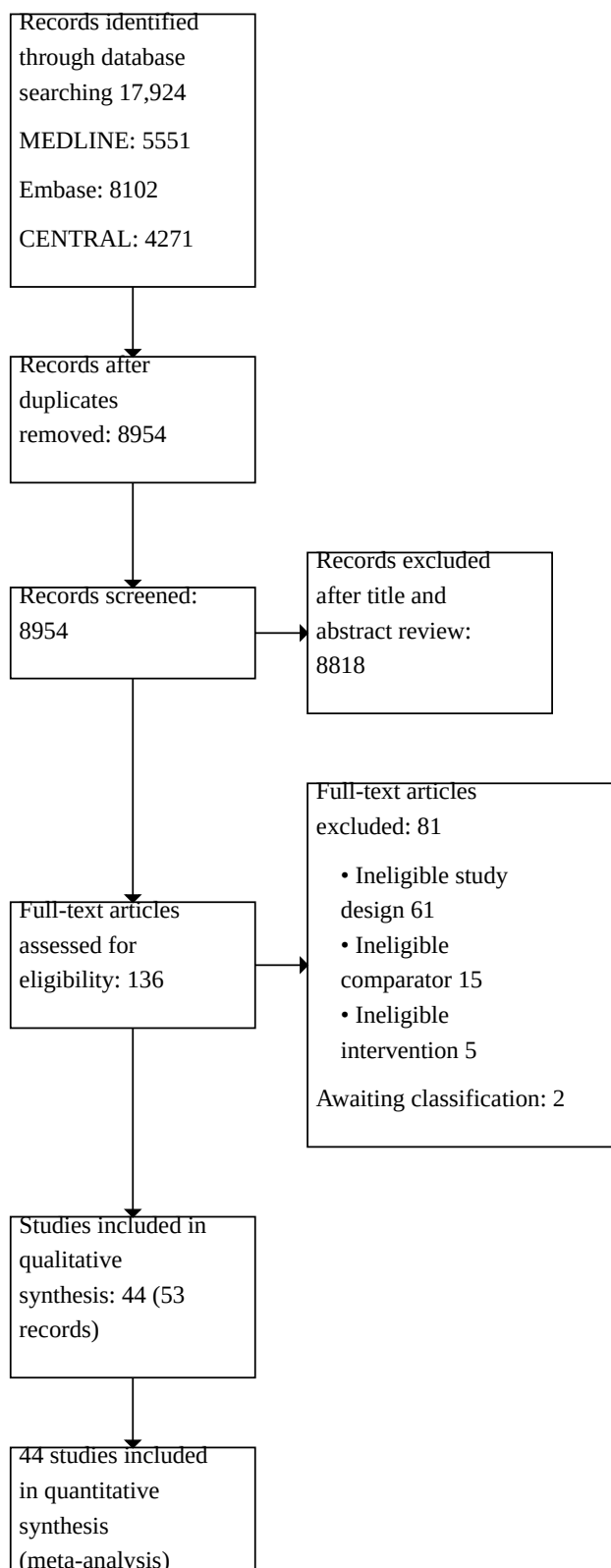


Figure 1. (Continued)

synthesis (meta-analysis)

Included studies

All 44 included trials were parallel-group RCTs, with a description of trial characteristics provided in the [Characteristics of included studies](#) tables.

Study design and setting

Forty-four studies with 3664 participants were identified. Out of these, 40 studies with 3134 participants compared PSG to CON, two studies with 140 participants compared PSG to NAV and two studies with 390 participants compared PSG to NAV and CON.

All 44 trials were parallel RCTs, 40 trials included two intervention arms, two trials included three intervention arms (Pfzner 2014; Yan 2014) and two trials included four intervention arms (Attard 2019; Zahn 2020).

The trials were conducted in Australia (Huijbregts 2016), Belgium (De Vloo 2017; Victor 2014; Vundelinckx 2013), Canada (Teeter 2019; Turgeon 2019; Zomar 2021), China (Gan 2015a, Liao 2024; Qiu 2017; Sun 2020; Yan 2014), France (Abane 2015; Abdel 2014; Ollivier 2016; Sariali 2019), Germany (Calliess 2017; Maus 2018; Pfzner 2014; Zahn 2020), Korea (Roh 2013), Italy (Cucchi 2019; Ferrara 2015; Giannotti 2019), Netherlands (Boonen 2013; Kosse 2017), Norway (Van Leeuwen 2018), Poland (Kotela 2014; Stolarczyk 2018), Portugal (Silva 2014; Vide 2017), Slovenia (Molicnik 2015), Thailand (Chareancholvanich 2013; Chotanaphuti 2014; Tammachote 2018), United States (Hamilton 2013; Lionberger 2014; Noble 2012; Woolson 2014), United Kingdom (Attard 2019; Hampton 2022).

Participants

The age and sex of the study groups have been presented in the [Characteristics of included studies](#) table.

The number of participants per trial ranged from 22 to 300. The mean age of participants was 68 years and ranged from 53 years to 82 years; 34 percent of participants were male, and 66 percent of participants were female. The average BMI of participants was 29, ranging from 17 to 44.5.

In general, groups were homogeneous for aetiology, with more than 90% of participants having OA. However, eligibility criteria were variable between studies, which may have produced heterogeneous groups with regard to underlying disease. Eleven studies included participants with severe arthritis only and the remaining studies simply included patients with arthritis that required TKA. Disease duration was not routinely recorded. Contraindications to MRI such as claustrophobia, presence of cardiac pacemakers and metalwork near the knee were standard exclusion criteria in MRI-based PSG studies. Other frequently stated contraindications were previous knee osteotomy, previous fractures around the knee, previous surgery on the knee, presence of infection and severe lower limb deformity.

Interventions

Forty out of the 44 RCTs compared PSG to CON in 2900 TKAs (1452 PSG, 1448 CON) (Abane 2015; Abdel 2014; Attard 2019; Boonen 2013; Calliess 2017; Chareancholvanich 2013; Chotanaphuti 2014; Cucchi 2019; De Vloo 2017; Ferrara 2015; Gan 2015; Giannotti 2019; Hamilton 2013; Hampton 2022; Huijbregts 2016; Kosse 2017; Kotela 2014; Liao 2024; Maus 2018; Molicnik 2015; Noble 2012; Parratte 2013; Pfzner 2014; Pietsch 2014; Qiu 2017; Randelli 2019; Roh 2013; Sariali 2019; Silva 2014; Stolarczyk 2018; Sun 2020; Tammachote 2018; Teeter 2019; Turgeon 2019; Van Leeuwen 2018; Victor 2014; Vide 2017; Vundelinckx 2013; Woolson 2014; Zomar 2021). Two studies with 140 TKAs compared PSG with NAV (70 in each group) (Lionberger 2014; Ollivier 2016). Two studies compared PSG with CON and NAV with 90 TKAs in Yan 2014; and 300 TKA in Zahn 2020).

Twenty-seven studies evaluated MRI-based PSGs (Abane 2015; Abdel 2014; Boonen 2013; Calliess 2017; Chareancholvanich 2013; De Vloo 2017; Ferrara 2015; Hampton 2022; Huijbregts 2016; Kosse 2017; Lionberger 2014; Maus 2018; Molicnik 2015; Noble 2012; Ollivier 2016; Parratte 2013; Pietsch 2014; Silva 2014; Stolarczyk 2018; Tammachote 2018; Teeter 2019; Turgeon 2019; Van Leeuwen 2018; Vide 2017; Vundelinckx 2013; Yan 2014; Zahn 2020), 15 studies evaluated CT-based PSGs (Attard 2019; Chotanaphuti 2014; Cucchi 2019; Gan 2015; Giannotti 2019; Hamilton 2013; Kotela 2014; Liao 2024; Qiu 2017; Randelli 2019; Roh 2013; Sariali 2019; Sun 2020; Woolson 2014; Zomar 2021), and two studies evaluated both CT- and MRI-based PSGs versus CON (Pfzner 2014; Victor 2014).

Posterior-stabilised prostheses were implanted in all patients in 28 studies (Abane 2015; Abdel 2014; Attard 2019; Chareancholvanich 2013; Chotanaphuti 2014; Cucchi 2019; Ferrara 2015; Gan 2015; Giannotti 2019; Hamilton 2013; Kosse 2017; Liao 2024; Molicnik 2015; Ollivier 2016; Parratte 2013; Pfzner 2014; Pietsch 2014; Randelli 2019; Roh 2013; Sariali 2019; Silva 2014; Sun 2020; Tammachote 2018; Teeter 2019; Victor 2014; Woolson 2014; Yan 2014; Zomar 2021). Cruciate retaining prostheses were implanted in all patients in four studies (Kotela 2014; Noble 2012; Van Leeuwen 2018; Victor 2014). In nine studies, the details of the prosthesis type (PS or CR) were not provided (Boonen 2013; Hampton 2022; Huijbregts 2016; Lionberger 2014; Qiu 2017; Stolarczyk 2018; Turgeon 2019; Vundelinckx 2013; Zahn 2020). One study used both PS and CR TKAs based on the severity of deformity and condition of cruciate ligaments intraoperatively (De Vloo 2017) and one other study used both PS and CR TKAs based on surgeon preference (Maus 2018).

Eleven studies used fixed-bearing implants (Abane 2015; Attard 2019; Boonen 2013; Chareancholvanich 2013; Chotanaphuti 2014; Ferrara 2015; Hamilton 2013; Kosse 2017; Pfzner 2014; Victor 2014; Vide 2017). Three studies used mobile-bearing implants (Abdel 2014; Cucchi 2019; Roh 2013). The remaining studies did not specify the type of bearing used. Twenty-three studies utilised cemented implants (Abane 2015; Abdel 2014; Attard 2019; Boonen

2013; Chareancholvanich 2013; Chotanaphuti 2014; Cucchi 2019; Giannotti 2019; Hamilton 2013; Hampton 2022; Huijbregts 2016; Kosse 2017; Molicnik 2015; Pfitzner 2014; Sariali 2019; Silva 2014; Sun 2020; Tammachote 2018; Teeter 2019; Van Leeuwen 2018; Victor 2014; Vide 2017; Woolson 2014) and only one study used cementless implants (Kotela 2014). The remaining studies did not specify the type of implant fixation used.

Thirteen studies were prospectively registered in ClinicalTrials.gov trial registry (Abane 2015; Attard 2019; Chareancholvanich 2013; Cucchi 2019; Lionberger 2014; Noble 2012; Sun 2020; Tammachote 2018; Teeter 2019; Turgeon 2019; Van Leeuwen 2018; Woolson 2014; Zomar 2021) and one in WHO International Clinical Trials Registry (Roh 2013). A flow chart in accordance with CONSORT guidelines was provided in 25 studies (Abane 2015; Abdel 2014; Attard 2019; Boonen 2013; Chareancholvanich 2013; Cucchi 2019; De Vloo 2017; Hampton 2022; Huijbregts 2016; Kosse 2017; Lionberger 2014; Maus 2018; Pfitzner 2014; Randelli 2019; Roh 2013; Sariali 2019; Stolarczyk 2018; Tammachote 2018; Teeter 2019; Van Leeuwen 2018; Victor 2014; Vide 2017; Woolson 2014; Yan 2014; Zomar 2021) and 24 studies described the power that the study employed for calculation of sample size (Attard 2019; Boonen 2013; Chareancholvanich 2013; Cucchi 2019; De Vloo 2017; Giannotti 2019; Hamilton 2013; Hampton 2022; Kosse 2017; Lionberger 2014; Maus 2018; Ollivier 2016; Randelli 2019; Sariali 2019; Sun 2020; Tammachote 2018; Teeter 2019; Turgeon 2019; Van Leeuwen 2018; Victor 2014; Woolson 2014; Yan 2014; Zahn 2020; Zomar 2021).

Outcomes

Major outcomes

1. Survival rate of the implant (any revision surgery to change a prosthetic component, i.e. the proportion of knees that undergo revision surgery). This was reported in nine studies (Boonen 2013; Hampton 2022; Kosse 2017; Kotela 2014; Sariali 2019; Sun 2020; Teeter 2019; Turgeon 2019; Woolson 2014).
2. Functional outcome measures with validated instruments (such as participant reported measures: Western Ontario & McMaster Universities Osteoarthritis Index (WOMAC), Knee Injury and Osteoarthritis Outcome Score (KOOS), International Knee Documentation Committee (IKDC) score, Oxford Knee score (OKS) and Knee Society Score (KSS) was reported in twenty-three studies (Abane 2015; Abdel 2014; Attard 2019; Boonen 2013; Calliess 2017; Cucchi 2019; Giannotti 2019; Hampton 2022; Huijbregts 2016; Kosse 2017; Kotela 2014; Maus 2018; Pfitzner 2014; Pietsch 2014; Randelli 2019; Stolarczyk 2018; Tammachote 2018; Teeter 2019; Turgeon 2019; Vundelinckx 2013; Woolson 2014; Yan 2014; Zomar 2021).
3. Radiographic lower limb alignment: recorded as the number of outliers for femorotibial coronal angle (FTCA) also called the hip-knee-ankle (HKA) angle measured following surgery. In trials where the investigators performed these measurements at more than one time point, the earliest postoperative measurement was used for analysis. "Outliers" were defined as alignments outside +/- 3° range for FTCA measured as a dichotomous variable. All studies used mechanical alignment in surgical aims for the implantation and orientation of the TKA. There were differences in how studies analysed this and postoperative imaging. This outcome was reported in thirty studies (Abane 2015; Abdel 2014; Boonen 2013; Chareancholvanich 2013; Chotanaphuti 2014; Gan 2015; Hamilton 2013; Hampton 2022; Huijbregts 2016; Kotela 2014; Liao 2024; Lionberger 2014; Maus

2018; Molicnik 2015; Noble 2012; Parratte 2013; Pfitzner 2014; Qiu 2017; Roh 2013; Sariali 2019; Sun 2020; Tammachote 2018; Teeter 2019; Turgeon 2019; Van Leeuwen 2018; Victor 2014; Vide 2017; Woolson 2014; Yan 2014; Zahn 2020).

Abane 2015 obtained radiographic outcomes at three months postoperatively by using radiographs. An HKA angle > 180° was indicative of valgus, whereas values < 180° indicated varus. Abdel 2014 did not note the time of postoperative radiographs to assess mechanical alignment. An HKA angle > 180° was indicative of valgus, whereas values < 180° indicated varus. The measurement was part of gait analysis and validity testing. Boonen 2013 followed up patients with x-rays at six weeks. Mechanical alignment of knees in the study was compared, with HKA angle > 180° indicative of valgus, whereas values < 180° indicated varus. Software accuracy was noted to be within 1 degree.

Chareancholvanich 2013 measured the radiological parameters using the digital tools of the picture archiving and communication system (PACS), four weeks postoperatively. Deviations of both component alignments were noted as positive if the LDFA or MPTA was < 90° (valgus position) and as negative if > 90° (varus position). Chotanaphuti 2014 reported on CT scans of the knee and a weight-bearing film of hip-knee-ankle six weeks postoperatively. The measurement used to determine component coronal alignment was the mechanical axis. Gan 2015 undertook postoperative radiographs and CT to assess the mechanical axis, which was the angle between the centres of the femoral head, knee, and ankle. Alignment of the femoral and tibial components were measured with respect to the mechanical axis in the frontal plane and the longitudinal axis of the shaft of the femur and tibia, respectively, for the two components in the sagittal plane. For the mechanical axis of the leg and frontal alignment, the following angles were measured: the hip-knee-ankle angle (mechanical axis of the leg), the frontal femoral component (FFC) angle, and the frontal tibial component (FTC) angle. For the sagittal plane, the following angles were measured: the lateral femoral component (LFC) angle and the lateral tibial component (LTC) angle. The LFC angle was measured between the anterior cortex of the distal femur and the shield of the femoral component. The LTC angle was measured in relation to the posterior tibial cortex. In addition, the rotational femoral angle (RFA) was sought. In Hamilton 2013, radiographic measurements involved the mechanical axis. A postoperative (unknown time point) lateral x-ray was used to evaluate the tibial component slope and femoral component flexion. Hip, knee and ankle (HKA) long-limb alignment radiographs were used to evaluate the mechanical axis of the extremity, the tibiofemoral angle, the varus/valgus of the femoral component to the mechanical and anatomic axis, and the varus/valgus of the tibial component. All measurements were made by a single observer who was blinded with regard to treatment allocation.

Hampton 2022 assessed radiological alignment parameters using the TraumaCad orthopaedic templating software on standardised postoperative radiographs at an unknown follow-up period. The authors noted that CT scans were not used to assess postoperative femoral rotation because of the increased cost associated with the scans and the additional radiation exposure to the patient. Huijbregts 2016 assessed postoperative HKA angle measurements using long leg radiographs and CT scans. In Kotela 2014, radiographs were assessed by an independent orthopaedic surgeon for two angle measurements. The second

set of measurements was carried out two weeks after the first. [Liao 2024](#) performed weight-bearing x-rays of double lower-limb extremities and short AP & LAT films of the affected knees, both preoperatively and postoperatively. Each patient had a postoperative CT of the surgical side, but the study did not describe who performed radiographic analysis and the blinding process.

[Lionberger 2014](#) review used radiographs to calculate all angles, including the mechanical axis. Ninety-day coronal long-leg standing radiographs were obtained in [Maus 2018](#). In [Molicnik 2015](#), full-length, standing, anterior-posterior radiographs were taken on all patients preoperatively and postoperatively before discharge, which averaged postoperative day four to six upon the condition of full extension being achieved.

[Noble 2012](#) did not describe their method or timing of postoperative radiographic analysis. [Parratte 2013](#) noted that radiographic analysis included pre and postoperative full-length standing x-rays, an ML view of the knee and a skyline view at 30° of flexion. The radiological evaluation was used intra-operatively with dedicated software (Centricity WebO) and included an analysis of the following angles: HKA, frontal femoral component angle (FFC), frontal tibial component angle (FTC), lateral femoral component angle (LFC) and lateral tibial component angle (LTC) as well as the tibial slope. [Pfitzner 2014](#) utilised both radiographs and CT to analyse postoperative mechanical alignment with measurements. [Qiu 2017](#) only noted that a group of surgeons had been responsible for postoperative radiographic analysis and did not comment on method or timing. In [Sariali 2019](#), CT scanning was the modality of choice for analysis and was performed at three months. Pre and postoperative CT scans were used along with the KNEE-PLAN® software for angular measurements. [Roh 2013](#) utilised x-rays at one month postoperatively. The mechanical axis was measured twice with the same by one senior radiologist and the average value of the two measurements was obtained. [Tammachote 2018](#) commented upon analysis being performed with x-rays at three months post-surgery.

[Teeter 2019](#) analysis differed from the other studies in that migration at each time point was measured using model-based RSA software (MBRSA, RSAcore, Leiden, The Netherlands). [Turgeon 2019](#) analysis was performed using radiographs at six months postoperatively. The mechanical axis of the lower limb, the femoral component alignment, and the tibial component alignment in the coronal plane were measured from the 3-foot standing AP radiograph. The authors reported that measurements were completed digitally using an IMPAX 6.6.1 system (Agfa HealthCare N.V.). [Victor 2014](#) performed both radiograph and CT scan analysis. [Woolson 2014](#) utilised OsiriX (version 3.5.1) software that incorporated a custom program with the use of MATLAB (MathWorks, Natick, Massachusetts). [Yan 2014](#) did not describe their timings or methodology for analysis of radiographic outcomes or outliers. [Zahn 2020](#) utilised computer software for radiograph analysis but did not comment on timing, methods or number of observers.

4. Pain: mean overall pain measured on a visual analogue scale (VAS) or numerical rating scale (NRS), or other measures of pain was measured in ten studies ([Abdel 2014](#); [Boonen 2013](#); [Cucchi 2019](#); [Kosse 2017](#); [Kotela 2014](#); [Pietsch 2014](#); [Stolarczyk 2018](#); [Turgeon 2019](#); [Van Leeuwen 2018](#); [Vundelinckx 2013](#)).

5. Participant-reported global assessment (e.g. better, not better, worse, dichotomised to better and not better or worse) was not measured or reported in any of the studies included.

6. Total adverse events (including infection, thrombosis, palsy, death, etc.) were reported in thirteen studies: [Abdel 2014](#); [Boonen 2013](#); [Hampton 2022](#); [Huijbregts 2016](#); [Kosse 2017](#); [Kotela 2014](#); [Pietsch 2014](#); [Qiu 2017](#); [Turgeon 2019](#); [Vundelinckx 2013](#); [Woolson 2014](#); [Yan 2014](#); [Zomar 2021](#).

7. Re-operation rate (not involving implant change) was reported in ten studies: [Boonen 2013](#); [Hampton 2022](#); [Huijbregts 2016](#); [Kosse 2017](#); [Kotela 2014](#); [Pietsch 2014](#); [Sariali 2019](#); [Turgeon 2019](#); [Vundelinckx 2013](#); [Woolson 2014](#).

Minor outcomes

1. Quality of life was reported in three studies ([Abdel 2014](#); [Huijbregts 2016](#); [Teeter 2019](#)).
2. Operative time was reported in twenty-five studies ([Abane 2015](#); [Boonen 2013](#); [Chareancholvanich 2013](#); [Chotanaphuti 2014](#); [Cucchi 2019](#); [Ferrara 2015](#); [Gan 2015](#); [Hamilton 2013](#); [Hampton 2022](#); [Kosse 2017](#); [Kotela 2014](#); [Lionberger 2014](#); [Maus 2018](#); [Noble 2012](#); [Pietsch 2014](#); [Randelli 2019](#); [Sariali 2019](#); [Sun 2020](#); [Tammachote 2018](#); [Teeter 2019](#); [Turgeon 2019](#); [Van Leeuwen 2018](#); [Woolson 2014](#); [Yan 2014](#); [Zomar 2021](#)).
3. Blood loss (including transfusion rate) was reported in thirteen studies ([Abane 2015](#); [Attard 2019](#); [Boonen 2013](#); [Chareancholvanich 2013](#); [Gan 2015](#); [Giannotti 2019](#); [Hampton 2022](#); [Kotela 2014](#); [Maus 2018](#); [Noble 2012](#); [Pietsch 2014](#); [Sun 2020](#); [Teeter 2019](#)).
4. Range of motion was reported in eight studies ([Lionberger 2014](#); [Pietsch 2014](#); [Sun 2020](#); [Tammachote 2018](#); [Van Leeuwen 2018](#); [Vundelinckx 2013](#); [Woolson 2014](#); [Yan 2014](#)).
5. Length of hospital stay was reported in thirteen studies ([Abane 2015](#); [Attard 2019](#); [Chareancholvanich 2013](#); [Ferrara 2015](#); [Kotela 2014](#); [Maus 2018](#); [Noble 2012](#); [Turgeon 2019](#); [Van Leeuwen 2018](#); [Vide 2017](#); [Vundelinckx 2013](#); [Woolson 2014](#); [Zomar 2021](#)).
6. Incision length was reported in three studies ([Lionberger 2014](#); [Noble 2012](#); [Vundelinckx 2013](#)).
7. Radiographic outliers in alignment for individual prosthetic components included: femoral component coronal alignment, tibial component coronal alignment, femoral component sagittal alignment, tibial component sagittal alignment and femoral component rotational alignment. "Outliers" were defined as alignments outside +/- 3° range for all individual component alignments ([Victor 2014](#)); we measured these as dichotomous variables. Seventeen studies reported these radiographic outliers ([Abane 2015](#); [Boonen 2013](#); [Chareancholvanich 2013](#); [Ferrara 2015](#); [Hamilton 2013](#); [Huijbregts 2016](#); [Kotela 2014](#); [Ollivier 2016](#); [Pfitzner 2014](#); [Qiu 2017](#); [Roh 2013](#); [Sariali 2019](#); [Tammachote 2018](#); [Van Leeuwen 2018](#); [Victor 2014](#); [Woolson 2014](#); [Yan 2014](#)).

Follow-up

Duration of follow-up was reported in 28 studies ([Abane 2015](#); [Abdel 2014](#); [Attard 2019](#); [Boonen 2013](#); [Cucchi 2019](#); [De Vloo 2017](#); [Giannotti 2019](#); [Hampton 2022](#); [Huijbregts 2016](#); [Kosse 2017](#); [Kotela 2014](#); [Maus 2018](#); [Ollivier 2016](#); [Pfitzner 2014](#); [Pietsch 2014](#); [Randelli 2019](#); [Sariali 2019](#); [Stolarczyk 2018](#); [Sun 2020](#); [Tammachote 2018](#); [Teeter 2019](#); [Turgeon 2019](#); [Van Leeuwen 2018](#); [Vundelinckx 2013](#); [Woolson 2014](#); [Yan 2014](#); [Zahn 2020](#); [Zomar 2021](#)). Most

studies were short term with a mean follow-up duration of 11.5 months (range 5 days to 60 months). Short follow-up duration was justifiable for most studies as these only evaluated radiological alignment of component positioning. Two studies had mid-term follow-up (two-five years) for implant survivorship and functional outcomes (Hampton 2022; Boonen 2013).

Funding

Twenty-three trials reporting their funding source (Attard 2019, Calliess 2017, Chareancholvanich 2013, Cucchi 2019, Hamilton 2013, Huijbregts 2016, Kosse 2017, Lionberger 2014, Maus 2018, Molicnik 2015, Noble 2012, Ollivier 2016, Qiu 2017, Randelli 2019, Roh 2013, Sun 2020, Tammachote 2018, Teeter 2019, Turgeon 2019, Vundelinckx 2013, Woolson 2014, Zahn 2020, Zomar 2021).

Eighteen trials did not report their funding source (Abane 2015, Abdel 2014, Boonen 2013, Chotanaphuti 2014, De Vloo 2017, Ferrara 2015, Giannotti 2019, Hampton 2022, Kotela 2014, Pfitzner 2014, Pietsch 2014, Sariali 2019, Silva 2014, Stolarczyk 2018, Van Leeuwen 2018, Victor 2014, Vide 2017, Yan 2014).

Seven trials had funding from industry (Abane 2015, Attard 2019, Kosse 2017, Maus 2018, Noble 2012, Sariali 2019, Turgeon 2019).

Excluded studies

Out of the 83 studies excluded after full-text assessment, 61 had an ineligible study design, 15 had an ineligible comparator and five had an ineligible intervention.

Fifteen studies were excluded because they had an ineligible comparator, i.e. they compared one PSG group with another PSG group (Asada 2014; Bollars 2023; Ensini 2014; Gan 2015a; Geng 2024; Lee 2024; Londhe 2024; NCT05980442; Sarzaeem 2024; Schotanus 2016; Smolle 2023; Theeuwen 2024; Hommel 2017; Schotanus 2016; Timoncini 2012).

Five studies had an ineligible intervention; one study evaluated mechanical alignment using CON guides with kinematic alignment using PSG (Calliess 2017), and one study had kinematically and mechanically aligned total knee replacements (Dossett 2014). Three studies were not PSG-based studies but used preoperative imaging and rapid prototyping for TKA (Gan 2015a; Hafez 2006; Zhang 2016).

Risk of bias in included studies

The risk of bias table and summary (Figure 2, Figure 3) provide details of possible bias in included studies.

Figure 2. Risk of bias graph: review authors' judgements about each risk of bias item presented as percentages across all included studies

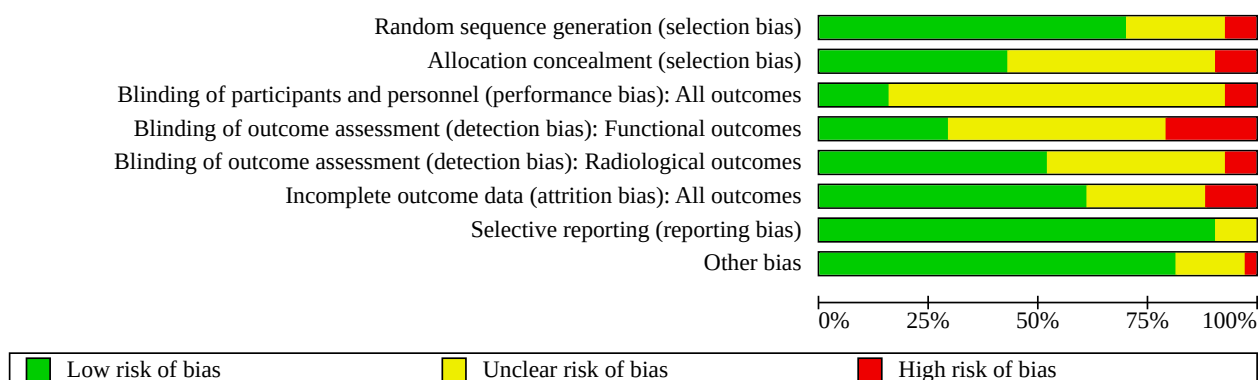


Figure 3. Risk of bias summary: review authors' judgements about each risk of bias item for each included study

	Random sequence generation (selection bias)	Allocation concealment (selection bias)	Blinding of participants and personnel (performance bias): All outcomes	Blinding of outcome assessment (detection bias): Functional outcomes	Blinding of outcome assessment (detection bias): Radiological outcomes	Incomplete outcome data (attrition bias): All outcomes	Selective reporting (reporting bias)	Other bias
Abane 2015	?	?	?	?	+	-	+	?
Abdel 2014	+	+	?	?	+	+	+	+
Attard 2019	+	?	?	?	?	?	+	-
Boonen 2013	+	+	+	+	+	+	+	?
Calliess 2017	?	-	-	-	-	+	+	+
Chareancholvanich 2013	+	?	?	+	+	+	+	+
Chotanaphuti 2014	?	?	?	+	+	+	+	+
Cucchi 2019	+	+	?	?	?	+	?	+
De Vloo 2017	+	+	?	?	?	?	+	+
Ferrara 2015	+	+	?	+	+	+	+	+
Gan 2015	?	?	?	?	+	+	?	+
Giannotti 2019	-	?	?	?	?	?	+	+
Hamilton 2013	?	-	?	+	?	+	+	+
Hampton 2022	+	?	+	+	+	?	?	+
Huijbregts 2016	+	+	?	?	+	?	+	+
Kosse 2017	+	+	?	-	?	+	+	+
Kotela 2014	?	-	?	-	+	-	+	+

Figure 3. (Continued)

Kotela 2014	?	+	?	+	+	+	+	+
Liao 2024	+	?	?	?	?	?	+	+
Lionberger 2014	?	?	?	?	+	+	+	+
Maus 2018	+	?	+	?	+	?	+	+
Molicnik 2015	?	?	?	+	?	+	+	+
Noble 2012	?	?	?	+	?	+	+	?
Ollivier 2016	+	+	?	+	?	+	+	+
Parratte 2013	+	+	+	+	+	+	?	+
Pfitzner 2014	+	+	?	?	+	+	+	+
Pietsch 2014	+	+	?	+	?	+	+	+
Qiu 2017	+	?	?	?	?	?	+	+
Randelli 2019	+	+	?	?	?	+	+	?
Roh 2013	+	?	?	?	?	+	+	+
Sariali 2019	+	+	+	+	+	+	+	+
Silva 2014	?	?	?	?	+	+	+	+
Stolarczyk 2018	+	+	?	+	+	+	+	+
Sun 2020	+	?	+	?	?	?	+	+
Tammachote 2018	+	+	?	?	+	+	+	+
Teeter 2019	+	+	?	?	?	?	+	+
Turgeon 2019	+	?	+	+	+	+	+	+
Van Leeuwen 2018	+	+	+	?	?	+	+	?
Victor 2014	+	+	?	+	+	+	+	+
Vide 2017	+	?	?	?	+	+	+	?
Vundelinckx 2013	+	+	?	+	+	?	+	+
Woolson 2014	+	+	?	+	+	+	+	?
Yan 2014	+	?	?	+	+	+	+	+
Zahn 2020	+	?	?	?	?	?	+	+
Zomar 2021	+	?	+	+	+	+	+	+

Allocation

Nineteen studies adequately generated a random sequence and concealed the sequence until allocation; thus, we considered these studies at low risk of selection bias (Abdel 2014; Boonen 2013; Cucchi 2019; De Vloo 2017; Ferrara 2015; Huijbregts 2016; Kosse 2017; Ollivier 2016; Pfitzner 2014; Pietsch 2014; Randelli 2019; Sariali 2019; Stolarczyk 2018; Tammachote 2018; Teeter 2019; Van Leeuwen 2018; Victor 2014; Woolson 2014).

Nineteen studies did not report their method of randomisation or allocation concealment, and we judged their risk of selection bias as unclear (Abane 2015; Attard 2019; Chareancholvanich 2013; Chotanaphuti 2014; Gan 2015; Hampton 2022; Liao 2024;

Lionberger 2014; Maus 2018; Molicnik 2015; Noble 2012; Roh 2013; Sun 2020; Turgeon 2019; Vide 2017; Yan 2014; Zahn 2020; Zomar 2021; Silva 2014), while six studies reported inadequate methods and were judged at high risk of selection bias (Calliess 2017; Giannotti 2019; Hamilton 2013; Kotela 2014; Qiu 2017; Vundelinckx 2013).

Blinding

Three studies were at high risk of performance bias due to lack of participant blinding (Calliess 2017; Maus 2018; Zomar 2021). Thirty-four studies were at unclear risk of performance bias due to the interventions in studies precluding blinding of participants to treatment allocation (Abane 2015; Abdel 2014; Attard 2019;

Chareancholvanich 2013; Chotanaphuti 2014; Cucchi 2019; De Vloo 2017; Ferrara 2015; Gan 2015; Giannotti 2019; Hamilton 2013; Huijbregts 2016; Kosse 2017; Kotela 2014; Liao 2024; Lionberger 2014; Molicnik 2015; Noble 2012; Ollivier 2016; Pfitzner 2014; Pietsch 2014; Qiu 2017; Randelli 2019; Roh 2013; Silva 2014; Stolarczyk 2018; Tammachote 2018; Teeter 2019; Victor 2014; Vide 2017; Vundelinckx 2013; Woolson 2014; Yan 2014; Zahn 2020). Seven studies were at low risk of performance bias as they reported participant blinding by performing preoperative MRI or CT scans (Boonen 2013; Hampton 2022; Parratte 2013; Soriali 2019; Sun 2020; Turgeon 2019; Van Leeuwen 2018) in both groups.

For self-reported outcomes (pain, function, treatment success, adverse events), participants were adequately blinded and there was a low risk of detection bias in 13 studies (Boonen 2013; Chareancholvanich 2013; Chotanaphuti 2014; Ferrara 2015; Hamilton 2013; Hampton 2022; Noble 2012; Ollivier 2016; Parratte 2013; Pietsch 2014; Soriali 2019; Turgeon 2019; Victor 2014). Due to the lack of participant blinding, we deemed the risk of detection biases to be high in nine studies (Calliess 2017; Kosse 2017; Kotela 2014; Molicnik 2015; Stolarczyk 2018; Vundelinckx 2013; Woolson 2014; Yan 2014; Zomar 2021). Twenty-two studies did not report if participants were blinded to treatment, and we were unclear if there was a risk of detection bias (Abane 2015; Abdel 2014; Attard 2019; Cucchi 2019; De Vloo 2017; Gan 2015; Giannotti 2019; Huijbregts 2016; Liao 2024; Lionberger 2014; Maus 2018; Pfitzner 2014; Qiu 2017; Randelli 2019; Roh 2013; Silva 2014; Sun 2020; Tammachote 2018; Teeter 2019; Van Leeuwen 2018; Vide 2017; Zahn 2020).

In 23 studies, there was low risk of detection bias for assessor-reported outcomes (radiological outcomes, survival rate of the implant, re-operation), as assessors were effectively blinded to the treatment assignment (Abane 2015; Abdel 2014; Boonen 2013; Chareancholvanich 2013; Chotanaphuti 2014; Ferrara 2015; Gan 2015; Hampton 2022; Huijbregts 2016; Kotela 2014; Lionberger 2014; Parratte 2013; Pfitzner 2014; Soriali 2019; Silva 2014; Stolarczyk 2018; Tammachote 2018; Turgeon 2019; Victor 2014; Vide 2017; Vundelinckx 2013; Woolson 2014; Yan 2014), while 18 studies were at unclear risk (Attard 2019; Cucchi 2019; De Vloo 2017; Giannotti 2019; Hamilton 2013; Kosse 2017; Liao 2024; Molicnik 2015; Noble 2012; Ollivier 2016; Pietsch 2014; Qiu 2017; Randelli 2019; Roh 2013; Sun 2020; Teeter 2019; Van Leeuwen 2018; Zahn 2020) and three studies at high risk of detection bias for assessor-reported outcomes, as outcome assessors were not blinded (Calliess 2017; Maus 2018; Zomar 2021).

Incomplete outcome data

Most of the studies included in this analysis had a low risk of attrition bias (< 20% dropout rate) (Abdel 2014; Boonen 2013; Calliess 2017; Chareancholvanich 2013; Chotanaphuti 2014; Cucchi 2019; Ferrara 2015; Gan 2015; Hamilton 2013; Kosse 2017; Lionberger 2014; Molicnik 2015; Ollivier 2016; Parratte 2013; Pfitzner 2014; Pietsch 2014; Randelli 2019; Roh 2013; Soriali 2019; Silva 2014; Stolarczyk 2018; Tammachote 2018; Turgeon 2019; Victor 2014; Vide 2017; Woolson 2014; Yan 2014).

Attrition bias was found to be at high risk in five studies, due to high attrition (> 20% dropout rate) or unequal dropout rates between the intervention and control arms (Abane 2015; Kotela 2014; Noble 2012; Van Leeuwen 2018; Zomar 2021).

Twelve studies were at unclear risk of attrition bias as there were either incomplete outcome data or studies did not clearly specify dropout rates (Attard 2019; De Vloo 2017; Giannotti 2019; Hampton 2022; Huijbregts 2016; Liao 2024; Maus 2018; Qiu 2017; Sun 2020; Teeter 2019; Vundelinckx 2013; Zahn 2020).

Selective reporting

Five studies were at unclear risk of reporting bias (Hampton 2022; Kosse 2017; Sun 2020; Turgeon 2019; Zomar 2021) as some of their outcome measures were poorly reported. None of the studies were at high risk of reporting bias. Most of the studies included in this analysis were not prospectively registered but all the outcome measures mentioned in the methodology were reported; hence, they were at low risk of selective reporting bias.

Other potential sources of bias

Most of the studies were at low risk of other bias. Attard 2019 was deemed to have a high risk of other potential bias as the outcomes are likely to be influenced by surgeon performance with one type of implant over another, uneven randomised groups and performance issues of theatre staff when recording time as an outcome.

Seven studies with unclear risk of other potential sources of bias were: Abane 2015 (unable to provide accuracy data for some outcome measurements and the same observer used who was blinded to instrumentation); Boonen 2013 (mismatch of target and posterior slope may have influenced outcomes); Noble 2012 (single surgeon data); Randelli 2019 (inadequately powered to investigate more than one plane of alignment and limited follow-up period); Van Leeuwen 2018 (underpowered, no interobserver agreement); Vide 2017 (lack of blinding and no power analysis or sample size calculation performed); Woolson 2014 (possible impaired performance due to familiarity with implants).

Effects of interventions

See: **Summary of findings 1** Patient-specific cutting guides compared to conventional instrumentation for total knee arthroplasty; **Summary of findings 2** Patient-specific cutting guides compared to computer-assisted navigation

Patient-specific cutting guides (PSGs) versus conventional cutting guides (CON)

Major outcomes

See **Summary of findings 1**.

Survival rate of the implant (participants that did not require revision of components)

Nine studies (Boonen 2013; Hampton 2022; Kosse 2017; Kotela 2014; Soriali 2019; Sun 2020; Teeter 2019; Turgeon 2019; Woolson 2014) reported data on survival of the implant that were eligible for analysis, with 342 participants in the PSG group and 347 in the CON group. Compared to conventional instrumentation, PSGs may result in little to no difference in survival of the implant. At 26 months, 7/347 participants (20 per 1000) in the conventional instrumentation group reported survival of the implant (risk of revision) compared to 4/342 participants (16 per 1000) in the PSG group (RR 0.79 (95% CI 0.25 to 2.52); $I^2 = 0\%$; 689 participants; 9 studies; low-certainty evidence downgraded for bias and imprecision). The mean follow-up period was 26 months (range 10 months to 60 months). Concerning Boonen 2013, ten-year

follow-up results and survivability were published in a sequential study by [Theeuwes 2024](#), although we were unable to incorporate this into the meta-analysis due to the presentation of the data. At 10-year follow-up, there was no significant difference in clinical outcomes or revision rate between PSI and CI for TKA found. This follow-up study included 129 patients (losing 23% of the original patient cohort in Boonen) noting a mean follow-up of 10.1 years. No statistically significant differences between the PSG and CI groups were found for any of the PROMs and revision rates were comparable, totalling six in the PSG group and three in the CI group ($P = 0.29$) ([Analysis 1.1](#)).

Function

Function was reported in 23 studies ([Abane 2015](#); [Abdel 2014](#); [Attard 2019](#); [Boonen 2013](#); [Calliess 2017](#); [Cucchi 2019](#); [Giannotti 2019](#); [Hampton 2022](#); [Huijbregts 2016](#); [Kosse 2017](#); [Kotela 2014](#); [Maus 2018](#); [Pfitzner 2014](#); [Pietsch 2014](#); [Randelli 2019](#); [Stolarczyk 2018](#); [Tammachote 2018](#); [Teeter 2019](#); [Turgeon 2019](#); [Vundelinckx 2013](#); [Woolson 2014](#); [Yan 2014](#); [Zomar 2021](#)). Time points varied between six weeks to five years postoperatively, with the majority of these studies reporting short-term outcomes and only two studies reporting intermediate-term outcomes ([Boonen 2013](#); [Hampton 2022](#)). Numerous scales were used to assess function, including Knee Society Score (KSS), Oxford Knee Score (OKS), Western Ontario and McMaster University Osteoarthritis Index (WOMAC) and Knee injury and Osteoarthritis Outcome Score (KOOS). Compared to conventional instrumentation, PSGs may result in little to no improvement in function. The mean function (KSS, OKS, WOMAC, KOOS) was 15.1 points with conventional instrumentation and 13.44 points with PSG (SMD -0.11, 95% CI -0.25 to 0.03, back-translated to MD 1.66 points lower (95% CI 3.77 points lower, 0.45 points higher; $I^2 = 56\%$; 23 studies, 1913 participants; low-certainty evidence downgraded for bias and indirectness) at the short term (up to two years) ([Analysis 1.2](#)).

Knee Society Score (KSS):

KSS were reported in 14 studies ([Abane 2015](#); [Abdel 2014](#); [Boonen 2013](#); [Giannotti 2019](#); [Hampton 2022](#); [Kosse 2017](#); [Kotela 2014](#); [Maus 2018](#); [Pfitzner 2014](#); [Pietsch 2014](#); [Stolarczyk 2018](#); [Teeter 2019](#); [Woolson 2014](#); [Yan 2014](#)). Most studies reported the data for the two components separately; hence, analysis of KSS knee and KSS functional components were performed separately. Time points for reported scores ranged from two months to five years, with the majority of studies reporting short-term scores (up to two years). [Hampton 2022](#) was the only study to report intermediate-term (two-five years) KSS knee scores and [Boonen 2013](#) was the only study to report intermediate-term KSS function. Five studies did not report standard deviations and these were imputed from a representative study. There were no significant differences for the KSS knee component (short term) from twelve studies ([Abane 2015](#); [Abdel 2014](#); [Giannotti 2019](#); [Hampton 2022](#); [Kosse 2017](#); [Kotela 2014](#); [Maus 2018](#); [Pfitzner 2014](#); [Pietsch 2014](#); [Stolarczyk 2018](#); [Woolson 2014](#); [Yan 2014](#)) (MD -0.66, 95% CI -2.03 to 0.7).

KSS functional component data from twelve studies ([Abane 2015](#); [Abdel 2014](#); [Boonen 2013](#); [Kosse 2017](#); [Kotela 2014](#); [Maus 2018](#); [Pfitzner 2014](#); [Pietsch 2014](#); [Stolarczyk 2018](#); [Teeter 2019](#); [Woolson 2014](#); [Yan 2014](#)) showed no clinically significant difference between PSG and CON groups (MD -1.33, 95% CI -3.64 to 0.99).

Oxford Knee Score (OKS):

Nine studies reported Oxford Knee Scores ([Abane 2015](#); [Attard 2019](#); [Boonen 2013](#); [Cucchi 2019](#); [Hampton 2022](#); [Huijbregts 2016](#); [Randelli 2019](#); [Turgeon 2019](#); [Yan 2014](#)). SDs were not reported in three studies and were imputed from a representative study. Five studies reported short-term OKS ([Abane 2015](#); [Attard 2019](#); [Boonen 2013](#); [Hampton 2022](#); [Huijbregts 2016](#); [Yan 2014](#)) showing little to no difference in final scores (MD 0.29, 95% CI -0.92 to 1.49, $P = 0.64$). Only two studies reported intermediate-term OKS ([Boonen 2013](#); [Hampton 2022](#)), again showing little to no difference in final scores (MD 0.20, 95% CI -3.32 to 3.72). Three studies reported change in OKS scores ([Cucchi 2019](#); [Randelli 2019](#); [Turgeon 2019](#)) with no clinically significant difference found between groups (MD -0.04, 95% CI -3.95 to 3.88).

Western Ontario and McMaster University Osteoarthritis Index (WOMAC):

A total of seven studies reported WOMAC scores ([Boonen 2013](#); [Calliess 2017](#); [Kotela 2014](#); [Pfitzner 2014](#); [Teeter 2019](#); [Zomar 2021](#)). Two studies did not report standard deviations which were imputed from representative studies. Three studies ([Calliess 2017](#); [Kotela 2014](#); [Tammachote 2018](#)) reported subscores or modified versions with different scales. There was no clinically significant difference between groups (SMD -0.19, 95% CI -0.52 to 0.13).

Knee injury and Osteoarthritis Outcome Score (KOOS):

Only two studies reported the KOOS score ([Abdel 2014](#); [Vundelinckx 2013](#)). Both studies reported individual KOOS subscales but only [Vundelinckx 2013](#) reported the total score. There was no clinically significant difference in any of the subscale scores between groups; KOOS pain (MD 4.99, 95% CI -2.85 to 12.82), KOOS symptoms (MD 1.77, 95% CI -2.42 to 5.96), KOOS ADL (MD 4.29, 95% CI -1.57 to 10.14), KOOS sports (MD 3.53, 95% CI -4.10 to 11.16), KOOS QoL (MD 5.05, 95% CI -3.73 to 13.83).

Radiographic outliers in Femorotibial Coronal Alignment (FTCA)

Twenty-nine studies reported on outliers in FTCA ([Abane 2015](#); [Abdel 2014](#); [Boonen 2013](#); [Chareancholvanich 2013](#); [Chotanaphuti 2014](#); [Gan 2015](#); [Hamilton 2013](#); [Hampton 2022](#); [Huijbregts 2016](#); [Kotela 2014](#); [Liao 2024](#); [Lionberger 2014](#); [Maus 2018](#); [Molicnik 2015](#); [Noble 2012](#); [Parratte 2013](#); [Pfitzner 2014](#); [Qiu 2017](#); [Roh 2013](#); [Sariali 2019](#); [Sun 2020](#); [Tammachote 2018](#); [Teeter 2019](#); [Turgeon 2019](#); [Van Leeuwen 2018](#); [Victor 2014](#); [Vide 2017](#); [Woolson 2014](#); [Yan 2014](#); [Zahn 2020](#)). PSGs may result in little to no difference in precision in lower limb alignment as measured by the proportion of outliers for femorotibial coronal angle (FTCA). At up to two years, 266/1208 participants (220 per 1000) in the conventional instrumentation group reported radiographic outlier events compared to 234/1204 participants (189 per 1000) in the PSG group (RR 0.86, 95% CI 0.68 to 1.10, $I^2 = 49\%$; 29 studies, 2412 participants; low-certainty evidence downgraded for bias and imprecision) ([Analysis 1.3](#)).

Pain

Mean overall pain was measured on a visual analogue scale (VAS) or numerical rating scale (NRS), or other measures of pain. Ten studies ([Abdel 2014](#); [Boonen 2013](#); [Cucchi 2019](#); [Kosse 2017](#); [Kotela 2014](#); [Pietsch 2014](#); [Stolarczyk 2018](#); [Turgeon 2019](#); [Van Leeuwen 2018](#); [Vundelinckx 2013](#)) with 715 participants reported pain scores, of which 357 participants were in the PSG group and 358 participants in the CON group. Low-certainty evidence downgraded for bias and indirectness (multiple varied scales) showed that PSGs may result

in little to no difference in pain. Mean pain was 17.4 points in the CON group and 15.8 points in the PSG group (SMD -0.09, 95% CI -0.23 to 0.06, back-translated to MD 1.52 points lower (3.88 lower, 1.01 higher; $I^2 = 0\%$, 10 studies, 715 participants) (Analysis 1.4).

Global assessment (participant-reported)

None of our included studies comparing PSG versus CON provided data on global assessment.

Total adverse events

Fourteen studies (Abdel 2014; Boonen 2013; Hampton 2022; Liao 2024; Kosse 2017; Kotela 2014; Pietsch 2014; Turgeon 2019; Vundelinckx 2013; Woolson 2014; Yan 2014; Zomar 2021) with 1169 participants have reported on the total number of adverse events following surgery. Low-certainty evidence downgraded for bias and imprecision showed that PSGs may result in little to no difference in adverse events. Total adverse events (infection, thrombosis) were reported in 60/590 participants (102 per 1000) in the conventional instrumentation group compared to 61/579 participants (99 per 1000) in the PSG group (RR 0.97, 95% CI 0.68 to 1.39, $I^2 = 5\%$, 14 studies, 1169 participants). Total adverse events included infection, thrombosis, death (Analysis 1.5).

Abdel 2014 reported two flexion contractures not requiring intervention, and one postoperative patella subluxation. Boonen 2013 reported two participants with haematoma, 16 participants with clinically poor range of movement under 90 degrees, two participants with persistent pain, one participant with persistent effusion, one participant with pulmonary embolus, one participant with deep infection, and one participant with a malaligned tibial component. Hampton 2022 reported two participants with leaking wounds, one participant with calf ulcer, two participants with postoperative urinary retention, and one participant with pulmonary embolus. Kosse 2017 reported three participants with knee flexion deficits, three participants with knee extension deficits, and two participants with prolonged effusion. Kotela 2014 reported one participant with delayed wound healing, one participant with hospital-acquired pneumonia, two participants with postoperative haematoma with delayed wound healing, and one participant with knee joint contracture. Liao 2024 reported one participant with poor wound healing requiring debridement. Pietsch 2014 reported two participants with anterior femoral notching. Turgeon 2019 reported one participant with extensor mechanism pain, two participants with superficial wound infections, one participant with arthrofibrosis, and two participants with deep infection. Vundelinckx 2013 reported one participant with flexion contracture, one participant with wound breakdown, and four participants with delayed wound healing. Woolson 2014 reported one participant with wound dehiscence, one participant with early tibial loosening, and two participants with deaths unrelated to surgery at three and four months postoperatively. Yan 2014 reported two participants with superficial wound infections, one participant with deep vein thrombosis, and one participant with an exacerbation of gout in a replaced knee. Zomar 2021 reported one participant with medial collateral ligament injury, one participant with pulmonary embolus, three participants with deep infections of the knee, one participant with chronic deep vein thrombosis, and three deaths postoperatively unrelated to surgery.

Re-operation rate (not involving implant change)

Ten studies (Boonen 2013; Hampton 2022; Huijbregts 2016; Kosse 2017; Kotela 2014; Pietsch 2014; Soriali 2019; Turgeon 2019; Vundelinckx 2013; Woolson 2014) with 843 participants reported on re-operation rates not involving revision of any implants. Low-certainty evidence downgraded for bias and imprecision showed that PSGs may result in little to no difference in re-operation rates. Re-operations were reported in 21/424 participants (50 per 1000) in the conventional instrumentation group compared to 16/419 participants (44 per 1000) in the PSG group (RR 0.87, 95% CI 0.45 to 1.68, $I^2 = 0\%$, 10 studies, 843 participants) Analysis 1.6.

Minor outcomes

Quality of life (QoL)

Six studies with 448 participants reported QoL scores as the EQ-5D score (with differing scales between studies) (Abdel 2014; Boonen 2013; Teeter 2019; Turgeon 2019; Van Leeuwen 2018; Zomar 2021). Low-certainty evidence from three studies showed that PSGs may lead to little or no difference in quality of life between groups on SF-12-MCS (MD -0.16, 95% CI -2.51 to 2.84) Analysis 1.7 (Abdel 2014; Huijbregts 2016; Teeter 2019).

Operative time

Low-certainty evidence from 25 studies with 2033 participants showed that PSGs may lead to little or no difference in operating time compared to CON (MD -2.79, 95% CI -5.66 to 0.08, $I^2 = 96\%$) (Analysis 1.8) (Abane 2015; Boonen 2013; Chareancholvanich 2013; Chotanaphuti 2014; Cucchi 2019; Ferrara 2015; Gan 2015; Hamilton 2013; Hampton 2022; Kosse 2017; Kotela 2014; Liao 2024; Maus 2018; Noble 2012; Pietsch 2014; Randelli 2019; Soriali 2019; Sun 2020; Tammachote 2018; Teeter 2019; Turgeon 2019; Van Leeuwen 2018; Woolson 2014; Yan 2014; Zomar 2021).

Blood loss

Moderate-certainty evidence from 14 studies with 1189 participants showed that use of PSGs probably leads to reduced surgical blood loss compared to CON (MD -66.47, 95% CI -93.30 to -39.64, $I^2 = 80\%$) (Abane 2015; Attard 2019; Boonen 2013; Chareancholvanich 2013; Gan 2015; Giannotti 2019; Hampton 2022; Kotela 2014; Liao 2024; Maus 2018; Noble 2012; Pietsch 2014; Sun 2020; Teeter 2019). Low-certainty evidence from four studies with 211 participants showed that use of PSGs may lead to a reduction in haematocrit change postoperatively compared to CON (SMD -0.88, 95% CI -1.63, -0.13, $I^2 = 83\%$) (Ferrara 2015; Molicnik 2015; Vide 2017; Woolson 2014). Low-certainty evidence from seven studies with 502 participants showed that the use of PSGs may result in a lower transfusion rate compared to CON (RR 0.7, 95% CI 0.5 to 1.0) (Analysis 1.9, Analysis 1.10, Analysis 1.11) (Chareancholvanich 2013; Ferrara 2015; Pietsch 2014; Tammachote 2018; Vide 2017; Vundelinckx 2013; Woolson 2014).

Range of motion

Low-certainty evidence from eight studies with 586 participants showed that the use of PSGs may not lead to better or worse postoperative knee flexion compared to CON (MD -0.42, 95% CI -2.07 to 1.23) (Lionberger 2014; Pietsch 2014; Sun 2020; Tammachote 2018; Van Leeuwen 2018; Vundelinckx 2013; Woolson 2014; Yan 2014). Low-certainty evidence from three studies with 258 participants showed that the use of PSGs may not lead to better or worse postoperative knee extension compared to CON

(MD 0.26, 95% CI -0.28 to 0.80) (Tammachote 2018; Van Leeuwen 2018; Vundelinckx 2013) (Analysis 1.12, Analysis 1.13).

Length of hospital stay

Low-certainty evidence from 13 studies with 1062 participants showed that use of PSGs may lead to a slightly reduced or similar length of hospital stay compared to CON (MD -0.23, 95% CI -0.46 to 0.00) (Abane 2015; Attard 2019; Chareancholvanich 2013; Ferrara 2015; Kotela 2014; Maus 2018; Noble 2012; Turgeon 2019; Van Leeuwen 2018; Vide 2017; Vundelinckx 2013; Woolson 2014; Zomar 2021) (Analysis 1.14).

Incision length

Two studies reported incision length (Noble 2012; Vundelinckx 2013); however, no standard deviations were reported, and further analysis was not possible.

Vundelinckx 2013 found no statistically significant differences in incision length in either flexion or extension in the control versus PSG group. Mean incision length in extension was 12.8 cm versus 13.6 cm, whilst for extension it was 17.4 cm and 17.8 cm respectively. Noble 2012 did not find significant differences between incision length between groups (mean 136 mm, range 120-160 versus mean 151.8 mm, range 120-175, $P = 0.14$).

Radiographic outliers in alignment for individual prosthetic components

See Additional Table 1.

Outliers in Femoral Coronal Alignment

Moderate-certainty evidence from 17 studies with 1495 participants showed that PSGs probably result in little to no difference in outliers for femoral coronal alignment (RR 0.77, 95% CI 0.53 to 1.12, $I^2 = 35\%$) (Abane 2015; Boonen 2013; Chareancholvanich 2013; Ferrara 2015; Hamilton 2013; Huijbregts 2016; Kotela 2014; Liao 2024; Pfitzner 2014; Qiu 2017; Roh 2013; Soriali 2019; Tammachote 2018; Van Leeuwen 2018; Victor 2014; Woolson 2014; Yan 2014).

Outliers in Tibial Coronal Alignment

Moderate-certainty evidence from 17 studies with 1495 participants showed that PSGs probably result in little to no difference in outliers for tibial coronal alignment (RR 1.29, 95% CI 0.84 to 2.00, $I^2 = 29\%$) (Abane 2015; Boonen 2013; Chareancholvanich 2013; Ferrara 2015; Hamilton 2013; Huijbregts 2016; Kotela 2014; Liao 2024; Pfitzner 2014; Qiu 2017; Roh 2013; Soriali 2019; Tammachote 2018; Van Leeuwen 2018; Victor 2014; Woolson 2014; Yan 2014).

Outliers in Femoral Sagittal Alignment

Moderate-certainty evidence from 11 studies with 1075 participants showed that PSGs probably result in little to no difference in outliers for femoral sagittal alignment (RR 1.03, 95% CI 0.83 to 1.26, $I^2 = 36\%$) (Abane 2015; Boonen 2013; Ferrara 2015; Hamilton 2013; Huijbregts 2016; Kotela 2014; Liao 2024; Roh 2013; Van Leeuwen 2018; Victor 2014; Yan 2014). Soriali 2019 reported on outliers in femoral sagittal alignment; however, the data were not clear from their results and therefore the study was not included in the analysis.

Outliers in Tibial Sagittal Alignment

Moderate-certainty evidence from 14 studies with 1233 participants showed that PSGs probably result in little to no difference in the number of outliers for tibial sagittal alignment (RR 1.17, 95% CI 0.82 to 1.68, $I^2 = 61\%$) (Abane 2015; Boonen 2013; Ferrara 2015; Hamilton 2013; Huijbregts 2016; Kotela 2014; Liao 2024; Pfitzner 2014; Qiu 2017; Roh 2013; Van Leeuwen 2018; Victor 2014; Woolson 2014; Yan 2014). Soriali 2019 reported on outliers in tibial sagittal alignment; however, the data were not clear from their results and were therefore not included in the analysis.

Outliers in Femoral Rotational Alignment

Low-certainty evidence from ten studies with 922 participants showed that PSGs may result in decreased numbers of outliers for femoral rotational alignment compared to CON (RR 0.63, 95% CI 0.41 to 0.99) (Chotanaphuti 2014; Huijbregts 2016; Liao 2024; Pfitzner 2014; Roh 2013; Soriali 2019; Tammachote 2018; Van Leeuwen 2018; Victor 2014; Woolson 2014) (Analysis 1.16; Analysis 1.17; Analysis 1.18; Analysis 1.19; Analysis 1.20).

PSG versus computer-assisted surgical NAVigation (NAV)

Major outcomes

See Summary of findings 2

Survival rate (any revision surgery to change a prosthetic component)

None of our included studies comparing PSG versus NAV included data on survival rates of prostheses.

Function

Two studies (Lionberger 2014; Yan 2014) with 120 participants had reported function using the Knee Society Score (KSS). Low-certainty evidence showed that PSGs may result in little to no difference in function (0 to 100, 100 best function) (MD 5.00 points higher, 95% CI 1.31 lower to 11.31 higher). The evidence was downgraded for bias and imprecision. There was no clinically significant improvement in KSS Clinical Score (0 to 100, 100 best score) (MD 2.25, 95% CI -0.89 to 5.39; $I^2 = 0\%$, 2 studies, 120 participants) (Analysis 2.1).

Radiographic outliers in FemoroTibial Coronal Alignment (FTCA)

Three studies (270 participants) reported on outliers in FTCA (Lionberger 2014; Yan 2014; Zahn 2020). Outlier events occurred in 28 of 135 participants (20.7%) in the PSG group compared to 13 of 135 participants (9.6%) in the NAV group. Low-certainty evidence (downgraded for bias and imprecision) showed that use of PSGs may result in worse precision in lower limb alignment compared to NAV (RR 2.04, 95% CI 0.87 to 4.82; $I^2 = 44\%$, 3 studies, 270 participants) (Analysis 2.2).

Pain

None of our included studies comparing PSG versus NAV provided data on pain measures.

Global assessment (participant-reported)

None of our included studies comparing PSG versus NAV provided data on global assessment.

Total adverse events

None of our included studies comparing PSG versus NAV provided data on total adverse events.

Re-operation rate (not involving implant change)

None of our included studies comparing PSG versus NAV provided data on re-operation rates.

Minor outcomes

None of our included studies provided data on QoL, blood loss or length of hospital stay.

Operative time

Low-certainty evidence from two studies with 120 participants showed that PSGs may result in shorter surgical times compared to NAV (MD -21.54, 95% CI -27.93 to -15.16) ([Analysis 2.3](#)) ([Lionberger 2014](#); [Yan 2014](#)).

Range of motion

Low-certainty evidence from two studies with 120 participants showed that PSGs may result in no difference in flexion postoperatively compared to NAV (MD 1.12, 95% CI -5.05 to 7.28) ([Analysis 2.5](#)) ([Lionberger 2014](#); [Yan 2014](#)).

Incision length

[Lionberger 2014](#) reported the mean incision length in the PSG group as 12.4 cm with 12.8 cm in the NAV group. They did not report standard deviations or standard errors.

Radiographic outliers in alignment for individual prosthetic components

See additional [Table 2](#). Very low-certainty evidence from two studies with 133 participants ([Ollivier 2016](#); [Yan 2014](#)) showed that we are uncertain whether PSGs result in little to no difference in outliers in femoral coronal alignment (RR 1.09, 95% CI 0.30 to 3.96); in tibial coronal alignment (RR 1.79, 95% CI 0.42 to 7.55); in femoral sagittal alignment (RR 3.19, 95% CI 0.57 to 17.77); or in tibial sagittal alignment compared to NAV (RR 2.96, 95% CI 0.90 to 9.70) ([Analysis 2.7](#); [Analysis 2.10](#); [Analysis 2.9](#); [Analysis 2.8](#)).

None of our included studies comparing PSG with NAV had data on outliers in femoral rotational alignment.

Subgroup analysis of imaging modality used for PSG production compared to conventional cutting guides

A subgroup analysis of radiological outliers using CT-based and MRI-based PSGs was performed using a fixed-effects model ([Analysis 3.5](#); [Analysis 3.6](#); [Analysis 3.7](#); [Analysis 3.8](#); [Analysis 3.9](#)).

Outliers in FemoroTibial Coronal Alignment (FTCA)

Ten studies including 627 participants showed that the use of CT-based PSGs resulted in minimal or no difference in outlier rates compared to CON (RR 0.91, 95% CI 0.68 to 1.22) ([Chotanaphuti 2014](#); [Hamilton 2013](#); [Kotela 2014](#); [Pfitzner 2014](#); [Qiu 2017](#); [Roh 2013](#); [Sariali 2019](#); [Sun 2020](#); [Victor 2014](#); [Woolson 2014](#)).

Eighteen studies with 1568 participants showed that MRI-based PSGs resulted in minimal or no difference in outlier rates compared to CON (RR 0.93, 95% CI 0.77 to 1.13) ([Abane 2015](#); [Abdel 2014](#); [Boonen 2013](#); [Chareancholvanich 2013](#); [Hampton 2022](#); [Huijbregts 2016](#); [Maus 2018](#); [Molicnik 2015](#); [Noble 2012](#); [Pfitzner 2014](#); [Tammachote 2018](#); [Teeter 2019](#); [Turgeon 2019](#); [Van Leeuwen 2018](#); [Victor 2014](#); [Vide 2017](#); [Yan 2014](#); [Zahn 2020](#)).

Outliers in Femoral Coronal Alignment

Eight studies with 460 participants showed that CT-based PSGs may result in lower outlier rates for femoral coronal alignment compared to CON (RR 0.58, 95% CI 0.37 to 0.90) ([Hamilton 2013](#); [Kotela 2014](#); [Pfitzner 2014](#); [Qiu 2017](#); [Roh 2013](#); [Sariali 2019](#); [Victor 2014](#); [Woolson 2014](#)).

Ten studies with 928 participants showed that MRI-based PSGs resulted in minimal or no difference in outlier rates compared to CON (RR 0.87, 95% CI 0.62 to 1.23) ([Abane 2015](#); [Boonen 2013](#); [Chareancholvanich 2013](#); [Ferrara 2015](#); [Huijbregts 2016](#); [Pfitzner 2014](#); [Tammachote 2018](#); [Van Leeuwen 2018](#); [Victor 2014](#); [Yan 2014](#)).

Outliers in Tibial Coronal Alignment

Eight studies with 460 participants showed that CT-based PSGs may lead to higher outlier rates compared to CON (RR 1.62, 95% CI 0.99 to 2.63) ([Hamilton 2013](#); [Kotela 2014](#); [Pfitzner 2014](#); [Qiu 2017](#); [Roh 2013](#); [Sariali 2019](#); [Victor 2014](#); [Woolson 2014](#)).

Ten studies with 928 participants showed that MRI-based PSGs may or may not result in higher outlier rates compared to CON (RR 1.23, 95% CI 0.78 to 1.95) ([Abane 2015](#); [Boonen 2013](#); [Chareancholvanich 2013](#); [Ferrara 2015](#); [Huijbregts 2016](#); [Pfitzner 2014](#); [Tammachote 2018](#); [Van Leeuwen 2018](#); [Victor 2014](#); [Yan 2014](#)).

Outliers in Femoral Sagittal Alignment

Four studies with 268 participants showed that CT-based PSGs resulted in minimal or no difference in outlier rates compared to CON (RR 1.09, 95% CI 0.79 to 1.50) ([Hamilton 2013](#); [Kotela 2014](#); [Roh 2013](#); [Victor 2014](#)).

Seven studies with 701 participants showed that MRI-based PSGs resulted in minimal or no difference in outlier rates (RR 1.01, 95% CI 0.85 to 1.20) ([Abane 2015](#); [Boonen 2013](#); [Ferrara 2015](#); [Huijbregts 2016](#); [Van Leeuwen 2018](#); [Victor 2014](#); [Yan 2014](#)).

Outliers in Tibial Sagittal Alignment

Seven studies with 380 participants showed that CT-based PSGs may or may not result in higher outlier rates compared to CON (RR 1.28, 95% CI 0.91 to 1.8) ([Hamilton 2013](#); [Kotela 2014](#); [Pfitzner 2014](#); [Qiu 2017](#); [Roh 2013](#); [Victor 2014](#); [Woolson 2014](#)).

Eight studies with 746 participants showed that MRI-based PSGs resulted in minimal or no difference in outlier rates (RR 1.13, 95% CI 0.90 to 1.44) ([Abane 2015](#); [Boonen 2013](#); [Ferrara 2015](#); [Huijbregts 2016](#); [Pfitzner 2014](#); [Van Leeuwen 2018](#); [Victor 2014](#); [Yan 2014](#)).

Outliers in Femoral Rotational Alignment

Five studies with 287 participants showed that PSGs may result in lower outlier rates for femoral rotational alignment compared to CON (RR 0.54, 95% CI 0.33 to 0.87) ([Pfitzner 2014](#); [Roh 2013](#); [Sariali 2019](#); [Victor 2014](#); [Woolson 2014](#)).

Five studies with 362 participants showed that MRI-based PSGs resulted in lower or no difference in outlier rates (RR 0.80, 95% CI 0.51 to 1.27) ([Huijbregts 2016](#); [Pfitzner 2014](#); [Tammachote 2018](#); [Van Leeuwen 2018](#); [Victor 2014](#)).

DISCUSSION

Summary of main results

This systematic review included 44 RCTs with 3664 participants, comparing patient-specific cutting guides with conventional instrumentation for total knee arthroplasty in 40 trials and with computer-assisted navigation in two trials. Two trials compared patient-specific cutting guides to both conventional instrumentation and computer-assisted navigation.

There are several main findings.

Firstly, the use of patient-specific cutting guides leads to no benefit in survival of implant, pain, function, or increased adverse events and re-operation rates when compared to conventional instrumentation or to computer-assisted navigation. The certainty of evidence was low and there were limited studies with intermediate or long-term follow-up.

Secondly, with low-certainty evidence, radiological outcomes show no meaningful difference between patient-specific cutting guides and conventional instrumentation in the proportion of outliers for overall limb alignment, femoral or tibial component coronal or sagittal alignment. There was low-certainty evidence that use of patient-specific cutting guides may lead to lower outlier rates for femoral rotation compared to conventional instrumentation. Clinically, low-certainty evidence that may suggest improved accuracy in only one out of six measured radiological alignment outcomes using patient-specific cutting guides cannot justify their use over conventional instrumentation.

Thirdly, patient-specific cutting guides may reduce surgical blood loss (MD -63.12, 95% CI -95.02 to -31.22), lower transfusion events (RR 0.7, 95% CI 0.5 to 1.0) and lead to a shorter length of hospital stay compared to conventional instrumentation (MD -0.23, 95% CI -0.46 to 0.00). The certainty of evidence for these outcomes was low, downgraded for detection bias and imprecision. The clinical relevance of these findings is uncertain given the unstated use of anti-fibrinolytics and two studies reporting very high blood loss.

Fourthly, the use of patient-specific guides may result in worse precision in overall limb alignment (FTCA) (RR 2.04, 95% CI 0.87 to 4.82) compared with computer-assisted navigation but the certainty of evidence for this was low. No differences were observed for the other alignment parameters. Clinically, the use of computer-assisted navigation resulted in increased surgical time compared to patient-specific cutting guides (MD -21.54, 95% CI -27.93 to -15.16).

Finally, subgroup analysis of the imaging modality utilised for patient-specific cutting guides production showed that CT-based guides produced fewer outliers compared to conventional instrumentation for femoral component rotational (RR 0.54, 95% CI 0.33 to 0.87) and femoral component coronal alignment (RR 0.58, 95% CI 0.37 to 0.90). However, outliers in tibial coronal alignment were increased when using CT-based patient-specific cutting guides (RR 1.62, 95% CI 0.99 to 2.63). No other differences were found between groups based on imaging modalities used for patient-specific cutting guides. The certainty of evidence was low.

Overall completeness and applicability of evidence

Using broad search criteria and a systematic search of databases, we have, to the best of our knowledge, identified all eligible

RCTs comparing patient-specific cutting guides to conventional instrumentation or computer-assisted navigation, with the inclusion of 44 studies in the meta-analysis. Overall, studies included participants and treatments that are reflective of modern clinical practice and, hence, the evidence generated from this review can be broadly applied. There was some variation in the eligibility criteria of trials according to the age of participants and pathophysiology. A number of studies included only primary OA, while others included both primary and secondary causes of OA such as inflammatory arthritis and post-traumatic arthritis. None of the review authors were involved in the conduct of the included trials.

When comparing patient-specific cutting guides to conventional instrumentation, a number of major outcomes, including survival of implant, pain, global assessment, adverse events and re-operation rate were only reported in few studies. Survival of the implant was reported in nine trials; however, the length of follow-up was short, with only two studies reporting intermediate-term survival and none reporting long-term survival; hence, it is difficult to draw clinically meaningful conclusions about this outcome. When analysing the harms of patient-specific cutting guides (adverse events, re-operation rate), it is likely that our estimates are unreliable due to a combination of low event rates, inadequately sized trials and only a few trials reporting on serious adverse events.

A large proportion of studies assessed functional outcomes by patient-reported outcome scores. Due to the considerable variability in the outcome scores utilised between studies, we performed an additional ad hoc subgroup analysis for each of the commonly used individual scores. As each individual scoring system assesses different aspects of function, we hoped that this would aid in clinical interpretation of this outcome.

The analysis was comprehensive, particularly when determining the precision of radiological alignment when patient-specific cutting guides were compared to conventional instrumentation. Radiological alignment of components was analysed in three planes – coronal, sagittal and rotational, in addition to overall limb alignment. The number of outliers in radiological alignment was reported as this is clinically more relevant in TKA alignment. Reducing variance (or improving precision) in achieving a desired target is what most clinicians are aiming to achieve.

The reduction in surgical blood loss using patient-specific cutting guides may be statistically significant as there were also differences in transfusion requirements. However, only a few studies reported data on transfusion requirements, with considerable variability between these studies. Furthermore, the transfusion rates from a few of these studies ([Chareancholvanich 2013](#); [Tammachote 2018](#)) are considerably higher than expected and may not reflect modern day management of surgical blood loss where the use of peri-operative tranexamic acid is routine.

When analysing patient-specific cutting guides compared to computer-assisted navigation, there were only four included studies with significantly fewer participants and, as such, the strength of these findings may be considered less robust. A number of major outcome parameters, including survival of implant, pain, global assessment, adverse events and re-operation rate, were not reported by any of the included studies. Therefore, we cannot draw

any conclusions regarding the clinical outcomes of patient-specific cutting guides compared to computer-assisted navigation.

Concerning outliers for femorotibial coronal alignment (FTCA) at the short term (< two-year follow-up) with three studies (270 participants), PSGs may result in overall worse precision of bony cuts aligned in this plane than computer navigation (20.7% PSG versus 9.6%), but given the low number of studies and patient numbers, we cannot draw any certain conclusions on this.

From the studies included in this Cochrane review, we note the difference in sample sizes and length of follow-up of many level one RCTs in comparison to national arthroplasty registries. This poses difficulty in considering important ‘harmful outcomes’ such as revision rates from the studies, which often do not have the longevity in outcomes or follow-up to make meaningful conclusions about revisions. It is important to acknowledge this and bring in additional registry data to comment upon the established study findings and interpretations.

Our efforts to collect missing data from the authors of published studies were unsuccessful after repeated attempts.

Quality of the evidence

The quality of evidence was assessed using the GRADE approach and ranged from moderate to low certainty. Due to the lack of participant blinding in the majority of included studies, we deemed the risk of performance and detection bias to be high and have therefore downgraded the certainty of our findings for all of our results to reflect this.

Patient-specific cutting guides versus conventional instrumentation

We downgraded the evidence for survival of the implant and for radiographic lower limb alignment to low certainty due to risk of performance and detection biases and imprecision. We downgraded the evidence for function and pain to low certainty due to risk of performance and detection biases and for serious indirectness, as different scales were used across studies resulting in potential skewing of data. We downgraded the evidence for adverse events and re-operation rates to low certainty, due to performance and detection biases and serious imprecision as there were wide 95% confidence intervals with very few events and likely reporting bias. We downgraded the evidence on outliers for femoral rotation due to serious imprecision, as the confidence interval crossed the line of no difference. No studies reported global assessment.

Concerning minor outcomes, the quality of evidence for operation time, blood loss, haematocrit, transfusion requirement, length of stay, range of motion and incision length was downgraded to low certainty due to risk of performance and detection biases and imprecision.

Patient-specific cutting guides versus computer-assisted navigation

We downgraded the evidence for function to low certainty, due to risk of performance and detection biases and for serious indirectness as, out of the 41 studies, only two studies provided these data. We downgraded the evidence to low certainty for radiographic alignment parameters due to risk of performance and detection biases and for serious imprecision as there were wide confidence intervals due to few events. Alignment measured

by outliers in FTCA, femoral coronal alignment, tibial coronal alignment, femoral sagittal alignment, and tibial sagittal alignment were downgraded due to serious imprecision as there were wide CIs with few events and the confidence interval crossed the line of no difference. Concerning minor outcomes, the quality of evidence for operation time, blood loss, haematocrit, transfusion requirement, length of stay, range of motion and incision length was downgraded to low certainty due to risk of performance and detection biases and imprecision.

Potential biases in the review process

We identified, to our knowledge, all relevant trials that met the eligibility criteria of our review through a comprehensive and systematic search of the major electronic databases and trial registries. Up to three separate reviewers were involved in the study selection process and performed independent risk of bias analysis. None of the review authors have been involved in the conduct of the included trials.

The risk of publication bias is likely to be low as there were few studies that exhibited selective reporting. Numerous studies did not report SDs for continuous data and attempts to contact the authors were made to retrieve these data. Imputed SDs from the most representative studies with the least bias were utilised in accordance with Cochrane guidelines. We did not identify any ongoing trials comparing patient-specific cutting guides to CON or NAV.

Agreements and disagreements with other studies or reviews

We identified 13 previously published meta-analyses comparing patient-specific cutting guides with conventional cutting guides for TKA (Alcelik 2017; Cavaignac 2015; Fu 2015; Gong 2019; Huijbregts 2016; Jiang 2015; Shen 2015; Kizaki 2019; Lin 2020; Thienpont 2017; Voleti 2014; Mannan 2016; Russell 2014).

Seven of these studies included studies that were not randomised controlled trials (Cavaignac 2015; Fu 2015; Kizaki 2019; Russell 2014; Shen 2015; Thienpont 2017; Voleti 2014). Several of the meta-analyses evaluated radiological alignment only (Alcelik 2017; Cavaignac 2015; Jiang 2015; Mannan 2016; Russell 2014). We did not identify any meta-analysis comparing patient-specific cutting guides with computer-assisted surgical navigation.

Voleti 2014 performed a meta-analysis of nine trials, five of which were RCTs, and four of which were retrospective studies. They included four trials that appear in this review (Boonen 2013; Chareancholvanich 2013; Noble 2012; Vundelinckx 2013). Dossett 2014 was excluded from our review as it evaluated kinematic alignment with the use of patient-specific cutting guides and mechanical alignment with conventional guides, making a direct comparison inappropriate. The authors found no difference in coronal or sagittal alignment outliers and operative times when comparing patient-specific guides to conventional guides, which is in agreement with our review. However, contrary to our findings, they reported no statistically significant difference in blood loss and transfusion requirements.

Russell 2014 analysed five RCTs and two conference abstract data sets, including five studies that appear in this review (Chareancholvanich 2013; Hamilton 2013; Noble 2012; Roh 2013; Victor 2014). Chen 2014 was not included in our analysis as it was

a non-randomised study. Data from the Hamadouche trial were later published as [Abane 2015](#) which is included in our analysis. They conclude that patient-specific cutting guides did not offer a significant advantage over conventional guides with regard to postoperative mechanical alignment of the limb, which agrees with our findings.

[Jiang 2015](#) performed a meta-analysis of 18 studies, including seven studies that appear in this review ([Boonen 2013](#); [Chareancholvanich 2013](#); [Chotanaphuti 2014](#); [Hamilton 2013](#); [Noble 2012](#); [Roh 2013](#); [Victor 2014](#)). They concluded that there were no statistical differences in outliers of mechanical axis, femoral component coronal, tibial component coronal and femoral component rotation between patient-specific guides compared with conventional instrumentation, which agrees with our findings.

[Fu 2015](#) analysed 10 studies, eight of which were included in our review ([Boonen 2013](#); [Chareancholvanich 2013](#); [Chotanaphuti 2014](#); [Hamilton 2013](#); [Noble 2012](#); [Pietsch 2014](#); [Roh 2013](#); [Victor 2014](#)). [Ng 2012](#) was not included in our analysis as it is a retrospective study. The authors found no difference between patient-specific cutting guides and conventional instrumentation in terms of overall limb alignment and alignment of the femoral component in the coronal and sagittal planes, which is in keeping with our findings. They found a statistically significant increase in outliers for tibial component alignment in the coronal and sagittal plane when using patient-specific cutting guides, whereas our analysis found no difference. Contrary to our findings, they also found that operative time was significantly shorter with patient-specific cutting guides than with conventional instrumentation. The authors found insufficient data for analysis of blood loss or patient-reported outcomes.

[Cavaignac 2015](#) analysed 15 studies, including six RCTs, all of which appear in our review ([Chareancholvanich 2013](#); [Hamilton 2013](#); [Roh 2013](#); [Victor 2014](#); [Woolson 2014](#)). They found that patient-specific cutting guides did not improve the mechanical axis as measured by HKA and did not reduce the risk of outliers in mechanical axis alignment when compared to conventional instrumentation, which is consistent with our findings. The study found that patient-specific cutting guides reduced the risk of outliers in femoral component coronal alignment when looking at all studies (RCTs, quasi-RCTs, and studies with historic cohorts), but no difference between the two groups when only the data from RCTs were analysed. They concluded that using patient-specific cutting guides does not provide superior accuracy compared to conventional instrumentation during TKA.

[Mannan 2016](#) analysed six RCTs, all of which were included in our review ([Chotanaphuti 2014](#); [Pfitzner 2014](#); [Roh 2013](#); [Silva 2014](#); [Victor 2014](#); [Woolson 2014](#)). They concluded that patient-specific cutting guides increased accuracy of femoral component rotation by reducing the risk of outliers and these findings are supported by our review.

[Huijbregts 2016](#) analysed 21 RCTs and quasi-RCTs, 18 of which were included in this review ([Abane 2015](#); [Abdel 2014](#); [Boonen 2013](#); [Chareancholvanich 2013](#); [Chotanaphuti 2014](#); [Ferrara 2015](#); [Hamilton 2013](#); [Huijbregts 2016](#); [Kotela 2014](#); [Molicnik 2015](#); [Noble 2012](#); [Pietsch 2014](#); [Roh 2013](#); [Silva 2014](#); [Victor 2014](#); [Vundelinckx 2013](#); [Woolson 2014](#); [Yan 2014](#)). [Ng 2012](#) was a retrospective study while [Chen 2014](#) was a non-randomised study, and both were excluded from our analysis. The authors found that use of patient-

specific cutting guides did not reduce the outlier risk ratio for overall limb alignment and femoral component alignment when compared to conventional instrumentation, which is in keeping with our review. However, they report that the proportion of tibial component coronal and sagittal plane outliers was higher in the patient-specific cutting guides group, while this review found no difference. They found no difference in blood loss or transfusion rates, whereas this review found a clinically meaningful reduction in surgical blood loss. The authors reported no difference in overall complications, but a reduction in length of hospital stay in the patient-specific cutting guides group, which agrees with this review.

[Alcelik 2017](#) analysed 12 RCTs, all of which were included in our analysis ([Abane 2015](#); [Abdel 2014](#); [Boonen 2013](#); [Chareancholvanich 2013](#); [Chotanaphuti 2014](#); [Hamilton 2013](#); [Kotela 2014](#); [Pfitzner 2014](#); [Roh 2013](#); [Victor 2014](#); [Woolson 2014](#); [Yan 2014](#)). They concluded that there was no significant benefit from using patient-specific cutting guides compared to conventional instrumentation in primary knee arthroplasty to aid in the overall mechanical axis or positioning of either the tibial or femoral components, which is in agreement with this review. They found sagittal plane tibial component positioning was worse in the patient-specific cutting guides group, whilst we found no meaningful difference.

[Thienpont 2017](#) performed a meta-analysis of 20 RCTs and 24 observational studies, 18 of which appear in this review ([Abane 2015](#); [Abdel 2014](#); [Boonen 2013](#); [Chareancholvanich 2013](#); [Chotanaphuti 2014](#); [Ferrara 2015](#); [Hamilton 2013](#); [Kotela 2014](#); [Molicnik 2015](#); [Noble 2012](#); [Pietsch 2014](#); [Roh 2013](#); [Silva 2014](#); [Victor 2014](#); [Vide 2017](#); [Vundelinckx 2013](#); [Woolson 2014](#); [Yan 2014](#)). Two of the quoted RCTs were duplicate publications of previous studies. They concluded that the risk of mechanical axis malalignment and femoral component coronal plane malalignment was significantly lower in TKAs using patient-specific cutting guides, while the risk of tibial component malalignment both in coronal and sagittal plane was significantly higher with patient-specific cutting guides. We did not find any significant difference between patient-specific cutting guides compared to conventional instrumentation regarding mechanical axis alignment or individual component alignment in coronal and sagittal planes. They reported patient-specific cutting guides yielded relatively less blood loss and lower transfusion requirements, which is consistent with our findings. To evaluate function, they analysed the Knee Society Functional Score, which was in favour of patient-specific cutting guides; the Knee Society Knee Score was similar between groups. In this review, we found no clinically meaningful difference between groups for either score.

[Kizaki 2019](#) included 38 studies, 24 of which were RCTs and 14 of which were non-RCTs. They included 23 studies that are in this review ([Abane 2015](#); [Abdel 2014](#); [Boonen 2013](#); [Chareancholvanich 2013](#); [Chotanaphuti 2014](#); [Ferrara 2015](#); [Noble 2012](#); [Hamilton 2013](#); [Huijbregts 2016](#); [Kosse 2017](#); [Kotela 2014](#); [Maus 2018](#); [Molicnik 2015](#); [Pietsch 2014](#); [Pfitzner 2014](#); [Roh 2013](#); [Stolarczyk 2018](#); [Tammachote 2018](#); [Van Leeuwen 2018](#); [Vide 2017](#); [Vundelinckx 2013](#); [Woolson 2014](#); [Yan 2014](#)), but did not include five eligible studies at the time of their search ([Calliess 2017](#); [De Vloo 2017](#); [Qiu 2017](#); [Silva 2014](#); [Victor 2014](#)) and their conclusions are largely in agreement with this review. They reported that patient-specific cutting guides do not improve patient-reported outcome

measures, surgery time, and complication rates as compared to conventional TKA. They found that patient-specific cutting guides reduce surgical blood loss with a small effect; however, unlike our review, this was not enough to reduce transfusion rates.

Gong 2019 included 17 studies that appear in this review (Abane 2015; Boonen 2013; Chareancholvanich 2013; Hamilton 2013; Huijbregts 2016; Kotela 2014; Kosse 2017; Maus 2018; Pietsch 2014; Roh 2013; Silva 2014; Van Leeuwen 2018; Victor 2014; Vide 2017; Vundelinckx 2013; Woolson 2014; Yan 2014) and drew similar conclusions to this review. This meta-analysis included 23 randomised controlled trials; however, several of the quoted studies were duplicate publications of previous studies. They reported that patient-specific cutting guides have advantages in axial alignment of the femoral component and peri-operative blood loss relative to conventional instrumentation, which is in keeping with our findings. No significant differences were found between patient-specific cutting guides and conventional instrumentation with respect to the number of outliers of the other alignment parameters, which is again supported by our review. They found that there was a small but significant reduction in operative time with patient-specific cutting guides, which differs from our findings.

Lin 2020 analysed 29 RCTs, 24 of which appear in this review (Abane 2015; Abdel 2014; Boonen 2013; Chareancholvanich 2013; Chotanaphuti 2014; Cucchi 2019; De Vloo 2017; Hamilton 2013; Huijbregts 2016; Kosse 2017; Kotela 2014; Maus 2018; Molcnik 2015; Noble 2012; Pfizner 2014; Pietsch 2014; Roh 2013; Silva 2014; Van Leeuwen 2018; Victor 2014; Vide 2017; Vundelinckx 2013; Woolson 2014; Yan 2014). Several of the included studies are duplicate publications of previous studies, the results of which are included in this review. There were five eligible studies at the time of their search that were not included (Calliess 2017; Ferrara 2015; Qiu 2017; Stolarczyk 2018; Tammachote 2018). The authors reported that the use of patient-specific cutting guides when compared to conventional instrumentation did not improve the alignment of the mechanical axis (FTCA) or offer any significant benefit regarding risk of complication and function using the Oxford Knee Score, which is in agreement with our review. They also supported the findings from our review that patient-specific cutting guides resulted in reduced blood loss. However, they reported that patient-specific cutting guides resulted in a statistically improved KSS and a reduction in surgical time, which was not found in this review.

Shen 2015 performed a meta-analysis of 14 studies consisting of seven randomised controlled trials and seven non-randomised controlled trials, six of which appear in this review (Boonen 2013; Chareancholvanich 2013; Chotanaphuti 2014; Hamilton 2013; Parratte 2013; Victor 2014). They found that the use of patient-specific implants was not likely to improve the accuracy of component placement compared with conventional instrumentation. No statistical differences with respect to the outliers of mechanical axis, coronal femoral component, sagittal femoral component, femoral component rotation placement, operative time, blood loss, and length of hospital stay between groups was identified.

McAuliffe 2019 analysed 83,823 total knee replacement procedures from the Australian Orthopaedic Association National Joint Replacement Registry. The primary aim of the study was to compare the revision rate, or described time to first revision, in TKA cases amongst three groups; Image-Derived Instrumentation (IDI), computer navigation, or 'other' TKA. All cases were performed

for osteoarthritis. Overall, IDI TKA did not demonstrate an overall difference in early to mid-term revision rates compared with standard implantation methods. The five-year cumulative per cent revision for IDI was 3.3% versus 2.4% for the computer-navigated group, and 2.5% for the 'other' or 'standard' TKA group. Concerning IDI, posterior stabilised TKA using IDI had a significantly higher rate of revision at over three months, as did patients under 65 years in the IDI group. Finally, the risk of patella revision was more likely in the IDI group. The inference from these results suggest that IDI should be used with caution in these groups and further study is recommended. This registry data and analysis meets our meta-analysis in that there is no benefit in survival of the implant to follow-up times between the two groups.

AUTHORS' CONCLUSIONS

Implications for practice

This review of 44 randomised controlled trials including 3664 participants indicates that patient-specific cutting guides provide little to no benefit over conventional cutting guides or computer-assisted surgical navigation in primary total knee arthroplasty.

There was no benefit in terms of early survival of the implant, adverse clinical outcomes including venothrombotic events and infection, function and pain. The precision of radiological alignment of prosthetic components using patient-specific guides was not superior to conventional instrumentation, except for femoral component rotation. Use of patient-specific guides may lead to a clinically meaningful reduction in surgical blood loss and a slightly shorter hospital stay compared to conventional instrumentation.

Implications for research

There is currently no reliable long-term data assessing the benefits and harms of PSGs. Long-term survival of primary TKA using PSGs cannot be clearly defined from this analysis and further research in this area is required. Intermediate to long-term data on the clinical and functional outcomes of PSGs remain sparse and larger studies in this area are required to draw robust conclusions. Overall, outcomes from the included studies were of low certainty evidence. Further studies using well-designed randomised controlled trials with standardised outcome tools are required to minimise bias and improve the certainty of evidence. Long-term follow-up of participants in trials or data from prospectively designed registries is important to provide more accurate estimates of the risk of adverse events.

Furthermore, the individual expertise and caseload of surgeons using PSG may be an area for further study. It could be theorised from the randomised controlled trials reviewed that the performing surgeons were significantly 'high-volume' surgeons utilising PSG which may not necessarily offer or reveal advantages in ease of use over junior or lower-volume surgeon counterparts.

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- Managing Editor (managed the central editorial service): Jessica R. Thomas, Central Editorial Service;
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* Indicates the major publication for the study

CHARACTERISTICS OF STUDIES

Characteristics of included studies [ordered by study ID]

Abane 2015

Study characteristics	
Methods	Study Design: parallel RCT Country: France Setting: Hôpital Cochin, Paris, France and Universitaire de Clermont Ferrand, France Length of Follow-up: 3 months
Participants	Inclusion Criteria - age between 18 and 85 years - primary and secondary osteoarthritis of knee - ability to understand the information given - entitlement to social security Exclusion Criteria - active or suspected sepsis - tumour around the knee - previous TKA - presence of hardware that could cause artefacts on MRI - any contraindication to MRI - any extraarticular deformity that required osteotomy around the knee in conjunction with TKA - social circumstances which could impair follow-up Number of participants randomised: 140 - Intervention: Patient-specific cutting guides N = 70 - Control: Conventional instrumentation N = 7 Number of participants who received allocated intervention: 140 Number of dropouts: 14/140 (10%) - Intervention: 11/70 (15.7%) - Control: 3/70 (4.3%) BMI - Control group: 28.6 - Intervention group: 28.8 Preoperative mechanical axis

Abane 2015 (Continued)

- Control group: 174.1
- Intervention group: 176.1

Interventions	Control group: Conventional instrumentation Intervention group: Patient-specific cutting guides Type of implant: Posterior stabilised Bearing design: Fixed bearing Mode of fixation: Cemented Patellar resurfacing: performed in all patients Preop imaging for designing PSG: MRI Implant Brand: Genesis II PSG Brand: Visionaire Intraoperative Data • Number of times PSG was abandoned: two (tibia alone) • Reasons for abandonment: fitted poorly at the time of operation • Size correlation between planned component size and actual component used: • Same Size: Femur - 49/70, Tibia - 50/70, Insert thickness - 53/70 • Within One Size: Femur - 21/70, Tibia - 20/70, Insert thickness - 12/70 • Within two sizes: Femur - 0/70, Tibia - 0/70, Insert thickness - 5/70 • Any other relevant intraoperative data: Nil Surgeon experience and pre-training: Adequate - Quote from the study stated: "Each had performed at least ten TKAs using the PSCG system, and was familiar with Genesis II TKA".
Outcomes	Outcomes assessed: Intraoperative, radiographs at 3 months follow-up Imaging modality used for postoperative measurement: Radiograph Outcomes used in this review: 1. Functional outcome scores 2. Radiographic alignment of component 3. Number of outliers in radiographic alignment 4. Blood loss 5. Length of hospital stay 6. Operative time 7. Total adverse events Outcomes not used in this review: Nil
Notes	No SDs available for KSS, OKS, blood loss, operative time, length of stay. SDs were imputed from relevant studies.
Risk of bias	
Bias	Authors' judgement Support for judgement

Abane 2015 (Continued)

Random sequence generation (selection bias)	Unclear risk	Randomisation procedure not reported
Allocation concealment (selection bias)	Unclear risk	Allocation concealment not reported
Blinding of participants and personnel (performance bias) All outcomes	Unclear risk	No mention of blinding procedures to participants and personnel
Blinding of outcome assessment (detection bias) Functional outcomes	Unclear risk	Not reported
Blinding of outcome assessment (detection bias) Radiological outcomes	Low risk	Quote from the study stated: "... measurements were carried out by a single independent observer (LA) who was not involved in the surgical procedures, and who was blinded to the instrumentation used."
Incomplete outcome data (attrition bias) All outcomes	High risk	11 participants from the intervention group and 3 participants from the control group were excluded from the final analysis. Dropout rate: Intervention group: 11/70 (15.7%) Control group: 3/70 (4.3%) Comment: Due to unequal dropout rate, there is high risk of attrition bias.
Selective reporting (reporting bias)	Low risk	All the outcome measures mentioned in the methodology were reported.
Other bias	Unclear risk	Quote from study: "One or more authors received benefits from a commercial party related directly or indirectly to the subject of this article".

Abdel 2014
Study characteristics

Methods	Study Design: parallel RCT Country: France Setting: Aix-Marseille University, Marseille, France Length of Follow-up: 3 months
Participants	Inclusion Criteria • severe, symptomatic knee arthritis • aged between 50 and 85 years • willing to wait 6 to 8 weeks before surgery • were accepting of new technology Exclusion Criteria • metallic hardware within 10 cm of the knee

Patient-specific cutting guides for total knee arthroplasty (Review)

Abdel 2014 (Continued)

- any other previous knee surgery that could lead to an artifact effect on imaging
- prior THA

Number of people randomised: 40

- Intervention: 20
- Control: 20

Number of dropouts: 0

BMI

- Control group: 30
- Intervention group: 28.3

Preoperative mechanical axis

- Control group: Varus 16 patients, valgus 2 patients, neutral 2 patients
- Intervention group: Varus 14 patients, valgus 4 patients, neutral 2 patients

Interventions	Control group: Conventional instrumentation Intervention group: Patient-specific instrumentation Type of implant: Posterior stabilised Bearing design: Mobile bearing Mode of fixation: Cemented Patellar resurfacing: Performed in all patients. Quote from study stated: "Patella was resurfaced systematically in both groups". Preop imaging for designing PSG: MRI Implant Brand: Negev LPS-Flex mobile PSG Brand: not reported Intraoperative Data • Number of times PSG was abandoned: 0 • Reasons for abandonment: Not applicable • Size correlation between planned component size and actual component used: Not reported • Any other relevant intraoperative data: In the intervention group, "25% of cases required modification based on intraoperative judgement by the operative surgeon....three patients required an additional distal femoral resection of 2 mm and two patients required additional tibial resection of 2 mm". Surgeon experience and pre-training: probably adequate - quote from the study stated: "... patients operated ... by the 2 senior authors were included".
Outcomes	Outcomes assessed at: 3 months postoperatively Imaging modality used for postoperative measurement: Radiograph and CT scan Outcomes used in this review: 1. Functional outcome scores 2. Radiographic alignment of components 3. Number of outliers in radiographic alignment 4. Total adverse events

Abdel 2014 (Continued)

5. Quality of Life scores

Outcomes not used in this review:

1. Gait performance analysis

Notes	There is another report published in Knee Surgery, Sports Traumatology, Arthroscopy, in which they have reported the radiological outcomes.
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Risk of bias

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Low risk	Quote from the study: "Randomisation of patients into one of the two groups was done by the Hospital Informatics Department with the use of a systematic sampling method."
Allocation concealment (selection bias)	Low risk	Quote from the study: "The randomisation protocol was not revealed to the authors, who received the information regarding the group to which the patient was assigned in numbered, sealed envelopes."
Blinding of participants and personnel (performance bias) All outcomes	Unclear risk	Not reported
Blinding of outcome assessment (detection bias) Functional outcomes	Unclear risk	Not reported
Blinding of outcome assessment (detection bias) Radiological outcomes	Low risk	Quote from the study: "The radiological evaluation was performed according to a previously described method by two independent observers (one orthopaedic fellow and one staff radiologist) blind to the type of technique used intra-operatively".
Incomplete outcome data (attrition bias) All outcomes	Low risk	All the participants contributed to the outcomes reported in the trial.
Selective reporting (reporting bias)	Low risk	All the outcome measures mentioned in the methodology were reported.
Other bias	Low risk	None detected

Attard 2019
Study characteristics

Methods	Study Design: RCT Country: UK Setting: Edinburgh Length of Follow-up: 1 year
Participants	Inclusion Criteria: Patients diagnosed with osteoarthritis of the knee which was sufficiently symptomatic to require TKA

Patient-specific cutting guides for total knee arthroplasty (Review)

Attard 2019 (Continued)

Exclusion Criteria: Patients were excluded if they showed signs of inflammatory arthritis, ligament problems, significant knee deformities, or if they required complex bone augmentation procedures.

Number of participants randomised:

- Intervention group: 23
- Control group: 9

Number of participants who received allocated intervention:

- Intervention group: 23
- Control group: 9

Number of dropouts:

- Unclear – not reported

Age: Mean (range)

Intervention group: 70.7 (21)

Control group: 70.1 (35)

Sex (Male/Female):

- Intervention group: 5/4
- Control group: 10/13

Duration of disease: Not reported

Concurrent treatment: Not reported

BMI:

- Intervention group: 27.19 (14.66)
- Control group: 30.8 (15.2)

Preoperative mechanical axis

- Intervention group: NA
- Control group: NA

Interventions

Intervention group: Patient-specific instrumentation (single use)

Control group: Conventional instrumentation (reusable)

Type of implant: GMK Sphere, Medacta

Bearing design: Fixed bearing

Mode of fixation: Cement

Patellar resurfacing: NA

Preop imaging for designing PSG: CT

Implant Brand: Medacta

PSG Brand: MyKnee PSI Medacta

Intraoperative Data: Complications

Outcomes

Imaging modality used for postoperative measurement: NA

Outcomes used in this review: Complications, EBV, OKS, surgical time

Attard 2019 (Continued)

Outcomes not used in this review: Radiological data

Notes	Data from conventional reusable (CVR) group used as control group. Data from PSI single use (PSS) used as intervention group. Standard deviations missing for relevant data and, therefore, SDs were imputed from relevant studies (most weighted).
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Risk of bias

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Low risk	The patients were randomised into four separate groups by block stratification.
Allocation concealment (selection bias)	Unclear risk	Not reported
Blinding of participants and personnel (performance bias) All outcomes	Unclear risk	Not reported
Blinding of outcome assessment (detection bias) Functional outcomes	Unclear risk	Not reported
Blinding of outcome assessment (detection bias) Radiological outcomes	Unclear risk	Not reported
Incomplete outcome data (attrition bias) All outcomes	Unclear risk	Loss to follow-up was not reported.
Selective reporting (reporting bias)	Low risk	All outcomes discussed in methods reported
Other bias	High risk	CONSORT diagram has discrepancy between numbers excluded and numbers analysed.

Boonen 2013

Study characteristics

Methods	Study Design: parallel RCT Country: Netherlands Setting: Zuyderland Medical Centre, Koningsbosch, Netherlands and St. Anna Hospital, Geldrop, Netherlands Length of Follow-up: mean 44 months (24 to 57 months) Theeuwen: 10-year follow-up study, mean 10.1 years
Participants	Inclusion Criteria: disabling osteoarthritis of the knee who were candidates for primary unilateral TKA

Boonen 2013 (Continued)

Exclusion Criteria:

- presence of metal near the knee, ankle or hip joint
- patients with contraindications to MRI scans
- those who had previously undergone knee surgery (except for arthroscopic meniscectomy)
- those with indications for TKA other than osteoarthritis
- those with active infection near the knee or with active systemic infection
- those who were unable to comply with the postoperative rehabilitation protocol
- were unwilling to participate

Number of participants randomised: 180

- Intervention: N = 90
- Control: N = 90

Number of participants who received allocated intervention: 180

Number of dropouts: 12

- Intervention: N = 4/90 (4.4%)
- Control: N = 8/90 (8.9%)

Number of dropouts at 10-year follow-up:

- Intervention: 28 (8 refused, 14 deceased, 6 revisions) 31.1%
- Control: 23 (7 refused, 13 deceased, 3 revisions) 25.5%

Age

- Intervention group: 69 ± 8.0
- Control group: 65 ± 8.8

Sex (Male/Female)

- Intervention group: 34/56
- Control group: 40/50

Duration of disease: Not reported

Concurrent treatment: Not reported

BMI (Mean)

- Intervention group: 30.3
- Control group: 29.5

Preoperative mechanical axis: Not reported

Interventions

Intervention group: Patient-specific instrumentation

Control group: Conventional instrumentation

Type of implant: not reported

Bearing design: not reported

Mode of fixation: Cemented

Patellar resurfacing: Performed when deemed necessary

Preop imaging for designing PSG: MRI

Implant Brand: Vanguard

Boonen 2013 (Continued)

PSG Brand: Signature

Intraoperative Data

· Number of times PSG was abandoned: 0. Quote from study: "No PMPG procedure had to be converted to a traditional instrumented TKA"

· Reasons for abandonment: Not applicable

· Size correlation between planned component size and actual component used: Not reported

· Any other relevant intraoperative data: Not reported

Surgeon experience and pre-training: Quote from study: "All (surgeons) had extensive experience with conventional TKA and had undertaken at least 100 TKAs using PMPG technology".

Outcomes

Outcomes assessed: perioperative, 3 months, 1 year and 2 years, 5 years and 10 years

Imaging modality used for postoperative measurement: Radiograph

Outcomes used in this review:

1. Functional outcome measures
2. Radiological alignment of components
3. Number of outliers in radiographic alignment
4. Pain measure (VAS)
5. Quality of Life measure (EQ -5D)
6. Operative time
7. Blood loss
8. Total adverse events
9. Re-operation rate
- 10.Length of hospital stay
- 11.10 year-follow up: revision

Outcomes not used in this review: Nil

Notes

Results of 5-year (intermediate) implant survival, KSS; OKS; WOMAC; VAS and the EQ-5D published as another study (Schonatus 2019).

10-year revision rates in Theeuwen study

Risk of bias

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Low risk	Quote from study: "Randomisation involved an online random number generator and was stratified per hospital".
Allocation concealment (selection bias)	Low risk	Quote from study: "The surgeon enrolling the patients in the trial was unaware of the type of treatment that patients would receive".
Blinding of participants and personnel (performance bias) All outcomes	Low risk	Quote from the study: "Patients were blinded to the type of alignment method used patients in this group (control group) also underwent MRI scanning to maintain patient blinding".
Blinding of outcome assessment (detection bias) Functional outcomes	Low risk	Quote from the study: "... questionnaires was completed by an independent physician who was blinded to the type of instrumentation used".

Boonen 2013 (Continued)

Blinding of outcome assessment (detection bias) Radiological outcomes	Low risk	Quote from the study: "All radiographic outcome measures were performed by 2 independent reviewers in both centres, blinded to the type of alignment method used".
Incomplete outcome data (attrition bias) All outcomes	Low risk	Quote from study: "Four radiographs in group 1 (intervention group) and 8 radiographs in group 2 (control) group were of unacceptable quality not included in the analysis". Loss to follow-up - Intervention group: 8 patients - Control group: 9 patients Comment: Possible low risk of attrition – no evidence of differential dropout rate in the two arms
Selective reporting (reporting bias)	Low risk	All the outcome measures mentioned in the methodology were reported.
Other bias	Unclear risk	Quote from study: "The author or one or more of the authors have received or will receive benefits for personal or professional use from a commercial party related directly or indirectly in the subject of this article".

Calliess 2017

Study characteristics

Methods	Study Design: RCT Country: Germany Setting: Hannover Medical School, Hannover, Germany Length of Follow-up: 12 months
Participants	Inclusion Criteria: progression in osteoarthritic changes, loss of function, and disabling knee pain - preoperative varus–valgus deformity was $\leq 10^\circ$ - mLDFA was 86°–94° - mMPTA was 86°–94° Exclusion Criteria : - body mass index (BMI) > 40 - history of infection in the knee - post-traumatic osteoarthritis - bony osteotomy - chronic polyarthritis - underlying neurological disease with mobility impairment Number of participants randomised: - Intervention group: 100

Calliess 2017 (Continued)

- Control group: 100
- Number of participants who received allocated intervention: Not reported
- Number of dropouts:
- Intervention group: 0
 - Control group: 1 patient was lost to follow-up after 12 months.
- Age
- Intervention group: 70 ± 8
 - Control group: 67 ± 8
- Sex (Male/Female)
- Intervention group: 43:57
 - Control group: 39:61
- Duration of disease: Not reported
- Concurrent treatment: Not reported
- BMI
- Intervention group: 30 ± 5
 - Control group: 30 ± 4
- Preoperative mechanical axis
- Intervention group: 2 ± 5
 - Control group: 3 ± 4

Interventions	Intervention group: Patient-specific instrumentation Control group: Conventional instrumentation Type of implant: cruciate retaining Bearing design: Not reported Mode of fixation: Not reported Patellar resurfacing: Not reported Preop imaging for designing PSG: MRI scan Implant Brand: Triathlon System, Stryker, Mahwah, NJ, USA PSG Brand: ShapeMatch Technology, Otismed Stryker Intraoperative Data · Number of times PSG was abandoned: 1 patient was intraoperatively converted to a posterior stabilised prosthesis · Reasons for abandonment: due to the accidental resection of the PCL while the tibial cut was being performed. · Size correlation between planned component size and actual component used: Not reported · Any other relevant intraoperative data: Ligament balance was evaluated using a soft tissue tensioner (Stryker, Mahwah, NJ, USA), and balancing was performed according to Whiteside's algorithms if necessary.
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Calliess 2017 (Continued)

Surgeon experience and pre-training: consultant teams

Outcomes	Outcomes assessed at KSS and WOMAC scores were recorded on follow-up after 12 months Imaging modality used for postoperative measurement: weight-bearing radiographs Outcomes used in this review: 1. Survival rate of implant (revision surgery to change a prosthetic component) 1. Functional outcome measures - WOMAC, Knee society score (12 months) 2. Radiological alignment of components Outcomes not used in this review: 1. Operative time 2. Total adverse events 3. Re-operation rate 4. Length of hospital stay 5. Global assessment (participant-reported)	
Notes		
Risk of bias		
Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Unclear risk	Quote from study: "Between 2012 and 2013, 200 patients underwent TKA and were randomly assigned to 2 groups". Comment: Randomisation method not mentioned
Allocation concealment (selection bias)	High risk	Quote from study: "Two different consultant teams performed the surgeries, acquired the patients, and assigned them to a single group." Comment: As the surgical team was involved in allocation, probably the allocation was not concealed.
Blinding of participants and personnel (performance bias) All outcomes	High risk	Quote from study: "Due to high overall costs, patient blinding was not performed in this study."
Blinding of outcome assessment (detection bias) Functional outcomes	High risk	Quote from study: "The KSS and WOMAC scores were recorded on follow-up after 12 months". Comment: Highly likely that these outcome assessments were not blinded
Blinding of outcome assessment (detection bias) Radiological outcomes	High risk	Quote from study: "radiological evaluations were performed by one senior consultant and one resident..... In the KA group, the extent of post-operative limb alignment and deviation from initial plan was evaluated". Comment: Highly likely that these outcome assessments were not blinded
Incomplete outcome data (attrition bias) All outcomes	Low risk	Quote from study: "... in the MA group (control group), 1 patient was lost to follow-up after 12 months". Comment: There was no attrition reported during the study period (12 months).

Calliess 2017 (Continued)

Selective reporting (re-reporting bias)	Low risk	All the outcome measures mentioned in the methodology were reported.
Other bias	Low risk	None detected

Chareancholvanich 2013
Study characteristics

Methods	Study Design: RCT Country: Thailand Setting: Siriraj Hospital, Mahidol University, Bangkok Length of Follow-up: Not reported
Participants	Inclusion Criteria: patients undergoing unilateral posterior-stabilised TKR for the treatment of osteoarthritis (OA) Exclusion Criteria: Patients with flexion contracture > 30°, previous complex knee surgery, retained metal hardware at the hip, knee or ankle, extra-articular deformity of the knee, contraindications to MRI, and those who refused consent Number of participants randomised: 83 - Intervention group: 43 - Control group: 40 Number of participants who received allocated intervention: - Intervention group: 40 (all 43 received TKA but 3 did not have PSG due to claustrophobia) - Control group: 40 Number of dropouts: - Intervention group: 0 - Control group: 0 Age - Intervention group: 69.5 - Control group: 70.3 Sex (Male/Female) - Intervention group: 6 M 34 F - Control group: 4 M 36 F Duration of disease: Not reported Concurrent treatment: Not reported BMI - Intervention group: 27.7 (20.2 to 44.15) - Control group: 28.0 (22 to 39.6) Preoperative mechanical axis

Chareancholvanich 2013 (Continued)

- Intervention group: 168.7 (160 to 188)
- Control group: 168.4 (160 to 186)

Interventions	Intervention group: Patient-specific instrumentation Control group: Conventional instrumentation Type of implant: PS Bearing design: fixed bearing Mode of fixation: Cemented Patellar resurfacing: No Preop imaging for designing PSG: MRI Implant Brand: NexGen LPS-Flex fixed bearing (Zimmer Warsaw, Indiana) PSG Brand: Patient Specific Instruments system, Zimmer, Warsaw, Indiana Intraoperative Data • Number of times PSG was abandoned: Not reported • Reasons for abandonment: • Size correlation between planned component size and actual component used: Not reported • Any other relevant intraoperative data: Surgeon experience and pre-training: "same experienced knee surgeon".
Outcomes	Outcomes assessed at: four weeks postoperatively Imaging modality used for postoperative measurement: Anteroposterior hip-to-ankle CT scanograms Outcomes used in this review: 1. Radiological alignment of components 2. Number of outliers in radiographic alignment 3. Operative time 4. Length of hospital stay 5. Blood loss 6. Incision length Outcomes not used in this review: Nil
Notes	SDs for blood loss, operative time, length of stay and incision length not reported. SDs imputed from other relevant studies.

Risk of bias

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Low risk	Quote from study: "The patients were randomised to each treatment arm via the blocks-of-four method".
Allocation concealment (selection bias)	Unclear risk	Not reported

Chareancholvanich 2013 (Continued)

Blinding of participants and personnel (performance bias) All outcomes	Unclear risk	Not reported
Blinding of outcome assessment (detection bias) Functional outcomes	Low risk	Functional outcome measures not reported in this study
Blinding of outcome assessment (detection bias) Radiological outcomes	Low risk	Quote from study: "Two independent blinded investigators measured the radiological parameters".
Incomplete outcome data (attrition bias) All outcomes	Low risk	3 patients in the intervention group did not receive the allocated intervention after randomisation. However, all the patients who received the allocated intervention contributed to the outcome measurement. Comment: Probably low risk
Selective reporting (reporting bias)	Low risk	All the outcome measures mentioned in the methodology were reported.
Other bias	Low risk	None detected

Chotanaphuti 2014

Study characteristics

Methods	Study Design: RCT Country: Thailand Setting: Department of Orthopedic Surgery, Phramongkutklao Hospital and College of Medicine, Bangkok, Thailand Length of Follow-up:
Participants	Inclusion Criteria: patients who underwent primary total knee arthroplasty; no femoral nails/bone plates that extend into the knee, i.e. within 8 cm of joint line; no metal device that could cause CT scatter about the knee and no deformities greater than 15° of fixed varus, valgus or flexion contracture Exclusion Criteria: patients with previous ipsilateral distal femoral or high tibial osteotomies, ankylosis of the hip joint on the side to be treated, inflammatory arthritis and previous patellectomy Number of participants randomised: 80 - Intervention group: 40 - Control group: 40 Number of participants who received allocated intervention: - Intervention group: 40 - Control group: 40 Number of dropouts: Not reported - Intervention group - Control group

Chotanaphuti 2014 (Continued)

Age

- Intervention group: 69.7
- Control group: 69.3

Sex (Male/Female) 70 women and 10 men in total

- Intervention group
- Control group

Duration of disease: Not reported

Concurrent treatment: Not reported

BMI

- Intervention group: 25
- Control group: 25

Preoperative mechanical axis: Not reported

- Intervention group
- Control group

Interventions	Type of implant: posterior stabilised Bearing design: Fixed Mode of fixation: cemented Patellar resurfacing: all cases Preop imaging for designing PSG: CT Implant Brand: Sigma™ High Performance Instrument; DePuy, Warsaw, Ind.) PSG Brand: TruMatch™ Personalized Solutions; DePuy, Warsaw, Ind Intraoperative Data · Number of times PSG was abandoned: 0 · Reasons for abandonment: · Size correlation between planned component size and actual component used: femur - 38 of 40 knees, tibia - 36 of 40 knees · Any other relevant intraoperative data: Surgeon experience and pre-training: Not reported
Outcomes	Outcomes assessed at 6 weeks postoperatively Imaging modality used for postoperative measurement: CT scan of the knee and a weight-bearing film of hip-knee-ankle Outcomes used in this review: 1. Radiological alignment of components 2. Number of outliers in radiographic alignment 3. Operative time 4. Blood loss Outcomes not used in this review: Nil

Chotanaphuti 2014 (Continued)

Notes

Risk of bias

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Unclear risk	Quote from the study: "were assigned to one of the two groups (1:1)" Comment: Method of random sequence generation was not reported.
Allocation concealment (selection bias)	Unclear risk	Not reported
Blinding of participants and personnel (performance bias) All outcomes	Unclear risk	This was an open-label study. However, it is unclear whether there was any performance bias.
Blinding of outcome assessment (detection bias) Functional outcomes	Low risk	Functional outcome measures not reported in this study
Blinding of outcome assessment (detection bias) Radiological outcomes	Low risk	Quote from study: "All patients received a CT scan of the knee evaluated by two hip & knee fellows who did not know the procedures".
Incomplete outcome data (attrition bias) All outcomes	Low risk	All the patients who received the allocated intervention contributed to the outcome measurement.
Selective reporting (reporting bias)	Low risk	All the outcome measures mentioned in the methodology were reported.
Other bias	Low risk	None detected

Cucchi 2019

Study characteristics

Methods	Study Design: RCT Country: Germany Setting: University Hospital Bonn Length of Follow-up: 2 months
Participants	Inclusion Criteria: Indication for total knee replacement Exclusion Criteria: Metal devices within 8 cm from the knee, articular surfaces or any fixed deformities greater than 15 ° in varus, valgus, flexion or tibial slope Number of participants randomised: 24

Cucchi 2019 (Continued)

- Intervention group: 13
- Control group: 11

Number of participants who received allocated intervention:

- Intervention group: 12
- Control group: 10

Number of dropouts:

- Intervention group: 1 (different prosthesis)
- Control group: 1 (refused surgery)

Age

- Intervention group: 74.25
- Control group: 69

Sex (Male/Female)

- Intervention group: F:M ratio 0.75/0.25
- Control group: F:M ratio 0.8/0.2

Duration of disease: Not reported

Concurrent treatment: Not reported

BMI

- Intervention group: 28.7
- Control group: 27.9

Preoperative mechanical axis: Not recorded

Interventions	Intervention group: Patient-specific instrumentation Control group: Conventional instrumentation Type of implant: Bearing design: Mobile bearing Mode of fixation: Cemented Patellar resurfacing: Yes Preop imaging for designing PSG: CT Implant Brand: P.F.C. Sigma, DePuy PSG Brand: TRUMATCH (DePuy Orthopaedics Inc, Warsaw, IN) Intraoperative Data: Tourniquet time, surgical time
Outcomes	Outcomes assessed Imaging modality used for postoperative measurement: CT Outcomes used in this review: OKS, VAS, femoral rotation, Tourniquet Time (TT), surgical time, change in component size, recuts CT alignment assessment: Femoral component rotation to the surgical/clinical transepicondylar axis

Cucchi 2019 (Continued)

Blood loss

Notes

Risk of bias

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Low risk	Block randomisation (4 per block) was performed to allocate patients to control or treatment groups.
Allocation concealment (selection bias)	Low risk	Concealed sequence in sequentially numbered sealed envelopes until interventions assigned
Blinding of participants and personnel (performance bias) All outcomes	Unclear risk	Not reported
Blinding of outcome assessment (detection bias) Functional outcomes	Unclear risk	Not reported
Blinding of outcome assessment (detection bias) Radiological outcomes	Unclear risk	Not reported
Incomplete outcome data (attrition bias) All outcomes	Low risk	None lost to follow-up
Selective reporting (reporting bias)	Unclear risk	Outlier data not reported
Other bias	Low risk	None

De Vloo 2017
Study characteristics

Methods	Study Design: RCT Country: Belgium Setting: AZ Klina, Department of Orthopedic Surgery and Traumatology, Augustijnslei 100, 2930 Brasschaat, Belgium KU Leuven, Department of Mechanical Engineering, Biomechanics Section, Celestijnenlaan 300C, 3001 Leuven, Belgium Length of Follow-up: Not reported
Participants	Inclusion Criteria: Painful and disabled knee joint resulting from osteoarthritis. One or more compartments are involved, as assessed by X-ray. High need to obtain pain relief and improve function Exclusion Criteria: Active infection in the knee; Body Mass Index (BMI) ≥ 38; age less than 50 years old; general infection; distant foci of infections which may spread to the implant site; failure of previous joint replacement; pregnancy; previous major knee surgery, except for arthroscopic meniscectomy;

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De Vloo 2017 (Continued)

high sensitivity to radiation; metal within 10 cm of knee joint; presence of pacemaker, neuro-stimulator or cerebral shunt; claustrophobia

Number of participants randomised: 50

- Intervention group: 25
- Control group: 25

Number of participants who received allocated intervention:

- Intervention group: 24
- Control group: 25

Number of dropouts:

- Intervention group: 0
- Control group: 0

Excluded from analysis due to motion artefact in CT scan postop

- Intervention group: 4
- Control group: 1

Age

- Intervention group: 72.72
- Control group: 72.28

Sex (Male/Female)

- Intervention group: 14 females and 11 males
- Control group: 15 females and 10 males

Duration of disease: Not reported

Concurrent treatment: Not reported

BMI

- Intervention group: 29.24
- Control group: 29.92

Preoperative mechanical axis not reported

- Intervention group
- Control group

Interventions

Intervention group: Patient-specific instrumentation

Control group: Conventional instrumentation

Type of implant: posterior-stabilised (PS) or a cruciate retained (CR) Vanguard knee prosthesis

- Intervention group: CR, fixed bearing 14, PS, fixed bearing 6
- Control group: : CR, fixed bearing 14, PS, fixed bearing 10

Bearing design: Fixed

Mode of fixation:

Patellar resurfacing: Not reported

Preop imaging for designing PSG: MR

De Vloo 2017 (Continued)

Implant Brand: Vanguard Knee System

PSG Brand: Signature MRI total knee system (Biomet, Inc., Warsaw, Indiana, United States)

Intraoperative Data

- Number of times PSG was abandoned
- Reasons for abandonment
- Size correlation between planned component size and actual component used
- PSI/intervention group: F100%, T79%
- CON/Control group: F64% as compared to PSI plan; T56% as compared to PSI plan
- Any other relevant intraoperative data

Surgeon experience and pre-training: single surgeon with experience since April 2011 with the Signature patient-specific

Outcomes

Imaging modality used for postoperative measurement:

All patients received MRI for PSI and preop CT. Postop CT was superimposed on preop CT and component alignment measured

Outcomes used in this review: Radiological alignment of components

Outcomes not used in this review: Nil

Notes
Risk of bias

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Low risk	Four surgeries were planned per day. Randomisations were done the day of surgery by an independent person who tossed a coin to define the two patients that would be operated using conventional instrumentation.
Allocation concealment (selection bias)	Low risk	All patients received MRI for PSI and preop CT. Group was decided on the day of surgery.
Blinding of participants and personnel (performance bias) All outcomes	Unclear risk	Not reported
Blinding of outcome assessment (detection bias) Functional outcomes	Unclear risk	Functional outcomes not analysed
Blinding of outcome assessment (detection bias) Radiological outcomes	Unclear risk	Not reported
Incomplete outcome data (attrition bias) All outcomes	Unclear risk	Except for 4 patients (con 1, PSG 4) with motion artefacts in postop CT for alignment, all other patients' outcomes presented

De Vloo 2017 (Continued)

Selective reporting (reporting bias)	Low risk	None
Other bias	Low risk	None

Ferrara 2015

Study characteristics

Methods	Study Design: RCT Country: Italy Setting: Department of Radiological Sciences and Department of Orthopaedics and Traumatology, Catholic University of Sacred Heart, "Agostino Gemelli" Hospital, Rome, Italy Length of Follow-up: Not reported
Participants	Inclusion Criteria: primary knee osteoarthritis; age 50 to 85 years; failure of nonoperative treatments; no contraindications to surgery, acceptance of a relatively new method; and willingness to wait an additional 7 to 8 weeks required for the instrumentation to be prepared Exclusion Criteria: previous surgery or trauma of the limb to be treated; coronal deformity ($> 10^\circ$ deviation); general contraindications to MRI examination Number of participants randomised: 30 - Intervention group: 15 - Control group: 15 Number of participants who received allocated intervention: - Intervention group: 15 - Control group: 15 Number of dropouts: - Intervention group - Control group Age - Intervention group: 75.3 ± 6.7 - Control group: 74.5 ± 7.2 Sex (Male/Female) - Intervention group: 6/9 - Control group: 7/8 Duration of disease: Not reported Concurrent treatment: Not reported BMI - Intervention group: 28.4 ± 3.6 - Control group: 28.1 ± 3.8 Preoperative mechanical axis: Not reported

Ferrara 2015 (Continued)

- Intervention group
- Control group

Interventions	Intervention group: Patient-specific instrumentation	
	Control group: Conventional instrumentation	
	Type of implant: posterior stabilised	
	Bearing design	
	Mode of fixation: cemented	
	Patellar resurfacing: None in PSG group	
	Preop imaging for designing PSG	
	Implant Brand: NexGen LPS; Zimmer, Warsaw, Indiana	
	PSG Brand: Not reported	
	Intraoperative Data	
	· Number of times PSG was abandoned: 0	
	· Reasons for abandonment	
	· Size correlation between planned component size and actual component used: 100%	
	· Any other relevant intraoperative data	
	Surgeon experience and pre-training: more than 10 years of experience in TKA	
Outcomes	Outcomes assessed at	
	Imaging modality used for postoperative measurement: CT at 6 to 8 weeks postop	
	Outcomes used in this review:	
	1. Radiological alignment of components	
	2. Operative time	
	3. Blood loss	
	4. Length of hospital stay	
	Outcomes not used in this review: Nil	
Notes		
Risk of bias		
Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Low risk	Quote from the study: "Patients were randomised into 2 groups by the informatics department of the authors' hospital, using a systematic sampling method".
Allocation concealment (selection bias)	Low risk	Quote from the study: "randomisation protocol was not revealed to the authors numbered and sealed envelopes"

Ferrara 2015 (Continued)

Blinding of participants and personnel (performance bias) All outcomes	Unclear risk	Not reported
Blinding of outcome assessment (detection bias) Functional outcomes	Low risk	Functional outcome measures not reported in this study
Blinding of outcome assessment (detection bias) Radiological outcomes	Low risk	Quote from the study: "Another radiologist, blinded to the type of operative technique used, evaluated the positioning of both prosthetic components".
Incomplete outcome data (attrition bias) All outcomes	Low risk	All the patients who received the allocated intervention contributed to the outcome measurement.
Selective reporting (reporting bias)	Low risk	All the outcome measures mentioned in the methodology were reported.
Other bias	Low risk	None detected

Gan 2015
Study characteristics

Methods	Study Design: parallel RCT Country: China Setting: Kunming General Hospital of Chengdu Military Command of PLA, Kunming, Yunnan, China Length of Follow-up: Not recorded
Participants	Inclusion Criteria: - Serious knee osteoarthritis Exclusion Criteria: - Infections, revision surgery, severe instability Number of participants randomised: 70 - Intervention: N = 35 - Control: N = 35 Number of participants who received allocated intervention: 70 Number of dropouts: 0 - Intervention: N = 0/35 (0%) - Control: N = 0/35 (0%) Number of dropouts at follow-up: - Intervention: 0, 0% - Control: 0, 0%

Gan 2015 (Continued)

Age

- Intervention group: 68.5 ± 4.8
- Control group: 67.8 ± 3.4

Sex (Male/Female)

- Intervention group: 10/25
- Control group: 9/26

Duration of disease: Not reported

Concurrent treatment: Not reported

BMI (Mean): Not reported

Preoperative mechanical axis:

- Intervention group deviation: $6.1 \text{ degrees} \pm 4.2$
- Control group: $6.5 \text{ degrees} \pm 4.4$

Interventions

Intervention group: Patient-specific instrumentation

Control group: Conventional instrumentation

Type of implant: not reported

Bearing design: not reported

Mode of fixation: not reported

Patellar resurfacing: not reported

Preop imaging for designing PSG: CT

Implant Brand: not reported

PSG Brand: Somos 14120, DSM Desotech Inc.

Intraoperative Data

- Number of times PSG was abandoned: 0
- Reasons for abandonment: Not applicable
- Size correlation between planned component size and actual component used: Not reported
- Any other relevant intraoperative data: Not reported

Surgeon experience and pre-training: Quote from study: "All procedures were performed using a medial parapatellar arthrotomy... All operations were performed by the same surgeon."

Outcomes

Outcomes assessed at: immediate postoperative review

Imaging modality used for postoperative measurement: Radiograph and CT

Outcomes used in this review:

1. Number of outliers in radiographic alignment
2. Operative time
3. Blood loss

Outcomes not used in this review:

1. Functional outcome measures

Gan 2015 (Continued)

2. Radiological alignment of components
3. Pain measure (VAS)
4. Quality of Life measure (EQ -5D)
5. Total adverse events
6. Re-operation rate
7. Length of hospital stay
8. Revision rate

Notes	Other parameters of radiographic alignment were used to calculate differences in groups, including frontal femoral component angle (FFC), frontal tibial component angle (FTC), lateral femoral component angle (LFC) and lateral tibial component angle (LTC)
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Risk of bias

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Unclear risk	Process not explained
Allocation concealment (selection bias)	Unclear risk	Process not explained
Blinding of participants and personnel (performance bias) All outcomes	Unclear risk	Process not explained
Blinding of outcome assessment (detection bias) Functional outcomes	Unclear risk	Process not explained
Blinding of outcome assessment (detection bias) Radiological outcomes	Low risk	Surgeon blinded to technique
Incomplete outcome data (attrition bias) All outcomes	Low risk	No incomplete data for length of follow-up noted
Selective reporting (reporting bias)	Unclear risk	Incomplete outcome reporting
Other bias	Low risk	None detected

Giannotti 2019

Study characteristics

Methods	Study Design: RCT Country: Italy Setting: Department of Orthopaedic and Trauma Surgery, University of Pisa Length of Follow-up: 2 months
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Giannotti 2019 (Continued)

Participants

Inclusion Criteria: Patients more than 50 years old; diagnosis of primary knee osteoarthritis; values of anatomical lateral distal femoral angle between 78° and 84° (aLDFA); lateral proximal tibial angle (LP-TA) between 88° and 95°; femorotibial angle (FTA) between 174° and 182°

Exclusion Criteria: Severe lower limb malalignment; previous history of knee fractures or inflammatory arthritis; patellofemoral osteoarthritis with Iwano score of 2 or more; the presence of anterior knee pain; the presence of medial or lateral compartment constraint or loosening detected at physical examination

Number of participants randomised: 40

- Intervention group: 20
- Control group: 20

Number of participants who received allocated intervention:

- Intervention group: 20
- Control group: 20

Number of dropouts:

- Unclear – not reported

Age:

· Intervention group: 71 (IC 95% 70.5 to 77.3)

· Control group: 73 (IC 95% 67.2 to 74.3)

Sex (Male/Female)

- Intervention group: F:M ratio 16:4
- Control group: F:M ratio 10:10

Duration of disease: Not reported

Concurrent treatment: Not reported

BMI: Not reported

Preoperative mechanical axis

· Intervention group: not recorded

· Control group: not recorded

Interventions

Intervention group: Patient-specific instrumentation

Control group: Conventional instrumentation

Type of implant:

Bearing design: Fixed bearing

Mode of fixation: Cemented

Patellar resurfacing: No

Preop imaging for designing PSG: CT

Implant Brand: Medacta GMK

PSG Brand: Medacta GMK Efficiency

Outcomes

Outcomes assessed:

Giannotti 2019 (Continued)

Imaging modality used for postoperative measurement: XR

Outcomes used in this review:

Blood loss, KSS, FTA

Outcomes **not used in this review:**

Radiological outliers

Notes	Standard deviations of outcomes not reported in this study and, therefore, SDs were imputed from relevant studies (most weighted)
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Risk of bias

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	High risk	From April 2015 to September 2016, 40 patients (40 knees) were included in the randomisation process. Unclear description of randomisation process
Allocation concealment (selection bias)	Unclear risk	Not reported
Blinding of participants and personnel (performance bias) All outcomes	Unclear risk	Not reported
Blinding of outcome assessment (detection bias) Functional outcomes	Unclear risk	Not reported
Blinding of outcome assessment (detection bias) Radiological outcomes	Unclear risk	Not reported
Incomplete outcome data (attrition bias) All outcomes	Unclear risk	Not reported
Selective reporting (reporting bias)	Low risk	All the outcome measures mentioned in the methodology were reported.
Other bias	Low risk	None

Hamilton 2013

Study characteristics

Methods	Study Design: RCT Country: USA Setting: Anderson Orthopaedic Research Institute, Alexandria, Virginia Length of Follow-up: Not reported
Participants	Inclusion Criteria: non-inflammatory osteoarthritis scheduled for unilateral TKA

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Hamilton 2013 (Continued)

Exclusion Criteria: "previous ipsilateral hip, knee, or ankle replacement, a body mass index greater than 41 kg/m²; or a flexion contracture greater than 20°"

Number of participants randomised: 78

- Intervention group
- Control group

Number of participants who received allocated intervention:

- Intervention group: 26
- Control group: 26

Number of dropouts: 26

- Intervention group: 6 + 4
- Control group: 6

Age

- Intervention group: 68.1
- Control group: 67.6

Sex (Male/Female)

- Intervention group: 14:12
- Control group: 7:19

Duration of disease: Not reported

Concurrent treatment: Not reported

BMI

- Intervention group: 30.9
- Control group: 31.1

Preoperative mechanical axis

- Intervention group: 2.8 valgus (8.9 varus to 18.5 valgus)
- Control group: 3.4 valgus (7 varus to 20 valgus)

Interventions

Intervention group: Patient-specific instrumentation

Control group: Conventional instrumentation

Type of implant: Posterior Stabilised

Bearing design: fixed bearing

Mode of fixation: Cemented

Patellar resurfacing: All cases

Preop imaging for designing PSG: CT

Implant Brand: PFC total knee prosthesis (DePuy Orthopaedics, Warsaw, IN)

PSG Brand: TruMatch

Intraoperative Data

- Number of times PSG was abandoned: 0
- Reasons for abandonment

Hamilton 2013 (Continued)

· Size correlation between planned component size and actual component used: NA

· Any other relevant intraoperative data:

- Intervention group: Number of knees requiring recuts 23; number of recuts 27
- Control group: Number of knees requiring recuts 10; number of recuts 11

Surgeon experience and pre-training: single fellowship trained surgeon

Outcomes	Outcomes assessed: Not reported Imaging modality used for postoperative measurement: Long alignment radiograph Outcomes used in this review: 1. Radiological alignment of components 2. Number of outliers in radiographic alignment 3. Operative time Outcomes not used in this review: Nil
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Notes

Risk of bias

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Unclear risk	Quote from study: "Each patient was randomly assigned to either the control groupor the study group". Method used for random sequence generation was not reported.
Allocation concealment (selection bias)	High risk	Not reported The baseline demographic data showed a significantly higher percentage of males in the intervention group (54% vs 27%). Comment- Possibility of selection bias is high.
Blinding of participants and personnel (performance bias) All outcomes	Unclear risk	Not reported
Blinding of outcome assessment (detection bias) Functional outcomes	Low risk	Functional outcome measures not reported in this study
Blinding of outcome assessment (detection bias) Radiological outcomes	Unclear risk	Quote from study: "Limb and component alignment were measured for all cases on radiographs postoperatively". Comment: Whether the blinding of the assessor was performed is unclear
Incomplete outcome data (attrition bias) All outcomes	Low risk	Quote from study: "Of the 78 patients initially enrolled in the study, 26 were excluded from the study after randomisation". Comment: However, an equal number of participants in each arm (26) completed the trial.
Selective reporting (reporting bias)	Low risk	All the outcome measures mentioned in the methodology were reported.

Hamilton 2013 (Continued)

Other bias	Low risk	None detected
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Hampton 2022

Study characteristics

Methods	Study Design: RCT Country: UK Setting: Sheffield Teaching Hospitals NHS Foundation Trust Length of Follow-up: 5 years
Participants	Inclusion Criteria: Patients with primary or secondary osteoarthritis, or inflammatory arthritis, awaiting primary total knee replacement surgery, were invited to participate. Exclusion Criteria: Patients were excluded in the presence of active or suspected infection, a significant joint contracture identified by uncorrectable varus or valgus deformity or a fixed flexion deformity of > 20° on clinical examination or where known co-existent medical problems precluded recovery to independent mobility, such as neurologic conditions. Number of participants randomised: - Intervention group: 45 - Control group: 45 Number of participants who received allocated intervention: - Intervention group: 43 - Control group: 45 Number of dropouts: - Intervention group: 7 - Control group: 6 Age: Intervention group: 68 (±3) Control group: 71 (±3) Sex (Male/Female %): - Intervention group: 38/62 - Control group: 34/66 Duration of disease: Not reported Concurrent treatment: Not reported BMI (range): - Intervention group: 30.9 (21-36) - Control group: 30.3 (23-35) Preoperative mechanical axis: - Intervention group: Not reported

Hampton 2022 (Continued)

- Control group: Not reported

Interventions	Intervention group: Patient-specific instrumentation Control group: Conventional instrumentation Type of implant: Bearing design: Not reported Mode of fixation: Cement Patellar resurfacing: Not reported Preop imaging for designing PSG: MRI Implant Brand: NexGen Knee System (Zimmer, Warsaw, IN) PSG Brand: PSI (Zimmer Inc, UK) Intraoperative Data: Tourniquet time, estimated blood loss, and any intraoperative adverse events were recorded.
Outcomes	Imaging modality used for postoperative measurement: Xray Outcomes used in this review: The primary outcome measure was based on coronal femoral alignment. Secondary outcome measures included the changes in the Oxford knee score (OKS), International Knee Society Score (IKSS), and the Euroqol-5D (EQ-5D) to assess the quality of life, recorded preoperatively and at 1-year and 5-year post-surgery. Additional radiologic outcomes included restoration of an overall mechanical neutral axis of the lower limb, reduction of outliers, sagittal alignment of the femoral and tibial components, and coronal alignment of the tibial component.
Notes	Quantitative values for outliers in coronal, sagittal and rotational alignment for tibial and femoral components were not reported. No values provided for EQ-5D. No standard deviations for length of stay

Risk of bias

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Low risk	After exclusions, 88 patients (90 knees) consented to inclusion in the trial and were randomised to the treatment method by block randomisation.
Allocation concealment (selection bias)	Unclear risk	Not reported
Blinding of participants and personnel (performance bias) All outcomes	Low risk	Investigators were blinded to the randomisation block size, and patients were blinded to the treatment arm throughout the study. All patients underwent MRI scans irrespective of their designated treatment arm.
Blinding of outcome assessment (detection bias) Functional outcomes	Low risk	Patients were followed up by dedicated arthroplasty research nurses at each hospital, who were also blinded to the patient's treatment arm.
Blinding of outcome assessment (detection bias) Radiological outcomes	Low risk	The measurements were performed independently by 2 orthopaedic surgeons who were not involved in the surgical procedures and who were also blinded to the type of surgery performed.
Incomplete outcome data (attrition bias)	Unclear risk	Dropout rate PSI group, N = 7 (15.5%), control group N = 6 (13.3%). Not unacceptably high for 5-year follow-up

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Hampton 2022 (Continued)

All outcomes

Selective reporting (reporting bias)	Unclear risk	All outcomes discussed in methodology were reported.
Other bias	Low risk	Not detected

Huijbregts 2016
Study characteristics

Methods	Study Design: RCT Country: Australia Setting: The Joint Studio, Hollywood Medical Centre, 85 Monash Avenue, Nedlands, WA 6009, Australia. Length of Follow-up: 1 year
Participants	Inclusion Criteria: all adult patients with end-stage rheumatoid or osteoarthritis requiring TKA Exclusion Criteria: previous osteotomy or fracture, non-correctable valgus deformity, metal in close proximity of the knee joint, intracranial or cardiac metal implants preventing preoperative MRI scans and claustrophobia Number of participants randomised: 140 - Intervention group: 75 - Control group: 65 Number of participants who received allocated intervention: - Intervention group: 71 - Control group: 65 Number of dropouts: - Intervention group: 2 - Control group: 0 Age - Intervention group: 66.7 - Control group: 69 Sex (Male/Female) - Intervention group: 29/40 - Control group: 32/32 Duration of disease: Not reported Concurrent treatment: Not reported BMI – Not reported - Intervention group - Control group Preoperative mechanical axis – Not reported

Huijbregts 2016 (Continued)

- Intervention group
- Control group

Interventions	Intervention group: Patient-specific instrumentation Control group: Conventional instrumentation Type of implant: Genesis II or Legion systems (Smith & Nephew) Bearing design Mode of fixation: Cemented Patellar resurfacing: • Intervention group: 48 • Control group: 42 Preop imaging for designing PSG: MRI Implant Brand: Genesis II or Legion systems (Smith & Nephew) PSG Brand: Visionaire Smith & Nephew Intraoperative Data · Number of times PSG was abandoned: None One case of external rotation of the tibial cutting guide was deemed insufficient and was adjusted. · Reasons for abandonment · Size correlation between planned component size and actual component used · Any other relevant intraoperative data Surgeon experience and pre-training: Two experienced, fellowship-trained, surgeons
Outcomes	Outcomes assessed at approximately six weeks postoperatively for alignment; functional outcome – preoperative, 3 months, 1 year Imaging modality used for postoperative measurement: CT Perth protocol Outcomes used in this review: 1. Functional outcome measures 2. Radiological alignment of components 3. Number of outliers in radiographic alignment 4. Total adverse events 5. Operative time Outcomes not used in this review: Nil
Notes	SDs for functional outcomes (OKS) and QoL not reported. SDs imputed from other relevant studies

Risk of bias

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Low risk	Quote from study: "A website was used for computer-generated permuted block lists of randomisation".

Huijbregts 2016 (Continued)

Allocation concealment (selection bias)	Low risk	Quote from study: "The randomisation sequence and block size were concealed from the surgeons".
Blinding of participants and personnel (performance bias) All outcomes	Unclear risk	Not reported
Blinding of outcome assessment (detection bias) Functional outcomes	Unclear risk	Not reported
Blinding of outcome assessment (detection bias) Radiological outcomes	Low risk	Quote from study: "A protocol-trained CT technician who was blinded to the form of treatment performed the measurements".
Incomplete outcome data (attrition bias) All outcomes	Unclear risk	Number of patients who did not have complete functional outcome scores at 12 months follow-up: - Intervention group: 11/75 (14.7%) - Control group: 7/65 (10.8%) However, dropouts were different for each of the functional outcomes measured and data for radiological outcomes did not indicate a significant difference in dropouts.
Selective reporting (reporting bias)	Low risk	All the outcome measures mentioned in the methodology were reported.
Other bias	Low risk	None detected

Kosse 2017
Study characteristics

Methods	Study Design: RCT Country: The Netherlands Setting: Department of Orthopedics, Sint Maartenskliniek, Nijmegen, The Netherlands Length of Follow-up: 1 year
Participants	Inclusion Criteria: Patients with non-inflammatory knee osteoarthritis scheduled for a unilateral primary TKA. Patients were included if they were 40–70 years of age at the time of inclusion; had no large deformities (knee flexion < 90°, fixed flexion > 10°, noncorrectable varus/valgus > 10°, extension deficits > 30°); and had sufficient femoral and tibial bone stock. Exclusion Criteria: BMI > 35; rheumatoid arthritis; active, local, or systemic infections; previous hip and/or knee replacement in the last 6 months; sensitive to materials in the implants and/or cutting blocks; and contra-indications for MRI. Number of participants randomised: - Intervention group: 21 - Control group: 21 Number of participants who received allocated intervention:

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Kosse 2017 (Continued)

- Intervention group: 21
- Control group: 21

Number of dropouts:

- Intervention group: 0
- Control group: 0

Age:

Intervention group: 62.7 ± 4.5

Control group: 63.4 ± 4.2

Sex (Male/Female):

- Intervention group: 8:13
- Control group: 12:9

Duration of disease: Not reported

Concurrent treatment: Not reported

BMI:

Intervention group: 28.1 ± 3.3

Control group: 27.8 ± 3.1

Preoperative mechanical axis

• Intervention group: 2.6 varus ± 6.5

• Control group: 3.4 varus ± 5.8

Interventions	Intervention group: Patient-specific instrumentation Control group: Conventional instrumentation Type of implant: Bearing design: Fixed bearing Mode of fixation: Cement Patellar resurfacing: Yes Preop imaging for designing PSG: MRI Implant Brand: Genesis II PS (Smith & Nephew, Memphis, Tennessee) PSG Brand: Visionaire (Smith & Nephew) Intraoperative Data: operation duration (time from first incision until closing the skin), estimated blood loss, ligament releases, component sizes, and intraoperative complications
Outcomes	Outcomes assessed: Imaging modality used for postoperative measurement: Xray & CT Outcomes used in this review: Varus–valgus laxity, hip–knee–ankle angle (HKA; mechanical axis). Femoral and tibial component rotation. Frontal and sagittal alignment of the femoral and tibial components and radiolucency (> 2 mm).

Kosse 2017 (Continued)

The Knee Society Score (KSS), VAS pain, VAS Satisfaction, Knee injury and Osteoarthritis Outcome score (KOOS), Patella score (Kujala), and the University of California Los Angeles activity score (UCLA). The anterior-posterior laxity was measured. All (serious) adverse device events that occurred during the study period were recorded.

Outcomes not used in this review:

Radiological outliers

Notes

No radiological outliers. Standard deviations for clinical and functional outcomes not reported; therefore, SDs imputed from relevant studies (most weighted)

Risk of bias

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Low risk	Block randomisation lists were computer-generated by an independent researcher; block size was ten with equal allocations to the PSI and CI groups.
Allocation concealment (selection bias)	Low risk	The randomisation sequence was concealed from the patient prior to enrolment until 12 months postoperatively.
Blinding of participants and personnel (performance bias) All outcomes	Unclear risk	This study was a single-centre, patient-blinded, randomised controlled trial.
Blinding of outcome assessment (detection bias) Functional outcomes	High risk	Due to logistical reasons, it was not possible to blind the research nurse, which might have biased the clinical assessments.
Blinding of outcome assessment (detection bias) Radiological outcomes	Unclear risk	One researcher, blinded to group allocation, performed all radiological assessments.
Incomplete outcome data (attrition bias) All outcomes	Low risk	No dropouts
Selective reporting (reporting bias)	Low risk	All outcomes discussed in methodology reported
Other bias	Low risk	None

Kotela 2014

Study characteristics

Methods

Study Design: RCT

Country: Poland

Setting: Department of Orthopedic Surgery and Traumatology, Central Research Hospital of Ministry of Interior, Wołoska 137, 02-507, Warszawa, Poland

Length of Follow-up: 1 year

Kotela 2014 (Continued)

Participants

Inclusion Criteria: primary end-stage knee osteoarthritis, primary uncemented TKA, cruciate retaining, age >18 years, provision of written informed consent

Exclusion Criteria: nonosteoarthritic conditions as reason for operation (e.g. rheumatoid arthritis), co-morbidities that might affect bone quality (e.g. osteoporosis), history of previous surgery or trauma to the affected knee within 12 months preceding the study, advanced osteoarthritis and other conditions of large joints precluding reliable assessment of the operated knee joint, cognitive disorders resulting in suboptimal co-operation of the patient, a different operator than the person determined by the protocol, failure of patient to present for a follow-up examination, incomplete medical records

Number of participants randomised: 112

- Intervention group: 52
- Control group: 60

Number of participants who received allocated intervention:

- Intervention group
- Control group

Number of dropouts:

- Intervention group
- Control group

Age

- Intervention group: 66.1 ± 8.4
- Control group: 68.6 ± 9.9

Sex (Male/Female)

- Intervention group: 16/33
- Control group: 13/33

Duration of disease: Not reported

Concurrent treatment: Not reported

BMI

- Intervention group: 30.0 ± 4.6
- Control group: 29.6 ± 5.6

Preoperative mechanical axis

- Intervention group: 188.2 ± 7.0
- Control group: 188.9 ± 7.1

Interventions

Intervention group: Patient-specific instrumentation

Control group: Conventional instrumentation

Type of implant: cruciate retaining

Bearing design

Mode of fixation: uncemented

Patellar resurfacing: No

Preop imaging for designing PSG: CT

Implant Brand: Vanguard™ Complete Knee System; Biomet Inc. Warsaw, IN, USA

Kotela 2014 (Continued)

PSG Brand: CT-based PSI (Signature™ Personalized Patient Care System; Biomet Inc, Warsaw, IN, USA)

Intraoperative Data

- Number of times PSG was abandoned: No conversion to traditional instrumentation
- Reasons for abandonment
- Size correlation between planned component size and actual component used:
Femur: 44/49 (89.8%), tibia 37/49 (75.5%), insert 39/49 (79.6%)
- Any other relevant intraoperative data:

In 26.5% of patients, the position of the tibial guide had to be changed at least once after the pins had been placed; the figure for the femoral guide was 4%

Surgeon experience and pre-training: Not reported

Outcomes	Outcomes assessed: Imaging modality used for postoperative measurement: x-ray-based Outcomes used in this review: 1. Survival rate of implant (revision surgery to change a prosthetic component) 2. Functional outcome measures 3. Radiological alignment of components 4. Number of outliers in radiographic alignment 5. Operative time 6. Total adverse events 7. Re-operation rate 8. Global assessment (participant-reported) 9. Blood loss 10. Length of hospital stay Outcomes not used in this review: nil
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Notes

Risk of bias

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Unclear risk	Quote from study: "Participants were divided into two groups using a simple randomisation procedure."
Allocation concealment (selection bias)	High risk	Quote from study: "There was some room for bias in the study, as the first author was a member of the surgical team in all cases he would know the group they were assigned to".
Blinding of participants and personnel (performance bias) All outcomes	Unclear risk	Not reported
Blinding of outcome assessment (detection bias) Functional outcomes	High risk	Quote from study: "There was some room for bias in the study, as the first author was a member of the surgical team in all cases and cared for the patients after the procedure.... he would know the group they were assigned to during the clinical examination".

Kotela 2014 (Continued)

Blinding of outcome assessment (detection bias) Radiological outcomes	Low risk	Quote from study: "Radiographs were assessed by an independent orthopaedic surgeon for two angle measurements. The observer was blinded to surgical technique."
Incomplete outcome data (attrition bias) All outcomes	High risk	Dropouts: 17/112 (15.2%) - Intervention group: 3/52 (5.8%) - Control group: 14/60 (23.3%) Comment: Due to unequal dropout rates, there is high risk of attrition bias.
Selective reporting (reporting bias)	Low risk	All the outcome measures mentioned in the methodology were reported.
Other bias	Low risk	None detected

Liao 2024
Study characteristics

Methods	Study Design: RCT Country: China Setting: Third affiliated hospital of Southern Medical University, Guangzhou, Guangdong, China Length of Follow-up: NA
Participants	Inclusion Criteria: age > 40 and < 80 y; met the diagnostic criteria of osteoarthritis of American Academy of Orthopaedic Surgeons (AAOS); consent to participate in this study; met TKA indications Exclusion Criteria: patients who declined to participate in this study; inflammatory arthritis; revision TKA; active infections; arthrocentesis in recent 6 months, poor body condition that might not withstand the surgery Number of participants randomised: 102 patients, 107 knees - Intervention group: 54 - Control group: 53 Number of participants who received allocated intervention: - Intervention group: 54 - Control group: 53 Number of dropouts: - Intervention group: 0 - Control group: 0 Age: Intervention group: 68.6 +/- 7.1 Control Group: 67.8 +/- 7.7 Sex (Male/Female): - Intervention Group: 9/24 - Control Group: 12/9

Patient-specific cutting guides for total knee arthroplasty (Review)

Liao 2024 (Continued)

	Duration of disease: Not reported Concurrent treatment: Not reported BMI: Intervention Group: 26.1 +/- 3.4 Control Group: 26.3 +/-3.7 Preoperative mechanical Axis: Intervention Group: 8.2 varus Control Group: 9.1 varus
Interventions	Intervention Group: Patient specific Instrumentation Control Group: Conventional Instrumentation Type of implant: Attune, De Puy Bearing design: Posterior-Stabilised, mobile bearing Mode of fixation: Cemented Patella resurfacing: None in all cases Pre-op imaging for designing PSG: CT Implant brand: Attune, De Puy PSG Brand: Selective Laser Sintering (SLS) Technology 3D printer (Longwood Valley Technology, Beijing, China) Intraoperative data
Outcomes	Outcomes assessed: Imaging modality used for postoperative measurement: - Weightbearing long-leg AP & lateral radiographs - CT Outcomes used in this review: -Surgical time -Radiological outliers -Radiographic alignment (HKA, FCA, TCA, JLCA, PCO, FSA, TSA, PCA) Outcomes not used in this review: Nil
Notes	All surgeries performed by one surgeon with more than 5 years' experience Blinding: computer-generated permuted blocks of four, block randomisation - assessors unaware of block size
Risk of bias	
Bias	Authors' judgement Support for judgement

Liao 2024 (Continued)

Random sequence generation (selection bias)	Low risk	Computer-generated blocked randomisation
Allocation concealment (selection bias)	Unclear risk	Computer-generated blocked randomisation
Blinding of participants and personnel (performance bias) All outcomes	Unclear risk	Assessors unaware of block size during computer-generated block randomisation
Blinding of outcome assessment (detection bias) Functional outcomes	Unclear risk	Not reported
Blinding of outcome assessment (detection bias) Radiological outcomes	Unclear risk	Not reported
Incomplete outcome data (attrition bias) All outcomes	Unclear risk	Not reported
Selective reporting (reporting bias)	Low risk	All the outcome measures mentioned in the methodology were reported.
Other bias	Low risk	None

Lionberger 2014

Study characteristics

Methods	Study Design: RCT Country: USA Setting: Houston, Texas Length of Follow-up: 3 months
Participants	Inclusion Criteria: primary end-stage knee osteoarthritis, flexion contracture of less than 15° and ligament stability of no more than 2° instability in varus/valgus stress, age > 18 years Exclusion Criteria: Patients with any medical condition or personal circumstance that would prevent completion of follow-up visits (3 weeks, 6 weeks, and 3 months) post-operation or with any preoperative systemic infections; uncontrolled diabetes; or other conditions that interfere with the wound healing process or that prohibit the use of MRI. Additional exclusion criteria included individuals with fixed alignment deformities greater than 7° of varus or valgus or prior TKAs or hardware that may confound complexity of a standard TKA. Number of participants randomised: 63 - Intervention group: 33 - Control group: 30 Number of participants who received allocated intervention: - Intervention group: 32

Patient-specific cutting guides for total knee arthroplasty (Review)

Lionberger 2014 (Continued)

- Control group: 30

Number of dropouts:

- Intervention group: 2
- Control group: 0

Age: Not reported

Sex (Male/Female): Not reported

Duration of disease: Not reported

Concurrent treatment: Not reported

BMI: Not reported

Preoperative mechanical axis: Not reported

Interventions	Intervention group: Patient-specific instrumentation
	Control group: Navigation
	Type of implant: Not reported
	Bearing design: Not reported
	Mode of fixation: Not reported
	Patellar resurfacing: No
	Preop imaging for designing PSG: CT
	Implant Brand: Vanguard™ Complete Knee System; Biomet Inc. Warsaw, IN, USA
	PSG Brand: CT-based PSI (Signature™ Personalized Patient Care System; Biomet Inc, Warsaw, IN, USA)
	Intraoperative Data
	· Number of times PSG was abandoned: No conversion to traditional instrumentation

Outcomes	Outcomes assessed at; postoperative, 3 weeks, 6 weeks, 3 months
	Imaging modality used for postoperative measurement: x-ray-based
	Outcomes used in this review:
	1. Incision length
	2. Functional outcome measures
	3. Radiological alignment of components
	4. Number of outliers in radiographic alignment
	5. Operative time
	6. Range of motion
	Outcomes not used in this review : Cost comparison analysis

Notes

Risk of bias

Bias	Authors' judgement	Support for judgement
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Lionberger 2014 (Continued)

Random sequence generation (selection bias)	Unclear risk	Not reported. Quote from the study: "Sixty patients were randomised to"
Allocation concealment (selection bias)	Unclear risk	Not reported
Blinding of participants and personnel (performance bias) All outcomes	Unclear risk	Not reported
Blinding of outcome assessment (detection bias) Functional outcomes	Unclear risk	Not reported
Blinding of outcome assessment (detection bias) Radiological outcomes	Low risk	Quote from study: "Radiographs were read and calculated by three blinded interpreters".
Incomplete outcome data (attrition bias) All outcomes	Low risk	According to CONSORT diagram, two patients were lost to follow-up in one of the arms. The rest of all patients contributed to outcome measurements.
Selective reporting (reporting bias)	Low risk	All the outcome measures mentioned in the methodology were reported.
Other bias	Low risk	None

Maus 2018
Study characteristics

Methods	Study Design: RCT Country: Germany Setting: Multicentre randomised controlled trial Length of Follow-up: 90 days
Participants	Inclusion Criteria: Clinical and radiological signs of osteoarthritis of the knee with failed nonoperative treatment; no indication for a unicompartmental implant or joint-preserving surgery; over 18 years; no previous bone surgery to the index knee; no previous total joint replacement of the index leg; no foreign metal in the index leg Exclusion Criteria: A contraindication for MRI (i.e. pacemaker); non-compliance; and claustrophobia Number of participants randomised: - Intervention group: 79 - Control group: 78 Number of participants who received allocated intervention: - Intervention group: 59 - Control group: 66 Number of dropouts:

Patient-specific cutting guides for total knee arthroplasty (Review)

Maus 2018 (Continued)

- Intervention group: 2
- Control group: 1

Age:

- Intervention group: 68.1 ± 8.5
- Control group: 71.5 ± 8.1

Sex (Male/Female)

- Intervention group: 26/33
- Control group: 23/43

Duration of disease: Not reported

Concurrent treatment: Not reported

BMI

- Intervention group: 31.8 ± 6.1
- Control group: 30.6 ± 5.3

Preoperative mechanical axis

- Intervention group: 175.0 ± 6.4
- Control group: 176.4 ± 6.8

Interventions

Intervention group: Patient-specific instrumentation

Control group: Conventional instrumentation

Type of implant:

Bearing design: Surgeon-dependent

Mode of fixation: Surgeon-dependent

Patellar resurfacing: Surgeon-dependent

Preop imaging for designing PSG: MRI

Implant Brand: Each surgeon used the implant type with which he was most experienced.

In two centres, the e.motion® implant was used; two centres used the VEGA system® and two centres used the Columbus® knee system (all implants by Aesculap AG, Tuttlingen, Germany)

PSG Brand: Imprint®, Aesculap AG, Tuttlingen, Germany

Intraoperative Data: The operation duration and the quantity of blood loss were registered intraoperatively.

Outcomes

Imaging modality used for postoperative measurement: X-ray & CT

Outcomes used in this review:

For the primary outcome, the mechanical leg alignment (MLA) was measured, the cases with $MLA > 183^\circ$ (valgus) or $< 177^\circ$ (varus) were considered outliers. For secondary outcomes, the following angles were additionally measured pre- and postoperatively: mechanical Medial Distal Femoral Angle (mMD-FA), mechanical Medial Proximal Tibial Angle (mMPTA), and anatomical Posterior Proximal Tibial Angle (aPPTA). Further secondary outcomes used to assess clinical results were the Knee Society Score (KSS) and the Knee injury and Osteoarthritis Outcome score (KOOS).

Outcomes **not used in this review:**

Maus 2018 (Continued)

Femoral component rotation, tibial sagittal alignment, outlier data for parameters other than MLA

Notes	Only MLA outlier data available	
<i>Risk of bias</i>		
Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Low risk	The randomisation was made centrally with a random computer generator via the website. Each of the recruiting surgeons in all six centres used an individual password to assess the randomisation system.
Allocation concealment (selection bias)	Unclear risk	The surgeon received the information on which group the patient was allocated to and a case number was provided under which all relevant study data of the patient were anonymously collected.
Blinding of participants and personnel (performance bias) All outcomes	High risk	Not possible
Blinding of outcome assessment (detection bias) Functional outcomes	Unclear risk	Not reported
Blinding of outcome assessment (detection bias) Radiological outcomes	High risk	The assessors were not blinded.
Incomplete outcome data (attrition bias) All outcomes	Unclear risk	PSI N = 2 (3.4%), control N = 1 (1.5%)
Selective reporting (reporting bias)	Low risk	All the outcome measures mentioned in the methodology were reported.
Other bias	Low risk	None detected

Molicnik 2015
Study characteristics

Methods	Study Design: parallel RCT Country: Slovenia Setting: University Clinical Centre Maribor Length of Follow-up: not reported
Participants	Inclusion Criteria - advanced osteoarthritis Exclusion Criteria - BMI above 40 kg/m² - preoperative mechanical axis above 20° of varus and 10° of valgus

Patient-specific cutting guides for total knee arthroplasty (Review)

Molicnik 2015 (Continued)

Number of people randomised: 38

- Intervention: 19
- Control: 19

Number of dropouts: 0

Baseline Characteristics

BMI

- Control group: 33.3 ± 5.5
- Intervention group: 31.9 ± 5.3

Preoperative mechanical axis

- Control group: 3.6 ± 2.4
- Intervention group: 4.2 ± 2.0

Interventions	Control group: Standard instrumentation Intervention group: Patient matched instrumentation Type of implant: Posterior stabilised Bearing design: not specified Mode of fixation: cemented Patellar resurfacing: not done in any patients Preop imaging for designing PSG: MRI Implant Brand: Vanguard (Biomet) PSG Brand: Signature (Biomet) Intraoperative Data • Number of times PSG was abandoned: Nil • Reasons for abandonment: Not applicable • Size correlation between planned component size and actual component used: Not reported • Any other relevant intraoperative data: Not reported Surgeon experience and pre-training: Unclear - Quote from study stated: "All procedures were performed by one experienced surgeon".
Outcomes	Outcomes assessed at: postoperatively before discharge (on average 4-6th day) or after full extension was achieved Imaging modality used for postoperative measurement: Radiograph Outcomes used in this review: 1. FTCA: Femoro tibial coronal angle (mechanical alignment) 2. Number of outliers in FTCA 3. Operative duration Outcomes not used in this review: 1. Difference in haemoglobin level

Molicnik 2015 (Continued)

Notes Operative duration is mentioned as dichotomous data (as < 90 min and > 90 min)

Risk of bias

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Unclear risk	Quote from the study: "were randomly assigned to one of two groups (1:1)"
Allocation concealment (selection bias)	Unclear risk	Not reported
Blinding of participants and personnel (performance bias) All outcomes	Unclear risk	Not reported
Blinding of outcome assessment (detection bias) Functional outcomes	High risk	This was an open-label study and the outcome measures used were patient-reported.
Blinding of outcome assessment (detection bias) Radiological outcomes	Unclear risk	Not reported
Incomplete outcome data (attrition bias) All outcomes	Low risk	All the participants contributed to the outcomes reported in the trial.
Selective reporting (reporting bias)	Low risk	All the outcome measures mentioned in the methodology were reported.
Other bias	Low risk	None detected

Noble 2012
Study characteristics

Methods	Study Design: RCT Country: United States of America Setting: Imperial Medical Group, LLP Lake Charles, Louisiana and Smith & Nephew Inc. Length of Follow-up
Participants	Inclusion Criteria: Not reported Exclusion Criteria: Not reported Number of participants randomised: 38 - Intervention group: 19 - Control group: 19 Number of participants who received allocated intervention: 29 - Intervention group: 15

Patient-specific cutting guides for total knee arthroplasty (Review)

Noble 2012 (Continued)

- Control group: 14

Number of dropouts:

- Intervention group:
- Control group: 4

Age

- Intervention group: 65.4
- Control group: 68

Sex (Male/Female)

- Intervention group: 8/7
- Control group: 6/8

Duration of disease: Not reported

Concurrent treatment: Not reported

BMI

- Intervention group: 29.8
- Control group: 31.2

Preoperative mechanical axis not reported

- Intervention group
- Control group

Interventions

Intervention group: Patient-specific instrumentation

Control group: Conventional instrumentation

Type of implant: cruciate retaining

Bearing design: Not reported

Mode of fixation: Not reported

Patellar resurfacing: Not reported

Preop imaging for designing PSG: MRI

Implant Brand: Legion Total Knee System (Smith & Nephew, Inc.)

PSG Brand: Visionaire (Smith & Nephew, Inc.)

Intraoperative Data

- Number of times PSG was abandoned: 0
- Reasons for abandonment:
- Size correlation between planned component size and actual component used: Not reported
- Any other relevant intraoperative data:
- Surgeon experience and pre-training: Not reported

Outcomes

Outcomes assessed not reported

Imaging modality used for postoperative measurement: Not reported

Noble 2012 (Continued)

Outcomes used in this review:

1. Radiological alignment of components – FTCA (femoro-tibial coronal angle)
2. Number of outliers in radiographic alignment
3. Operative time
4. Length of hospital stay
5. Blood loss
6. Incision length

Outcomes not used in this review: Nil

Notes	Trial identifier NCT01107769 (prospectively registered) SDs for blood loss, length of stay and incision length not reported. SDs imputed from other relevant studies
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Risk of bias

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Unclear risk	Quote from study: "Thirty-eight primary TKA patients were randomly assigned to 1 of 2 groups (1:1)" but the method used for random sequence generation was not reported.
Allocation concealment (selection bias)	Unclear risk	Not reported
Blinding of participants and personnel (performance bias) All outcomes	Unclear risk	Not reported
Blinding of outcome assessment (detection bias) Functional outcomes	Low risk	Functional outcome measures not reported in this study
Blinding of outcome assessment (detection bias) Radiological outcomes	Unclear risk	Not reported
Incomplete outcome data (attrition bias) All outcomes	High risk	Dropouts: 9/38 (23.7%) Quote from study: "... 4 of 19 STD (control group) patients were preoperatively withdrawn from the study before operation..... 5 patients were excluded from study because of protocol deviation (BMI > 40 kg/m ²)".
Selective reporting (reporting bias)	Low risk	All important outcome measures mentioned in the methodology section of the paper and the trial registration document were reported.
Other bias	Unclear risk	The study was sponsored by Smith & Nephew Inc.

Ollivier 2016

Study characteristics

Patient-specific cutting guides for total knee arthroplasty (Review)

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Ollivier 2016 (Continued)

Methods	Study Design: RCT Country: France Setting: Department of Orthopaedics and Traumatology, Versailles Hospital Centre, 78150 Le Chesnay, France Length of Follow-up: 3 months
Participants	Inclusion Criteria: patients aged 18–85 years old the day of surgery who had varus deformity and were undergoing TKA for primary osteoarthritis Exclusion Criteria: Patients with valgus knee or any history of previous surgery (arthroscopy excepted) or trauma to the affected knee, and patients who declined to participate in the trial Number of participants randomised: 80 • PSG group: 40 • NAV group: 40 Number of participants who received allocated intervention: • PSG group: 40 • NAV group: 40 Number of dropouts: • PSG group: • NAV group: Age • PSG group: 76 ± 7.1 • NAV group: 72.2 ± 17.4 Sex (Male/Female) • PSG group: 28/12 • NAV group: 28/12 Duration of disease: Not reported Concurrent treatment: Not reported BMI • PSG group: 26.3 ± 6.9 (18–39) • NAV group: 27 ± 6.1 (17–41) Preoperative mechanical axis • PSG group: 173.3 ± 6.2 (164–179) • NAV group: 172 ± 5.2 (158–178)
Interventions	Type of implant: PS Bearing design: Fixed Mode of fixation Patellar resurfacing Preop imaging for designing PSG

Ollivier 2016 (Continued)

Implant Brand: Legion Primary™ Total Knee System Posterior-Stabilized (Smith and Nephew®, Andover, USA)

PSG Brand: Visionaire™ Personalized Patient Care System (Smith and Nephew®, Andover, MA, USA)

NAVIGATION System: CAS Navitrack™ Orthosoft System® (Zimmer®, Warsaw, USA)

Intraoperative Data

- Number of times PSG was abandoned where navigation was used instead: 7
- Reasons for abandonment:
 - FTCA (femoro-tibial coronal angle) - differing from the initial target position by more than 3°
 - Tibial Posterior slope differing by more than 5°
 - Femoral rotation differing by more than 5°
 - Size correlation between planned component size and actual component used: Not reported
 - Any other relevant intraoperative data

Surgeon experience and pre-training: Not reported

Outcomes

Outcomes assessed at three months

Imaging modality used for postoperative measurement: full weight-bearing radiographs. Rotation of the femoral component was assessed with axial CT images.

Outcomes used in this review:

1. Radiological alignment of components
2. Number of outliers in radiographic alignment

Outcomes not used in this review: Nil

Notes

Risk of bias

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Low risk	Quote from the study: "Knees were allocated according to a permuted block randomisation Program."
Allocation concealment (selection bias)	Low risk	Quote from the study: "The patients and the surgeons were notified a few days before the surgery".
Blinding of participants and personnel (performance bias) All outcomes	Unclear risk	This was a open-label study.
Blinding of outcome assessment (detection bias) Functional outcomes	Low risk	Functional outcome measures not reported in this study
Blinding of outcome assessment (detection bias) Radiological outcomes	Unclear risk	Not reported

Ollivier 2016 (Continued)

Incomplete outcome data (attrition bias) All outcomes	Low risk	Quote from study: "The use of the custom instrumentation was abandoned intraoperatively in seven knees". However, outcomes for these participants were reported separately.
Selective reporting (reporting bias)	Low risk	All the outcome measures mentioned in the methodology were reported.
Other bias	Low risk	None detected

Parratte 2013
Study characteristics

Methods	Study Design: parallel RCT Country: France Setting: Institute for Locomotion, Aix-Marseille University Length of Follow-up: Three months
Participants	Inclusion Criteria: - disabling knee osteoarthritis; ages between 50 and 85 years old; at an acceptable medical risk and failed nonoperative management; willingness to wait 4-6 weeks before surgery and the acceptance of the relatively new technology Exclusion Criteria: - metallic hardware within 10 cm of the knee or prior THA; previous surgery of the knee that could lead to artifacts Number of participants randomised: 40 - Intervention: 20 - Control: 20 Number of participants who received allocated intervention: 40 Number of dropouts: 0 - Intervention: 0/20 (0%) - Control: 0/20 (0%) Number of dropouts at follow-up: - Intervention: 0, 0% - Control: 0, 0% Age - Intervention group: not stated - Control group: not stated Sex (Male/Female) - Intervention group: not stated - Control group: not stated

Parratte 2013 (Continued)

Duration of disease: Not reported

Concurrent treatment: Not reported

BMI (Mean): Not reported

Preoperative mechanical axis:

- Intervention group deviation: Not reported
- Control group: Not reported

Interventions

Intervention group: Patient-specific instrumentation

Control group: Conventional instrumentation

Type of implant: Zimmer, NexGen

Bearing design: PS

Mode of fixation: cemented

Patellar resurfacing: not reported

Preop imaging for designing PSG: MRI

Implant Brand: Zimmer, NexGen

PSG Brand: Materialise, Belgium

Intraoperative Data

- Number of times PSG was abandoned: 0
- Reasons for abandonment: Not applicable
- Size correlation between planned component size and actual component used: Not reported
- Any other relevant intraoperative data: Not reported

Surgeon experience and pre-training: All surgeries performed by two senior surgeons

Outcomes

Outcomes assessed at: immediate postoperative and three-month review

Imaging modality used for postoperative measurement: Radiograph and CT

Outcomes used in this review:

1. Number of outliers in radiographic alignment

Outcomes not used in this review:

1. Functional outcome measures
2. Radiological alignment of components
3. Pain measure (VAS)
4. Quality of life measure (EQ -5D)
5. Total adverse events
6. Re-operation rate
7. Operative time
8. Blood loss
9. Length of hospital stay
10. Revision rate

Notes

Parratte 2013 (Continued)

1. Pre and postoperative cut differences between groups
2. FTC/LTC angle
3. Differences in postoperative rotation described in both groups as catagorised by single degrees

Notes

Risk of bias

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Low risk	Randomisation of patients into one of the two groups was done by the Hospital Informatics Department with the use of a systematic sampling method.
Allocation concealment (selection bias)	Low risk	Allocation concealed in numbered, sealed envelopes
Blinding of participants and personnel (performance bias) All outcomes	Low risk	Adequately blinded, randomisation not revealed to authors
Blinding of outcome assessment (detection bias) Functional outcomes	Low risk	Outcome not measured
Blinding of outcome assessment (detection bias) Radiological outcomes	Low risk	Two observers blinded
Incomplete outcome data (attrition bias) All outcomes	Low risk	No known incomplete outcome data
Selective reporting (reporting bias)	Unclear risk	Not all outcomes were reported
Other bias	Low risk	None detected

Pfitzner 2014

Study characteristics

Methods	Study Design: RCT Country: Germany Setting: Hospital Maerkisch Oderland, Section Wriezen, Wriezen, Germany and Center for Musculoskeletal Surgery- Orthopedic Department, Charite ´ -Universitaetsmedizin Berlin, Berlin, Germany Length of Follow-up: mean 3 months
Participants	Inclusion Criteria: diagnosis of primary osteoarthritis; no previous open knee procedures; an extension deficit less than 20; and coronal deformity less than 15 Exclusion Criteria: Patients with any metal near the knee that could have interfered with the imaging Number of participants randomised: 90

Patient-specific cutting guides for total knee arthroplasty (Review)

Pfizner 2014 (Continued)

- CT PSG group: 30
- MRI PSG group: 30
- Conventional group: 30

Number of participants who received allocated intervention:

- CT PSG group: 30
- MRI PSG group: 30
- Conventional group: 30

Number of dropouts:

- CT PSG group: 30
- MRI PSG group: 30
- Conventional group: 30

Age

- CT PSG group: 63
- MRI PSG group: 65
- Conventional group: 64

Sex (Male/Female)

- CT PSG group: 12/18
- MRI PSG group: 14/16
- Conventional group: 13/17

Duration of disease: Not reported

Concurrent treatment: Not reported

BMI

- CT PSG group: 30
- MRI PSG group: 30
- Conventional group: 31

Preoperative mechanical axis not reported

Interventions

Type of implant: cruciate substituting

Bearing design: Fixed

Mode of fixation: cemented

Patellar resurfacing: No

Preop imaging for designing PSG: CT and MRI according to group

Implant Brand:

- CT PSG group: Sigma1 Press Fit Condylar (PFC1) High Performance (HP) (DePuy)
- MRI PSG group: Journey Bicruciate Substituting (BCS) (Smith & Nephew)
- Conventional group: Journey Bicruciate Substituting (BCS) (Smith & Nephew)

PSG Brand:

- CT PSG group: TruMatch1; DePuy Orthopaedics, Warsaw, IN
- MRI PSG group: Visionaire1; Smith & Nephew, Inc, Memphis, TN, USA

Pfizzner 2014 (Continued)

Intraoperative Data

- Number of times PSG was abandoned: 0
- Reasons for abandonment
- Size correlation between planned component size and actual component used:
 - CT PSG group: Femur: 80%, Tibia: 70%
 - MRI PSG group: Femur: 83%, Tibia: 83%
 - Conventional group: Femur: 53%, Tibia: 60%
- Any other relevant intraoperative data:

Surgeon experience and pre-training: senior surgeon (HH) who had completed greater than 1000 primary TKAs and greater than 50 patient-specific instrumentation TKAs

Outcomes	Outcomes assessed at mean of 3 months Imaging modality used for postoperative measurement: standing long-leg radiograph, a lateral knee radiograph, and a CT scan to measure rotation of the components Outcomes used in this review: 1. Survival rate of implant (revision surgery to change a prosthetic component) 2. Functional outcome measures 3. Radiological alignment of components 4. Number of outliers in radiographic alignment 5. Operative time Outcomes not used in this review: Nil
Notes	Functional outcome data in this study compared between PSI (CT), PSI (MRI) and CON as three groups. PSI (CT) and CON used for analysis for this review. SD not available for functional outcomes – KSS, WOMAC. SDs imputed from relevant studies

Risk of bias

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Low risk	Quote from study: "An independent participant (KW) completed the randomisation process". Comment: Although the random sequence generation method was not reported, the use of sealed envelopes in concealing allocation suggests that the unpredictability of randomisation was preserved.
Allocation concealment (selection bias)	Low risk	Quote from study: "....and sealed envelopes were used".
Blinding of participants and personnel (performance bias) All outcomes	Unclear risk	This was an open-label study.
Blinding of outcome assessment (detection bias) Functional outcomes	Unclear risk	Not reported

Pfizner 2014 (Continued)

Blinding of outcome assessment (detection bias) Radiological outcomes	Low risk	Quote from study: "The radiologic measurements (radiographs and CT) were performed by an experienced investigator (TP) blinded to the patients' allocation".
Incomplete outcome data (attrition bias) All outcomes	Low risk	Quote from study: "After inclusion, no patients were lost to follow-up, and complete datasets were obtained for all patients".
Selective reporting (reporting bias)	Low risk	Not prospectively registered. However, all the outcome measures mentioned in the methodology were reported.
Other bias	Low risk	None detected

Pietsch 2014
Study characteristics

Methods	Study Design: RCT Country: Austria Setting: Department of Orthopaedic Surgery, General and Orthopaedic Hospital Stolzalpe, 8852 Stolzalpe, Austria Length of Follow-up: 12 weeks
Participants	Inclusion Criteria: OA Exclusion Criteria: Patients with rheumatoid arthritis due to higher expected blood loss and the fact that synovectomy limits the ability to perform minimally invasive surgery; patients with a history of a bleeding diathesis and those taking warfarin; patients who were not able to undergo MRI examination Number of participants randomised: - Intervention group: 40 - Control group: 40 Number of participants who received allocated intervention: 40 Number of dropouts: - Intervention group: 40 - Control group: 40 Age - Intervention group: 71.4 ± 6.6 - Control group: 69.2 ± 9.4 Sex (Male/Female) - Intervention group: 13/27 - Control group: 19/21 Duration of disease: Not reported Concurrent treatment: Not reported BMI

Pietsch 2014 (Continued)

- Intervention group: 29.0 ± 3.5
- Control group: 30.8 ± 4.9

Preoperative mechanical axis

- Intervention group: 6.0 ± 3.7 (0.7–17.9)
- Control group: 7.2 ± 4.9 (0.7–25.2)

Interventions

Intervention group: Patient-specific instrumentation

Control group: Conventional instrumentation

Type of implant: Posterior-Stabilised Implant

Bearing design: Fixed

Mode of fixation: cemented

Patellar resurfacing: Yes

Preop imaging for designing PSG: MRI

Implant Brand: NexGen, Zimmer, Warsaw, USA

PSG Brand: Patient Specific Instruments, Zimmer

Intraoperative Data

- Number of times PSG was abandoned: None
- Reasons for abandonment
- Size correlation between planned component size and actual component used: Not specified
- Any other relevant intraoperative data: None

Surgeon experience and pre-training: Two surgeons (MP and OD), each having performed more than 1000 primary TKAs including 400 minimally invasive surgeries before study initiation

Outcomes

Functional outcomes assessed postoperatively at week 2, 6 and 12

Blood loss: The volume in all drain bottles was recorded, and the drains were removed 48 h after surgery. Transfusion requirements noted

Imaging modality used for postoperative measurement: Not available

Outcomes used in this review:

1. Functional outcome measures
2. Operative time
3. Total adverse events

Outcomes not used in this review:

1. Radiological alignment of components
2. Number of outliers in radiographic alignment
3. Re-operation rate
4. Length of hospital stay

Notes
Risk of bias

Pietsch 2014 (Continued)

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Low risk	Quote from study: "Patients were randomised 8 weeks before surgery by sealed envelopes to receive either the custom-fit pin guide or standard procedure."
Allocation concealment (selection bias)	Low risk	Quote from study: "Patients were randomised 8 weeks before surgery by sealed envelopes".
Blinding of participants and personnel (performance bias) All outcomes	Unclear risk	This was an open-label study.
Blinding of outcome assessment (detection bias) Functional outcomes	Low risk	Quote from study: "Two independent, blinded observers (FP, ChZ) performed all measurements."
Blinding of outcome assessment (detection bias) Radiological outcomes	Unclear risk	Not reported
Incomplete outcome data (attrition bias) All outcomes	Low risk	All the patients who received the allocated intervention contributed to the outcome measurement.
Selective reporting (reporting bias)	Low risk	Not prospectively registered. However, all the outcome measures mentioned in the methodology were reported.
Other bias	Low risk	None

Qiu 2017
Study characteristics

Methods	Study Design: RCT Country: China Setting: · Department of Joints, Guizhou Orthopedic Hospital, Guiyang, China · Department of R and D, Arigin Medical Co., Ltd., Shanghai, China Length of Follow-up: Not reported
Participants	Inclusion Criteria: · end-stage knee osteoarthritis (OA) with no severe valgus/varus deformity (> 20 degrees) and no previous knee surgery history · patella with good mobility · failure of nonoperative treatments · enrolled patients who had completed surgery with no contraindications Exclusion Criteria

Patient-specific cutting guides for total knee arthroplasty (Review)

Qiu 2017 (Continued)

- tumour around the knee
- active or suspected sepsis
- age younger than 50 or older than 80

Number of participants randomised: 26

- Intervention group: 10
- Control group: 16

Number of participants who received allocated intervention:

Number of dropouts: Not reported

Age

- Intervention group: 67.57 ± 7
- Control group: 65.5 ± 6.58

Sex (Male/Female)

- Intervention group: 4M 6F
- Control group: 2M 14 F

Duration of disease: Not reported

Concurrent treatment: Not reported

BMI

- Intervention group
- Control group

Preoperative mechanical axis

- Intervention group: 8.97 ± 1.85
- Control group: 8.73 ± 1.67

Interventions

Intervention group: Patient-specific instrumentation

Control group: Conventional instrumentation

Type of implant: Not reported

Bearing design: Not reported

Mode of fixation: Not reported

Patellar resurfacing: Not reported

Preop imaging for designing PSG: CT

Implant Brand: Not reported

PSG Brand: Arigin 3D Pro and Arigin 3D surgical templating software

Intraoperative Data

- Number of times PSG was abandoned: Not reported
- Reasons for abandonment
- Size correlation between planned component size and actual component used: Not reported
- Any other relevant intraoperative data

Qiu 2017 (Continued)

Surgeon experience and pre-training: Not reported

Outcomes

Outcomes assessed not reported

Imaging modality used for postoperative measurement

Outcomes used in this review:

1. Radiological alignment of components
2. Number of outliers in radiographic alignment
3. Survival rate of implant (revision surgery to change a prosthetic component)
4. Total adverse events including infection, thrombosis, palsy, death, etc.
5. Functional outcome measures
6. Operative time
7. Length of hospital stay

Outcomes not used in this review: Nil

Notes

Risk of bias

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	High risk	Quote from study: "Twenty-eight patients were initially evaluated and 26 of them were randomly selected according to such inclusion criteria."
Allocation concealment (selection bias)	Unclear risk	Not reported
Blinding of participants and personnel (performance bias) All outcomes	Unclear risk	Not reported
Blinding of outcome assessment (detection bias) Functional outcomes	Unclear risk	Not reported
Blinding of outcome assessment (detection bias) Radiological outcomes	Unclear risk	Not reported
Incomplete outcome data (attrition bias) All outcomes	Unclear risk	Not reported
Selective reporting (reporting bias)	Low risk	None detected
Other bias	Low risk	None detected

Randelli 2019

Study characteristics

Patient-specific cutting guides for total knee arthroplasty (Review)

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Randelli 2019 (Continued)

Methods	Study Design: RCT Country: Italy Setting: IRCCS Policlinico San Donato, Milan, Italy Length of Follow-up: 2 months
Participants	Inclusion Criteria: Age 40-80; indication for TKA; surgical indication for ATUNNE posterior-stabilised, fixed-bearing TKA; patients signed written informed consent Exclusion Criteria: TKA revision surgery; metal devices within 8 cm from knee articular surfaces; fixed knee deformities greater than 15° in varus, valgus, flexion or tibial slope; knee ankyloses; previous tibial osteotomy surgery; serious comorbidity; active infections; pregnant or breastfeeding women; informed consent not accepted Number of participants randomised: 69 - Intervention group: 32 - Control group: 37 Number of participants who received allocated intervention: - Intervention group: 30 - Control group: 32 Number of drop-outs: - Intervention group: 2 - Control group: 5 Age - Intervention group: 70.46 - Control group: 70.34 Sex (Male/Female) - Intervention group: F:M ratio 0.69/0.31 - Control group: F:M ratio 0.81/0.19 Duration of disease: Not reported Concurrent treatment: Not reported BMI - Intervention group: 28.67 ± 3.93 - Control group: 29.20 ± 5.16 Preoperative mechanical axis - Intervention group: not recorded - Control group: not recorded
Interventions	Intervention group: Patient-specific instrumentation Control group: Conventional instrumentation

Randelli 2019 (Continued)

Type of implant: Attune posterior-stabilised, fixed bearing TKA

Bearing design: Fixed

Mode of fixation: Cemented

Patellar resurfacing: Not reported

Preop imaging for designing PSG: CT

Implant Brand: Attune

PSG Brand: TRUMATCH (DePuy Orthopaedics Inc, Warsaw, IN)

Intraoperative Data:

Tourniquet time; total surgical time (from incision to skin closure); femoral and tibial recuts; implant size; and complications of any kind were noted.

Outcomes	Imaging modality used for postoperative measurement: knee CT using a metal artifact reduction protocol Outcomes used in this review: Surgery: ischaemia time (tourniquet inflation to release), total surgical time (incision to closure), femoral recuts, tibial recuts Blood management: total blood loss, estimated RBC loss, perioperative haemoglobin reduction Clinical outcomes: OKS variation (preoperative – 2 months after surgery), VAS variation (preoperative – 2 months after surgery) CT alignment assessment: Femoral component rotation to the surgical transepicondylar axis, tibial component slope Device related outcomes (only in treatment group): perioperative modification of desired resections Complications
Notes	No radiological outlier data available; no blood loss quantitative data available

Risk of bias

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Low risk	Block randomisation was performed to allocate patients to treatment or control group. A random allocation list with 4 patients per block was generated by an investigator involved neither in patient enrolment nor in data analysis.
Allocation concealment (selection bias)	Low risk	The investigator concealed the sequence in sequentially numbered sealed envelopes until interventions were assigned.
Blinding of participants and personnel (performance bias) All outcomes	Unclear risk	Not reported
Blinding of outcome assessment (detection bias) Functional outcomes	Unclear risk	Not reported
Blinding of outcome assessment (detection bias)	Unclear risk	Not reported

Patient-specific cutting guides for total knee arthroplasty (Review)

Randelli 2019 (Continued)

Radiological outcomes

Incomplete outcome data (attrition bias) All outcomes	Low risk	Dropout rate of 13% (9/69). Treatment group: 3 patients declined surgery for personal reasons, 1 had problems with CT reconstructions, 1 had non-sterile tibial guide, 1 was lost to follow-up. Control group: 1 patient declined surgery for personal reasons, 1 lost to follow-up
Selective reporting (reporting bias)	Low risk	All the outcome measures mentioned in the methodology were reported.
Other bias	Unclear risk	Study only adequately powered to detect differences in femoral component rotation but not to investigate alignment on other planes.

Roh 2013

Study characteristics

Methods	Study Design: RCT Country: Korea Setting: Department of Orthopedic Surgery, Seoul National University Hospital, 101 Daehak-Ro, Jongno-Gu, Seoul 110-744, Korea Joint Reconstruction Center, Seoul NOW Hospital, Kyungki-Do, Korea Length of Follow-up: Not reported
Participants	Inclusion Criteria: only patients with primary osteoarthritis Exclusion Criteria: "any history of previous surgery trauma to the affected knee were excluded, as were patients who declined to participate in the trial" Number of participants randomised: 100 - Intervention group: 50 - Control group: 50 Number of participants who received allocated intervention: - Intervention group: 42 - Control group: 48 Number of dropouts: - Intervention group: 8 - Control group: 2 Age - Intervention group: 70 ± 7.2 - Control group: 70 ± 5.1 Sex (Male/Female) - Intervention group: 39/3 - Control group: 43/5 Duration of disease: Not reported

Roh 2013 (Continued)

	Concurrent treatment: Not reported BMI - Intervention group: 27 ± 4.2 - Control group: 27 ± 2.7 Preoperative mechanical axis - Intervention group: Varus 9.2 ± 4.1 - Control group: Varus 8.6 ± 5.2
Interventions	Intervention group: Patient-specific instrumentation Control group: Conventional instrumentation Type of implant: PS Bearing design: mobile bearing Mode of fixation: Patellar resurfacing: - Intervention group: Y 31 N 11 - Control group: Y 26 N 22 Preop imaging for designing PSG: CT Implant Brand: Vanguard PS Mobile Bearing Knee; Biomet Inc. PSG Brand: CT-based Signature™ Intraoperative Data - Number of times PSG was abandoned: 8 knees (16%) - Reasons for abandonment: discrepancies greater than 3° in the femoral component rotation compared with the conventional gap method - External rotation was excessive in six knees and insufficient in two knees - Among the knees with excessive femoral component rotation, two were accompanied by unacceptably decreased posterior slope of the tibia, and these were corrected by recutting the tibia with an extramedullary guide - Size correlation between planned component size and actual component used: NA - Any other relevant intraoperative data Surgeon experience and pre-training: one experienced surgeon
Outcomes	Outcomes assessed not reported Imaging modality used for postoperative measurement: Full weight-bearing plain radiographs for mechanical axis and CT for individual components and femoral rotation Outcomes used in this review: - Radiological alignment of components - Number of outliers in radiographic alignment Outcomes not used in this review: Nil

Roh 2013 (Continued)

Notes

Risk of bias

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Low risk	Quote from study: "Knees were allocated according to permuted block randomisation program".
Allocation concealment (selection bias)	Unclear risk	Quote from study: "... the patient and the surgeon were notified a few days before the surgery".
Blinding of participants and personnel (performance bias) All outcomes	Unclear risk	Open-label study
Blinding of outcome assessment (detection bias) Functional outcomes	Unclear risk	Not reported
Blinding of outcome assessment (detection bias) Radiological outcomes	Unclear risk	Not reported
Incomplete outcome data (attrition bias) All outcomes	Low risk	All the participants who were randomised contributed to outcome measurement. Although there was a dropout rate of 10%, a quote from the study stated: "We also performed an intention-to-treat analysis There were no significant differences between this analysis and the one described in the previous paragraph (per protocol analysis)".
Selective reporting (reporting bias)	Low risk	All the outcome measures mentioned in the methodology were reported.
Other bias	Low risk	None detected

Sariali 2019

Study characteristics

Methods	Study Design: RCT Country: France Setting: Hôpitaux Universitaires La Pitié Salpêtrière-Charles Foix, AP-HP, France Length of Follow-up: 2 years
Participants	Inclusion Criteria: Patients who underwent primary TKA by anteromedial approach with the same surgeon using a cemented postero-stabilised prosthesis Exclusion Criteria: Revision knee surgery Number of participants randomised: - Intervention group: 40 - Control group: 40

Sarali 2019 (Continued)

Number of participants who received allocated intervention:

- Intervention group: 40
- Control group: 40

Number of dropouts:

- Intervention group: 0
- Control group: 0

Age

- Intervention group: 67.7 ± 8.7 (65.0–70.4)
- Control group: 68.9 ± 9.2 (66.0–71.8)

Sex (Male/Female)

- Intervention group: 7/33
- Control group: 16/24

Duration of disease: Not reported

Concurrent treatment: Not reported

BMI

- Intervention group: 29.7 ± 5.3 (28.0–31.3)
- Control group: 29.5 ± 4.5 (28.0–30.9)

Preoperative mechanical axis

- Intervention group: 178.5 ± 8 (176.0–181.0)
- Control group: 179.3 ± 8.8 (176.6–182.0)

Interventions

Intervention group: Patient-specific instrumentation

Control group: Conventional instrumentation

Type of implant: PS

Bearing design: Not reported

Mode of fixation: Cemented

Patellar resurfacing: Not reported

Preop imaging for designing PSG: CT

Implant Brand: FIRST®, Symbios, Yverdon, Switzerland

PSG Brand: Knee-Plan® (Symbios, Switzerland)

Intraoperative Data: Operating time

Outcomes

Accuracy of restoration of component positioning angles and 3D knee alignment

The percentage of outliers for the components alignment and HKA angle which was outside the $-3^\circ/+3^\circ$ range

The clinical outcomes at two years of follow-up using the International Knee Society score

Notes

Sariali 2019 (Continued)

Risk of bias

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Low risk	Block randomisation was performed to allocate patients in the treatment (PSI) or control group.
Allocation concealment (selection bias)	Low risk	Sequence concealed in sequentially numbered sealed envelopes
Blinding of participants and personnel (performance bias) All outcomes	Low risk	All patients were blinded to the randomisation results.
Blinding of outcome assessment (detection bias) Functional outcomes	Low risk	All the assessments were fully blinded in relation to the groups.
Blinding of outcome assessment (detection bias) Radiological outcomes	Low risk	Radiographic assessor was blinded to treatment allocation.
Incomplete outcome data (attrition bias) All outcomes	Low risk	No patients lost to follow-up
Selective reporting (reporting bias)	Low risk	All the outcome measures mentioned in the methodology were reported.
Other bias	Low risk	None detected

Silva 2014

Study characteristics

Methods	Study Design: RCT Country: Portugal Setting: Hospital Militar D. Pedro V and Hospital da Boavista, Avenida da Boavista, Porto, Portugal Length of Follow-up: Not reported
Participants	Inclusion Criteria - primary osteoarthritis - without previous osteotomy and fractures around the knee - able to undergo magnetic resonance imaging Exclusion Criteria: Not reported Number of people randomised: 45 - Intervention group: 23 - Control group: 22

Silva 2014 (Continued)

	Number of dropouts: 0 Age (median) - Intervention group: 73 (67; 78) - Control group: 74 (70.5; 80) Sex (Male/Female) - Intervention group: Not reported - Control group: Not reported Duration of disease: Not reported Concurrent treatment: Not reported BMI: Not reported Preoperative mechanical axis: Not reported
Interventions	Intervention group: Patient-specific instrumentation Control group: Conventional instrumentation Type of implant: Posterior stabilised Bearing design: Not reported Mode of fixation: Cemented Patellar resurfacing: Not reported Preop imaging for designing PSG Implant Brand: Vanguard PSG Brand: Signature Intraoperative Data - Number of times PSG was abandoned: Not reported - Reasons for abandonment: Not reported - Size correlation between planned component size and actual component used: Not reported - Any other relevant intraoperative data: Not reported Surgeon experience and pre-training: Quote from study: "all surgeries were conducted by an experienced knee surgeon".
Outcomes	Outcomes assessed at: 1 week after surgery Imaging modality used for postoperative measurement: CT scan Outcomes used in this review: - Radiological alignment of components Outcomes not used in this review: - Amplitude and dispersion of component rotation around median
Notes	

Silva 2014 (Continued)

Risk of bias

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Unclear risk	Quote from study: "The patients were prospectively randomised into two groups".
Allocation concealment (selection bias)	Unclear risk	Not reported
Blinding of participants and personnel (performance bias) All outcomes	Unclear risk	Open-label study
Blinding of outcome assessment (detection bias) Functional outcomes	Unclear risk	Not reported
Blinding of outcome assessment (detection bias) Radiological outcomes	Low risk	Quote from study: "... reviewed by a musculoskeletal radiologist blind to the group the subject was in"
Incomplete outcome data (attrition bias) All outcomes	Low risk	All the participants contributed to the outcomes reported in the trial.
Selective reporting (reporting bias)	Low risk	Not prospectively registered. All the outcome measures mentioned in the methodology were reported.
Other bias	Low risk	None detected

Stolarczyk 2018

Study characteristics

Methods	Study Design: RCT Country: Poland Setting: Department of Orthopedics and Rehabilitation of Warsaw Medical University Length of Follow-up: 12 weeks
Participants	Inclusion Criteria: Patients all qualified for the TKA procedure due to primary knee osteoarthritis Exclusion Criteria: Not reported Number of participants randomised: - Intervention group: 30 - Control group: 30 Number of participants who received allocated intervention: - Intervention group: 30 - Control group: 30

Stolarczyk 2018 (Continued)

Number of dropouts:

- Intervention group: 1
- Control group: 1

Age:

- Intervention group: 70.2 +/- 5.9
- Control group: 69.6 +/- 7.1

Sex (Male/Female):

- Intervention group: 8/22
- Control group: 12/18

Duration of disease: Not reported

Concurrent treatment: Not reported

BMI:

- Intervention group: 30.4 +/- 4.4
- Control group: 31.6 +/- 5.4

Preoperative mechanical axis: Not reported

Interventions	Intervention group: Patient-specific instrumentation Control group: Conventional instrumentation Type of implant Bearing design: Not reported Mode of fixation: Not reported Patellar resurfacing: Not reported Preop imaging for designing PSG: MRI PSG Brand: Visionaire PSI (VISIONAIRE; Smith and Nephew, Memphis, TN) Intraoperative Data: Operation duration and length of surgical scar
Outcomes	Outcomes assessed: Imaging modality used for postoperative measurement: X-ray & CT Outcomes used in this review: The primary outcome was the rotational alignment of the femoral and tibial component. Secondary outcomes: KSS, VAS Tertiary endpoints included the assessment of surgery duration, the length of the surgical scar, and the duration of postoperative hospital stay. Outcomes not used in this review: Alignment outliers
Notes	Alignment outliers not used. Quantitative data of tertiary endpoints not provided

Stolarczyk 2018 (Continued)

Risk of bias

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Low risk	Randomisation was performed using a computer-generated random number list that was prepared by an investigator who had no clinical involvement in the trial.
Allocation concealment (selection bias)	Low risk	The sequence was concealed until all data were analysed.
Blinding of participants and personnel (performance bias) All outcomes	Unclear risk	The methodology of the study precluded the possibility of a blinded patient-surgeon relationship.
Blinding of outcome assessment (detection bias) Functional outcomes	High risk	The assessors were not blinded.
Blinding of outcome assessment (detection bias) Radiological outcomes	Low risk	The radiologist who assessed the degree of rotation in the femoral and tibial components was blinded concerning the technique used in a given patient.
Incomplete outcome data (attrition bias) All outcomes	Low risk	Number lost to follow-up intervention group: N = 1 (3.3%), control group: N = 1 (3.3%)
Selective reporting (reporting bias)	Low risk	All outcomes reported
Other bias	Low risk	None detected

Sun 2020
Study characteristics

Methods	Study Design: RCT Country: China Setting: Center for Joint Surgery, Southwest Hospital, Third Military Medical University (Army Medical University), Chongqing, China Length of Follow-up: 12 months
Participants	Inclusion Criteria: Patients with advanced OA of the knee; varus: deformity of no more than 15, flexion-contracture deformity of no more than 10, without extra-articular deformity of the knee; weight-bearing radiograph of the X-ray image showed the OA Kellgren-Lawrence classification grade IV; the treatment for the patient was TKA Exclusion Criteria: Patients with varus deformity of more than 15 flexion-contracture deformity of more than 10 of the knee; patients with traumatic arthritis and inflammatory arthritis; patients with huge bone defects around the knee; patients who had active infection around the knee; patients with knee valgus deformity

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Sun 2020 (Continued)

mity; patients with severe extra-articular deformity; patients who had previous knee surgery; patients who had periarticular soft tissue dysfunction and neuropathy; patients who had poor physical condition and could not withstand the operation; and patients who refused to be followed

Number of participants randomised:

- Intervention group: 40
- Control group: 40

Number of participants who received allocated intervention:

- Intervention group: 40
- Control group: 40

Number of dropouts:

- Unclear – not specifically reported

Age:

· Intervention group: 68.7 +/- 9.1

· Control group: 67.6 +/- 8.4

Sex (Male/Female):

- Intervention group: 7/33
- Control group: 8/32

Duration of disease: Not reported

Concurrent treatment: Not reported

BMI:

· Intervention group: 31.8 ± 6.1

· Control group: 30.6 ± 5.3

Preoperative mechanical axis

· Intervention group: varus 8.8 +/- 2.5

· Control group: varus 9.2 +/- 3.6

Interventions	Intervention group: Patient-specific instrumentation Control group: Conventional instrumentation Type of implant Bearing design: PS Mode of fixation: Cemented Patellar resurfacing: No Preop imaging for designing PSG: CT Implant Brand: LEGION Total Knee System, posterior stabilised prosthesis (Smith & Nephew, Memphis, USA) Intraoperative Data: Operation duration, drainage volume, drainage duration, ROM
Outcomes	Imaging modality used for postoperative measurement: X-ray & CT

Sun 2020 (Continued)

Outcomes used in this review:

HKA (Hip Knee Angle), PFA (Patella transverse axis-Femoral transepicondylar axis Angle), PCA (Posterior Condylar Angle), intramedullary guide position, depth of intramedullary guide, Hospital for Special Surgery Knee Score (HSS), American Knee Society Knee Score (AKS)

Outcomes not used in this review:

Alignment outliers other than HKA

Notes

Risk of bias

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Low risk	Patients assigned to two groups by random number table method, 40 cases in each group
Allocation concealment (selection bias)	Unclear risk	Not reported
Blinding of participants and personnel (performance bias) All outcomes	Low risk	Patients were unaware of the use of PSI during TKA.
Blinding of outcome assessment (detection bias) Functional outcomes	Unclear risk	Not reported
Blinding of outcome assessment (detection bias) Radiological outcomes	Unclear risk	Not reported
Incomplete outcome data (attrition bias) All outcomes	Unclear risk	Not specifically reported
Selective reporting (reporting bias)	Low risk	All outcomes reported
Other bias	Low risk	None detected

Tammachote 2018

Study characteristics

Methods	Study Design: RCT Country: Thailand Setting: Thammasat University Hospital, Faculty of Medicine, Thammasat University, Pathumthani, Thailand Length of Follow-up: 2 years
Participants	Inclusion Criteria: osteoarthritis or rheumatoid arthritis

Tammachote 2018 (Continued)

Exclusion Criteria: previous ipsilateral hip or ankle replacement; previous osteotomy with metallic ipsilateral tibial or femoral fractures

Number of participants randomised: 108

- Intervention group: 54
- Control group: 54

Number of participants who received allocated intervention:

- Intervention group: 53
- Control group: 54

Number of dropouts:

- Intervention group: 0 loss to follow-up; 2 discontinued intervention; 1 Infection; 1 periprosthetic fracture knee (not sure if it after knee replacement or before)
- Control group: 2 loss to follow-up (medical issues); 1 discontinued intervention – ipsilateral hip fracture

Age

- Intervention group: 59-87
- Control group: 61-89 years

Sex (Male/female)

- Intervention group: 12/42
- Control group: 15/39

Duration of disease: Not reported

Concurrent treatment: Not reported

BMI

- Intervention group: 25 ± 4
- Control group: 26 ± 5

Preoperative mechanical axis

- Intervention group: 169
- Control group: 167

Interventions

Intervention group: Patient-specific instrumentation

Control group: Conventional instrumentation

Type of implant: PS

Bearing design: fixed bearing

Mode of fixation: Cemented

Patellar resurfacing: No patella resurfacing

Preop imaging for designing PSG:

Implant Brand: Genesis II posterior-stabilized Smith and Nephew, Memphis, TN

PSG Brand: Visionaire system® (Smith and Nephew, Memphis, TN)

Intraoperative Data - Not reported

Tammachote 2018 (Continued)

- Number of times PSG was abandoned
 - Reasons for abandonment
 - Size correlation between planned component size and actual component used
 - Any other relevant intraoperative data:
- Surgeon experience and pre-training: experience of > 1500 TKAs

Outcomes	Outcomes assessed at radiological alignment of component 3 months after surgery Follow-up for 2 years (weeks 2, 6, 12 and 6 months then annually post-operation) Imaging modality used for postoperative measurement: Long leg knee standing X-rays Outcomes used in this review: · Functional outcome measures – WOMAC · Radiological alignment of component · Operative time · Postop transfusion · Knee range of motion · Number of outliers in radiographic alignment Outcomes not used in this review: Nil
Notes	Femoral component rotational alignment data published in Chanalithichai 2019 in the European Journal of Orthopaedic Surgery & Traumatology

Risk of bias

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Low risk	Quote from study: "A randomisation list was computer-generated using block of four randomisation".
Allocation concealment (selection bias)	Low risk	Quote from study: "Each randomisation card was placed in a sealed, opaque envelope that was opened by a study nurse after enrolment to reveal the surgical assignment".
Blinding of participants and personnel (performance bias) All outcomes	Unclear risk	Not specified
Blinding of outcome assessment (detection bias) Functional outcomes	Unclear risk	Not specified
Blinding of outcome assessment (detection bias) Radiological outcomes	Low risk	Quote from study: "Investigators who were arthroplasty fellowship trained and blinded to treatment allocation measured all of these (radiological) parameters".
Incomplete outcome data (attrition bias) All outcomes	Low risk	Loss to follow-up in control group 2 due to other medical issue In the intervention group, 1 did not receive PSG as the patient refused to wait, 2 discontinued intervention (1 infection, 1 periprosthetic fracture of knee)

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Tammachote 2018 (Continued)

Data analysed for 51 in each group.

Dropout rate = 5.6 % (3/54)

Selective reporting (reporting bias)	Low risk	Not prospectively registered. All the outcome measures mentioned in the methodology were reported.
Other bias	Low risk	None detected

Teeter 2019
Study characteristics

Methods	Study Design: RCT Country: Canada Setting: London Health Sciences Centre, London, Canada Length of Follow-up: 2 years
Participants	Inclusion Criteria: Any patient with osteoarthritis of the knee who was to undergo a unilateral primary TKA and had no metalwork in their knee as the result of a previous operation; only patients with a body mass index (BMI) < 40 kg/m² (to ensure clear RSA images); only patients who lived within 100 km of our centre (to ensure that they would be available for the multiple follow-up attendances) Exclusion Criteria: Patients with neuromuscular disorders or cognitive impairment; patients who were unable to undergo an MRI; and those who were pregnant or considering pregnancy Number of participants randomised: - Intervention group: 30 - Control group: 27 Number of participants who received allocated intervention: - Intervention group: 25 - Control group: 25 Number of dropouts: - Intervention group: 5 - Control group: 3 Age: - Intervention group: 69.0 +/- 8.4 - Control group: 69.4 +/- 8.4 Sex (Male/female): - Intervention group: 12/13 - Control group: 7/18 Duration of disease: Not reported Concurrent treatment: Not reported BMI: - Intervention group: 30.3 +/- 4.7

Patient-specific cutting guides for total knee arthroplasty (Review)

Teeter 2019 (Continued)

- Control group: 30.7 +/- 5.5

Preoperative mechanical axis: Not reported

Interventions	Intervention group: Patient-specific instrumentation Control group: Conventional instrumentation Type of implant: PS Bearing design: Fixed Mode of fixation: Cemented Patellar resurfacing: Yes Preop imaging for designing PSG: MRI, VISIONAIRE; Smith & Nephew Implant Brand: Legion Total Knee System, (Smith & Nephew) Intraoperative Data: Operation duration, blood loss, deviation from the preoperative plan, surgical waste	
Outcomes	Imaging modality used for postoperative measurement: X-ray Outcomes used in this review: HKA, HKA outliers, tibial slope, maximal total point motion, KSS function, WOMAC, EQ-5D, SF-12, UCLA activity score Outcomes not used in this review: Alignment outliers other than HKA	
Notes		
Risk of bias		
Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Low risk	Block randomisation was performed using concealed envelopes: the randomisation was carried out by the clinical study co-ordinators.
Allocation concealment (selection bias)	Low risk	Block randomisation was performed using concealed envelopes: the randomisation was carried out by the clinical study co-ordinators.
Blinding of participants and personnel (performance bias) All outcomes	Unclear risk	The nature of the instrumentation made it impossible to blind the surgeon to which technique was being used.
Blinding of outcome assessment (detection bias) Functional outcomes	Unclear risk	Not reported
Blinding of outcome assessment (detection bias) Radiological outcomes	Unclear risk	Not reported
Incomplete outcome data (attrition bias)	Unclear risk	Dropouts: Intervention 5/25 (20%), control 3/25 (12%)

Teeter 2019 (Continued)

All outcomes

Selective reporting (reporting bias)	Low risk	All outcomes reported
Other bias	Low risk	None

Turgeon 2019
Study characteristics

Methods	Study Design: RCT Country: Canada Setting: Concordia Joint Replacement Group Length of Follow-up: 2 years
Participants	Inclusion Criteria: Patients who were slated for primary knee replacement surgery for all diagnoses Exclusion Criteria: Patients with contraindications for magnetic resonance imaging (MRI); deformity of the femur preventing the use of the intramedullary guide used in the CI set; patients requiring constrained implants, if they were undergoing revision surgery or if they were scheduled for bilateral knee surgery Number of participants randomised: - Intervention group: 25 - Control group: 30 Number of participants who received allocated intervention: - Intervention group: 25 - Control group: 29 Number of dropouts: - Intervention group: 0 - Control group: 1 – clinical issue Age: - Intervention group: 63.8 ± 9.2 - Control group: 65.7 ± 9.2 Sex (Male/female): - Intervention group: 7/18 - Control group: 7/22 Duration of disease: Not reported Concurrent treatment: Not reported BMI: - Intervention group: 33.6 ± 7.2 - Control group: 32.0 ± 6.0 Preoperative mechanical axis:

Patient-specific cutting guides for total knee arthroplasty (Review)

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Turgeon 2019 (Continued)

- Intervention group: -5.2 ± 5.9
- Control group: -6.7 ± 7.5

Interventions	Intervention group: Patient-specific instrumentation Control group: Conventional instrumentation Type of implant Bearing design: Not reported Mode of fixation: Cemented Patellar resurfacing: Not reported Preop imaging for designing PSG: MRI, VISIONAIRE; Smith & Nephew Implant Brand: Legion Total Knee System, (Smith & Nephew) Intraoperative Data: Surgical time
Outcomes	Outcomes assessed: Imaging modality used for postoperative measurement: X-ray Outcomes used in this review: The mechanical axis of the lower limb, HKA outliers, the femoral component alignment, and the tibial component alignment in the coronal plane. Outcome scores: EQ-5D-3L, VAS, OKS, PCS, UCLA. Surgical time, length of hospital stay and perioperative complications were recorded. Outcomes not used in this review: Alignment outliers other than HKA

Notes

Risk of bias

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Low risk	Randomisation was done in a block design stratified by 1 of the 4 participating surgeons to ensure equal distribution between the 2 surgical techniques by surgeon.
Allocation concealment (selection bias)	Unclear risk	Not reported
Blinding of participants and personnel (performance bias) All outcomes	Low risk	Participants were blinded to group assignment and surgical technique. All participants underwent preoperative MRI assessment to ensure that blinding was maintained.
Blinding of outcome assessment (detection bias) Functional outcomes	Low risk	All observations were made by an observer who was blinded to patient group assignment and surgical technique.
Blinding of outcome assessment (detection bias) Radiological outcomes	Low risk	All observations were made by an observer who was blinded to patient group assignment and surgical technique.

Turgeon 2019 (Continued)

Incomplete outcome data (attrition bias) All outcomes	Low risk	Dropouts: Intervention N = 0, control group N = 1 (3.4%)
Selective reporting (reporting bias)	Low risk	All outcomes discussed in methodology reported
Other bias	Low risk	None detected

Van Leeuwen 2018

Study characteristics

Methods	Study Design: RCT Country: Norway Setting: Department of Orthopaedic Surgery, Betanien Hospital, Skien, Norway; 2 Division of Orthopaedic Surgery, Oslo University Hospital, Oslo, Norway; 3 Faculty of Medicine, University of Oslo, Oslo, Norway Length of Follow-up: 2 years
Participants	Inclusion Criteria: symptomatic knee osteoarthritis; only if having unilateral arthroplasty Exclusion Criteria: Marked bone loss, which could preclude adequate fixation of the device; non-cooperative subjects; neurologic and muscular disorders; severe vascular insufficiency of the affected limb; severe instability or deformity of the ligaments and/or surrounding soft tissue, which may preclude stability of the device; rheumatoid arthritis; and other systemic diseases and known metal allergy Number of participants randomised: 109 (excluded 15) - Intervention group: 44 - Control group: 50 Excluded (n = 15): - double randomisation: 2 - MRI artifacts: 1 - not operated: 5 - included by hospital but not participating in the study: 2 - operated but not included: 1 - declined to participate: 3 - PSPG guides delivered to wrong hospital: 1 Number of participants who received allocated intervention: - Intervention group: 44 - Control group: 50 Number of dropouts: (people who completed different outcome assessments) - Intervention group: - full-leg weight-bearing radiograph: 42

Van Leeuwen 2018 (Continued)

- computed tomography: 43
- questionnaires: 44
- Control group:
 - full-leg weight-bearing radiograph: 49
 - computed tomography: 47
 - questionnaires: 48

Age

- Intervention group: 67
- Control group: 64

Sex (Male/female)

- Intervention group: 14M/30F
- Control group: 18M/32F

Duration of disease: Not reported

Concurrent treatment: Not reported

BMI

- Intervention group: 31
- Control group: 29

Preoperative mechanical axis

- Intervention group
- Control group

Interventions

Intervention group: Patient-specific instrumentation

Control group: Conventional instrumentation

Type of implant: CR

Bearing design: Fixed

Mode of fixation: Cemented

Patellar resurfacing: Not reported

Preop imaging for designing PSG: MRI images

Implant Brand: Vanguard Complete Knee System, Biomet Inc, Warsaw, IN, USA

PSG Brand

Intraoperative data not reported

- Number of times PSG was abandoned
- Reasons for abandonment
- Size correlation between planned component size and actual component used
- Any other relevant intraoperative data

Surgeon experience and pre-training

Orthopaedic surgeons specialising in TKR performed the vast majority of operations; in 3 cases, a resident, under the guidance of a knee surgeon, performed the procedure

Van Leeuwen 2018 (Continued)

Outcomes	Outcomes assessed at radiological assessment 3 months postoperatively Questionnaire (KOOS, EQ-5D-3L, EQ VAS, pain score NRS-11) - At inclusion, after 3 months, and at the 1- and 2-year follow-up Imaging modality used for postoperative measurement: A full leg weight-bearing radiograph and a computed tomography (CT) scan according to the Perth protocol Outcomes used in this review: · Radiological alignment of component · Quality of life – EQ-5D · Operative time · Length of hospital stay · Knee range of motion · Number of outliers in radiographic alignment Outcomes not used in this review: Nil
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Notes	Perioperative data, functional outcomes and QoL data available from a supplementary file
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Risk of bias

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Low risk	After receiving written consent, block randomisation was obtained by variable block sizes.
Allocation concealment (selection bias)	Low risk	All patients were referred for a preoperative MRI (according to the Signature™ scanning protocol; Signature MRI Inc., Monrovia, CA, USA) so that they did not know which type of operation they would receive.
Blinding of participants and personnel (performance bias) All outcomes	Low risk	All patients were referred for a preoperative MRI (according to the Signature™ scanning protocol; Signature MRI Inc., Monrovia, CA, USA) so that they did not know which type of operation they would receive.
Blinding of outcome assessment (detection bias) Functional outcomes	Unclear risk	Not specified
Blinding of outcome assessment (detection bias) Radiological outcomes	Unclear risk	Not specified
Incomplete outcome data (attrition bias) All outcomes	High risk	Excluded (n = 15): double randomisation, 2; MRI artifacts, 1; not operated, 5; included by hospital not participating in the study, 2; operated but not included, 1; declined to participate, 3; PSPG guides delivered to wrong hospital, 1
Selective reporting (reporting bias)	Low risk	The trial is registered at ClinicalTrials.gov (NCT01696552). All outcomes reported
Other bias	Unclear risk	Total number of included patients was lower than planned (1 centre withdrew from the study). Therefore, the study was underpowered for the KOOS as we

Van Leeuwen 2018 (Continued)

calculated a need for 63 patients in each group. However, the study was sufficiently powered for radiological alignment.

Victor 2014

Study characteristics

Methods	Study Design: parallel RCT Country: Belgium Setting: AZ St-Lucas, Brugge, Belgium and Leuven University Hospitals, Leuven, Belgium Length of Follow-up: Not reported
Participants	Inclusion Criteria • end-stage knee osteoarthritis scheduled for TKA Exclusion Criteria • rheumatoid arthritis • previous osteotomy • fractures • retained hardware in the limb • claustrophobia Number of people randomised: 128 • Intervention: 64 • Control: 64 Number of dropouts: 3 • Intervention: 3 • Control: 0 Age (mean) • Intervention group: 67 • Control group: 64 Sex (Male/female) • Intervention group: 21/43 • Control group: 21/43 Duration of disease: Not reported Concurrent treatment: Not reported BMI • Control group: Not reported • Intervention group: Not reported Preoperative mechanical axis • Control group: $-3.5^{\circ} \pm 6.5$ • Intervention group: $-2.7^{\circ} \pm 7.0$

Victor 2014 (Continued)

Interventions

Control group: Conventional instrumentation

Intervention group: Patient-specific guides (from four different manufacturers) with surgical navigation as intraoperative control

- Subgroup 1: Signature (MRI-based)
- Subgroup 2: Trumatch (CT-based)
- Subgroup 3: Visionaire (MRI-based)
- Subgroup 4: PSI (MRI-based)

Type of implant: Posterior stabilised

Bearing design: Fixed bearing

Mode of fixation: Cemented

Patellar resurfacing: Performed in all participants

Preop imaging for designing PSG: Subgroups 1, 3 and 4 – MRI-based, subgroup 2 – CT-based

Implant Brand:

- Control group: Genesis II
- Intervention group
 - Subgroup 1: Vanguard
 - Subgroup 2: Sigma
 - Subgroup 3: Genesis II
 - Subgroup 4: Nex-Gen

PSG Brand: Mentioned above

Intraoperative Data

- Number of times PSG was abandoned: 14 patients (22%)
- Reasons for abandonment: Quote from the study stated: "In the PSG cohort, the suggested cuts were measured with the surgical navigation system (BrainLAB Knee 2.1 Universal; BrainLAB AG, Munich, Germany), after the block was carefully positioned and pinned in place. If the measured value exceeded the target alignment in any plane and any direction by more than 3°, the use of the PSG was abandoned for the affected bone and the cuts were made following the surgical navigation system".
- Size correlation between planned component size and actual component used: 79%; quote from the study stated: "In 13 patients, the implant size as determined by the PSG preoperative planning was incorrect. This occurred three times for the femur (5%) and 10 times for the tibia (16%)".
- Any other relevant intraoperative data: PSG procedure was modified in 18 patients (28%). Quote from the study stated: "A change in sizing was the most common reason for modifying the use of the PSG..... In nine patients in this cohort, the level of the cut was inappropriate and required intraoperative correction: five times at the level of the femur (8%) and four times at the level of the tibia (6%). Administrative errors not leading to abandoning or modifying the PSG were noted in the PSG cohort. In one patient, the name of the patient was incorrectly labelled on the guide. In another patient, the cutting guide was delivered for a femoral component size 7, whereas size 6 had been ordered".

Surgeon experience and pre-training: Adequate

Outcomes

Outcomes assessed at: Not reported

Imaging modality used for postoperative measurement: Radiograph and CT scan

Outcomes used in this review:

1. Radiological alignment of components
2. Number of outliers in radiographic alignment

Victor 2014 (Continued)

Outcomes not used in this review: Nil

Notes

Risk of bias

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Low risk	Quote from study: "Patients were randomised into two cohorts In the PSG cohort, PSG designs were from four different TKA suppliers (Subgroups PSG 1-4,)... this subgrouping was not randomised for logistic reasons; the use of the different systems was consequently consecutive". Comment – Although the random sequence generation method was not reported, the use of sealed envelopes in concealing allocation suggests that the unpredictability of randomisation was preserved.
Allocation concealment (selection bias)	Low risk	Quote from study: "...the nurse opened the sealed randomisation envelope."
Blinding of participants and personnel (performance bias) All outcomes	Unclear risk	Open-label study
Blinding of outcome assessment (detection bias) Functional outcomes	Low risk	Functional outcome measures were not reported in this study.
Blinding of outcome assessment (detection bias) Radiological outcomes	Low risk	Quote from study: "Radiographs and CT scans were measured by an investigator who was blinded to the allocation of the patient".
Incomplete outcome data (attrition bias) All outcomes	Low risk	Dropouts 3/128 (2.3%)
Selective reporting (reporting bias)	Low risk	Not prospectively registered. All the outcome measures mentioned in the methodology were reported.
Other bias	Low risk	None detected

Vide 2017
Study characteristics

Methods	Study Design: RCT Country: Portugal Setting: - Hospital Particular do Algarve, Hospital de Faro, Rua Alfredo Guisado, No. 14, Bloco 2, 60 E, 1500.030 Lisbon, Portugal - Hospital Particular do Algarve, Hospital de Faro, Urb. Casal de Gambelas, Lote 2, Gambelas, 8005.226 Faro, Portugal - Hospital Particular do Algarve, Urb. Casal de Gambelas, Lote 2, Gambelas, 8005.226 Faro, Portugal
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Vide 2017 (Continued)

Length of Follow-up: Not reported

Participants

Inclusion Criteria: end-stage knee osteoarthritis

Exclusion Criteria: previous osteotomies or fractures; retained hardware in the limb; contraindication for magnetic resonance (MR) imaging or inability to coordinate the customised cutting blocks construction and procedure execution

Number of participants randomised: 100

- Intervention group: 50
- Control group: 50

Number of participants who received allocated intervention:

- Intervention group: 50
- Control group: 50

Number of dropouts: (loss to FU)

- Intervention group: 3
- Control group: 2

Age

- Intervention group: 67.8
- Control group: 69.3

Sex (Male/female)

- Intervention group: M:F (15:32)
- Control group: M:F (15:33)

Duration of disease: Not reported

Concurrent treatment: Not reported

BMI

- Intervention group: 31.0
- Control group: 30.3

Preoperative mechanical axis

- Intervention group: 4.6° vr
- Control group: 3.5° vr

Interventions

Intervention group: Patient-specific instrumentation

Control group: Conventional instrumentation

Type of implant: CR

Bearing design: Fixed bearing

Mode of fixation: Cemented

Patellar resurfacing: No

Preop imaging for designing PSG: MR

Implant Brand: TC Plus® Total Knee System (Smith & Nephew Inc.)

PSG Brand: Visionaire® (Smith & Nephew Inc, Memphis, TN, USA)

Vide 2017 (Continued)

Intraoperative Data

- Number of times PSG was abandoned: 0
 - Reasons for abandonment
 - Size correlation between planned component size and actual component used: 100%
 - Any other relevant intraoperative data: Insert exchange occurred in five patients (10.6 %), all from a 9 to an 11 mm, due to removal of significant osteophytes
- Surgeon experience and pre-training: more than 15 years of experience with TKA and a current annual volume of around 100 procedures

Outcomes

Outcomes assessed not reported

Imaging modality used for postoperative measurement: long-leg standing AP digital

Outcomes used in this review:

- Radiological alignment of component
 - Operative time
 - Blood loss
 - Length of hospital stay
 - Number of outliers in radiographic alignment
- Outcomes not used in this review:
- Survival rate of implant (revision surgery to change a prosthetic component)
 - Functional outcome measures
 - Pain
 - Global assessment (participant-reported)
 - Total adverse events including infection, thrombosis, palsy, death, etc. (please mention)
 - Re-operation rate (not involving implant change)
 - Quality of life

Notes

Risk of bias

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Low risk	Quote from study: "Patients were randomised into two cohorts of 50 patients each, using the Rand formula from Microsoft Excel version 14.1.0".
Allocation concealment (selection bias)	Unclear risk	Open-label study
Blinding of participants and personnel (performance bias) All outcomes	Unclear risk	Not blinded

Vide 2017 (Continued)

Blinding of outcome assessment (detection bias) Functional outcomes	Unclear risk	Not analysed
Blinding of outcome assessment (detection bias) Radiological outcomes	Low risk	Deviations from the neutral mechanical axis—0° tibiofemoral angle (TFA), were registered independently and blindly by two authors (JPS, JV), with negative values for varus alignments and positive for valgus. The authors collecting the data relative to the coronal alignment, blood loss, transfusions and LOS, were unaware of the type of instrumentation used.
Incomplete outcome data (attrition bias) All outcomes	Low risk	None reported
Selective reporting (reporting bias)	Low risk	None
Other bias	Unclear risk	No power analysis or sample size calculation was performed.

Vundelinckx 2013

Study characteristics

Methods	Study Design: RCT Country: Belgium Setting: AZ Nikolaas, Sint-Niklaas, Belgium Length of Follow-up (mean): Control group – 215 days; intervention group – 200 days
Participants	Inclusion Criteria - patients undergoing a TKA Exclusion Criteria (only for intervention group) - severe obesity - claustrophobic patient Number of people randomised: 62 - Intervention group: 31 - Control group: 31 Number of dropouts: 4 Quote from the study: "4 patients refused further follow-up in the study because of different reasons.... Those patients were replaced by others, to maintain the same number of patients in both groups. This selection was consequently done". Age - Intervention group: 64.65 ± 8.23 - Control group: 68.19 ± 8.48 Sex (Male/female)

Vundelinckx 2013 (Continued)

- Intervention group: 15/ 16
- Control group: 11/ 20

Duration of disease: Not reported

Concurrent treatment: Not reported

BMI

- Intervention group: 27.61 ± 3.82
- Control group: 31.11 ± 5.25

Preoperative mechanical axis

- Intervention group: 7.0 ± 3.86
- Control group: 7.1 ± 4.12

Interventions	Intervention group: Patient-specific instrumentation Control group: Conventional instrumentation Type of implant: Not reported Bearing design: Not reported Mode of fixation: Not reported Patellar resurfacing: Not reported. Quote from study: "patellar component is not pre-operatively planned in the Visionaire software system". Preop imaging for designing PSG: MRI Implant Brand: Genesis II PSG Brand: Visionaire Intraoperative Data · Number of times PSG was abandoned: Not reported · Reasons for abandonment: Not applicable · Size correlation between planned component size and actual component used: femur 25/31 (80.6%); tibia 25/31 (80.6%); insert 28/31 (90.3%) · Any other relevant intraoperative data: Not reported Surgeon experience and pre-training: Quote from study: "All patients were operated by two experienced surgeons".
Outcomes	Outcomes assessed at: perioperative and mean follow-up of 200 days in intervention group and 215 days in control group Imaging modality used for postoperative measurement: Radiograph Outcomes used in this review: 1. Functional outcome measures 2. Radiological alignment of components 3. Pain 4. Global assessment (participant reported) 5. Operative time 6. Total adverse events 7. Re-operation rate

Vundelinckx 2013 (Continued)

8. Length of hospital stay
9. Blood loss
10. Knee range of motion
11. Incision length

Outcomes not used in this review: Nil

Notes	SDs for length of stay and incision length not reported. SDs imputed from other relevant studies
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Risk of bias

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	High risk	Quote from study: "The randomisation process was carried out in a 1:1 fashion; 1 patient allocated to the Visionaire group, consequently 1 patient to the control group, and so on... the only exclusion criteria[um] for allocation to the Visionaire group (intervention group) was a severe obesity". Quote from the study: "4 patients refused further follow-up in the study because of different reasons.... Those patients were replaced by others, to maintain the same number of patients in both groups. This selection was consequently done". Comment – As high BMI is known to result in poor outcomes following TKA, there is a probable high risk of selection bias.
Allocation concealment (selection bias)	High risk	See above
Blinding of participants and personnel (performance bias) All outcomes	Unclear risk	Open-label study
Blinding of outcome assessment (detection bias) Functional outcomes	High risk	Open-label study with inadequate allocation concealment and patient reported outcome measures
Blinding of outcome assessment (detection bias) Radiological outcomes	Low risk	Quote from study: "The clinical and radiological evaluation was performed by an independent physician who had not participated in the surgical procedure". Comment – Although it was not explicitly mentioned whether this assessor was blinded, the risk of detection bias was probably low.
Incomplete outcome data (attrition bias) All outcomes	Unclear risk	Quote from the study: "4 patients refused further follow-up in the study because of different reasons.... Those patients were replaced by others, to maintain the same number of patients in both groups. This selection was consequently done".
Selective reporting (reporting bias)	Low risk	Not prospectively registered. However, all the outcome measures mentioned in the methodology were reported.
Other bias	Low risk	None detected

Woolson 2014

Study characteristics

Methods	Study Design: RCT Country: USA Setting: Palo Alto Veterans Administration Hospital, Palo Alto Length of Follow-up: Minimum of 6 months (mean in intervention group 9.5 months & control group 10.8 months)
Participants	Inclusion Criteria - Arthritis of knee - Scheduled to undergo primary total knee arthroplasty Exclusion Criteria: Not reported Number of people randomised: 63 patients, 64 knees - Intervention group: N = 30 patients, 30 knees - Control group: N = 33 patients, 33 knees Number of dropouts: 15/63 - Intervention group: 8/30 - Control group: 7/33 In both the groups, the dropouts did not receive the allocated intervention Age: Not reported Sex: Not reported Duration of disease: Not reported Concurrent treatment: Not reported BMI: Not reported Preoperative mechanical axis: Not reported
Interventions	Intervention group: Patient-specific instrumentation Control group: Conventional instrumentation Type of implant: Posterior-cruciate-substituting prosthesis Bearing design: Not reported Mode of fixation: Cemented Patellar resurfacing: Performed in all patients Preop imaging for designing PSG: CT scan Implant Brand: Not reported PSG Brand: TruMatch Intraoperative Data - Number of times PSG was abandoned: 7/22 knees (32%) - Reasons for abandonment: Quote from study: "In three

Woolson 2014 (Continued)

knees, the surgeon abandoned (removed) the block before he made the proximal tibial cut because, after visually checking the proposed cut with an alignment rod, he judged that the cut would be malaligned if he used the custom block. In four knees, the proximal tibial osteotomy that had been made with a patient-specific cutting block was recut because malalignment (with regard to tibial slope in two, in the coronal plane in one, and in both planes in another) was noted after the osteotomy had been made. In these seven knees, the final proximal tibial cut was made with a standard extramedullary alignment guide. All twenty-two patient-specific femoral cutting blocks were used for the distal femoral osteotomy, but one was not used for the anterior and posterior condylar cuts because gross rotational malalignment would have occurred."

· Size correlation between planned component size and actual component used: 13/ 22 knees (59%)

Quote from study: "During the procedure, the patient-specific preoperative plan with regard to the size of the component was changed in nine study knees (41%). This change consisted of using a femoral component (three knees) or a tibial component (four knees), or both components (two knees), that was larger or smaller than the size that had been predicted preoperatively. Five (23%) of the twenty-two femoral components were downsized one or two sizes and six (27%) of the twenty-two tibial components were either downsized or increased in size (three each)".

· Any other relevant intraoperative data: Quote from study: "in twelve (55%) of the twenty-two study knees, the surgeon adjusted the extension space by increasing the amount of femoral or tibial bone, or both. Less distal femoral bone than planned was resected from one study knee when it appeared that excessive bone would have been resected with use of the patient-specific femoral cutting block.... In six knees (23%) in the control group, more bone (2 to 4 mm) was resected, from the distal part of the femur in three and from the proximal part of the tibia in three, after the initial cuts were made because of an inadequate extension space."

Surgeon experience and pre-training: Quote from study: "it could be argued that additional experience with patient-specific cutting blocks would have resulted in improved performance. However, we do not think that more practice using the cutting blocks would be likely to result in different outcomes, as the surgeon's ability to position the blocks correctly did not appear to improve after twenty cases had been performed."

Outcomes	Outcomes assessed at: perioperative and 6 months follow-up Imaging modality used for postoperative measurement: CT scan Outcomes used in this review: 1. Functional outcome measures 2. Radiological alignment of components 3. Number of outliers in radiographic alignment 4. Operative time 5. Total adverse events 6. Re-operation rate 7. Length of hospital stay 8. Postop haematocrit Outcomes not used in this review: Nil	
Notes	One patient scheduled for revision in the control group for tibial component loosening at 14 months follow-up - analysed as revision. SDs for KSS, haematocrit, length of stay and ROM. SDs imputed from other relevant studies.	
Risk of bias		
Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Low risk	Quote from study: "... were randomly assigned to a treatment group through the use of sealed envelopes"

Woolson 2014 (Continued)

Comment – Although the random sequence generation method was not reported, the use of sealed envelopes in concealing allocation suggests that the unpredictability of randomisation was preserved.		
Allocation concealment (selection bias)	Low risk	Quote from study: "... were randomly assigned to a treatment group through the use of sealed envelopes".
Blinding of participants and personnel (performance bias) All outcomes	Unclear risk	Open-label study
Blinding of outcome assessment (detection bias) Functional outcomes	High risk	Open-label study with patient reported outcome measures
Blinding of outcome assessment (detection bias) Radiological outcomes	Low risk	Quote from study: "An independent observer who was blinded to the method of component alignment evaluated each postoperative scan ..."
Incomplete outcome data (attrition bias) All outcomes	Low risk	No dropouts reported
Selective reporting (reporting bias)	Low risk	Not prospectively registered. All the outcome measures mentioned in the methodology were reported.
Other bias	Unclear risk	Quote from study: "Both the preoperative and postoperative CT scans and the patient-specific cutting blocks were funded by the manufacturer".

Yan 2014

Study characteristics

Methods	Study Design: parallel RCT Country: China Setting: Queen Mary Hospital, The University of Hong Kong, China Length of Follow-up: 3 months
Participants	Inclusion Criteria - indicated for primary TKA - physically and mentally willing and able to comply with the postoperative scheduled clinical and radiographic evaluations - agreed to sign an informed consent document specific to the study Exclusion Criteria - patients who were not suitable for MRI or whose MRI failed to satisfy the manufacturer's protocol - patients who had previous knee operations - extra-articular deformity in tibia or femur - associated medical diseases that could affect rehabilitation - active systemic infection or infection near the knee joint - current medical conditions that made them unfit for surgery

Yan 2014 (Continued)

- any conditions which in the opinion of the investigator suggested that the patient would not be compliant with the study protocol

Number of people randomised: 90

- Intervention 1: Navigation (NAV) N = 30
- Intervention 2: Patient-specific instruments (PSI) N = 30
- Control: Conventional Instrumentation (CON) N = 30

Number of dropouts: 0

Age

- NAV group: 66.7 ± 7.2
- PSI group: 67.5 ± 8
- CON group: 69.5 ± 8.4

Sex (M:F)

- NAV group: 4:26
- PSI group: 13:17
- CON group: 6:24

Duration of disease: Not reported

Concurrent treatment: Not reported

BMI: Not reported

Preoperative mechanical axis

- NAV group: 13.2 ± 5.6
- PSI group: 13.6 ± 6.6
- CON group: 13.5 ± 6.8

Interventions

Intervention group 1: Navigation

Intervention group 2: Patient-specific instrumentation

Control group: Conventional instrumentation

Type of implant: Posterior stabilised

Bearing design: Not reported

Mode of fixation: Not reported

Patellar resurfacing: Not reported

Preop imaging for designing PSG: MRI

Implant Brand: NexGen Legacy

PSG Brand: Not reported

Intraoperative Data

- Number of times PSG was abandoned: 0
- Reasons for abandonment: Not applicable
- Size correlation between planned component size and actual component used: Not reported
- Any other relevant intraoperative data: None

Yan 2014 (Continued)

Surgeon experience and pre-training: Quote from study: "Three experienced joint replacement surgeons (A, B, C) participated in the operations".

Outcomes	Outcomes assessed at: perioperative and 3 months Imaging modality used for postoperative measurement: Radiograph Outcomes used in this review: 1. Functional outcome measures 2. Radiological alignment of components 3. Number of outliers in radiological alignment 4. Operative time 5. Knee range of motion 6. Total adverse events Outcomes not used in this review: Nil
Notes	9 patients bilateral TKAs

Risk of bias

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Low risk	Quote from study: "Using computer software (Random Allocation Software Version 1.0.0, Isfahan, Iran), the knees were randomised in 1:1:1 ratio into one of the three groups — CON, NAV and patient-specific instruments (PSI)."
Allocation concealment (selection bias)	Unclear risk	Not reported
Blinding of participants and personnel (performance bias) All outcomes	Unclear risk	Open-label study
Blinding of outcome assessment (detection bias) Functional outcomes	High risk	An open-label study with patient-reported functional outcome scores
Blinding of outcome assessment (detection bias) Radiological outcomes	Low risk	Quote from study: "Two orthopaedic specialists, who did not participate in the surgery and were blinded to the randomisation, performed the radiological measurements"
Incomplete outcome data (attrition bias) All outcomes	Low risk	No dropouts reported
Selective reporting (reporting bias)	Low risk	Not prospectively registered. However, all the outcome measures mentioned in the methodology were reported.
Other bias	Low risk	None detected

Zahn 2020

Study characteristics

Patient-specific cutting guides for total knee arthroplasty (Review)

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Zahn 2020 (Continued)

Methods	Study Design: RCT Country: Germany Setting: Centre for Musculoskeletal Surgery Length of Follow-up: 3 months
Participants	Inclusion Criteria: 300 consecutive TKA patients with primary osteoarthritis were included between 2012 and 2015. Exclusion Criteria: Preoperative extension deficit > 20°, valgus or varus malalignment > 15° and previous surgery of the affected joint Number of participants randomised: - Intervention group PSI: 75 - Intervention group NAV: 75 - Control group IM: 75 - Control group EM: 75 Number of participants who received allocated intervention: - Intervention group PSI: 75 - Intervention group NAV: 75 - Control group IM: 75 - Control group EM: 75 Number of dropouts: Not reported Age: - Intervention group PSI: 66.9 (7.2) - Intervention group NAV: 67.2 (7.6) - Control group IM: 66.9 (7.8) - Control group EM: 67.5 (7.3) Sex (Male/female): - Intervention group PSI: 29/46 - Intervention group NAV: 29/46 - Control group IM: 33/42 - Control group EM: 32/43 Duration of disease: Not reported Concurrent treatment: Not reported BMI: - Intervention group PSI: 29.2 (2.7) - Intervention group NAV: 29.2 (2.8) - Control group IM: 28.8 (2.9) - Control group EM: 29.5 (2.7) Preoperative mechanical axis: - Intervention group PSI: 4.60 (4.6) - Intervention group NAV: 4.3 (4.3) - Control group IM: 4.21 (4.2)

Zahn 2020 (Continued)

- Control group EM: 4.2 (4.5)

Interventions	Intervention group: Patient-specific instrumentation, navigation Control group: Conventional instrumentation intramedullary and conventional instrumentation extramedullary (tibia) Type of implant Bearing design: Not reported Mode of fixation: Not reported Patellar resurfacing: Not reported Preop imaging for designing PSG: MRI, Visionaire; Smith & Nephew Navigation system: PI Galileo system Implant Brand: Patients with navigated TKA or extramedullary instrumentation were implanted with a TC Plus Primary (Smith & Nephew, Memphis, TN, USA). In PSI and intramedullary, the Journey II CR (Smith & Nephew, Memphis, TN, USA) was used. Intraoperative Data: Not reported
Outcomes	Outcomes used in this review: The mechanical tibiofemoral axis (mTFA), mTFA outliers, medial proximal tibial angle (mPTA), tibial slope Outcomes not used in this review: Radiological outcomes assessed only. Outlier data other than mTFA not included
Notes	We used the data from the conventional instrumentation IM group as the control.

Risk of bias

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Low risk	For randomisation, a consecutively numbered list with 300 positions was created. Each number was randomly assigned one operative technique using a spreadsheet. Subsequently, patients were allocated to one of the four operative techniques in ascending order of their operation date in the clinic.
Allocation concealment (selection bias)	Unclear risk	Not reported
Blinding of participants and personnel (performance bias) All outcomes	Unclear risk	Not reported
Blinding of outcome assessment (detection bias) Functional outcomes	Unclear risk	Not reported
Blinding of outcome assessment (detection bias) Radiological outcomes	Unclear risk	Not reported

Zahn 2020 (Continued)

Incomplete outcome data (attrition bias) All outcomes	Unclear risk	Dropout rates not reported
Selective reporting (reporting bias)	Low risk	All outcomes discussed in methods were reported.
Other bias	Low risk	None

Zomar 2021

Study characteristics

Methods	Study Design: RCT Country: Canada Setting: University Hospital, London Health Sciences Centre Length of Follow-up: 1 year
Participants	Inclusion Criteria: Participants were included if they were scheduled to undergo primary TKA, between the ages of 20 and 70 years, had a body mass index (BMI) greater than 30, had symptomatic osteoarthritis, and were English-speaking. Exclusion Criteria: Participants were excluded if they previously had open knee surgery, infection, or history of trauma to the affected knee; were scheduled to undergo simultaneous bilateral TKA; had underlying dementia or cognitive delay; inflammatory arthritis; chronic pain syndrome; or fibromyalgia. Number of participants randomised: - Intervention group: 97 - Control group: 97 Number of participants who received allocated intervention: - Intervention group: 86 - Control group: 87 Number of dropouts: - Intervention group: 0 - Control group: 2 Age: - Intervention group: 61.9 (6.7) - Control group: 61.7 (5.7) Sex: Male, n (%): - Intervention group: 26 (30) - Control group: 28 (32) Duration of disease: Not reported Concurrent treatment: Not reported BMI:

Zomar 2021 (Continued)

· Intervention group: 38.4 ± 6.2

· Control group: 39.5 ± 7.3

Preoperative mechanical axis: Not reported

Interventions	Intervention group: Patient-specific instrumentation Control group: Conventional instrumentation Type of implant Bearing design: PS Mode of fixation: Not reported Patellar resurfacing: Not reported Preop imaging for designing PSG: CT TRUMATCH Personalized Solutions Resection Guide (DePuy Synthes, Warsaw, IN) Implant Brand: Attune Total Knee Implants (DePuy Synthes, Warsaw, IN) Intraoperative Data: Length of stay, procedure time, tourniquet time
Outcomes	Outcomes assessed Imaging modality used for postoperative measurement: NA Outcomes used in this review: Length of stay, procedure time, tourniquet time European Quality of Life Scale (EQ-5D), the Western Ontario and McMaster Universities Osteoarthritis Index (WOMAC) Outcomes not used in this review: No radiological outcomes assessed
Notes	No standard deviations available for WOMAC, EQ-5D; therefore, SDs imputed from relevant studies

Risk of bias

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Low risk	We enrolled and randomised participants at the time of consultation for surgery, approximately six months before surgery. We stratified the randomisation by surgeon, and participants were allocated a one-to-one ratio to either PSI or SOC for their TKA using a computer-generated randomisation scheme.
Allocation concealment (selection bias)	Unclear risk	Not reported
Blinding of participants and personnel (performance bias) All outcomes	High risk	One limitation of this study was the lack of blinding to group allocation. It was not possible to blind any of the persons involved in the study to group allocation due to the required preoperative CT scan. It was not possible to blind the surgeon due to the nature of the surgical instrumentation.
Blinding of outcome assessment (detection bias)	High risk	It was also not possible to blind the research staff as they needed to arrange the CT appointment and manage the delivery of the guides in time for surgery.

Zomar 2021 (Continued)

Functional outcomes

Blinding of outcome assessment (detection bias) Radiological outcomes	High risk	It was also not possible to blind the research staff as they needed to arrange the CT appointment and manage the delivery of the guides in time for surgery.
Incomplete outcome data (attrition bias) All outcomes	High risk	Another limitation of the study was the number of participants who enrolled in the study and subsequently did not go on to have surgery within the study period. While waiting for surgery, a number of enrolled participants developed health concerns that either needed to be addressed before surgery, thus delaying their surgery, or permanently changed their health status, such that surgery was no longer appropriate. Loss to FU: Intervention N = 0, Control group N = 2 (2.3%)
Selective reporting (reporting bias)	Low risk	All outcomes discussed in the methods were reported.
Other bias	Low risk	None detected

AAOS: American Academy of Orthopaedic Surgeons

AKS: American Knee Society

aLDFA: anatomic Lateral Distal Femoral Angle

AP: Antero-Posterior

aPPTA: anatomic Posterior Proximal Tibial Angle

BMI: Body Mass Index

CON: conventional

CR: Cruciate Retaining

CT: Computerised Tomography

CVR: Conventional Reusable

EBV: Estimated Blood Volume

EM: Extra-Medullary

EQ-5D: EuroQol-5 Dimensions

FCA: Femoral Coronal Angle

FFC: Frontal Femoral Component angle

FSA: Femoral Sagittal Angle

FTA: FemoroTibial Angle

FTC: FemoroTibial Component angle

FTCA: FemoroTibial Coronal Angle

FU: follow-up

HKA: Hip-Knee-Ankle

HSS: Hospital for Special Surgery

IC: inclusion criteria

IKSS: International Knee Society Score

IM: Intra-Medullary

JLCA: Joint Line Convergence Angle

KOOS: Knee injury and Osteoarthritis Outcome Score

KSS: Knee Society Score

LFC: Lateral Femoral Component angle

LOS: length of stay

LPTA: Lateral Proximal Tibial Angle

LTC: Lateral Tibial Component angle

MLA: Mechanical Leg Alignment

mLDFA: mechanical Lateral Distal Femoral Angle

mMDFA: mechanical Medial Distal Femoral Angle

mMPTA: mechanical Medial Proximal Tibial Angle

MRI: Magnetic Resonance Imaging

mTFA: measured Tibio-Femoral Angle

NA: not available

NAV: navigation
 NRS: Numerical Rating Scale
 OA: OsteoArthritis
 OKS: Oxford Knee Score
 PCA: Posterior Condylar Angle
 PCL: Posterior Cruciate Ligament
 PCO: Posterior Condylar Offset
 PCS: Pain Catastrophising Scale
 PFA: Patella transverse axis-Femoral transepicondylar axis Angle
 PMPG: Patient-Matched Positioning Guides
 PS: Posterior-Stabilised
 PSG: Patient-Specific Guides
 PSI: Patient-Specific Implants
 PSPG: Patient-Specific Positioning Guides
 PSS: PSI-Single use
 QoL: Quality of Life
 RBC: Red Blood Cells
 RCT: randomised controlled trial
 ROM: Range Of Motion
 RSA: RadioStereometric Analysis
 SD: standard deviation
 SF-12: Short Form health survey 12-item
 SOC: standard of care
 STD: standard
 TCA: Tibial Coronal Alignment
 TFA: TibioFemoral Angle
 THA: Total Hip Arthroplasty
 TKA: Total Knee Arthroplasty
 TSA: Tibial Sagittal Alignment
 TT: Tourniquet Time
 UCLA: University of California Los Angeles
 VAS: visual analogue score
 WOMAC: Western Ontario and McMaster Universities Osteoarthritis Index

Characteristics of excluded studies *[ordered by study ID]*

Study	Reason for exclusion
Asada 2014	Ineligible comparator
Bollars 2023	Ineligible comparator
Calliess 2017a	Ineligible intervention
Dossett 2014	Ineligible intervention
Ensini 2014	Ineligible comparator
Gan 2015a	Ineligible intervention
Geng 2024	Ineligible comparator
Hafez 2006	Ineligible intervention
Hommel 2017	Ineligible comparator
Lee 2024	Ineligible comparator

Study	Reason for exclusion
Londhe 2024	Ineligible comparator
NCT05980442	Ineligible comparator
Sarzaeem 2024	Ineligible comparator
Schotanus 2016	Ineligible comparator
Smolle 2023	Ineligible comparators
Theeuwes 2024	Ineligible comparator
Timoncini 2012	Ineligible comparator
Van der Lelij 2023	Ineligible comparator
Yuan 2024	Ineligible comparator
Zhang 2016	Ineligible intervention

Characteristics of studies awaiting classification *[ordered by study ID]*

Musli 2016

Methods	Study Design: parallel RCT Country: Czech Republic Setting: Nemocnice České Budějovice, a.s., Ortopedické oddělení, České Budějovice Length of Follow-up: 1 year
Participants	Inclusion Criteria: Not recorded Exclusion Criteria: Not recorded Number of participants randomised: 23 - Intervention: Patient-specific cutting guides, N = Not recorded - Control: Conventional instrumentation, N = Not recorded Number of participants who received allocated intervention: 23 Number of dropouts: - Intervention: Not recorded - Control: Not recorded BMI - Control group: Not recorded - Intervention group: Not recorded Preoperative mechanical axis - Control group: Not recorded - Intervention group: Not recorded

Musli 2016 (Continued)

Interventions	Control group: Conventional instrumentation Intervention group: Patient-specific cutting guides Type of implant: Not recorded Bearing design: Fixed bearing Mode of fixation: Cemented Patellar resurfacing: Not recorded Preop imaging for designing PSG: CT Implant Brand: Zimmer Nex Gen PSG Brand: Zimmer Nex Gen Intraoperative Data: Not recorded
Outcomes	Outcomes assessed: Intraoperative, radiographs at 3 months follow-up Imaging modality used for postoperative measurement: Radiograph Outcomes used in this review: 1. Functional outcome scores 2. Radiographic alignment of component 3. Number of outliers in radiographic alignment 4. Total adverse events 5. Range of Motion (postoperative) Outcomes not used in this review: 1. Blood loss 2. Length of hospital stay 3. Operative time "At 1 post-operative year, the average Knee Society Score (KSS) was 85.5 points and the average functional score was 82.6 points. The satisfaction rate was 94% and, on a school rating system, the average mark was 1.3. The average postoperative ROM value was 116°. All patients were willing to undergo the surgery again. The only complication was thrombosis in one patient. Radiographic findings of knee alignment were optimal in 18, correct (up to 3° deviation) in three and incorrect (above 3° deviation) in two patients. Radiographic signs of loosening were not recorded."
Notes	Full text in Czech

Wang 2021

Methods	Study Design: parallel RCT Country: China Setting: Peking University, Third Hospital Length of Follow-up: Three months
Participants	Inclusion Criteria • patients diagnosed with primary knee osteoarthritis and scheduled for unilateral primary TKA

Wang 2021 (Continued)

- surgery and perioperative management performed by the same physician team

Exclusion Criteria

- severe varus or valgus deformity (varus or valgus > 20°)
- severe medical illness affecting surgery or anaesthesia
- patients or family members who did not understand the study or refused to sign the informed consent

Number of participants randomised: 80

- Intervention: Patient-specific cutting guides, N = 40
- Control: Computer assisted surgery, N = 40

Number of participants who received allocated intervention: 80

Number of dropouts: Not recorded

BMI

- Control group: 27.6 ± 3.7
- Intervention group: 27.6 ± 3.1

Sex:

- Control group: (Male/female) 6/34
- Intervention group: (Male/female) 5/35

Age:

- Control group: 68.6 ± 6
- Intervention group: 69 ± 6

Preoperative mechanical axis

- Control group: 190.00 ± 5.7
- Intervention group: 191.2 ± 6.4

Interventions	Control group: Computer assisted surgery. OrthoPilot computer navigation-assisted system and Columbus prosthesis (B. Braun, Germany) Intervention group: Patient-specific cutting guides Type of implant: Not recorded Bearing design: Not recorded Mode of fixation: Not recorded Patellar resurfacing: Not recorded Preop imaging for designing PSG: CT Implant Brand: B. Braun PSG Brand: A3 prosthesis (Beijing Aikang Yicheng Company), EOS P110 3D printer Intraoperative data: operation time
Outcomes	Outcomes assessed: Intraoperative, radiographs at 3 months follow-up Imaging modality used for postoperative measurement: Radiograph Outcomes used in this review:

Wang 2021 (Continued)

1. Radiographic alignment of component
2. Number of outliers in radiographic alignment
3. Operative time
4. Total adverse events

3. Operation time

The operation time of the CAS group was longer than that of the PSI group, and the difference was statistically significant [(81 ± 12) min vs. (52 ± 8) min, $t = 12.909$, $P < 0.01$].

Outcomes not used in this review:

1. Postoperative HKA angle: There was no significant difference in the postoperative HKA angle deviation between the CAS group and the PSI group ($2.0^\circ \pm 1.0^\circ$ vs. $2.6^\circ \pm 1.7^\circ$, $t' = -1.785$, $P = 0.084$). The postoperative HKA angle deviation rate of more than 3° in the CAS group was lower than that in the PSI group (10.0% vs. 27.5%, $\chi^2 = 4.201$, $P = 0.045$).
2. mLDFA: There was no significant difference in the postoperative mLDFA deviation between the CAS group and the PSI group ($1.3^\circ \pm 1.1^\circ$ vs. $1.6^\circ \pm 1.3^\circ$, $t = -1.101$, $P = 0.274$). There was also no significant difference in the postoperative mLDFA deviation rate of more than 3° between the two groups (5.0% vs. 10.0%, $\chi^2 = 0.721$, $P = 0.396$).
3. mMPPTA: There was no significant difference in the postoperative mMPPTA deviation between the CAS group and the PSI group ($1.3^\circ \pm 0.9^\circ$ vs. $1.6^\circ \pm 1.4^\circ$, $t = -0.958$, $P = 0.341$). There was also no significant difference in the postoperative mMPPTA deviation rate of more than 3° between the two groups (2.5% vs. 12.5%, $\chi^2 = 2.883$, $P = 0.201$).
4. sFCA: The sFCA deviation value in the CAS group was smaller than that in the PSI group ($1.7^\circ \pm 1.2^\circ$ vs. $2.5^\circ \pm 1.4^\circ$, $t = -2.615$, $P = 0.011$). The postoperative sFCA deviation rate of more than 3° in the CAS group was lower than that in the PSI group (5.0% vs. 20.0%, $\chi^2 = 4.114$, $P = 0.043$).
5. sTCA: The sTCA deviation value in the CAS group was smaller than that in the PSI group ($1.2^\circ \pm 0.9^\circ$ vs. $2.3^\circ \pm 1.8^\circ$, $t = -3.651$, $P < 0.01$). The postoperative sTCA deviation rate of more than 3° in the CAS group was lower than that in the PSI group (2.5% vs. 22.5%, $\chi^2 = 7.314$, $P = 0.007$).

Notes	Full text in Chinese
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CAS: Computer Assisted Surgery
CT: Computerised Tomography
FCA: Femoral Component Angle
HKA: Hip-Knee-Ankle
KSS: Knee Society Score
LDFA: Lateral Distal Femoral Angle
MPPTA: Medial Proximal Tibial Angle
PSI: Patient-Specific Implants
RCT: randomised controlled trial
ROM: Range Of Motion
TCA: Tibial Component Angle

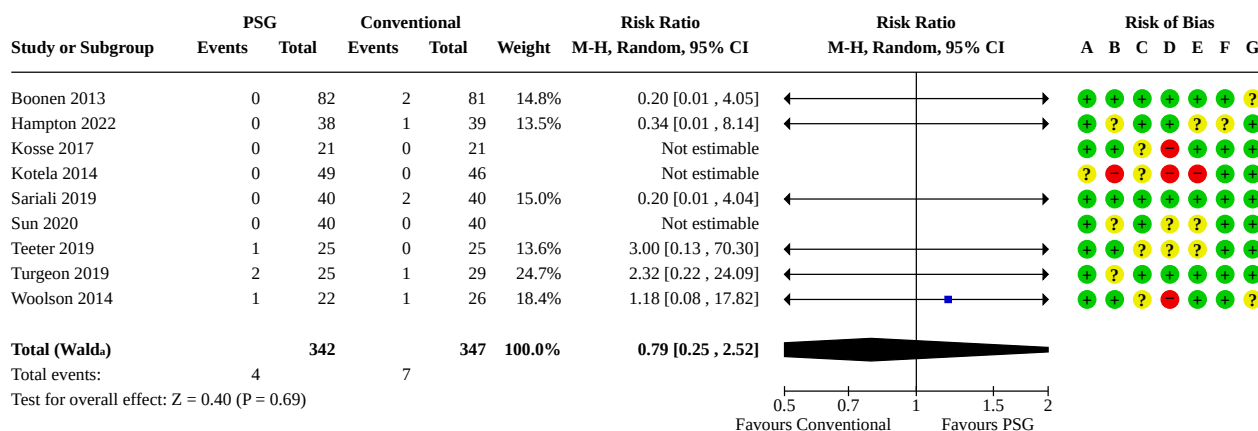
DATA AND ANALYSES

Comparison 1. Patient-specific cutting guides versus conventional instrumentation

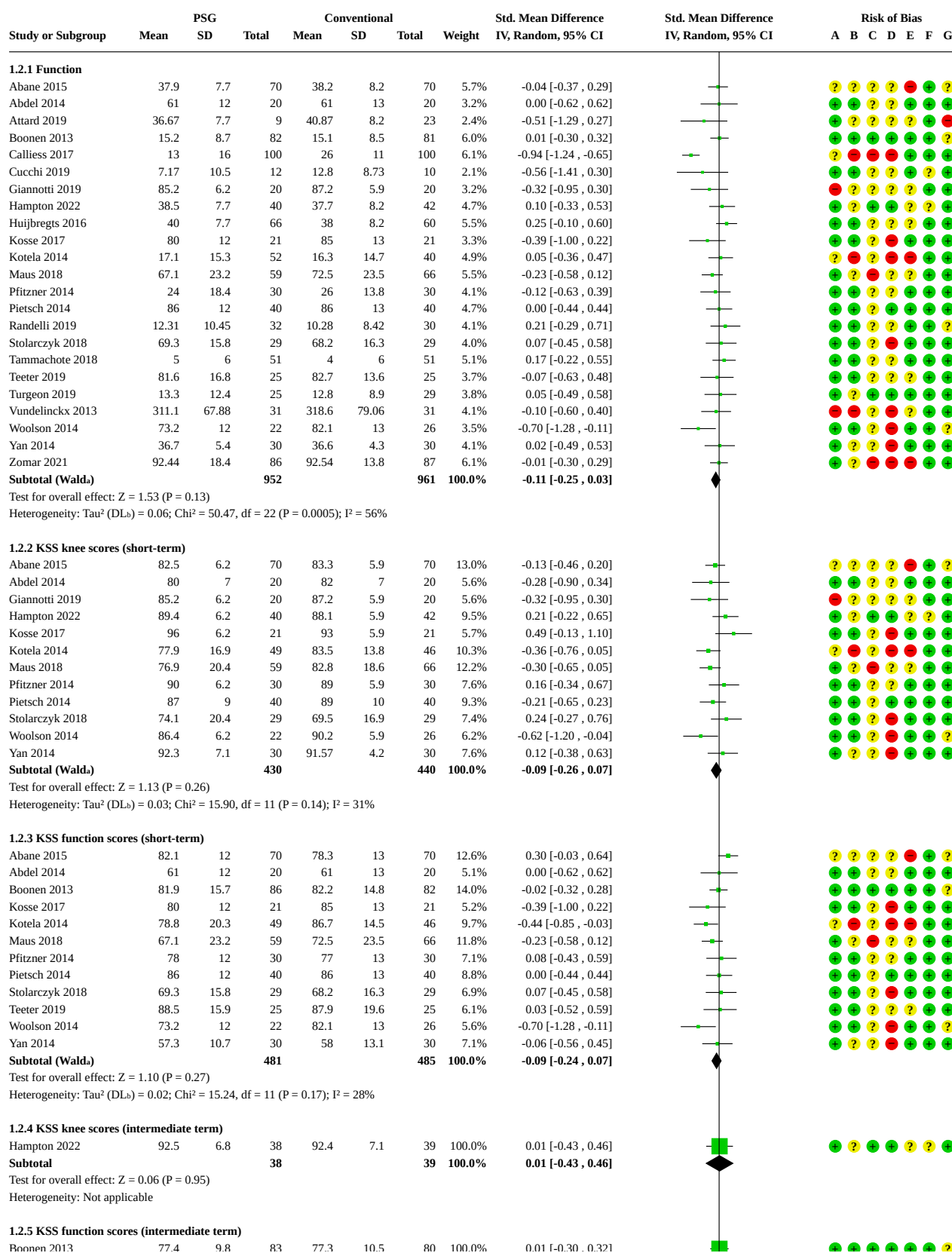
Outcome or subgroup title	No. of studies	No. of participants	Statistical method	Effect size
1.1 Survival of implant	9	689	Risk Ratio (M-H, Random, 95% CI)	0.79 [0.25, 2.52]

Outcome or subgroup title	No. of studies	No. of participants	Statistical method	Effect size
1.2 Function	24		Std. Mean Difference (IV, Random, 95% CI)	Subtotals only
1.2.1 Function	23	1913	Std. Mean Difference (IV, Random, 95% CI)	-0.11 [-0.25, 0.03]
1.2.2 KSS knee scores (short-term)	12	870	Std. Mean Difference (IV, Random, 95% CI)	-0.09 [-0.26, 0.07]
1.2.3 KSS function scores (short-term)	12	966	Std. Mean Difference (IV, Random, 95% CI)	-0.09 [-0.24, 0.07]
1.2.4 KSS knee scores (intermediate term)	1	77	Std. Mean Difference (IV, Random, 95% CI)	0.01 [-0.43, 0.46]
1.2.5 KSS function scores (intermediate term)	1	163	Std. Mean Difference (IV, Random, 95% CI)	0.01 [-0.30, 0.32]
1.2.6 Oxford Knee Score (short-term)	6	603	Std. Mean Difference (IV, Random, 95% CI)	0.04 [-0.12, 0.20]
1.2.7 Oxford Knee change score (short-term)	3	138	Std. Mean Difference (IV, Random, 95% CI)	0.02 [-0.35, 0.38]
1.2.8 Oxford Knee Score (intermediate-term)	2	240	Std. Mean Difference (IV, Random, 95% CI)	0.02 [-0.41, 0.45]
1.2.9 WOMAC (short-term)	7	840	Std. Mean Difference (IV, Random, 95% CI)	-0.19 [-0.52, 0.13]
1.2.10 WOMAC (intermediate-term)	1	163	Std. Mean Difference (IV, Random, 95% CI)	-0.35 [-0.66, -0.04]
1.2.11 KOOS total	1	62	Std. Mean Difference (IV, Random, 95% CI)	-0.10 [-0.60, 0.40]
1.2.12 KOOS pain	2	134	Std. Mean Difference (IV, Random, 95% CI)	0.35 [-0.01, 0.71]
1.2.13 KOOS symptoms	2	134	Std. Mean Difference (IV, Random, 95% CI)	0.18 [-0.16, 0.52]
1.2.14 KOOS ADL	2	134	Std. Mean Difference (IV, Random, 95% CI)	0.27 [-0.08, 0.61]
1.2.15 KOOS sports	2	134	Std. Mean Difference (IV, Random, 95% CI)	0.17 [-0.17, 0.51]
1.2.16 KOOS QOL	2	134	Std. Mean Difference (IV, Random, 95% CI)	0.33 [-0.01, 0.67]
1.3 Radiographic lower limb alignment measured by outliers in FTCA	29	2412	Risk Ratio (M-H, Random, 95% CI)	0.86 [0.68, 1.10]

Outcome or subgroup title	No. of studies	No. of participants	Statistical method	Effect size
1.4 Pain	10	715	Std. Mean Difference (IV, Random, 95% CI)	-0.09 [-0.23, 0.06]
1.4.1 VAS early	10	715	Std. Mean Difference (IV, Random, 95% CI)	-0.09 [-0.23, 0.06]
1.5 Total adverse events	14	1169	Risk Ratio (M-H, Random, 95% CI)	0.97 [0.68, 1.39]
1.6 Re-operation rate	10	843	Risk Ratio (M-H, Random, 95% CI)	0.87 [0.45, 1.68]
1.7 Quality of life	3	212	Mean Difference (IV, Random, 95% CI)	0.16 [-2.51, 2.84]
1.7.1 SF12 - M	3	212	Mean Difference (IV, Random, 95% CI)	0.16 [-2.51, 2.84]
1.8 Operative time	25	2033	Mean Difference (IV, Random, 95% CI)	-2.79 [-5.66, 0.08]
1.9 Blood loss	14	1189	Mean Difference (IV, Random, 95% CI)	-66.47 [-93.30, -39.64]
1.10 Haematocrit	4	211	Std. Mean Difference (IV, Random, 95% CI)	-0.88 [-1.63, -0.13]
1.11 Transfusion requirement	7	502	Risk Ratio (M-H, Random, 95% CI)	0.70 [0.50, 1.00]
1.12 Range of motion flexion	8	586	Mean Difference (IV, Random, 95% CI)	-0.42 [-2.07, 1.23]
1.12.1 New Subgroup	8	586	Mean Difference (IV, Random, 95% CI)	-0.42 [-2.07, 1.23]
1.13 Range of motion extension	3	258	Mean Difference (IV, Random, 95% CI)	0.26 [-0.28, 0.80]
1.14 Length of stay	13	1062	Mean Difference (IV, Random, 95% CI)	-0.23 [-0.46, -0.00]
1.14.2 Length of stay	13	1062	Mean Difference (IV, Random, 95% CI)	-0.23 [-0.46, -0.00]
1.15 Incision length	2	96	Std. Mean Difference (IV, Random, 95% CI)	Not estimable
1.16 Radiographic outliers in femur coronal alignment	17	1495	Risk Ratio (M-H, Random, 95% CI)	0.77 [0.53, 1.12]
1.17 Radiographic outliers in tibia coronal alignment	17	1495	Risk Ratio (M-H, Random, 95% CI)	1.29 [0.84, 2.00]
1.18 Radiographic outliers in femur sagittal alignment	11	1075	Risk Ratio (M-H, Random, 95% CI)	1.03 [0.83, 1.26]
1.19 Radiographic outliers in tibia sagittal alignment	14	1233	Risk Ratio (M-H, Random, 95% CI)	1.17 [0.82, 1.68]
1.20 Radiographic outliers in femur rotation	10	922	Risk Ratio (M-H, Random, 95% CI)	0.63 [0.41, 0.99]

Analysis 1.1. Comparison 1: Patient-specific cutting guides versus conventional instrumentation, Outcome 1: Survival of implant**Footnotes**^aCI calculated by Wald-type method.^bTau² calculated by DerSimonian and Laird method.**Risk of bias legend**

- (A) Random sequence generation (selection bias)
- (B) Allocation concealment (selection bias)
- (C) Blinding of participants and personnel (performance bias)
- (D) Blinding of outcome assessment (detection bias): Functional outcomes
- (E) Incomplete outcome data (attrition bias)
- (F) Selective reporting (reporting bias)
- (G) Other bias

Analysis 1.2. Comparison 1: Patient-specific cutting guides versus conventional instrumentation, Outcome 2: Function

Analysis 1.2. (Continued)

1.2.5 KSS function scores (intermediate term)

Boonen 2013	77.4	9.8	83	77.3	10.5	80	100.0%	0.01 [-0.30 , 0.32]		+	+	+	+	+	+	?
Subtotal			83			80	100.0%	0.01 [-0.30 , 0.32]								

Test for overall effect: $Z = 0.06$ ($P = 0.95$)

Heterogeneity: Not applicable

1.2.6 Oxford Knee Score (short-term)

Abane 2015	37.9	7.7	70	38.2	8.2	70	23.5%	-0.04 [-0.37 , 0.29]		?	?	?	?	?	+	?
Attard 2019	36.67	7.7	9	40.87	8.2	23	4.2%	-0.51 [-1.29 , 0.27]		+	+	+	?	?	+	+
Boonen 2013	15.2	8.7	82	15.1	8.5	81	27.4%	0.01 [-0.30 , 0.32]		+	+	+	+	+	+	?
Hampton 2022	38.5	7.7	40	37.7	8.2	42	13.8%	0.10 [-0.33 , 0.53]		+	?	?	?	?	?	?
Huijbregts 2016	40	7.7	66	38	8.2	60	21.0%	0.25 [-0.10 , 0.60]		+	+	?	?	?	?	+
Yan 2014	36.7	5.4	30	36.6	4.3	30	10.1%	0.02 [-0.49 , 0.53]		+	?	?	?	?	+	+
Subtotal (Wald)			297			306	100.0%	0.04 [-0.12 , 0.20]								

Test for overall effect: $Z = 0.50$ ($P = 0.62$)Heterogeneity: Tau^2 (DL_b) = 0.00; $\text{Chi}^2 = 3.58$, $\text{df} = 5$ ($P = 0.61$); $I^2 = 0\%$

1.2.7 Oxford Knee change score (short-term)

Cucchi 2019c	7.17	10.5	12	12.8	8.73	10	16.9%	-0.56 [-1.41 , 0.30]		+	+	?	?	+	?	+
Randelli 2019c	12.31	10.45	32	10.28	8.42	30	43.9%	0.21 [-0.29 , 0.71]		+	+	?	?	+	+	?
Turgeon 2019c	13.3	12.4	25	12.8	8.9	29	39.2%	0.05 [-0.49 , 0.58]		+	?	+	+	+	+	+
Subtotal (Wald)			69			69	100.0%	0.02 [-0.35 , 0.38]								

Test for overall effect: $Z = 0.09$ ($P = 0.93$)Heterogeneity: Tau^2 (DL_b) = 0.01; $\text{Chi}^2 = 2.29$, $\text{df} = 2$ ($P = 0.32$); $I^2 = 13\%$

1.2.8 Oxford Knee Score (intermediate-term)

Boonen 2013	23.5	9.4	83	21.6	8.4	80	56.9%	0.21 [-0.10 , 0.52]		+	+	+	+	+	+	?
Hampton 2022	40.8	6.9	38	42.5	7.4	39	43.1%	-0.24 [-0.68 , 0.21]		+	?	+	+	+	?	+
Subtotal (Wald)			121			119	100.0%	0.02 [-0.41 , 0.45]								

Test for overall effect: $Z = 0.09$ ($P = 0.93$)Heterogeneity: Tau^2 (DL_b) = 0.06; $\text{Chi}^2 = 2.59$, $\text{df} = 1$ ($P = 0.11$); $I^2 = 61\%$

1.2.9 WOMAC (short-term)

Boonen 2013	80.6	18.4	82	86	13.8	81	15.6%	-0.33 [-0.64 , -0.02]		+	+	+	+	+	+	?
Calliess 2017	13	16	100	26	11	100	15.8%	-0.94 [-1.24 , -0.65]		?	+	+	+	+	+	+
Kotela 2014	17.1	15.3	52	16.3	14.7	40	14.0%	0.05 [-0.36 , 0.47]		?	+	?	+	+	+	+
Pfizer 2014	24	18.4	30	26	13.8	30	12.6%	-0.12 [-0.63 , 0.39]		+	+	?	?	+	+	+
Tammachote 2018	5	6	51	4	6	51	14.4%	0.17 [-0.22 , 0.55]		+	+	?	?	+	+	+
Teeter 2019	81.6	16.8	25	82.7	13.6	25	11.9%	-0.07 [-0.63 , 0.48]		+	+	?	?	?	?	+
Zomar 2021	92.44	18.4	86	92.54	13.8	87	15.7%	-0.01 [-0.30 , 0.29]		+	?	+	+	+	+	+
Subtotal (Wald)			426			414	100.0%	-0.19 [-0.52 , 0.13]								

Test for overall effect: $Z = 1.17$ ($P = 0.24$)Heterogeneity: Tau^2 (DL_b) = 0.15; $\text{Chi}^2 = 31.49$, $\text{df} = 6$ ($P < 0.0001$); $I^2 = 81\%$

1.2.10 WOMAC (intermediate-term)

Boonen 2013	78.9	19.1	83	85.2	16.4	80	100.0%	-0.35 [-0.66 , -0.04]		+	+	+	+	+	+	?
Subtotal			83			80	100.0%	-0.35 [-0.66 , -0.04]								

Test for overall effect: $Z = 2.23$ ($P = 0.03$)

Heterogeneity: Not applicable

1.2.11 KOOS total

Vundelinckx 2013	311.1	67.88	31	318.6	79.06	31	100.0%	-0.10 [-0.60 , 0.40]		+	+	?	?	+	+	+
Subtotal			31			31	100.0%	-0.10 [-0.60 , 0.40]								

Test for overall effect: $Z = 0.40$ ($P = 0.69$)

Heterogeneity: Not applicable

1.2.12 KOOS pain

Abdel 2014	35	12	20	34	12	20	31.8%	0.08 [-0.54 , 0.70]		+	+	?	?	+	+	+
Van Leeuwen 2018	88	16	44	79	21	50	68.2%	0.47 [0.06 , 0.89]		+	+	+	?	+	+	?
Subtotal (Wald)			64			70	100.0%	0.35 [-0.01 , 0.71]								

Test for overall effect: $Z = 1.91$ ($P = 0.06$)Heterogeneity: Tau^2 (DL_b) = 0.00; $\text{Chi}^2 = 1.07$, $\text{df} = 1$ ($P = 0.30$); $I^2 = 6\%$

1.2.13 KOOS symptoms

Abdel 2014	37	10	20	37	8	20	30.1%	0.00 [-0.62 , 0.62]		+	+	?	?	+	+	+
Van Leeuwen 2018	82	16	44	78	15	50	69.9%	0.26 [-0.15 , 0.66]		+	+	+	?	+	+	?
Subtotal (Wald)			64			70	100.0%	0.18 [-0.16 , 0.52]								

Test for overall effect: $Z = 1.03$ ($P = 0.30$)Heterogeneity: Tau^2 (DL_b) = 0.00; $\text{Chi}^2 = 0.46$, $\text{df} = 1$ ($P = 0.50$); $I^2 = 0\%$

1.2.14 KOOS ADL

Abdel 2014	49	15	20	48	13	20	30.3%	0.07 [-0.55 , 0.69]		+	+	?	?	+	+	+
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Analysis 1.2. (Continued)

1.2.14 KOOS ADL

Abdel 2014	49	15	20	48	13	20	30.3%	0.07 [-0.55 , 0.69]
Van Leeuwen 2018	85	17	44	78	22	50	69.7%	0.35 [-0.06 , 0.76]
Subtotal (Wald)			64			70	100.0%	0.27 [-0.08 , 0.61]

Test for overall effect: $Z = 1.53$ ($P = 0.13$)Heterogeneity: Tau^2 (DL_{a}) = 0.00; $\text{Chi}^2 = 0.55$, $\text{df} = 1$ ($P = 0.46$); $I^2 = 0\%$

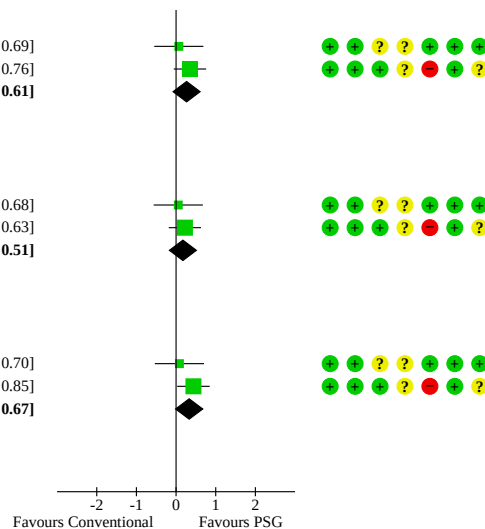
1.2.15 KOOS sports

Abdel 2014	26	18	20	25	17	20	30.1%	0.06 [-0.56 , 0.68]
Van Leeuwen 2018	46	24	44	40	29	50	69.9%	0.22 [-0.18 , 0.63]
Subtotal (Wald)			64			70	100.0%	0.17 [-0.17 , 0.51]

Test for overall effect: $Z = 0.99$ ($P = 0.32$)Heterogeneity: Tau^2 (DL_{a}) = 0.00; $\text{Chi}^2 = 0.19$, $\text{df} = 1$ ($P = 0.66$); $I^2 = 0\%$

1.2.16 KOOS QOL

Abdel 2014	28	13	20	27	10	20	30.4%	0.08 [-0.54 , 0.70]
Van Leeuwen 2018	76	21	44	66	24	50	69.6%	0.44 [0.03 , 0.85]
Subtotal (Wald)			64			70	100.0%	0.33 [-0.01 , 0.67]

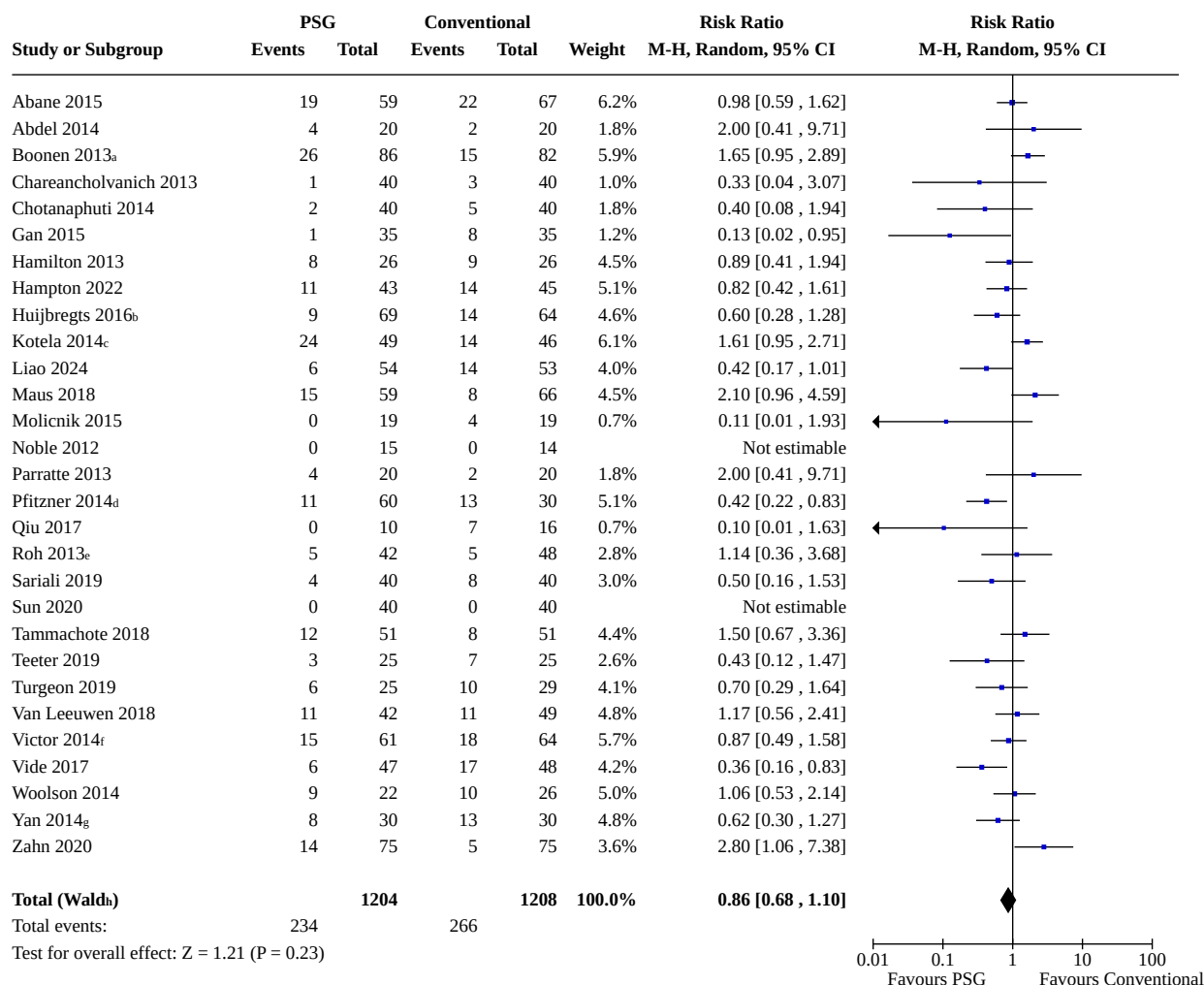
Test for overall effect: $Z = 1.89$ ($P = 0.06$)Heterogeneity: Tau^2 (DL_{a}) = 0.00; $\text{Chi}^2 = 0.87$, $\text{df} = 1$ ($P = 0.35$); $I^2 = 0\%$ Test for subgroup differences: $\text{Chi}^2 = 22.74$, $\text{df} = 15$ ($P = 0.09$), $I^2 = 34.0\%$ 

Footnotes

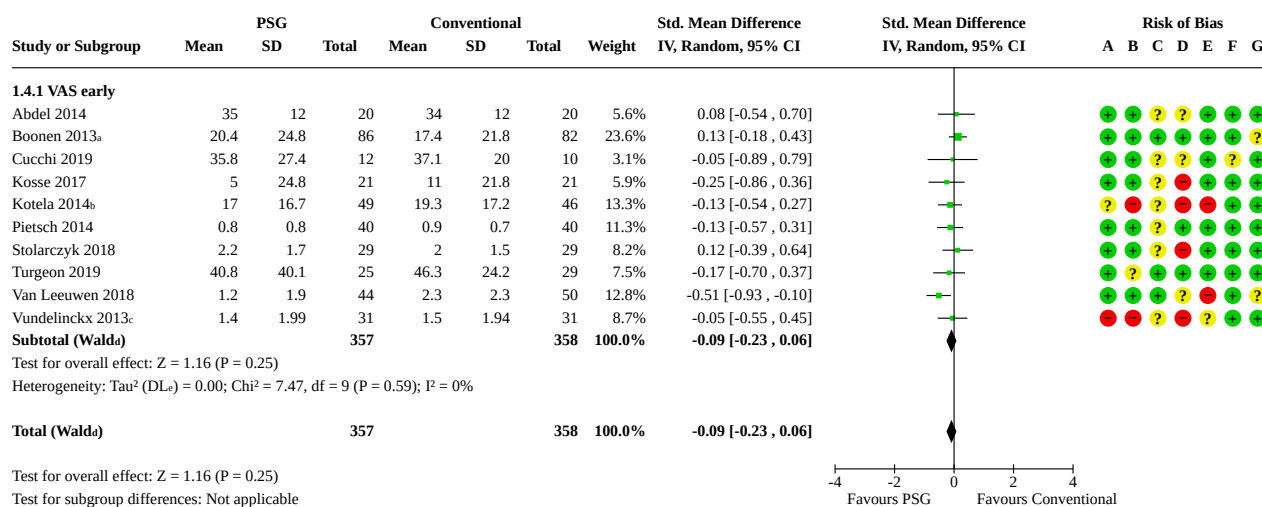
^aCI calculated by Wald-type method.^b Tau^2 calculated by DerSimonian and Laird method.^cChange in score

Risk of bias legend

- (A) Random sequence generation (selection bias)
- (B) Allocation concealment (selection bias)
- (C) Blinding of participants and personnel (performance bias)
- (D) Blinding of outcome assessment (detection bias): Functional outcomes
- (E) Incomplete outcome data (attrition bias)
- (F) Selective reporting (reporting bias)
- (G) Other bias

Analysis 1.3. Comparison 1: Patient-specific cutting guides versus conventional instrumentation, Outcome 3: Radiographic lower limb alignment measured by outliers in FTCAHeterogeneity: Tau² (DL_i) = 0.17; Chi² = 50.92, df = 26 (P = 0.002); I² = 49%**Footnotes**^aitt 86,82^bitt 69, 64^citt 49, 46^dCT and MRI PSG taken together^eitt 42, 48^fitt 61, 64^g81 patients 90 knees^hCI calculated by Wald-type method.ⁱTau² calculated by DerSimonian and Laird method.

Analysis 1.4. Comparison 1: Patient-specific cutting guides versus conventional instrumentation, Outcome 4: Pain

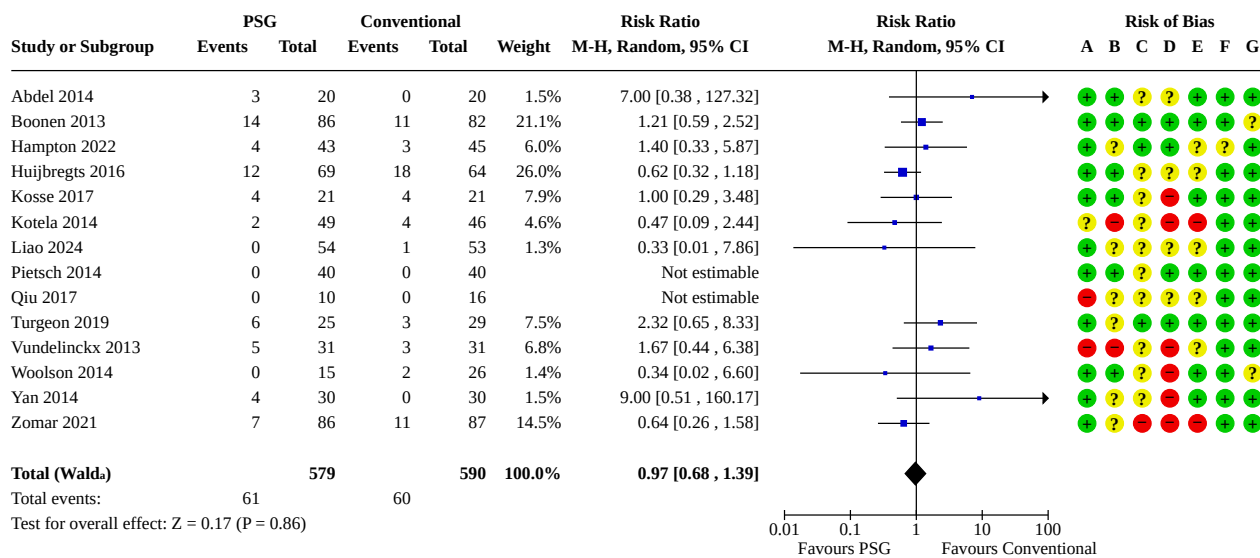


Footnotes

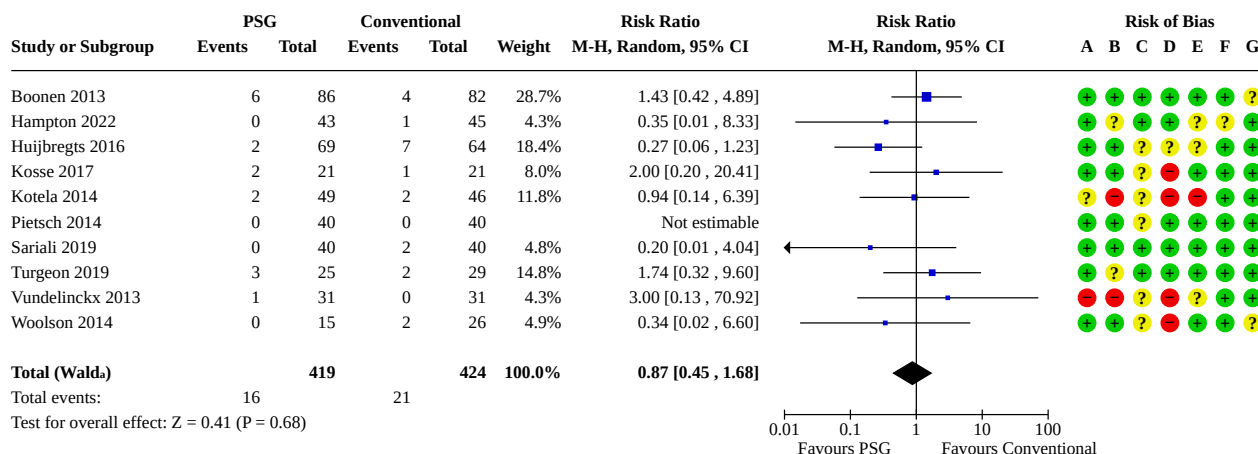
- ^askewed data from VAS
^bskewed data
^cVAS is skewed hence KOOS pain score selected
^dCI calculated by Wald-type method.
^eTau² calculated by DerSimonian and Laird method.

Risk of bias legend

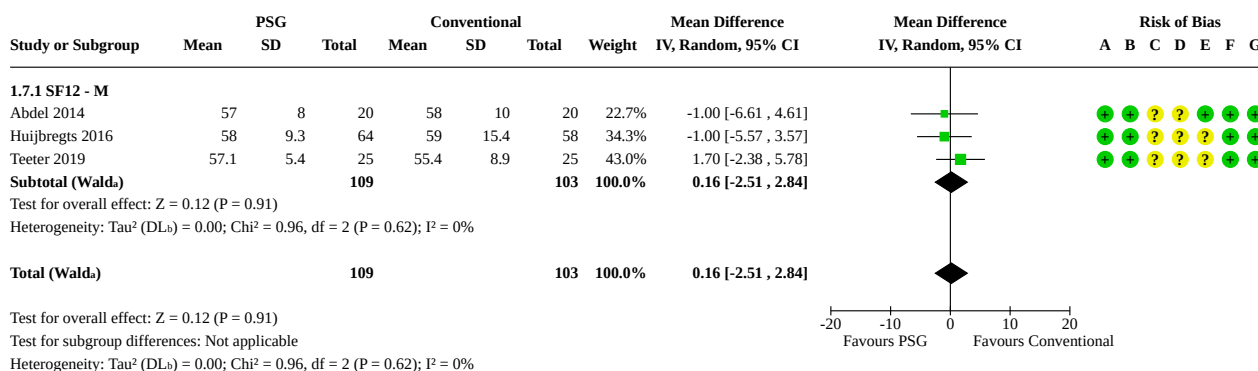
- (A) Random sequence generation (selection bias)
 (B) Allocation concealment (selection bias)
 (C) Blinding of participants and personnel (performance bias)
 (D) Blinding of outcome assessment (detection bias): Functional outcomes
 (E) Incomplete outcome data (attrition bias)
 (F) Selective reporting (reporting bias)
 (G) Other bias

Analysis 1.5. Comparison 1: Patient-specific cutting guides versus conventional instrumentation, Outcome 5: Total adverse events**Footnotes**^aCI calculated by Wald-type method.^bTau² calculated by DerSimonian and Laird method.**Risk of bias legend**

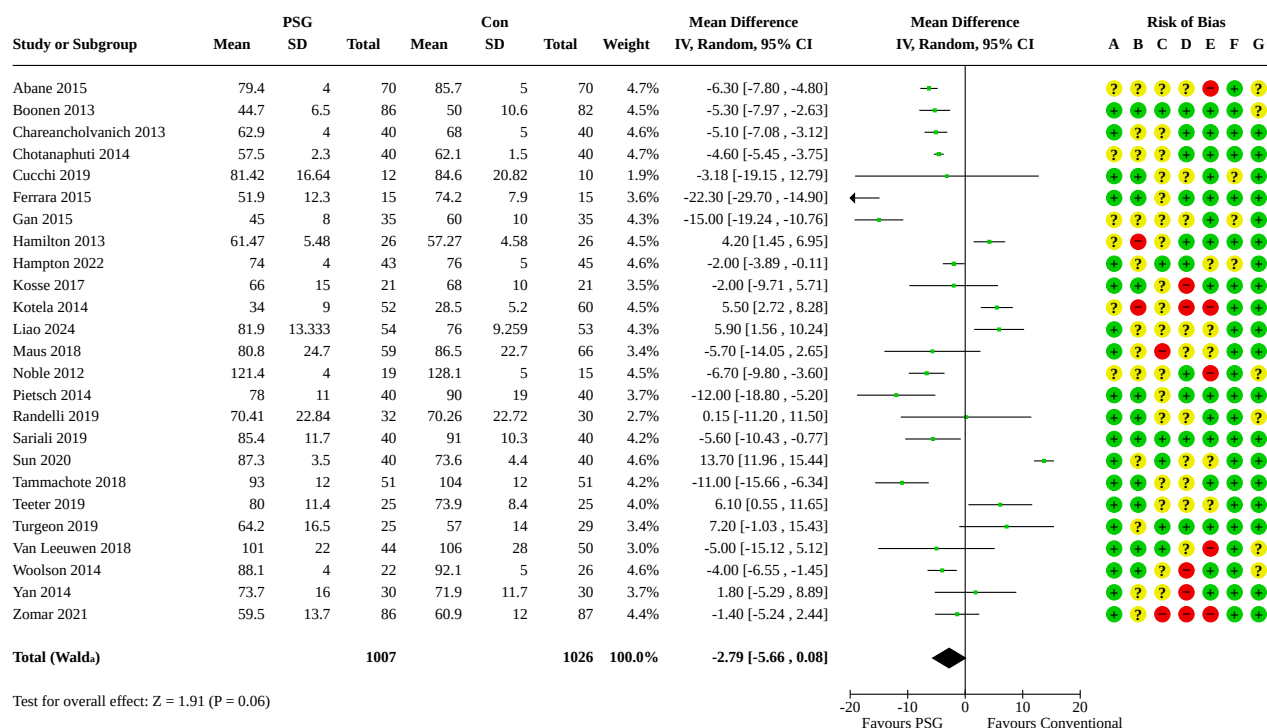
- (A) Random sequence generation (selection bias)
- (B) Allocation concealment (selection bias)
- (C) Blinding of participants and personnel (performance bias)
- (D) Blinding of outcome assessment (detection bias): Functional outcomes
- (E) Incomplete outcome data (attrition bias)
- (F) Selective reporting (reporting bias)
- (G) Other bias

Analysis 1.6. Comparison 1: Patient-specific cutting guides versus conventional instrumentation, Outcome 6: Re-operation rate**Footnotes**^aCI calculated by Wald-type method.^bTau² calculated by DerSimonian and Laird method.**Risk of bias legend**

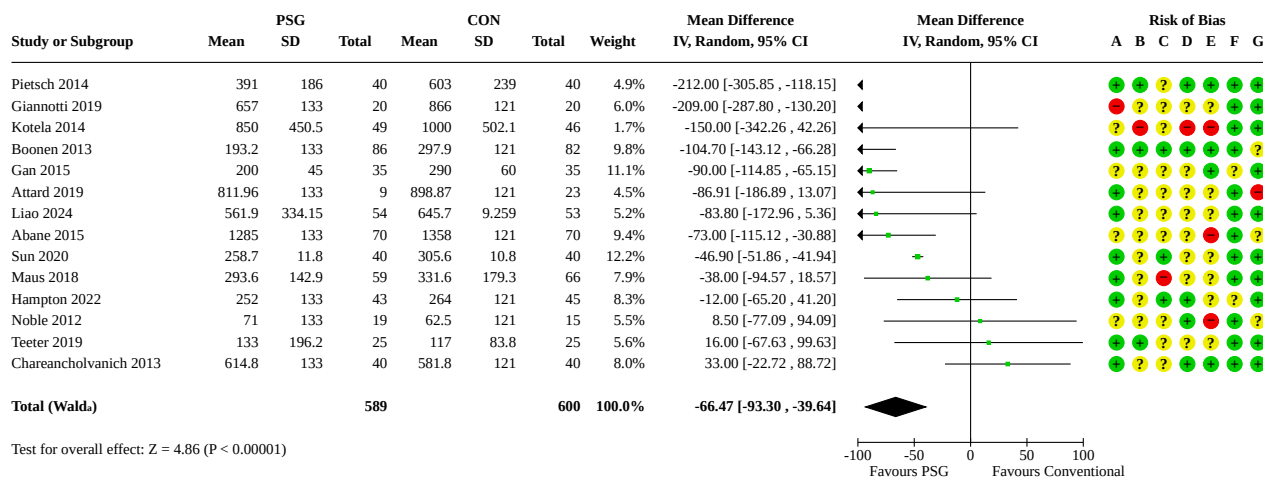
- (A) Random sequence generation (selection bias)
- (B) Allocation concealment (selection bias)
- (C) Blinding of participants and personnel (performance bias)
- (D) Blinding of outcome assessment (detection bias): Functional outcomes
- (E) Incomplete outcome data (attrition bias)
- (F) Selective reporting (reporting bias)
- (G) Other bias

Analysis 1.7. Comparison 1: Patient-specific cutting guides versus conventional instrumentation, Outcome 7: Quality of life**Footnotes**^aCI calculated by Wald-type method.^bTau² calculated by DerSimonian and Laird method.**Risk of bias legend**

- (A) Random sequence generation (selection bias)
- (B) Allocation concealment (selection bias)
- (C) Blinding of participants and personnel (performance bias)
- (D) Blinding of outcome assessment (detection bias): Functional outcomes
- (E) Incomplete outcome data (attrition bias)
- (F) Selective reporting (reporting bias)
- (G) Other bias

**Analysis 1.8. Comparison 1: Patient-specific cutting guides
versus conventional instrumentation, Outcome 8: Operative time****Footnotes**^aCI calculated by Wald-type method.^b Tau^2 calculated by DerSimonian and Laird method.**Risk of bias legend**

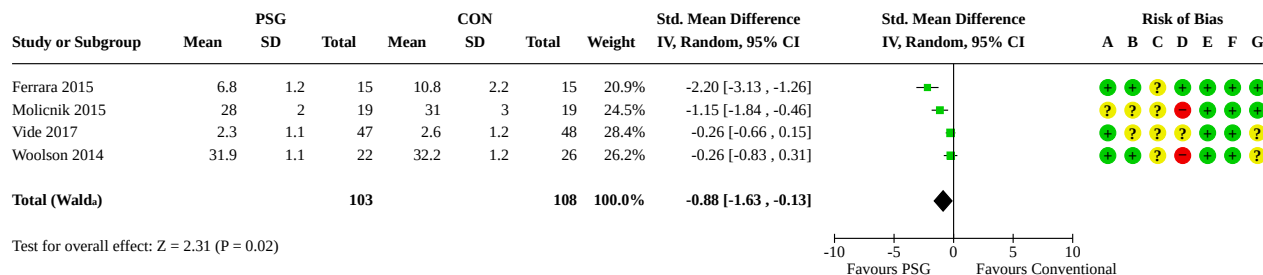
- (A) Random sequence generation (selection bias)
- (B) Allocation concealment (selection bias)
- (C) Blinding of participants and personnel (performance bias)
- (D) Blinding of outcome assessment (detection bias): Functional outcomes
- (E) Incomplete outcome data (attrition bias)
- (F) Selective reporting (reporting bias)
- (G) Other bias

**Analysis 1.9. Comparison 1: Patient-specific cutting guides
versus conventional instrumentation, Outcome 9: Blood loss****Footnotes**

aCI calculated by Wald-type method.

bTau² calculated by DerSimonian and Laird method.**Risk of bias legend**

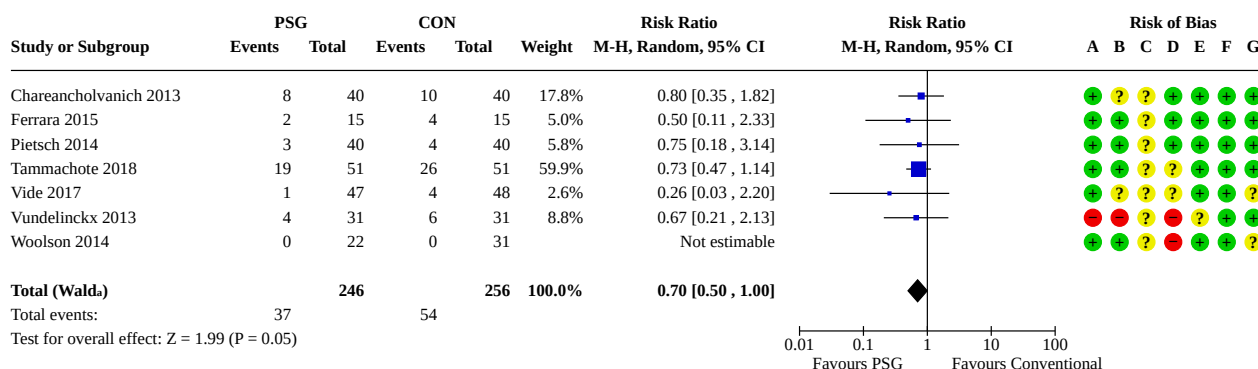
- (A) Random sequence generation (selection bias)
(B) Allocation concealment (selection bias)
(C) Blinding of participants and personnel (performance bias)
(D) Blinding of outcome assessment (detection bias): Functional outcomes
(E) Incomplete outcome data (attrition bias)
(F) Selective reporting (reporting bias)
(G) Other bias

**Analysis 1.10. Comparison 1: Patient-specific cutting guides
versus conventional instrumentation, Outcome 10: Haematocrit****Footnotes**

aCI calculated by Wald-type method.

bTau² calculated by DerSimonian and Laird method.**Risk of bias legend**

- (A) Random sequence generation (selection bias)
(B) Allocation concealment (selection bias)
(C) Blinding of participants and personnel (performance bias)
(D) Blinding of outcome assessment (detection bias): Functional outcomes
(E) Incomplete outcome data (attrition bias)
(F) Selective reporting (reporting bias)
(G) Other bias

Analysis 1.11. Comparison 1: Patient-specific cutting guides versus conventional instrumentation, Outcome 11: Transfusion requirement

Heterogeneity: Tau² (DL_b) = 0.00; Chi² = 1.20, df = 5 (P = 0.94); I² = 0%

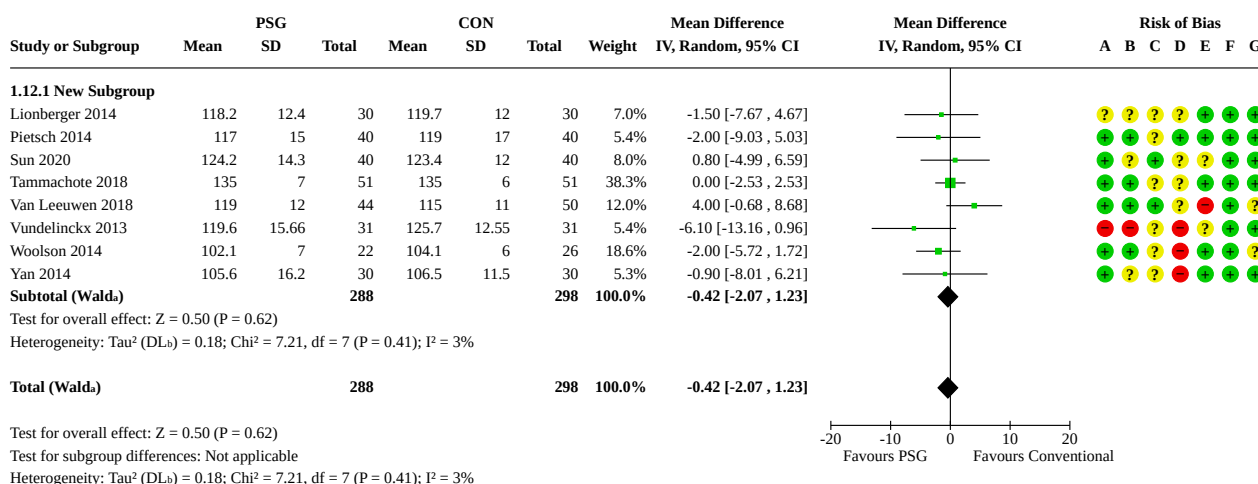
Footnotes

^aCI calculated by Wald-type method.

^bTau² calculated by DerSimonian and Laird method.

Risk of bias legend

- (A) Random sequence generation (selection bias)
- (B) Allocation concealment (selection bias)
- (C) Blinding of participants and personnel (performance bias)
- (D) Blinding of outcome assessment (detection bias): Functional outcomes
- (E) Incomplete outcome data (attrition bias)
- (F) Selective reporting (reporting bias)
- (G) Other bias

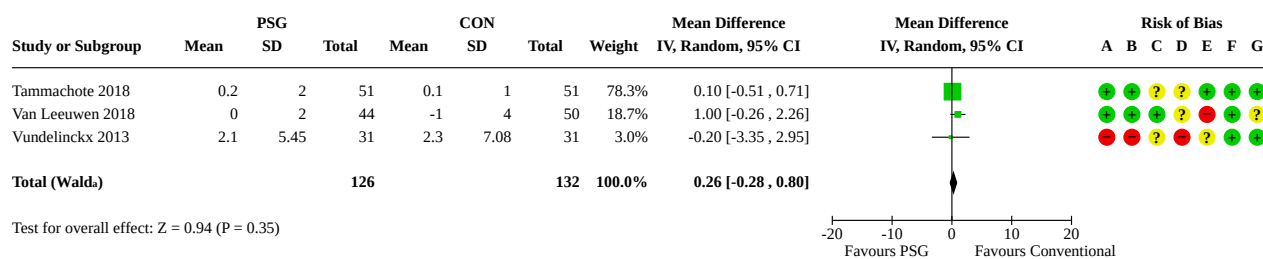
Analysis 1.12. Comparison 1: Patient-specific cutting guides versus conventional instrumentation, Outcome 12: Range of motion flexion**Footnotes**

^aCI calculated by Wald-type method.

^bTau² calculated by DerSimonian and Laird method.

Risk of bias legend

- (A) Random sequence generation (selection bias)
- (B) Allocation concealment (selection bias)
- (C) Blinding of participants and personnel (performance bias)
- (D) Blinding of outcome assessment (detection bias): Functional outcomes
- (E) Incomplete outcome data (attrition bias)
- (F) Selective reporting (reporting bias)
- (G) Other bias

Analysis 1.13. Comparison 1: Patient-specific cutting guides versus conventional instrumentation, Outcome 13: Range of motion extension**Footnotes**^aCI calculated by Wald-type method.^b Tau^2 calculated by DerSimonian and Laird method.**Risk of bias legend**

(A) Random sequence generation (selection bias)

(B) Allocation concealment (selection bias)

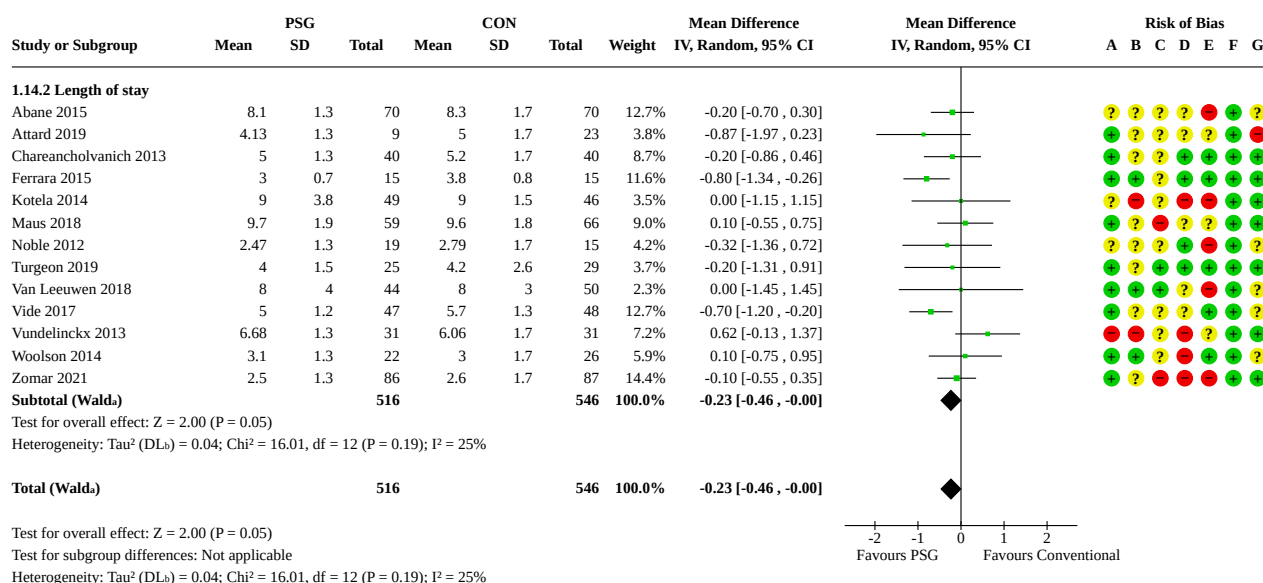
(C) Blinding of participants and personnel (performance bias)

(D) Blinding of outcome assessment (detection bias): Functional outcomes

(E) Incomplete outcome data (attrition bias)

(F) Selective reporting (reporting bias)

(G) Other bias

Analysis 1.14. Comparison 1: Patient-specific cutting guides versus conventional instrumentation, Outcome 14: Length of stay**Footnotes**^aCI calculated by Wald-type method.^b Tau^2 calculated by DerSimonian and Laird method.**Risk of bias legend**

(A) Random sequence generation (selection bias)

(B) Allocation concealment (selection bias)

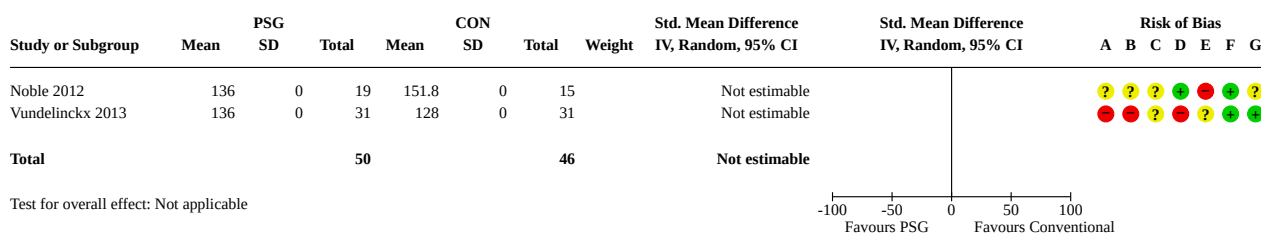
(C) Blinding of participants and personnel (performance bias)

(D) Blinding of outcome assessment (detection bias): Functional outcomes

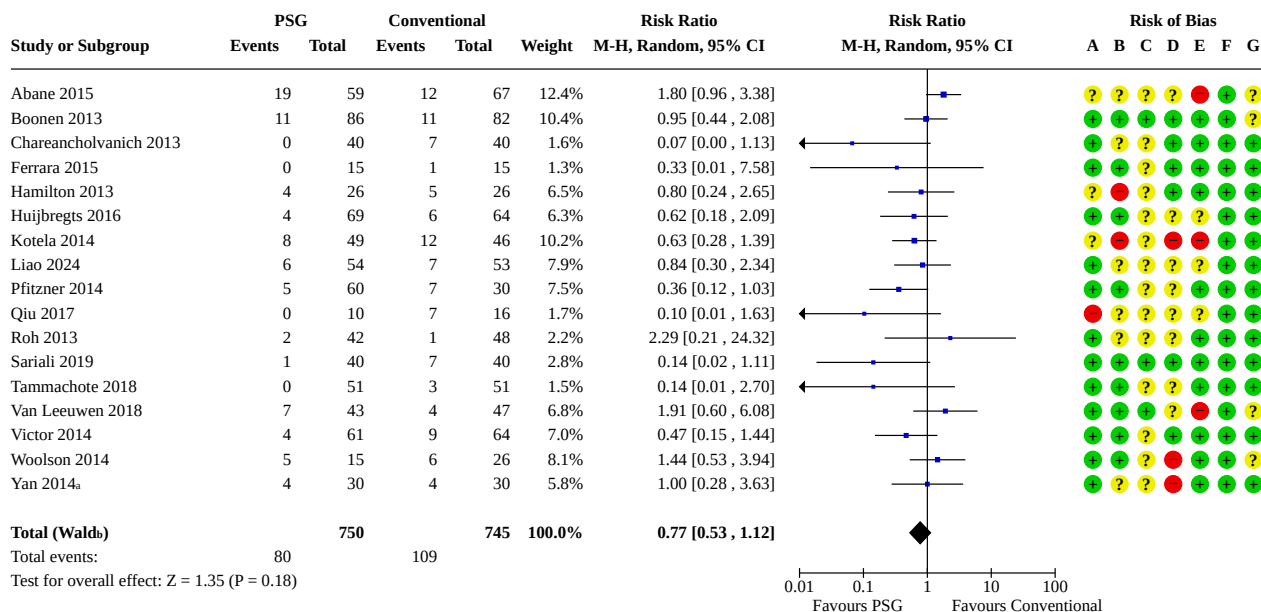
(E) Incomplete outcome data (attrition bias)

(F) Selective reporting (reporting bias)

(G) Other bias

Analysis 1.15. Comparison 1: Patient-specific cutting guides versus conventional instrumentation, Outcome 15: Incision length**Risk of bias legend**

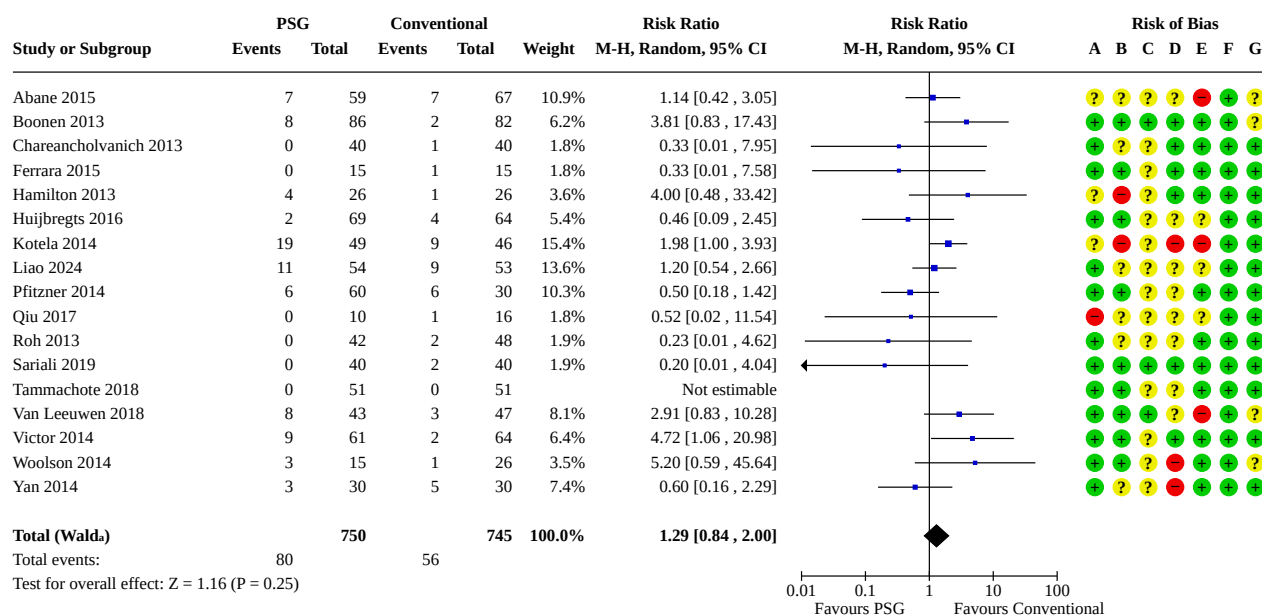
- (A) Random sequence generation (selection bias)
(B) Allocation concealment (selection bias)
(C) Blinding of participants and personnel (performance bias)
(D) Blinding of outcome assessment (detection bias): Functional outcomes
(E) Incomplete outcome data (attrition bias)
(F) Selective reporting (reporting bias)
(G) Other bias

Analysis 1.16. Comparison 1: Patient-specific cutting guides versus conventional instrumentation, Outcome 16: Radiographic outliers in femur coronal alignment**Footnotes**

- a90 knees 81 patients
bCI calculated by Wald-type method.
cTau² calculated by DerSimonian and Laird method.

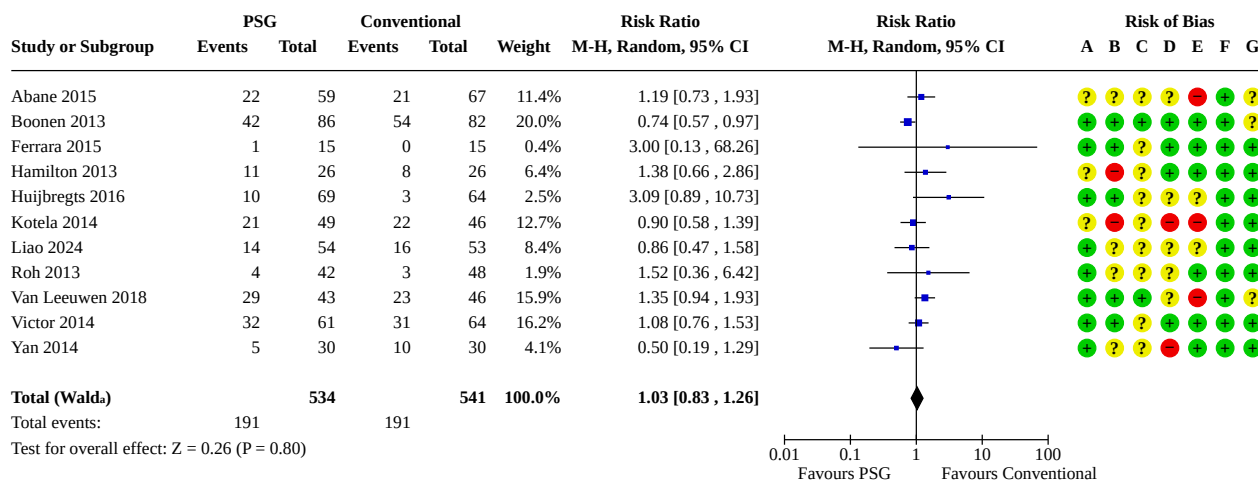
Risk of bias legend

- (A) Random sequence generation (selection bias)
(B) Allocation concealment (selection bias)
(C) Blinding of participants and personnel (performance bias)
(D) Blinding of outcome assessment (detection bias): Functional outcomes
(E) Incomplete outcome data (attrition bias)
(F) Selective reporting (reporting bias)
(G) Other bias

Analysis 1.17. Comparison 1: Patient-specific cutting guides versus conventional instrumentation, Outcome 17: Radiographic outliers in tibia coronal alignment**Footnotes**^aCI calculated by Wald-type method.^bTau² calculated by DerSimonian and Laird method.**Risk of bias legend**

- (A) Random sequence generation (selection bias)
- (B) Allocation concealment (selection bias)
- (C) Blinding of participants and personnel (performance bias)
- (D) Blinding of outcome assessment (detection bias): Functional outcomes
- (E) Incomplete outcome data (attrition bias)
- (F) Selective reporting (reporting bias)
- (G) Other bias

Analysis 1.18. Comparison 1: Patient-specific cutting guides versus conventional instrumentation, Outcome 18: Radiographic outliers in femur sagittal alignment



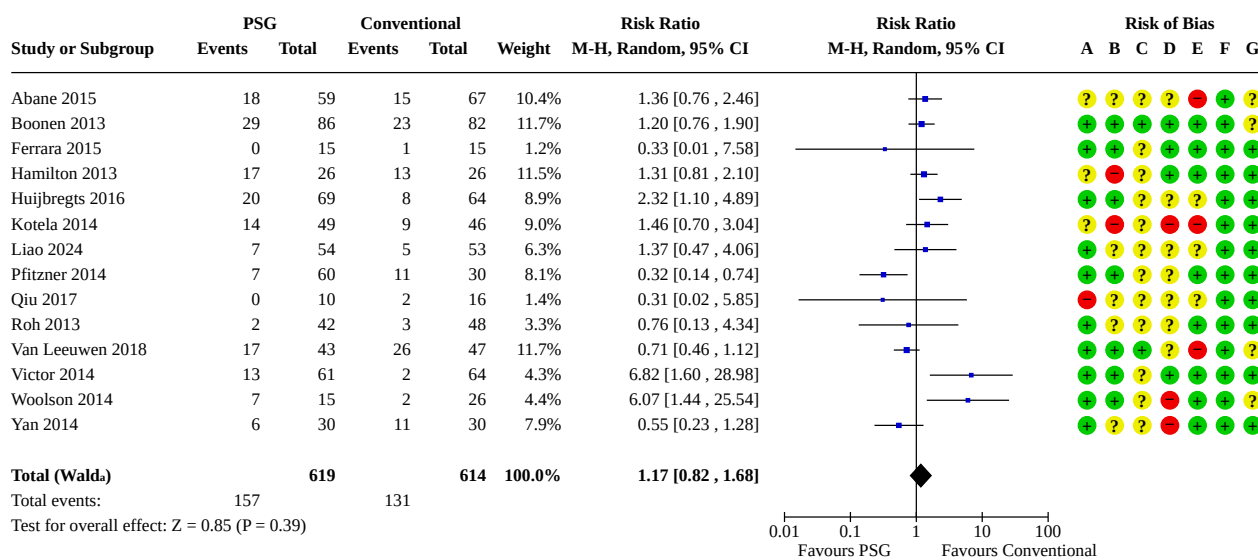
Footnotes

^aCI calculated by Wald-type method.

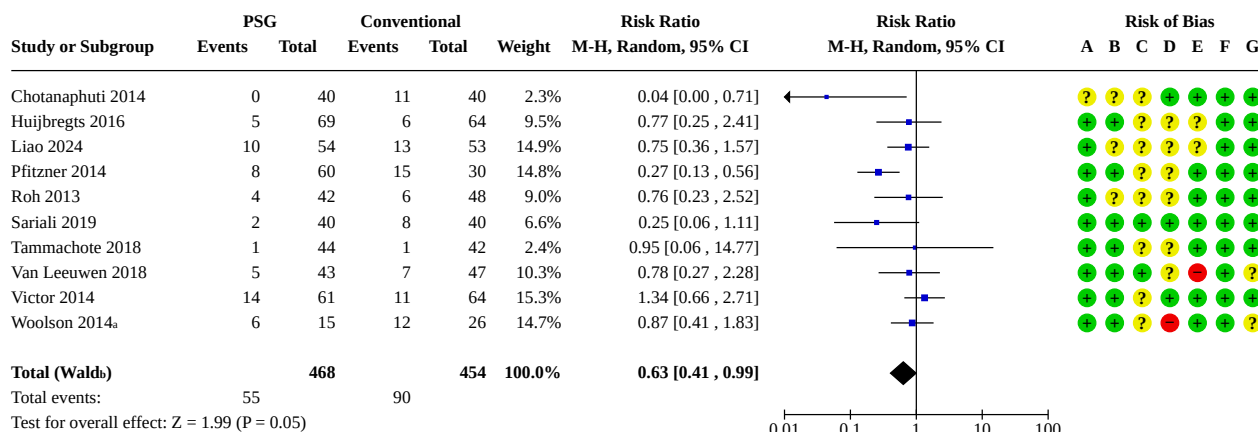
^bTau² calculated by DerSimonian and Laird method.

Risk of bias legend

- (A) Random sequence generation (selection bias)
- (B) Allocation concealment (selection bias)
- (C) Blinding of participants and personnel (performance bias)
- (D) Blinding of outcome assessment (detection bias): Functional outcomes
- (E) Incomplete outcome data (attrition bias)
- (F) Selective reporting (reporting bias)
- (G) Other bias

Analysis 1.19. Comparison 1: Patient-specific cutting guides versus conventional instrumentation, Outcome 19: Radiographic outliers in tibia sagittal alignment**Footnotes**^aCI calculated by Wald-type method.^bTau² calculated by DerSimonian and Laird method.**Risk of bias legend**

- (A) Random sequence generation (selection bias)
- (B) Allocation concealment (selection bias)
- (C) Blinding of participants and personnel (performance bias)
- (D) Blinding of outcome assessment (detection bias): Functional outcomes
- (E) Incomplete outcome data (attrition bias)
- (F) Selective reporting (reporting bias)
- (G) Other bias

Analysis 1.20. Comparison 1: Patient-specific cutting guides versus conventional instrumentation, Outcome 20: Radiographic outliers in femur rotation**Footnotes**^att n= 22 or n= 30^bCI calculated by Wald-type method.^cTau² calculated by DerSimonian and Laird method.**Risk of bias legend**

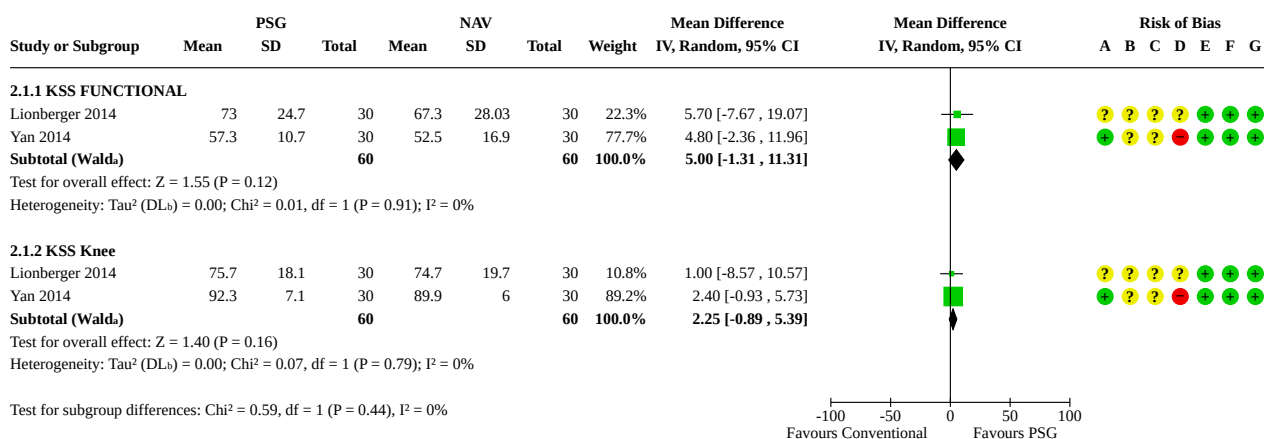
- (A) Random sequence generation (selection bias)
- (B) Allocation concealment (selection bias)
- (C) Blinding of participants and personnel (performance bias)
- (D) Blinding of outcome assessment (detection bias): Functional outcomes
- (E) Incomplete outcome data (attrition bias)
- (F) Selective reporting (reporting bias)
- (G) Other bias

Comparison 2. Patient-specific cutting guides versus computer-assisted navigation

Outcome or subgroup title	No. of studies	No. of participants	Statistical method	Effect size
2.1 Functional outcome measure	2		Mean Difference (IV, Random, 95% CI)	Subtotals only
2.1.1 KSS FUNCTIONAL	2	120	Mean Difference (IV, Random, 95% CI)	5.00 [-1.31, 11.31]
2.1.2 KSS Knee	2	120	Mean Difference (IV, Random, 95% CI)	2.25 [-0.89, 5.39]
2.2 Radiographic lower limb alignment measured by outliers in FTCA	3	270	Risk Ratio (M-H, Random, 95% CI)	2.04 [0.87, 4.82]
2.3 Operative time	2	120	Mean Difference (IV, Random, 95% CI)	-21.54 [-27.93, -15.16]
2.4 Range of motion	2	120	Mean Difference (IV, Fixed, 95% CI)	0.83 [-4.10, 5.75]

Outcome or subgroup title	No. of studies	No. of participants	Statistical method	Effect size
2.5 Range of motion flexion	2	120	Mean Difference (IV, Random, 95% CI)	1.12 [-5.05, 7.28]
2.6 Incision length	1	60	Mean Difference (IV, Fixed, 95% CI)	Not estimable
2.7 Radiographic outliers in femur coronal alignment	2	133	Risk Ratio (M-H, Random, 95% CI)	1.09 [0.30, 3.96]
2.8 Radiographic outliers in tibia coronal alignment	2	133	Risk Ratio (M-H, Random, 95% CI)	1.79 [0.42, 7.55]
2.9 Radiographic outliers in femur sagittal alignment	2	133	Risk Ratio (M-H, Random, 95% CI)	3.19 [0.57, 17.77]
2.10 Radiographic outliers in tibia sagittal alignment	2	133	Risk Ratio (M-H, Random, 95% CI)	2.96 [0.90, 9.70]

Analysis 2.1. Comparison 2: Patient-specific cutting guides versus computer-assisted navigation, Outcome 1: Functional outcome measure



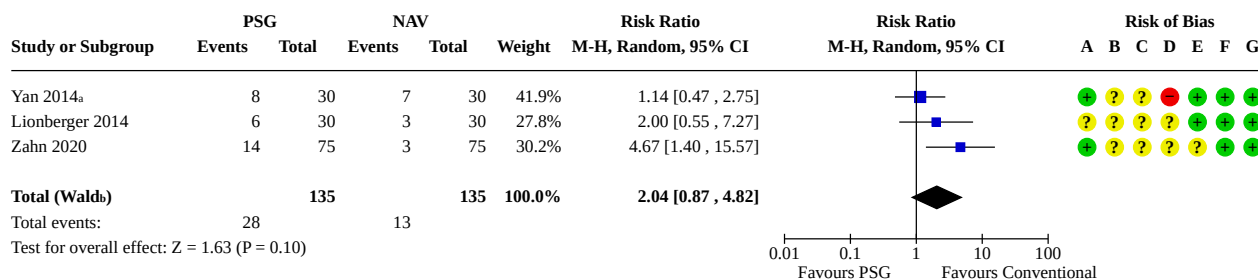
Footnotes

aCI calculated by Wald-type method.

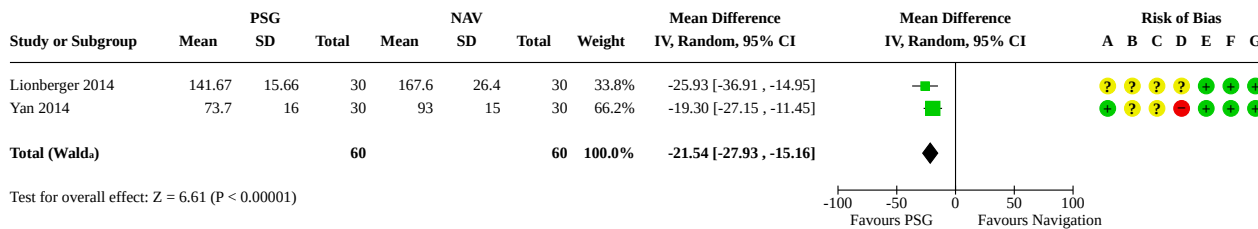
bTau² calculated by DerSimonian and Laird method.

Risk of bias legend

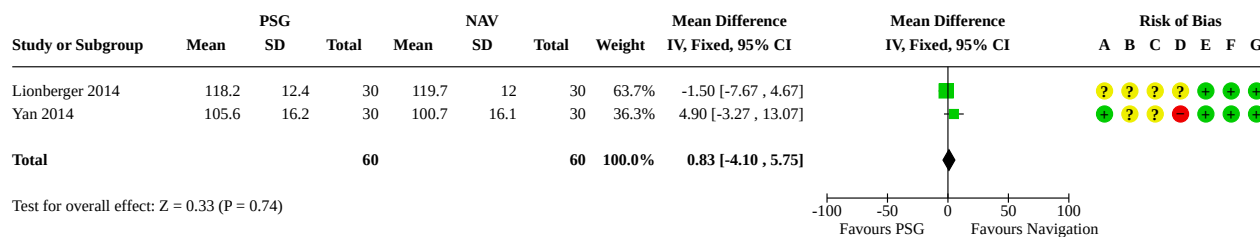
- (A) Random sequence generation (selection bias)
- (B) Allocation concealment (selection bias)
- (C) Blinding of participants and personnel (performance bias)
- (D) Blinding of outcome assessment (detection bias): Functional outcomes
- (E) Incomplete outcome data (attrition bias)
- (F) Selective reporting (reporting bias)
- (G) Other bias

Analysis 2.2. Comparison 2: Patient-specific cutting guides versus computer-assisted navigation, Outcome 2: Radiographic lower limb alignment measured by outliers in FTCA**Footnotes**^a81 patients 90 knees^bCI calculated by Wald-type method.^c Tau^2 calculated by DerSimonian and Laird method.**Risk of bias legend**

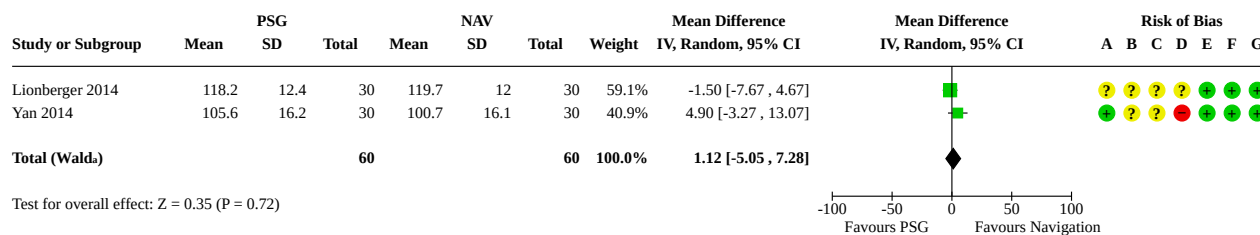
- (A) Random sequence generation (selection bias)
- (B) Allocation concealment (selection bias)
- (C) Blinding of participants and personnel (performance bias)
- (D) Blinding of outcome assessment (detection bias): Functional outcomes
- (E) Incomplete outcome data (attrition bias)
- (F) Selective reporting (reporting bias)
- (G) Other bias

Analysis 2.3. Comparison 2: Patient-specific cutting guides versus computer-assisted navigation, Outcome 3: Operative time**Footnotes**^aCI calculated by Wald-type method.^b Tau^2 calculated by DerSimonian and Laird method.**Risk of bias legend**

- (A) Random sequence generation (selection bias)
- (B) Allocation concealment (selection bias)
- (C) Blinding of participants and personnel (performance bias)
- (D) Blinding of outcome assessment (detection bias): Functional outcomes
- (E) Incomplete outcome data (attrition bias)
- (F) Selective reporting (reporting bias)
- (G) Other bias

**Analysis 2.4. Comparison 2: Patient-specific cutting guides
versus computer-assisted navigation, Outcome 4: Range of motion****Risk of bias legend**

- (A) Random sequence generation (selection bias)
- (B) Allocation concealment (selection bias)
- (C) Blinding of participants and personnel (performance bias)
- (D) Blinding of outcome assessment (detection bias): Functional outcomes
- (E) Incomplete outcome data (attrition bias)
- (F) Selective reporting (reporting bias)
- (G) Other bias

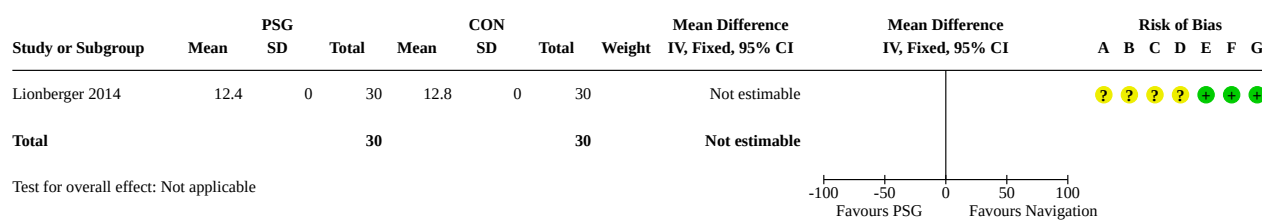
**Analysis 2.5. Comparison 2: Patient-specific cutting guides versus
computer-assisted navigation, Outcome 5: Range of motion flexion****Footnotes**

^aCI calculated by Wald-type method.

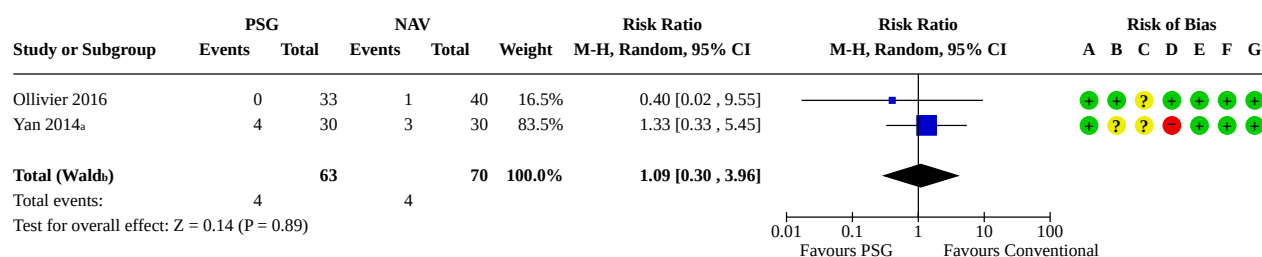
^b τ^2 calculated by DerSimonian and Laird method.

Risk of bias legend

- (A) Random sequence generation (selection bias)
- (B) Allocation concealment (selection bias)
- (C) Blinding of participants and personnel (performance bias)
- (D) Blinding of outcome assessment (detection bias): Functional outcomes
- (E) Incomplete outcome data (attrition bias)
- (F) Selective reporting (reporting bias)
- (G) Other bias

Analysis 2.6. Comparison 2: Patient-specific cutting guides versus computer-assisted navigation, Outcome 6: Incision length**Risk of bias legend**

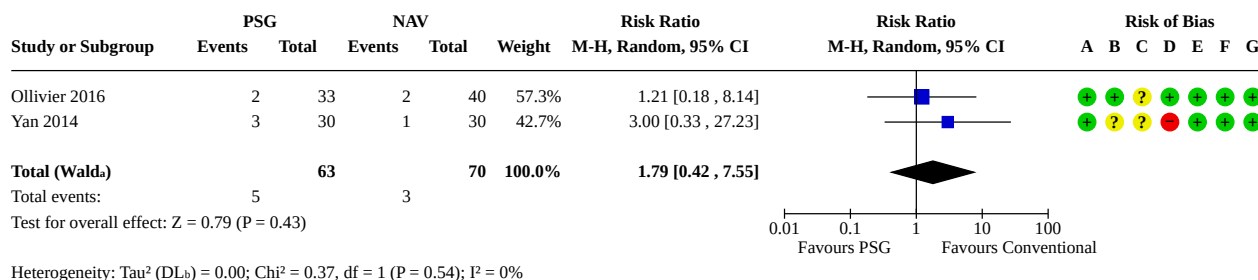
- (A) Random sequence generation (selection bias)
- (B) Allocation concealment (selection bias)
- (C) Blinding of participants and personnel (performance bias)
- (D) Blinding of outcome assessment (detection bias): Functional outcomes
- (E) Incomplete outcome data (attrition bias)
- (F) Selective reporting (reporting bias)
- (G) Other bias

Analysis 2.7. Comparison 2: Patient-specific cutting guides versus computer-assisted navigation, Outcome 7: Radiographic outliers in femur coronal alignment**Footnotes**

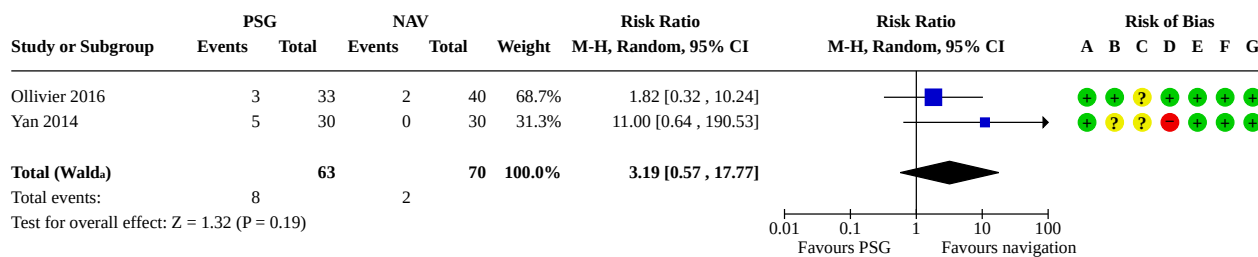
- ^a90 knees 81 patients
- ^bCI calculated by Wald-type method.
- ^cTau² calculated by DerSimonian and Laird method.

Risk of bias legend

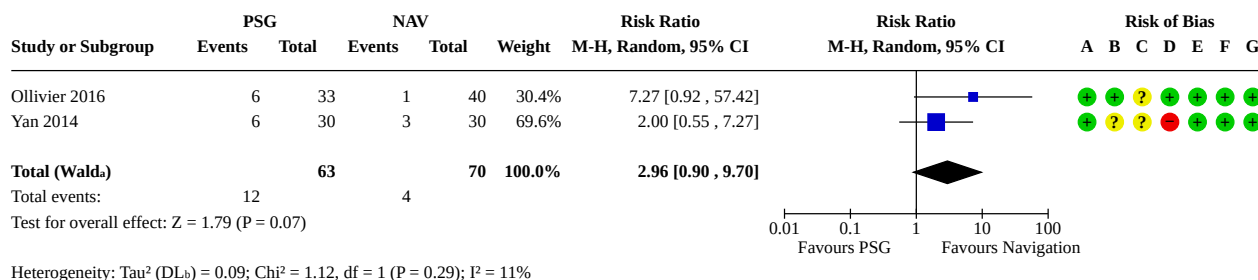
- (A) Random sequence generation (selection bias)
- (B) Allocation concealment (selection bias)
- (C) Blinding of participants and personnel (performance bias)
- (D) Blinding of outcome assessment (detection bias): Functional outcomes
- (E) Incomplete outcome data (attrition bias)
- (F) Selective reporting (reporting bias)
- (G) Other bias

Analysis 2.8. Comparison 2: Patient-specific cutting guides versus computer-assisted navigation, Outcome 8: Radiographic outliers in tibia coronal alignment**Footnotes**^aCI calculated by Wald-type method.^bTau² calculated by DerSimonian and Laird method.**Risk of bias legend**

- (A) Random sequence generation (selection bias)
- (B) Allocation concealment (selection bias)
- (C) Blinding of participants and personnel (performance bias)
- (D) Blinding of outcome assessment (detection bias): Functional outcomes
- (E) Incomplete outcome data (attrition bias)
- (F) Selective reporting (reporting bias)
- (G) Other bias

Analysis 2.9. Comparison 2: Patient-specific cutting guides versus computer-assisted navigation, Outcome 9: Radiographic outliers in femur sagittal alignment**Footnotes**^aCI calculated by Wald-type method.^bTau² calculated by DerSimonian and Laird method.**Risk of bias legend**

- (A) Random sequence generation (selection bias)
- (B) Allocation concealment (selection bias)
- (C) Blinding of participants and personnel (performance bias)
- (D) Blinding of outcome assessment (detection bias): Functional outcomes
- (E) Incomplete outcome data (attrition bias)
- (F) Selective reporting (reporting bias)
- (G) Other bias

Analysis 2.10. Comparison 2: Patient-specific cutting guides versus computer-assisted navigation, Outcome 10: Radiographic outliers in tibia sagittal alignment**Footnotes**^aCI calculated by Wald-type method.^b Tau^2 calculated by DerSimonian and Laird method.**Risk of bias legend**

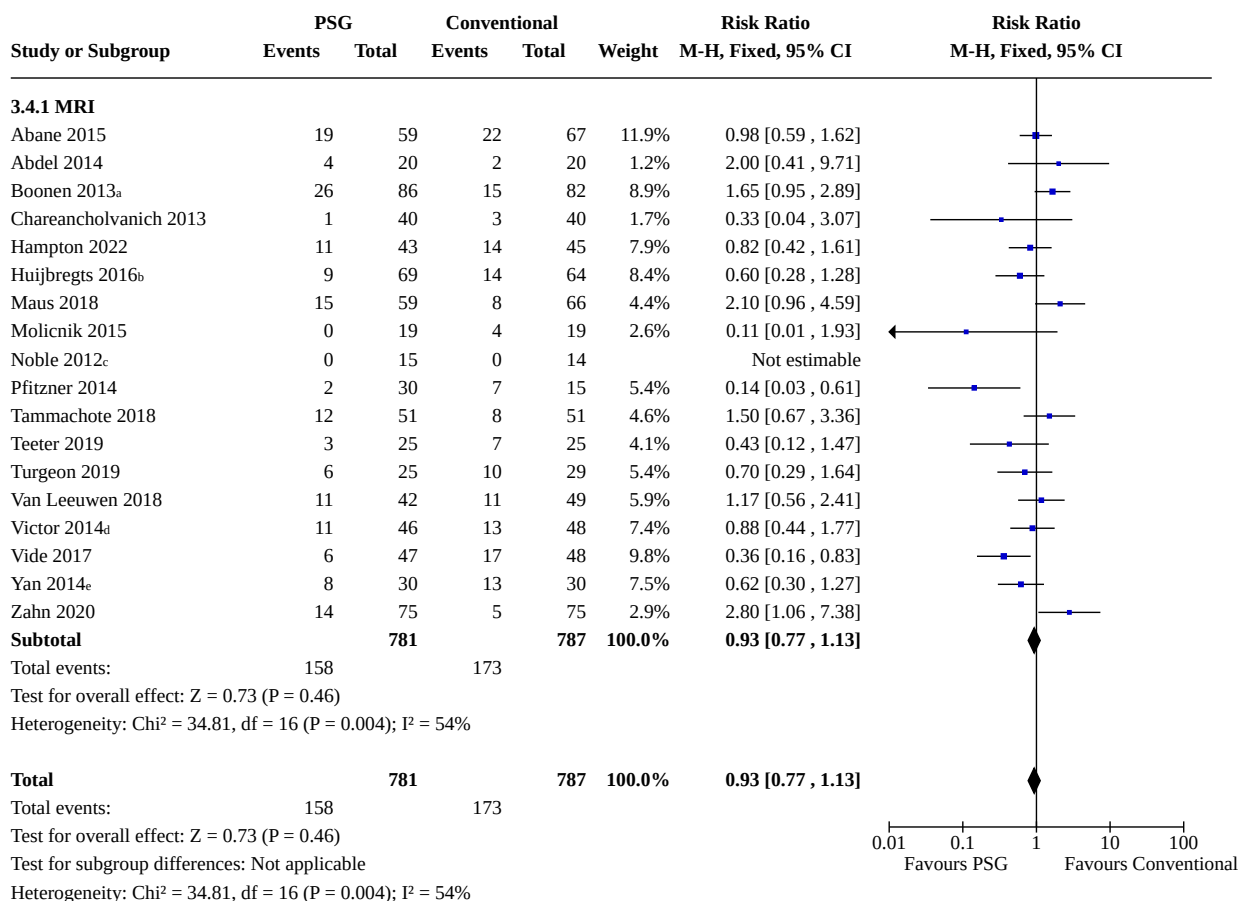
- (A) Random sequence generation (selection bias)
- (B) Allocation concealment (selection bias)
- (C) Blinding of participants and personnel (performance bias)
- (D) Blinding of outcome assessment (detection bias): Functional outcomes
- (E) Incomplete outcome data (attrition bias)
- (F) Selective reporting (reporting bias)
- (G) Other bias

Comparison 3. Subgroup: Patient-specific cutting guides versus conventional instrumentation

Outcome or subgroup title	No. of studies	No. of participants	Statistical method	Effect size
3.4 Alignment measured by outliers in FTCA	18	1568	Risk Ratio (M-H, Fixed, 95% CI)	0.93 [0.77, 1.13]
3.4.1 MRI	18	1568	Risk Ratio (M-H, Fixed, 95% CI)	0.93 [0.77, 1.13]
3.5 Outliers in femur coronal alignment	8	460	Risk Ratio (M-H, Fixed, 95% CI)	0.58 [0.37, 0.90]
3.5.1 CT	8	460	Risk Ratio (M-H, Fixed, 95% CI)	0.58 [0.37, 0.90]
3.6 Outliers in tibia coronal alignment	10	928	Risk Ratio (M-H, Fixed, 95% CI)	1.23 [0.78, 1.95]
3.6.1 MRI	10	928	Risk Ratio (M-H, Fixed, 95% CI)	1.23 [0.78, 1.95]
3.7 Outliers in femur sagittal alignment	7	701	Risk Ratio (M-H, Fixed, 95% CI)	1.01 [0.85, 1.20]
3.7.1 MRI	7	701	Risk Ratio (M-H, Fixed, 95% CI)	1.01 [0.85, 1.20]
3.8 Outliers in tibia sagittal alignment	8	746	Risk Ratio (M-H, Fixed, 95% CI)	1.13 [0.90, 1.44]
3.8.1 MRI	8	746	Risk Ratio (M-H, Fixed, 95% CI)	1.13 [0.90, 1.44]
3.9 Outliers in femur rotation	5	448	Risk Ratio (M-H, Fixed, 95% CI)	0.80 [0.51, 1.27]

Outcome or subgroup title	No. of studies	No. of participants	Statistical method	Effect size
3.9.1 MRI	5	448	Risk Ratio (M-H, Fixed, 95% CI)	0.80 [0.51, 1.27]

Analysis 3.4. Comparison 3: Subgroup: Patient-specific cutting guides versus conventional instrumentation, Outcome 4: Alignment measured by outliers in FTCA



Footnotes

^aitt 86.82

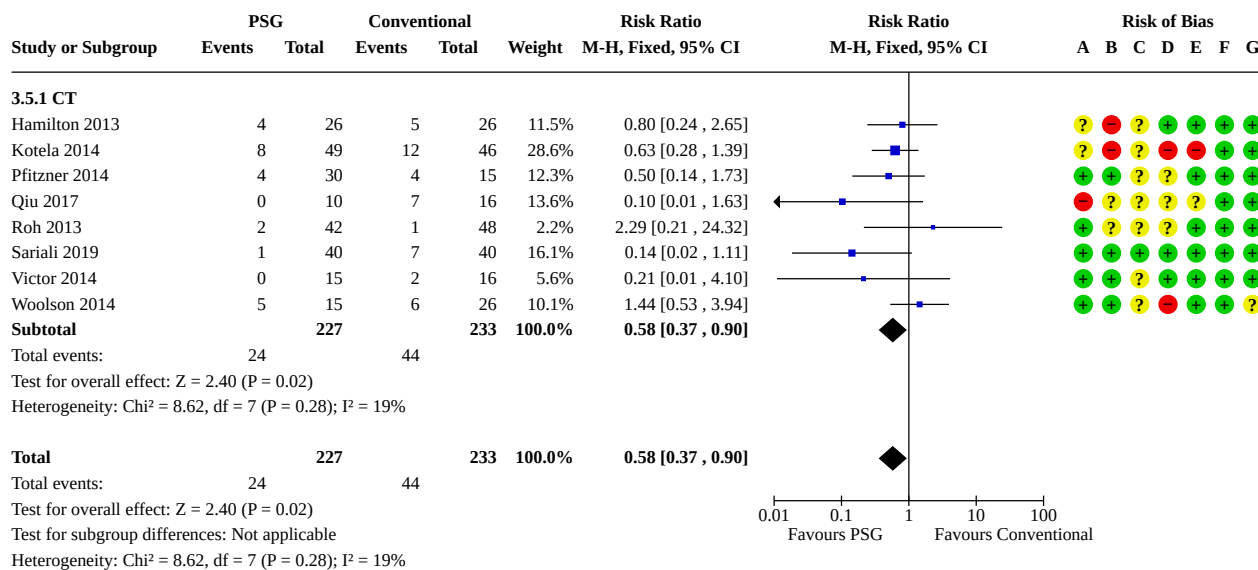
^bitt 69, 64

^citt 15, 14

^ditt 61, 64

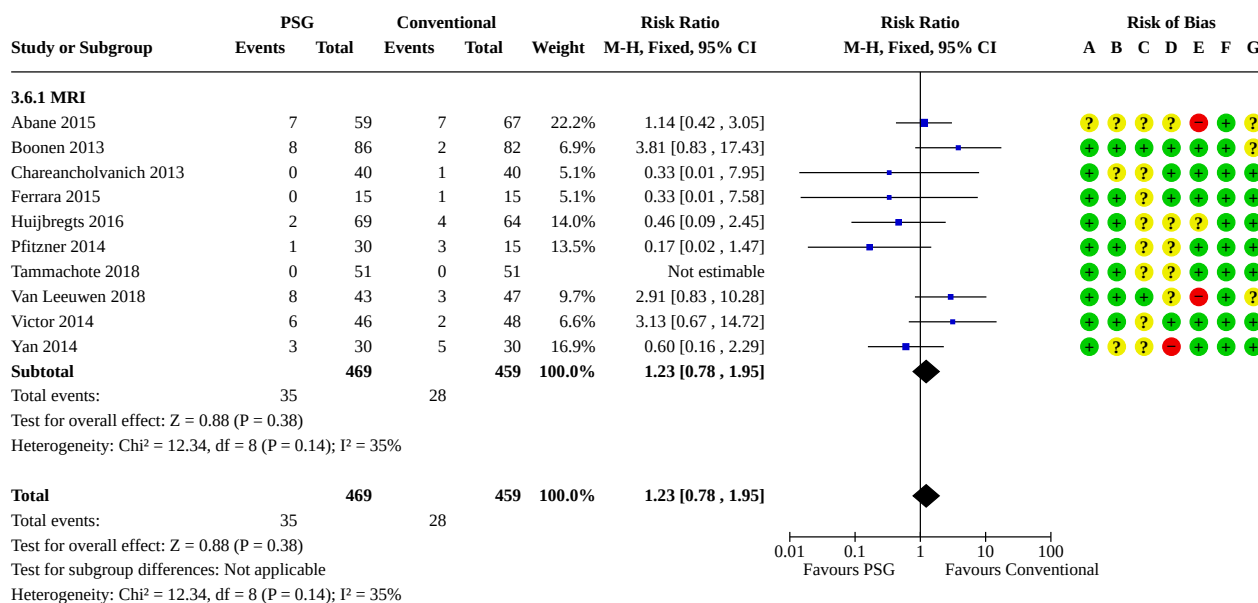
^e81 patients 90 knees

Analysis 3.5. Comparison 3: Subgroup: Patient-specific cutting guides versus conventional instrumentation, Outcome 5: Outliers in femur coronal alignment



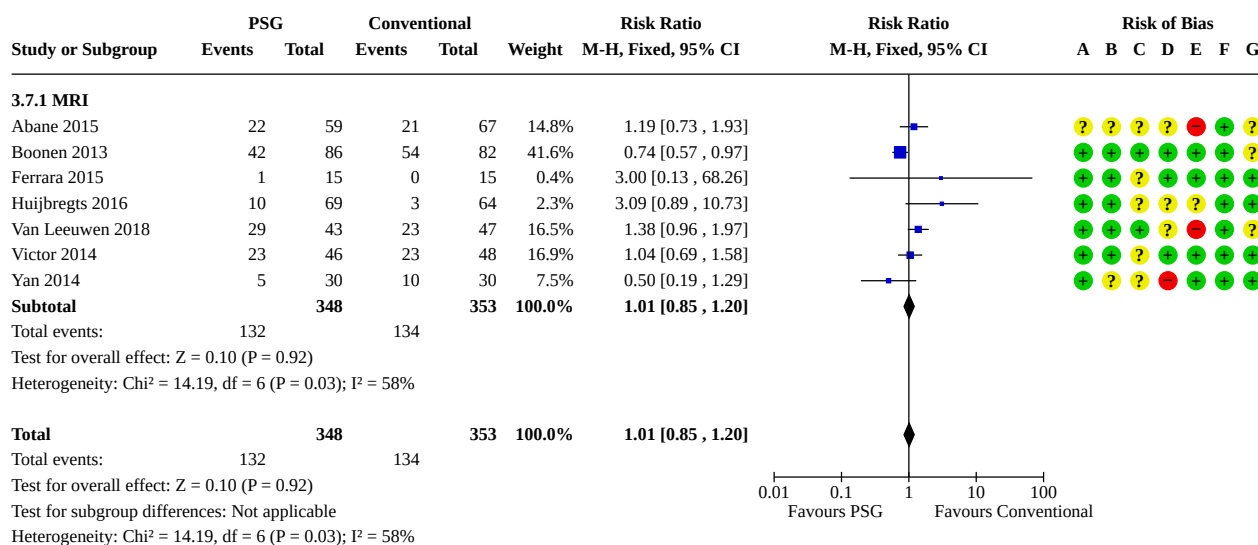
Risk of bias legend

- (A) Random sequence generation (selection bias)
- (B) Allocation concealment (selection bias)
- (C) Blinding of participants and personnel (performance bias)
- (D) Blinding of outcome assessment (detection bias): Functional outcomes
- (E) Incomplete outcome data (attrition bias)
- (F) Selective reporting (reporting bias)
- (G) Other bias

Analysis 3.6. Comparison 3: Subgroup: Patient-specific cutting guides versus conventional instrumentation, Outcome 6: Outliers in tibia coronal alignment**Risk of bias legend**

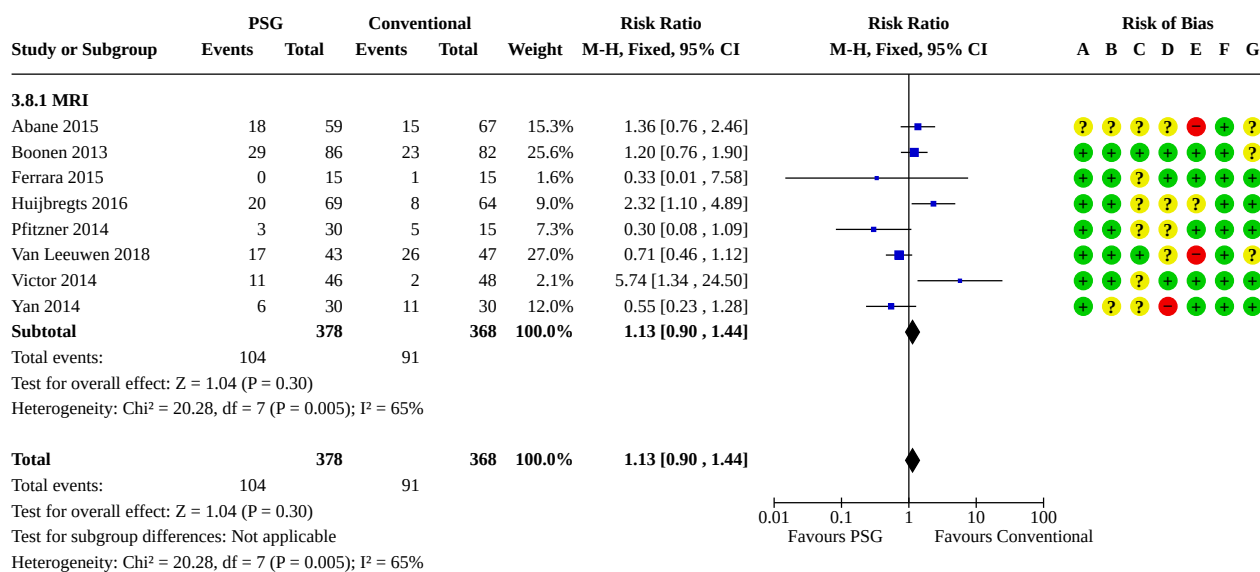
- (A) Random sequence generation (selection bias)
- (B) Allocation concealment (selection bias)
- (C) Blinding of participants and personnel (performance bias)
- (D) Blinding of outcome assessment (detection bias): Functional outcomes
- (E) Incomplete outcome data (attrition bias)
- (F) Selective reporting (reporting bias)
- (G) Other bias

Analysis 3.7. Comparison 3: Subgroup: Patient-specific cutting guides versus conventional instrumentation, Outcome 7: Outliers in femur sagittal alignment

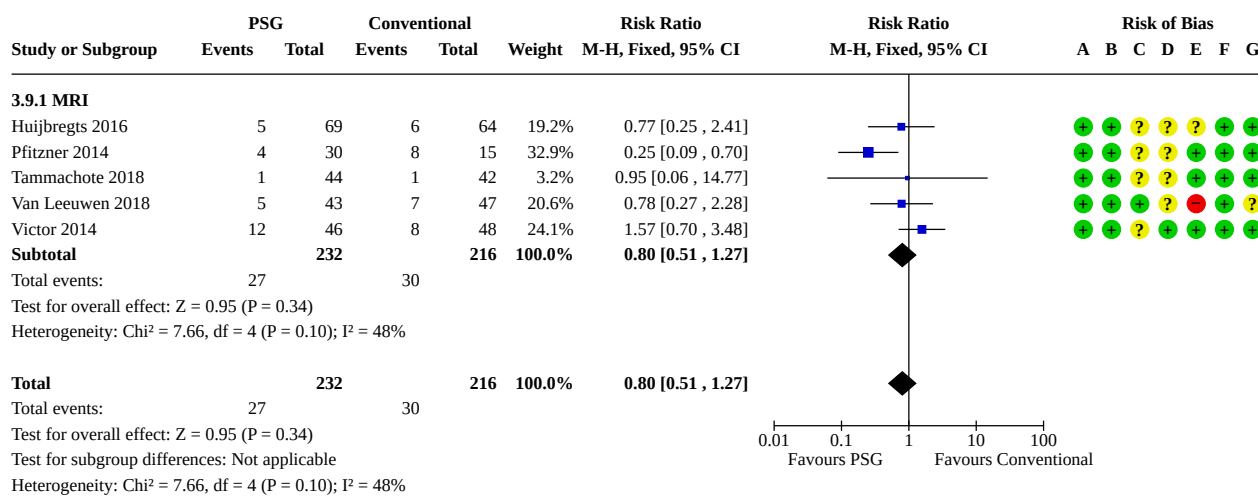


Risk of bias legend

- (A) Random sequence generation (selection bias)
- (B) Allocation concealment (selection bias)
- (C) Blinding of participants and personnel (performance bias)
- (D) Blinding of outcome assessment (detection bias): Functional outcomes
- (E) Incomplete outcome data (attrition bias)
- (F) Selective reporting (reporting bias)
- (G) Other bias

Analysis 3.8. Comparison 3: Subgroup: Patient-specific cutting guides versus conventional instrumentation, Outcome 8: Outliers in tibia sagittal alignment**Risk of bias legend**

- (A) Random sequence generation (selection bias)
- (B) Allocation concealment (selection bias)
- (C) Blinding of participants and personnel (performance bias)
- (D) Blinding of outcome assessment (detection bias): Functional outcomes
- (E) Incomplete outcome data (attrition bias)
- (F) Selective reporting (reporting bias)
- (G) Other bias

Analysis 3.9. Comparison 3: Subgroup: Patient-specific cutting guides versus conventional instrumentation, Outcome 9: Outliers in femur rotation**Risk of bias legend**

- (A) Random sequence generation (selection bias)
(B) Allocation concealment (selection bias)
(C) Blinding of participants and personnel (performance bias)
(D) Blinding of outcome assessment (detection bias): Functional outcomes
(E) Incomplete outcome data (attrition bias)
(F) Selective reporting (reporting bias)
(G) Other bias

ADDITIONAL TABLES**Table 1. Patient-specific cutting guides compared to conventional instrumentation – radiographic outliers**

Patient-specific cutting guides compared with conventional instrumentation for total knee arthroplasty						
Patient or population: Patients undergoing total knee arthroplasty						
Settings: Hospitals						
Intervention: Patient-specific cutting guides						
Comparison: Conventional instrumentation						
Outcomes	Anticipated absolute effects* (95% CI)		Relative effect (95% CI)	No of participants (studies)	Certainty of the evidence (GRADE)	Comments
	Conventional instrumentation	Patient-specific cutting guides				
Alignment measured by outliers in FTCA	22.0 %	20.4 % (15.6% to 25.1%)	RR 0.9 (0.71 to 1.14)	2195 (26 RCTs)	⊕⊕⊕⊕ MODERATE 1	PSGs probably do not improve precision in lower limb alignment. Relative change 10% less (29% less to 14% more)
Outliers in femur coro-	14.7%	10.6%	RR 0.76	1388 (16 RCTs)	⊕⊕⊕⊕	PSGs result in little to no difference in the precision of femoral coronal alignment.

Table 1. Patient-specific cutting guides compared to conventional instrumentation – radiographic outliers (Continued)

nal align- ment	(7.4% to 16.7%)	(0.50 to 1.13)			MODERATE 1	Relative change 24% less (50% less to 13% more)
Outliers in tibia coro- nal align- ment	6.8%	8.8% (5.4% to 14.5%)	RR 1.3 (0.79 to 2.14)	1388 (16 RCTs)	⊕⊕⊕⊖ MODERATE 1	PSGs result in little to no difference in the precision of tibial coronal alignment. Relative change 30% more (21% less to 114% more)
Outliers in femur sagit- tal align- ment	33.1%	36.9% (28.6% to 47.6%)	RR 1.06 (0.84 to 1.33)	1049 (11 RCTs)	⊕⊕⊕⊖ MODERATE 1	PSGs result in little to no difference in the precision of femoral sagittal alignment. Relative change 6% more (16% less to 33% more)
Outliers in tibia sagit- tal align- ment	22.5%	26.5% (18.4% to 40.2%)	RR 1.21 (0.82 to 1.79)	1388 (16 RCTs)	⊕⊕⊕⊖ MODERATE 1	PSGs result in little to no difference in the precision of tibial sagittal alignment. Relative change 21% more (18% less to 79% more)
Outliers in femur rota- tion	19.2%	10.9% (6.9% to 19.8%)	RR 0.61 (0.36 to 1.03)	815 (9 RCTs)	⊕⊕⊖⊖ LOW ^{1 2}	PSGs are likely to result in improved pre- cision of femoral rotational alignment. Relative change 39% less (64% less to 3% more)

*The risk in the intervention group (and its 95% confidence interval) is based on the assumed risk in the comparison group and the relative effect of the intervention (and its 95% CI).

CI: Confidence interval; RR: Risk ratio

GRADE Working Group grades of evidence

High quality: Further research is very unlikely to change our confidence in the estimate of effect.

Moderate quality: Further research is likely to have an important impact on our confidence in the estimate of effect and may change the estimate.

Low quality: Further research is very likely to have an important impact on our confidence in the estimate of effect and is likely to change the estimate.

Very low quality: We are very uncertain about the estimate.

¹ Downgraded due to lack of participant blinding and high risk of detection/performance bias

² Downgraded due to imprecision; the confidence interval crosses the line of no difference and does not rule out a small increase

Table 2. Patient-specific cutting guides compared to computer-assisted navigation – radiographic outliers

Patient-specific cutting guides compared with computer-assisted navigation					
Patient or population: Patients undergoing total knee arthroplasty					
Settings: Hospitals					
Intervention: Patient-specific cutting guides					
Comparison: Computer-assisted navigation					
Outcomes	Anticipated absolute ef- fects* (95% CI)	Relative effect (95% CI)	No of par- ticipants (studies)	Certain- ty of the evidence (GRADE)	Comments

Patient-specific cutting guides for total knee arthroplasty (Review)

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Table 2. Patient-specific cutting guides compared to computer-assisted navigation – radiographic outliers (Continued)

	Conven- tional in- strumen- tation	Pa- tient-spe- cific cut- ting guides				
Alignment measured by outliers in FT-CA	9.6%	20.7% (8.4% to 46.4%)	RR 2.04 (0.87 to 4.82)	270 (3 RCTs)	⊕⊕⊕⊕ VERY LOW 1 2 3	PSGs are likely to result in worse precision in lower limb alignment. Relative change 104% more (13% less to 382% more)
Outliers in femur coronal alignment	5.7%	6.3% (1.7% to 22.6%)	RR 1.09 (0.30 to 3.96)	133 (2 RCTs)	⊕⊕⊕⊕ VERY LOW 1 4 5	PSGs result in little to no difference in the precision of femoral coronal alignment. Relative change 9% more (70% less to 296% more)
Outliers in tibia coronal alignment	4.3%	7.9% (1.8% to 32.3%)	RR 1.79 (0.42 to 7.55)	133 (2 RCTs)	⊕⊕⊕⊕ VERY LOW 1 4 5	PSGs probably result in little to no difference in the precision of tibial coronal alignment. Relative change 79% more (58% less to 655 % more)
Outliers in femur sagittal alignment	2.9%	12.7% (1.6% to 50.8%)	RR 3.19 (0.57 to 17.77)	133 (2 RCTs)	⊕⊕⊕⊕ VERY LOW 1 3 4	PSGs probably result in decreased precision of femoral sagittal alignment. Relative change 219% more (43% less to 1677% more)
Outliers in tibia sagittal alignment	5.7%	16.9% (5.1% to 55.4%)	RR 2.96 (0.9 to 9.7)	133 (2 RCTs)	⊕⊕⊕⊕ VERY LOW 1 3 4	PSGs probably result in decreased precision of tibial sagittal alignment. Relative change 196% more (10% less to 870% more)
Outliers in femur rotation	Insufficient data for analysis					

*The risk in the intervention group (and its 95% confidence interval) is based on the assumed risk in the comparison group and the relative effect of the intervention (and its 95% CI).

CI: Confidence interval; RR: Risk ratio

GRADE Working Group grades of evidence

High quality: Further research is very unlikely to change our confidence in the estimate of effect.

Moderate quality: Further research is likely to have an important impact on our confidence in the estimate of effect and may change the estimate.

Low quality: Further research is very likely to have an important impact on our confidence in the estimate of effect and is likely to change the estimate.

Very low quality: We are very uncertain about the estimate.

¹ Downgraded due to lack of participant blinding and high risk of detection/performance bias

² Downgraded due to serious indirectness; of the 41 studies of this comparison, only three studies provided outlier data for lower limb alignment.

³ Downgraded due to serious imprecision; wide CI with few events. The confidence interval crosses the line of no difference and does not rule out a small decrease.

⁴ Downgraded due to serious indirectness; of the 41 studies of this comparison, only two studies provided outlier data for individual component alignment.

⁵Downgraded due to serious imprecision; wide CI with few events

APPENDICES

Appendix 1. CENTRAL search strategy

Database: EBM Reviews - Cochrane Central Register of Controlled Trials <December 2024>

Search Strategy:

-
- 1 osteoarthritis/
 - 2 osteoarthrit\$.tw.
 - 3 osteoarthro\$.tw.
 - 4 oa.tw.
 - 5 degenerative joint disease.tw.
 - 6 degenerative arthritis.tw.
 - 7 djd.tw.
 - 8 gonarthro\$.tw.
 - 9 or/1-8
 - 10 exp KNEE/
 - 11 knee\$.tw.
 - 12 femorotibia\$.tw.
 - 13 or/10-12
 - 14 9 and 13
 - 15 Knee Osteoarthritis/
 - 16 14 or 15
 - 17 exp Arthroplasty, Replacement, Knee/
 - 18 (knee\$ adj4 (replace\$ or arthroplast\$ or prosthesis\$ or endoprosthesis\$ or implant\$)).tw.
 - 19 (tka or tkr or ukr).tw.
 - 20 Knee Prosthesis/
 - 21 exp Arthroplasty, Replacement/ or exp Arthroplasty/
 - 22 Knee Prosthesis/ or "Prostheses and Implants"/
 - 23 or/17-22
 - 24 16 and 23
 - 25 limit 24 to yr="2010 -Current"

Appendix 2. MEDLINE (Ovid) search strategy

Database: Ovid MEDLINE(R) <1946 to January 21, 2025>

Search Strategy:

Patient-specific cutting guides for total knee arthroplasty (Review)

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- 1 osteoarthritis/
- 2 osteoarthritis.tw.
- 3 osteoarthritis.tw.
- 4 oa.tw.
- 5 degenerative joint disease.tw.
- 6 degenerative arthritis.tw.
- 7 djd.tw.
- 8 gonarthrosis.tw.
- 9 or/1-8
- 10 exp KNEE/
- 11 knee\$.tw.
- 12 femorotibia\$.tw.
- 13 or/10-12
- 14 9 and 13
- 15 Knee Osteoarthritis/
- 16 14 or 15
- 17 exp Arthroplasty, Replacement, Knee/
- 18 (knee\$ adj4 (replace\$ or arthroplast\$ or prosthesis\$ or endoprosthesis\$ or implant\$)).tw.
- 19 (tka or tkr or ukr).tw.
- 20 Knee Prosthesis/
- 21 exp Arthroplasty, Replacement/ or exp Arthroplasty/
- 22 Knee Prosthesis/ or "Prostheses and Implants"/
- 23 or/17-22
- 24 randomized controlled trial.pt.
- 25 controlled clinical trial.pt.
- 26 randomized.ab.
- 27 placebo.ab.
- 28 clinical trials as topic.sh.
- 29 randomly.ab.
- 30 trial.ti.
- 31 or/24-30
- 32 exp animals/ not humans.sh.
- 33 31 not 32
- 34 16 and 23 and 33
- 35 limit 34 to yr="2010 -Current"

Appendix 3. Embase search strategy

Database: Embase Classic+Embase <1947 to 2025 January 31>

Search Strategy:

1 osteoarthritis/

2 osteoarthritis.tw.

3 osteoarthro\$.tw.

4 oa.tw.

5 degenerative joint disease.tw.

6 degenerative arthritis.tw.

7 djd.tw.

8 gonarthro\$.tw.

9 or/1-8

10 exp KNEE/

11 knee\$.tw.

12 femorotibia\$.tw.

13 or/10-12

14 9 and 13

15 Knee Osteoarthritis/

16 14 or 15

17 exp Arthroplasty/

18 exp Joint Prosthesis/

19 exp "Prostheses and Implants"/

20 exp Arthroplasty, Replacement, Knee/

21 exp Knee Prosthesis/

22 exp knee/su

23 (knee\$ adj4 (replace\$ or arthroplast\$ or prosthe\$ or endoprosthe\$ or implant\$)).tw.

24 (tka or tkr or ukr).tw.

25 or/17-24

26 Randomized controlled trial/

27 Controlled clinical study/

28 random\$.ti,ab.

29 randomization/

30 intermethod comparison/

31 placebo.ti,ab.

32 (compare or compared or comparison).ti.

- 33 ((evaluated or evaluate or evaluating or assessed or assess) and (compare or compared or comparing or comparison)).ab.
- 34 (open adj label).ti,ab.
- 35 ((double or single or doubly or singly) adj (blind or blinded or blindly)).ti,ab.
- 36 double blind procedure/
- 37 parallel group\$1.ti,ab.
- 38 (crossover or cross over).ti,ab.
- 39 ((assign\$ or match or matched or allocation) adj5 (alternate or group\$1 or intervention\$1 or patient\$1 or subject\$1 or participant \$1)).ti,ab.
- 40 (assigned or allocated).ti,ab.
- 41 (controlled adj7 (study or design or trial)).ti,ab.
- 42 (volunteer or volunteers).ti,ab.
- 43 human experiment/
- 44 trial.ti.
- 45 or/26-44
- 46 (random\$ adj sampl\$ adj7 (cross section\$ or questionnaire\$1 or survey\$ or database\$1)).ti,ab. not (comparative study/ or controlled study/ or randomi?ed controlled.ti,ab. or randomly assigned.ti,ab.)
- 47 Cross-sectional study/ not (randomized controlled trial/ or controlled clinical study/ or controlled study/ or randomi?ed controlled.ti,ab. or control group\$1.ti,ab.)
- 48 (((case adj control\$) and random\$) not randomi?ed controlled).ti,ab.
- 49 (Systematic review not (trial or study)).ti.
- 50 (nonrandom\$ not random\$).ti,ab.
- 51 Random field\$.ti,ab.
- 52 (random cluster adj3 sampl\$).ti,ab.
- 53 (review.ab. and review.pt.) not trial.ti.
- 54 we searched.ab. and (review.ti. or review.pt.)
- 55 update review.ab.
- 56 (databases adj4 searched).ab.
- 57 (rat or rats or mouse or mice or swine or porcine or murine or sheep or lambs or pigs or piglets or rabbit or rabbits or cat or cats or dog or dogs or cattle or bovine or monkey or monkeys or trout or marmoset\$1).ti. and animal experiment/
- 58 Animal experiment/ not (human experiment/ or human/)
- 59 or/46-58
- 60 45 not 59
- 61 16 and 25 and 60
- 62 limit 61 to yr="2010 -Current"

HISTORY

Protocol first published: Issue 3, 2017

CONTRIBUTIONS OF AUTHORS

KM and SM conceived the idea for this systematic review.

CZ, KM, JJ and SM have co-ordinated the development and writing of this review.

CZ, JJ, VSK and KM performed the search for trials, extracted data from included studies, and performed quality assessment.

CZ and KM contacted the original authors for missing data and data input.

CZ, JJ, KM and VSK screened search results, selected articles for inclusion, assessed trial eligibility and assessed trial quality.

SM resolved disagreements in trial selection and assessed trial quality.

DC independently verified the extracted data and data in Review Manager.

SM and DC provided input and co-ordinated all stages of the review.

IAH supervised all stages of the review and acted as guarantor to ensure the review met the standards for publication.

DECLARATIONS OF INTEREST

Dr Darren Chen declares interests for MicroPort Orthopedics Inc and Stryker Corporation.

None of the other authors have any interests to declare that would be in conflict with this systematic review on patient-specific cutting guides.

SOURCES OF SUPPORT

Internal sources

- None, Other

There are no internal sources of support to declare.

External sources

- None, Other

There are no external sources of support to declare.

DIFFERENCES BETWEEN PROTOCOL AND REVIEW

With regard to radiological outcomes, our protocol had initially committed to assessing the FTCA (FemoroTibial Coronal Angle, i.e. mechanical axis) as the primary outcome and to measuring the outcome as a continuous variable in degrees.

During the analysis, we noted that different authors used different means of expressing the alignment (“x”° of varus/valgus or as a continuous measurement of overall axis 176, 184° etc.) and different target values were used for the component alignment. More clinically meaningful information is the precision in achieving the alignment within $\pm 3^\circ$ of the target value and the number of participants outside this range (outliers). Presenting our findings as outliers for all alignment parameters is more clinically meaningful; hence, the protocol was changed to reflect the same.

Initial protocol:

“Radiological alignment of component: we will accept the femorotibial coronal angle (FTCA) and other radiographic data points measured following surgery (any time within three months). If investigators perform the measurements at more than one time point, we will use the earliest postoperative measurement; we will measure the outcome as a continuous variable in degrees”.

After the change:

“Radiological alignment of component: we will accept the femorotibial coronal angle (FTCA) and other radiographic data points measured following surgery (any time within three months). If investigators perform the measurements at more than one time point, we will use the earliest postoperative measurement. The radiological alignment will be recorded as the number of knees in which the FTCA was within $\pm 3^\circ$ of the neutral/target value (INLIERS) and in the number of knees that it was outside $\pm 3^\circ$ of the neutral/target value (OUTLIERS)”.

At the time of conceptualisation of this review, we planned for a subgroup analysis between CR and PS knees as well as between patellar resurfacing versus non-resurfacing. These subgroup analyses may have been meaningful in studies reporting patient outcome measures, as implant design and patellar resurfacing may be potential confounders affecting functional outcome. Patient's BMI and severity of

preoperative deformity was also postulated to be a possible confounder for producing alignment outliers. We did not perform any of the subgroup analyses due to insufficient data.

For functional outcomes, we performed an additional ad hoc subgroup analysis for the commonly utilised individual patient-reported outcome scores including Knee Society Score (KSS), Oxford Knee Score (OKS), Western Ontario and McMaster University Osteoarthritis Index (WOMAC) and Knee injury and Osteoarthritis Outcome Score (KOOS). We noticed high variability in individual patient reported scores used between studies and, as each individual scoring system assesses different aspects of function, we believe that this subgroup analysis would aid in clinical interpretation of this outcome.